Day 1: Experiments 1 to 5

From fixed rules to machine learning: route client messages, sort client documents and recognise objects live.

THE FLOW

- **Ex 1 Rule Based Process**: keyword rules route client messages to a desk.
- **Ex 2 Text Recognition**: a model learns the desks from labelled examples.
- **Ex 3 Evaluate**: accuracy, precision, recall and where the model goes wrong.
- **Ex 4 Image Recognition**: a model sorts client documents (India, US, China samples).
- **Ex 5 Live Object Detection**: a ready model, then your own, on the webcam.

USE CASE FOR NUVAMA

Client messages and documents reach the custody desks every day; sending each to the right desk quickly cuts handling time and avoids missed deadlines.

TEXT RECOGNITION ANSWERS: PRESS `CTRL` + `SHIFT` + `Z`

In the sample labelling dialog the shortcut fills every query with its correct label; press it again to clear them.

USE THE AI: PASTE THE MODEL URL

After training, Export shows the model's URL: Copy it, click Use it in AI, paste the URL into **use deployed model** and press Enter. The model's labels then appear on the left: drag them into the empty slots and press Run.

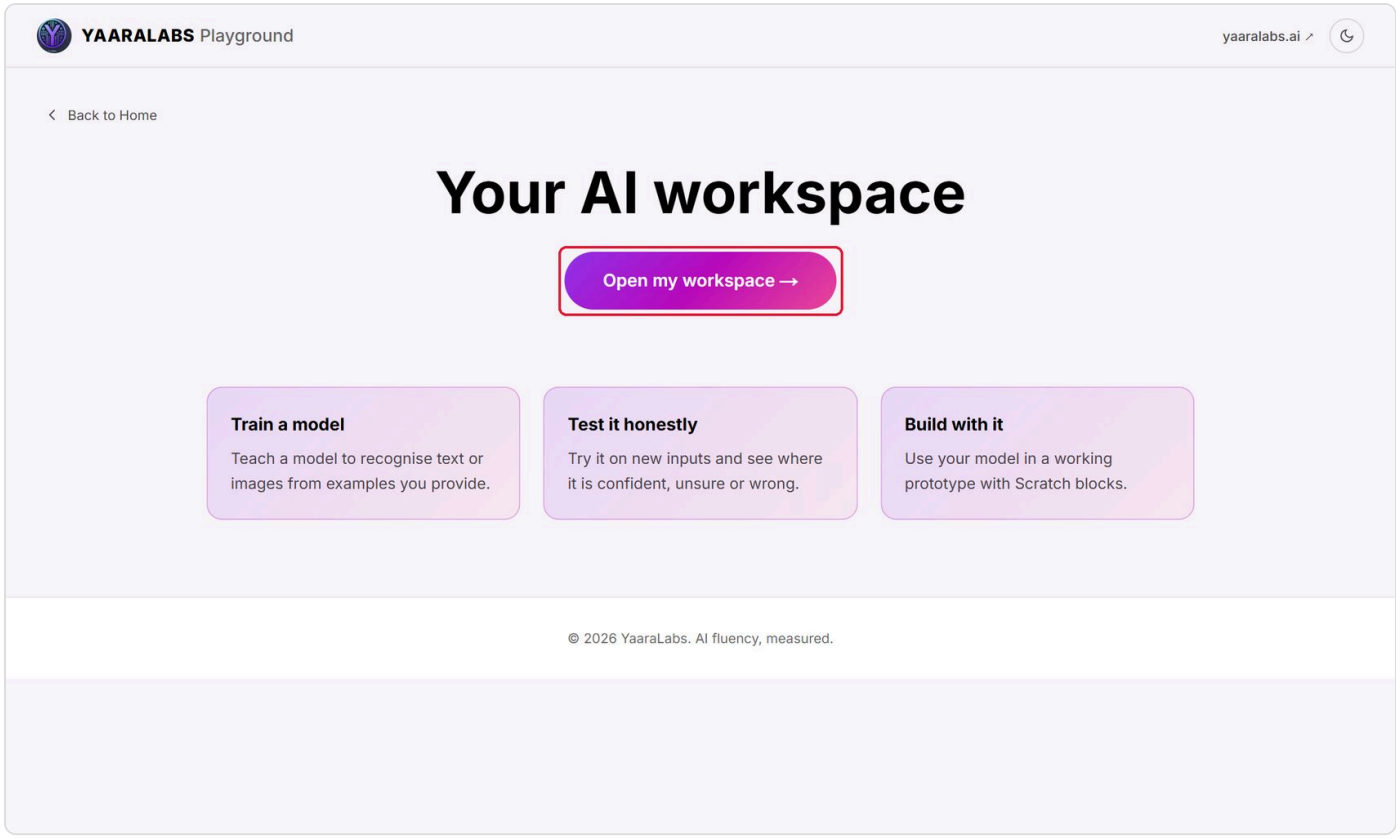
Red boxes mark what to click or type; numbers show the order.

Experiment 1: Rule Based Process

Build keyword rules with blocks that send each client message to the right desk, then make them smarter.

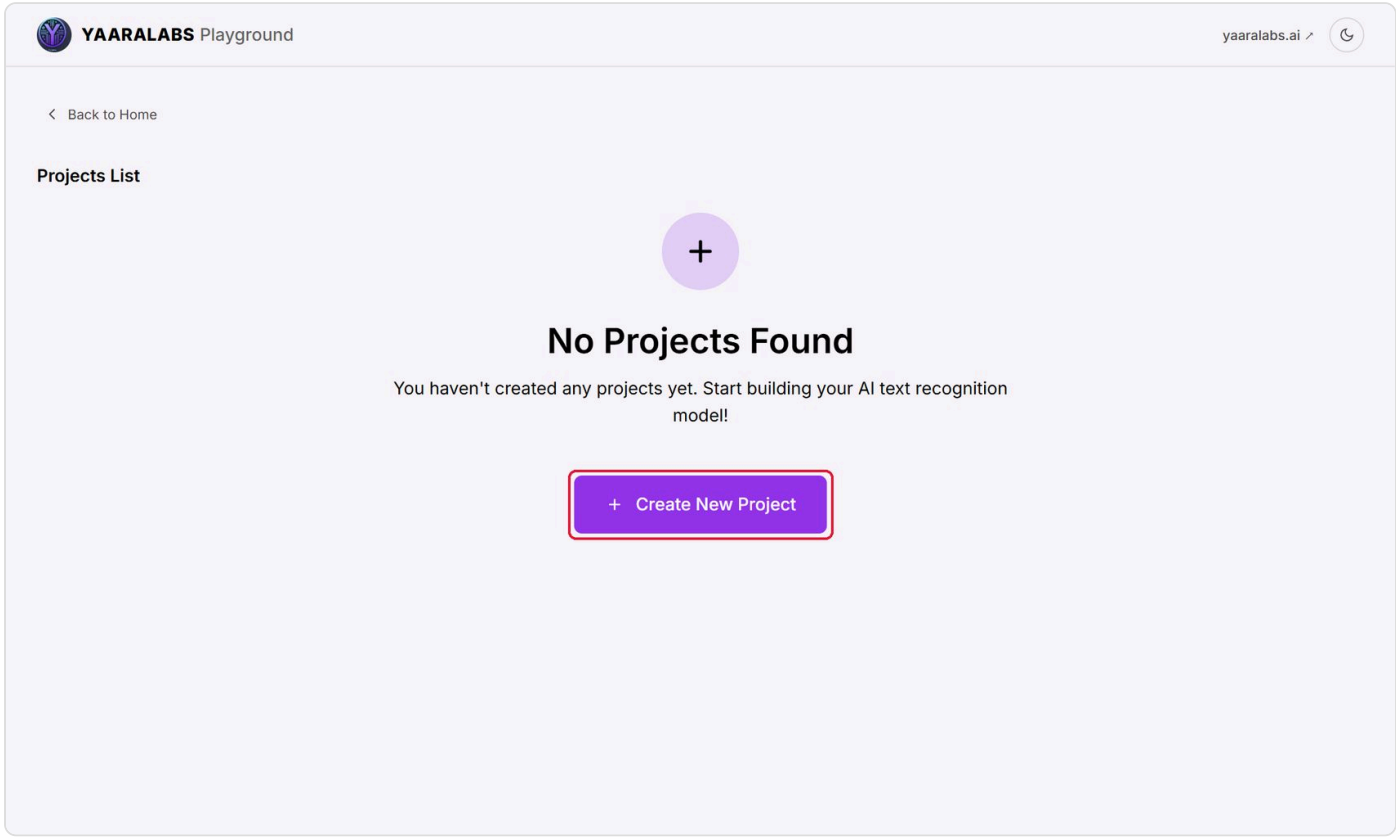
1 Open the workspace

Open the site and click Open my workspace.



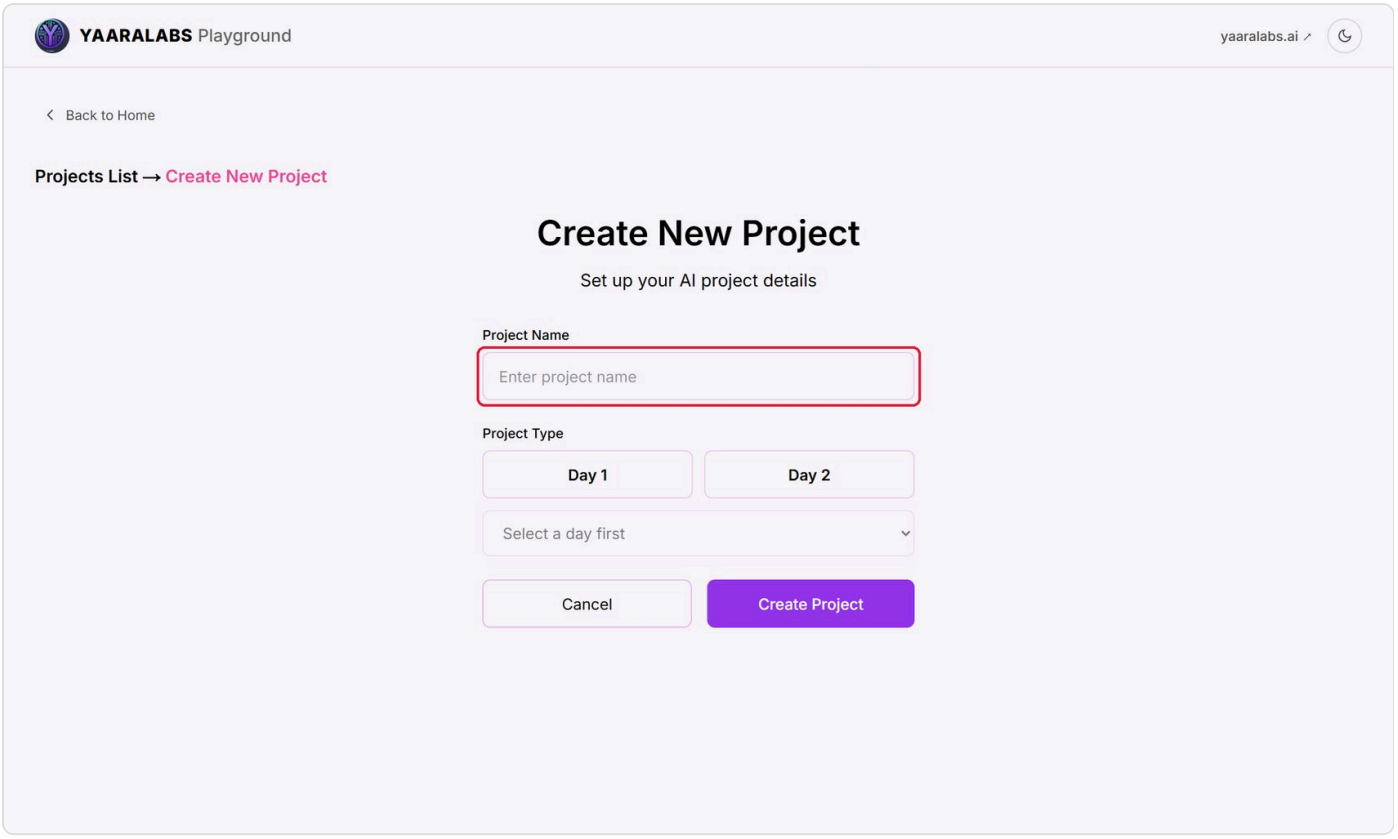
2 Create a new project

Your workspace opens (empty at first): click Create New Project.



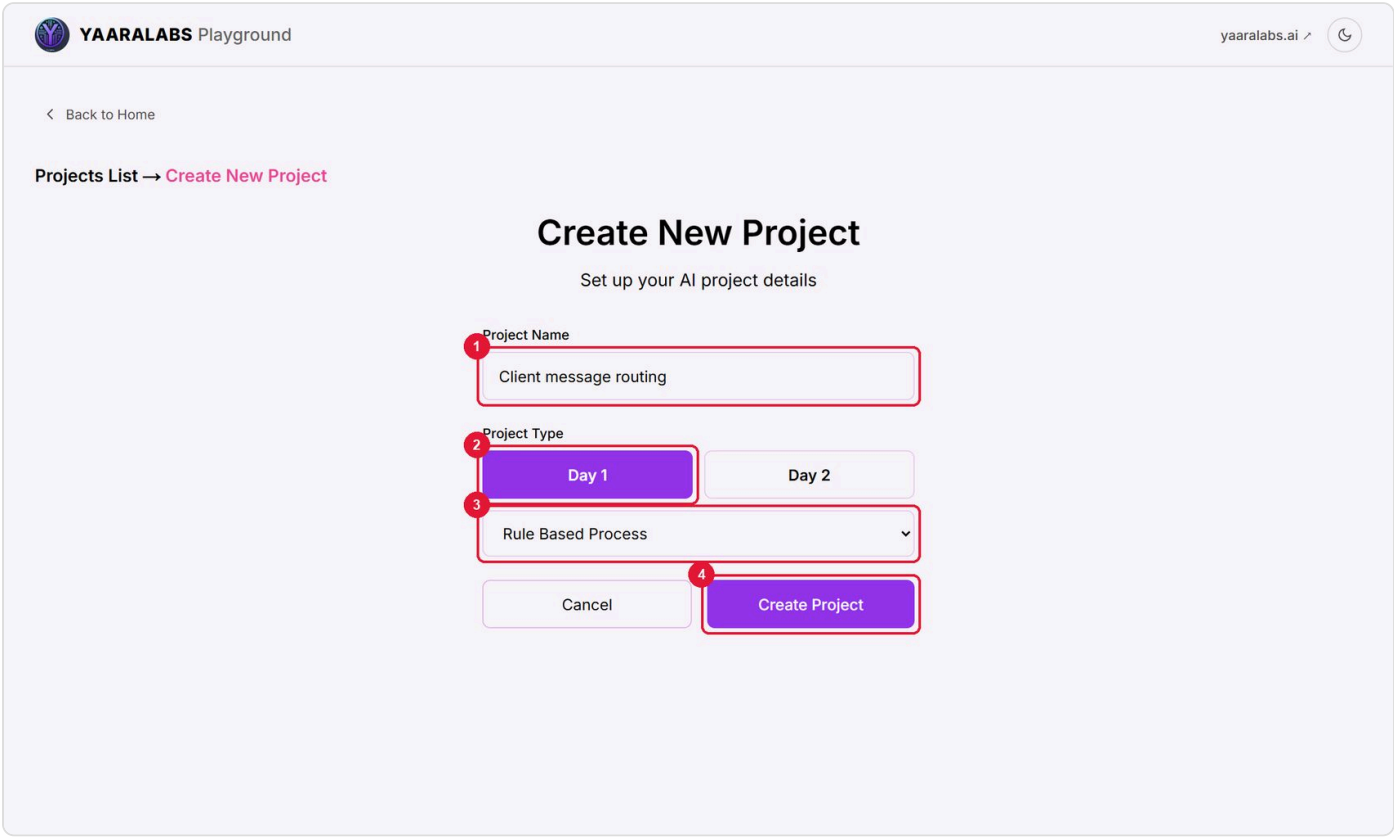
3 The new project form

A form opens: you give the project a name, a day and a type.



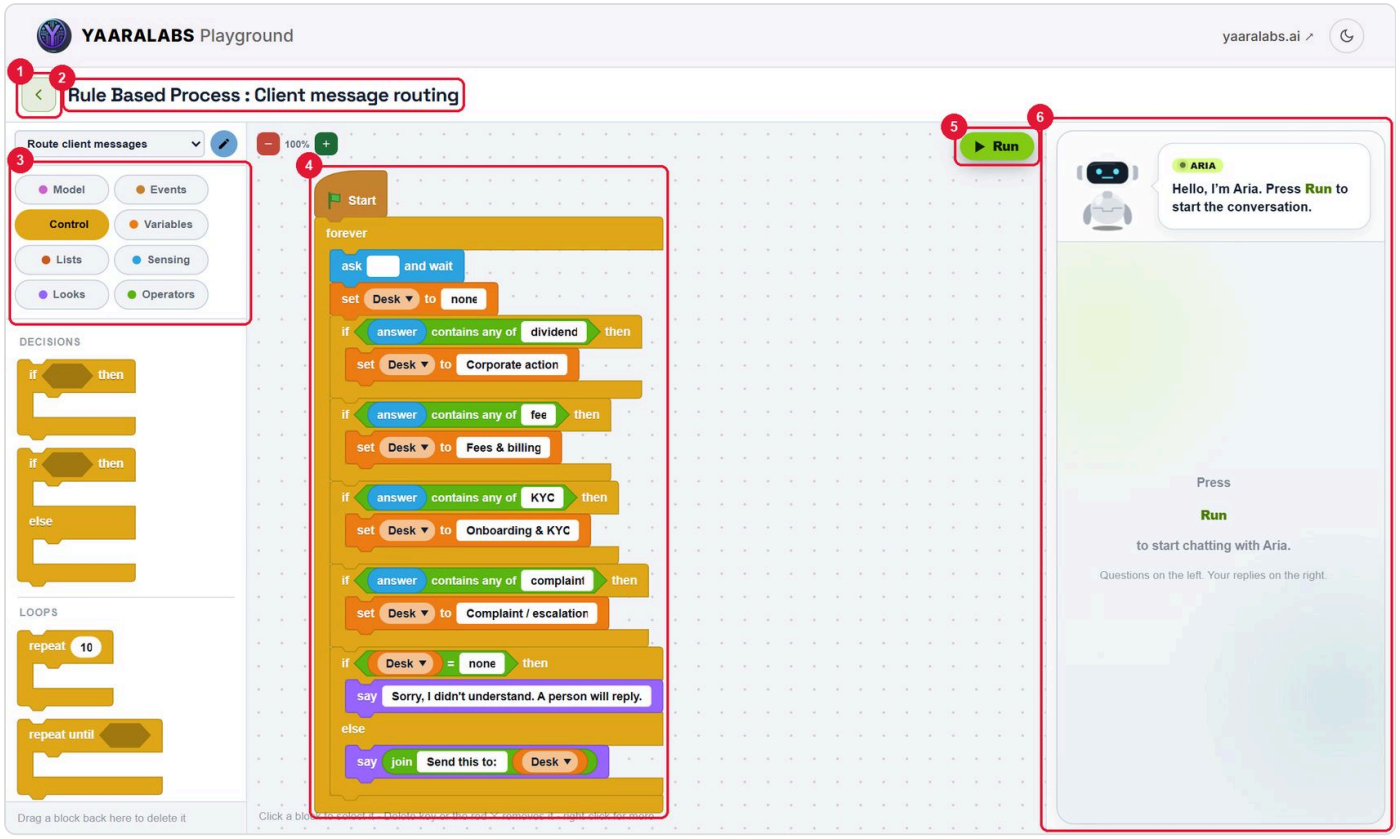
4 Name it and choose Rule Based Process

Type Client message routing, pick Day 1, choose Rule Based Process, then click Create Project.



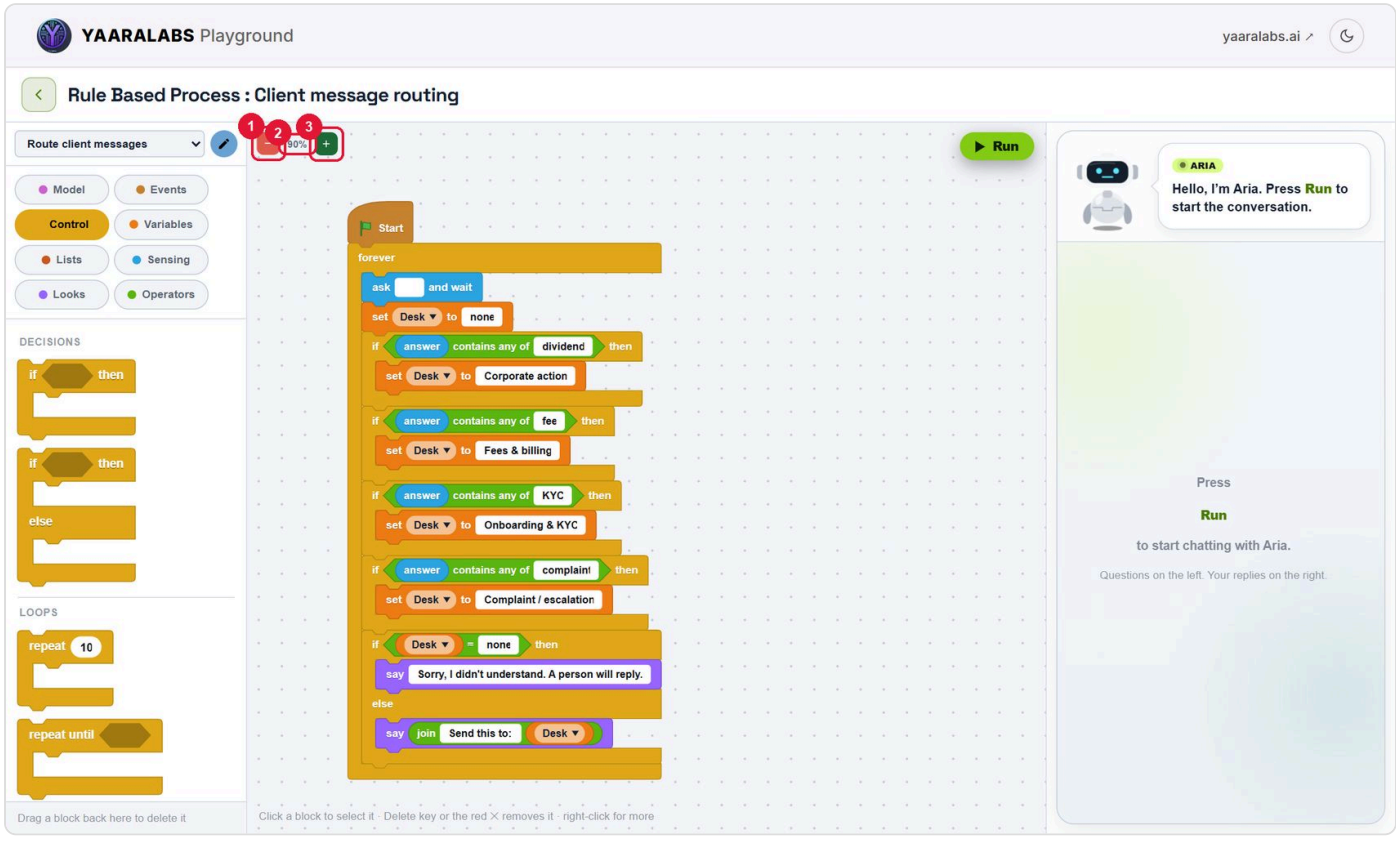
5 The block editor opens

The project opens in the block editor: blocks on the left, your script in the middle, the chat with Aria on the right.



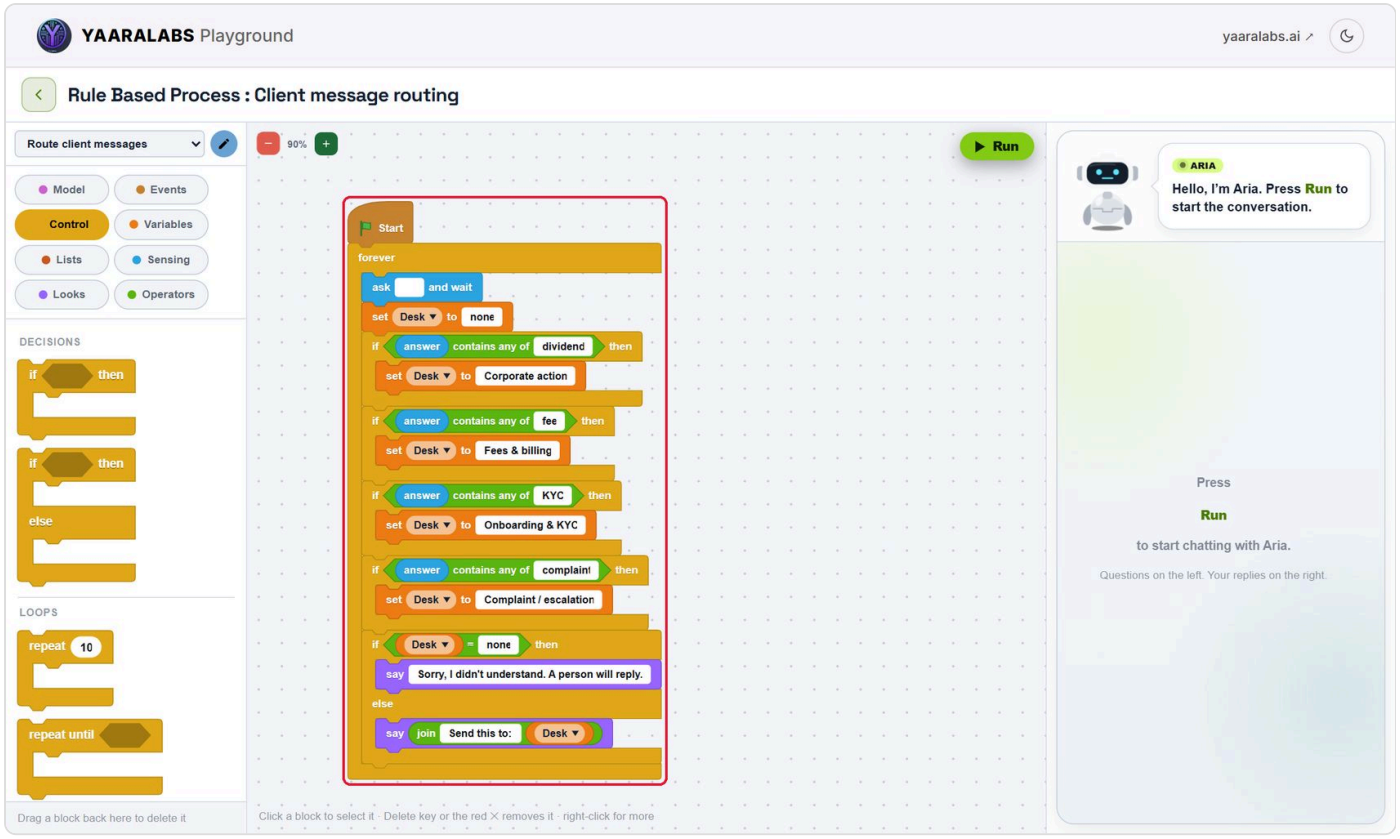
6 Zoom out to see the whole script

Click − (zoom out) at the top of the canvas until the whole script fits; + zooms back in.



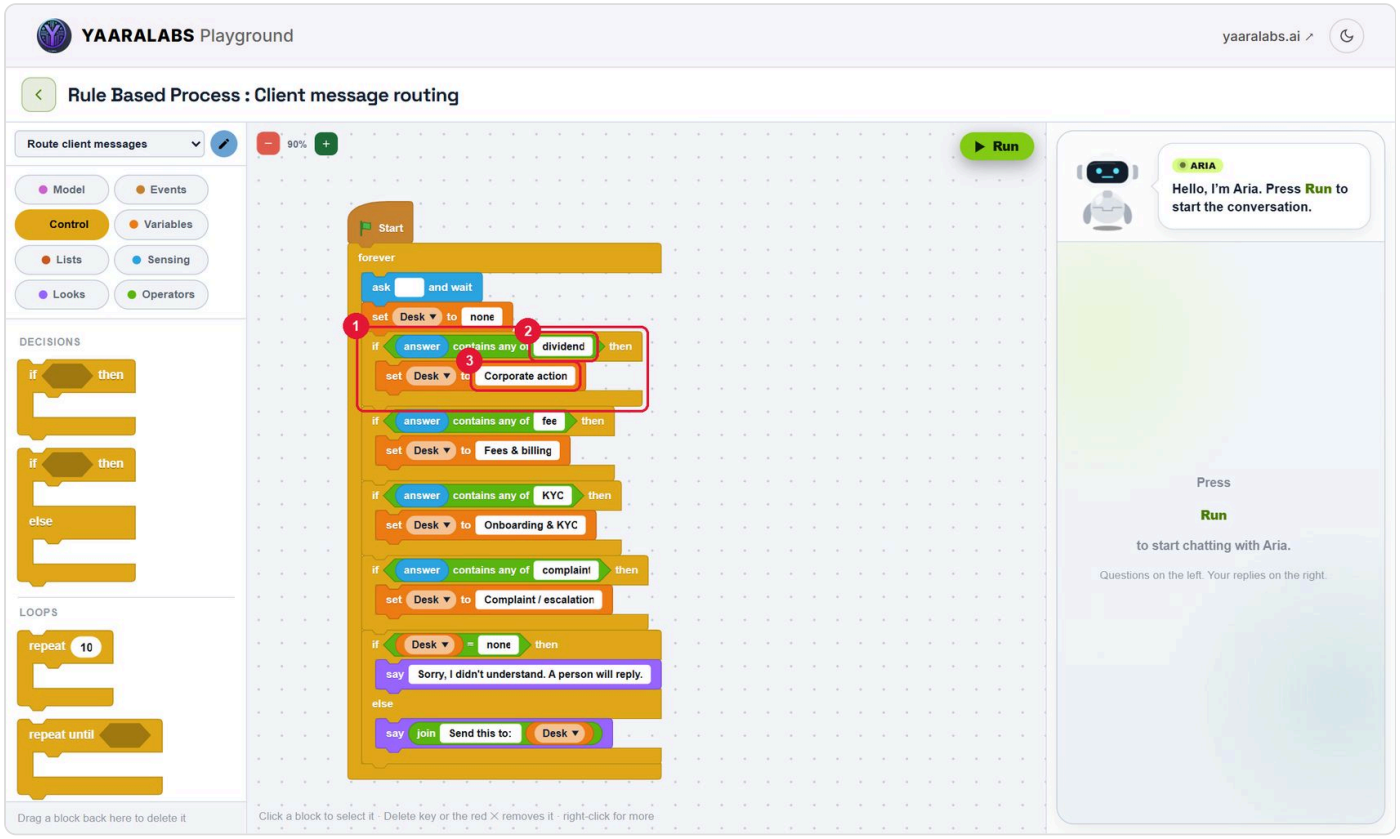
7 The starter script: Route client messages

Forever: wait for a message, set Desk to none, each if checks a keyword and sets the Desk; if Desk is still none Aria says Sorry, else Send this to: Desk.



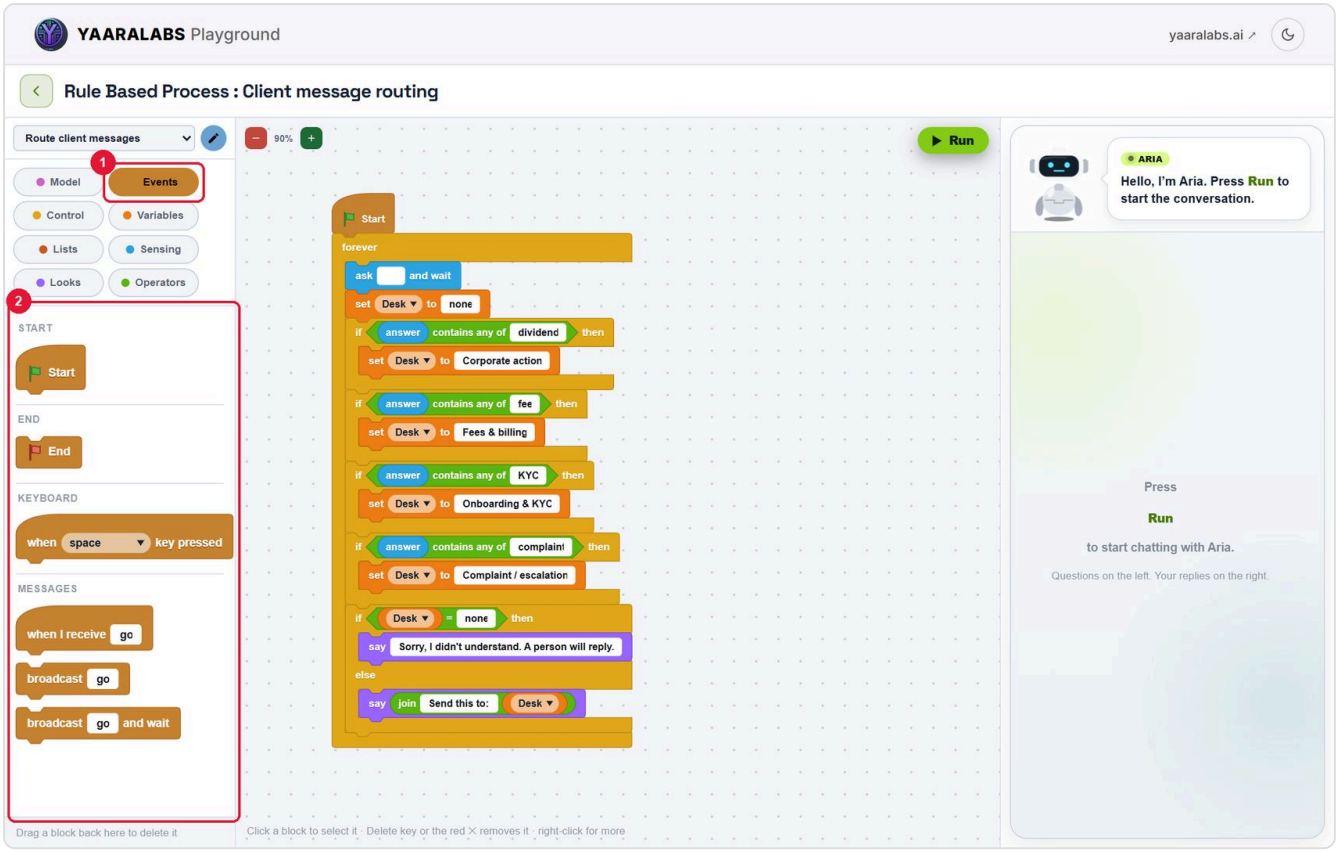
8 One rule = one keyword and one desk

Each rule reads: if answer contains any of dividend then set Desk to Corporate action. The white boxes are where you type.



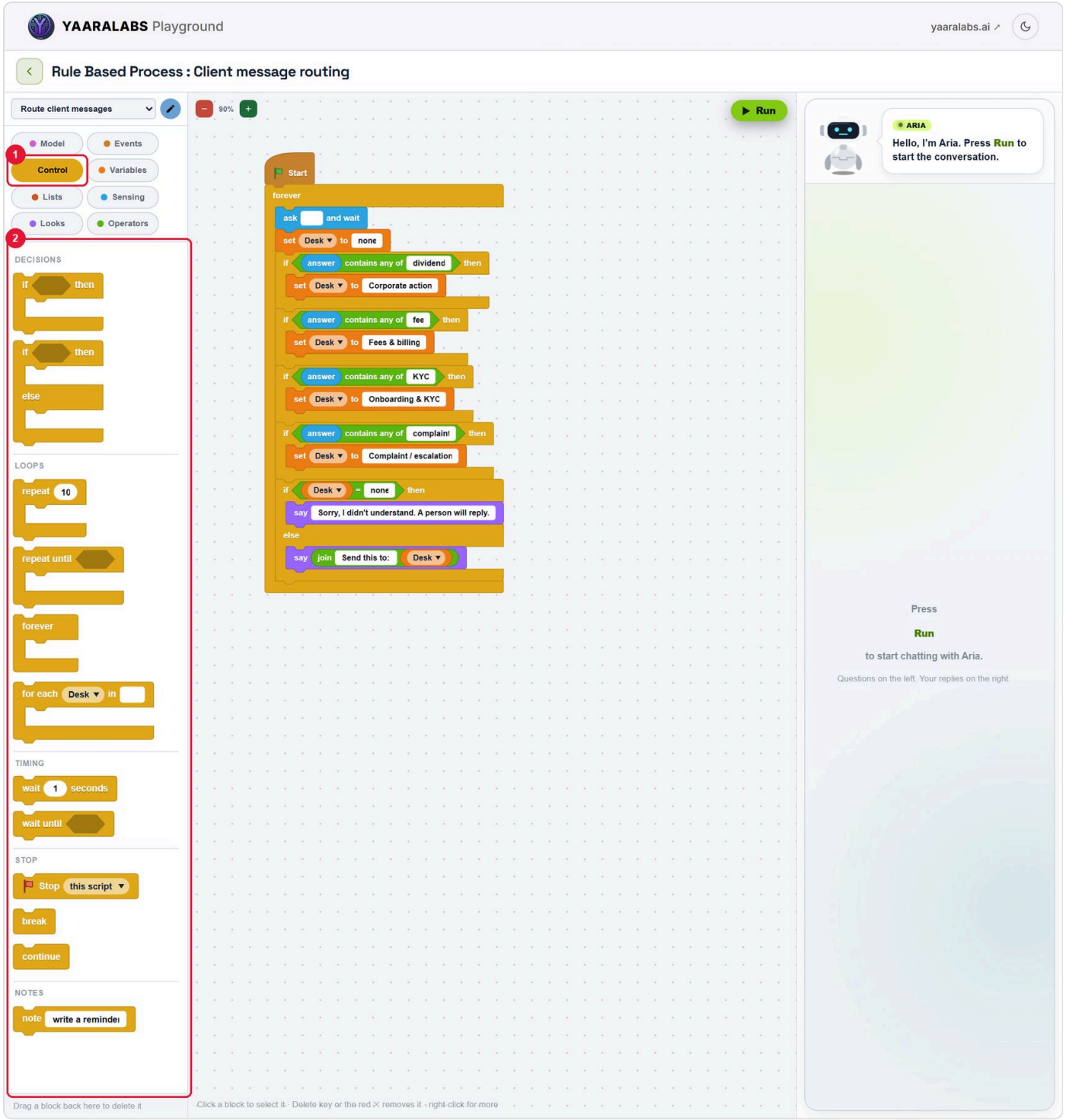
9 Blocks: Events

Click Events: Start blocks that begin a script when you press Run.



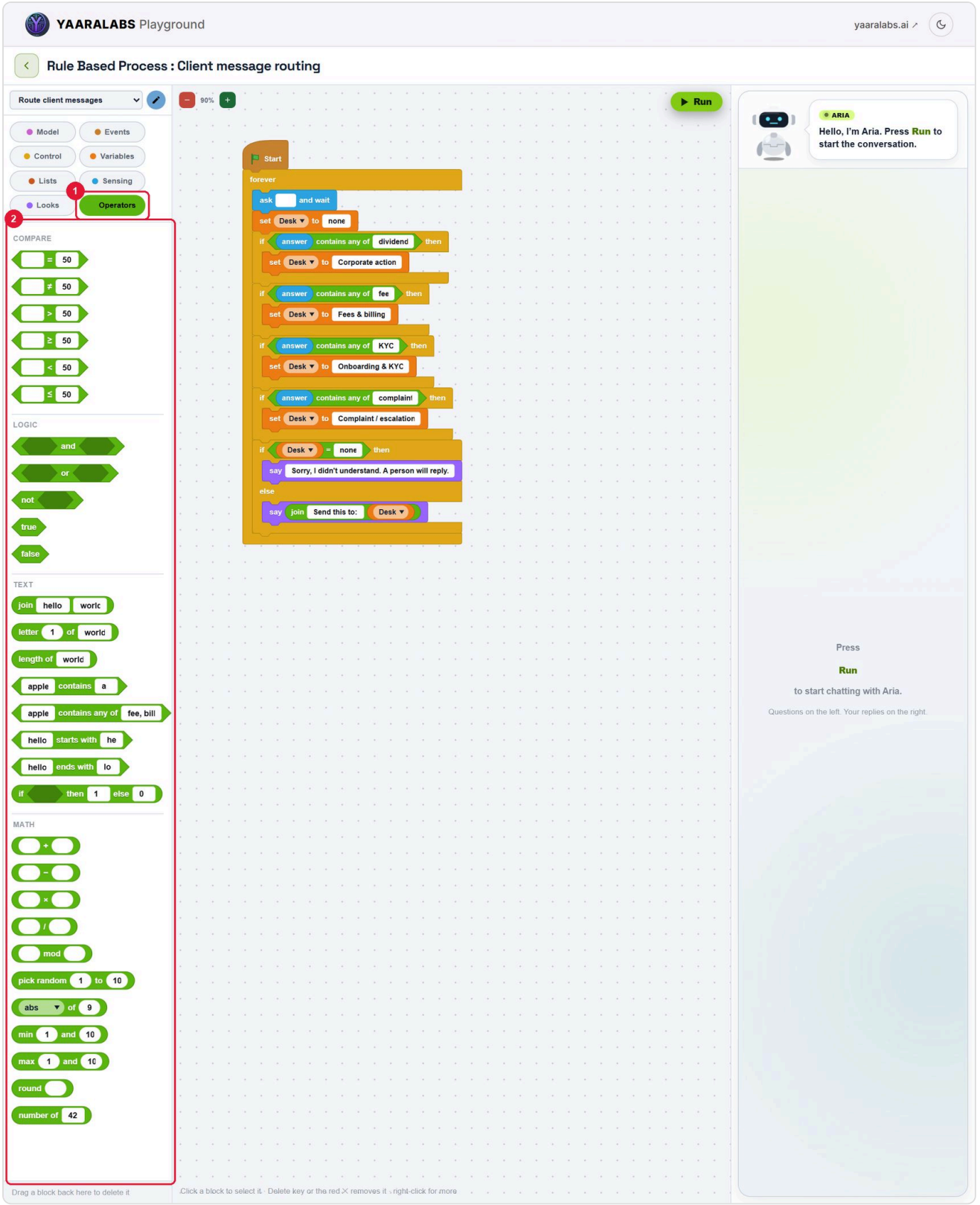
10 Blocks: Control

Click Control: forever, repeat, if … then and if … then … else: the blocks that make decisions.



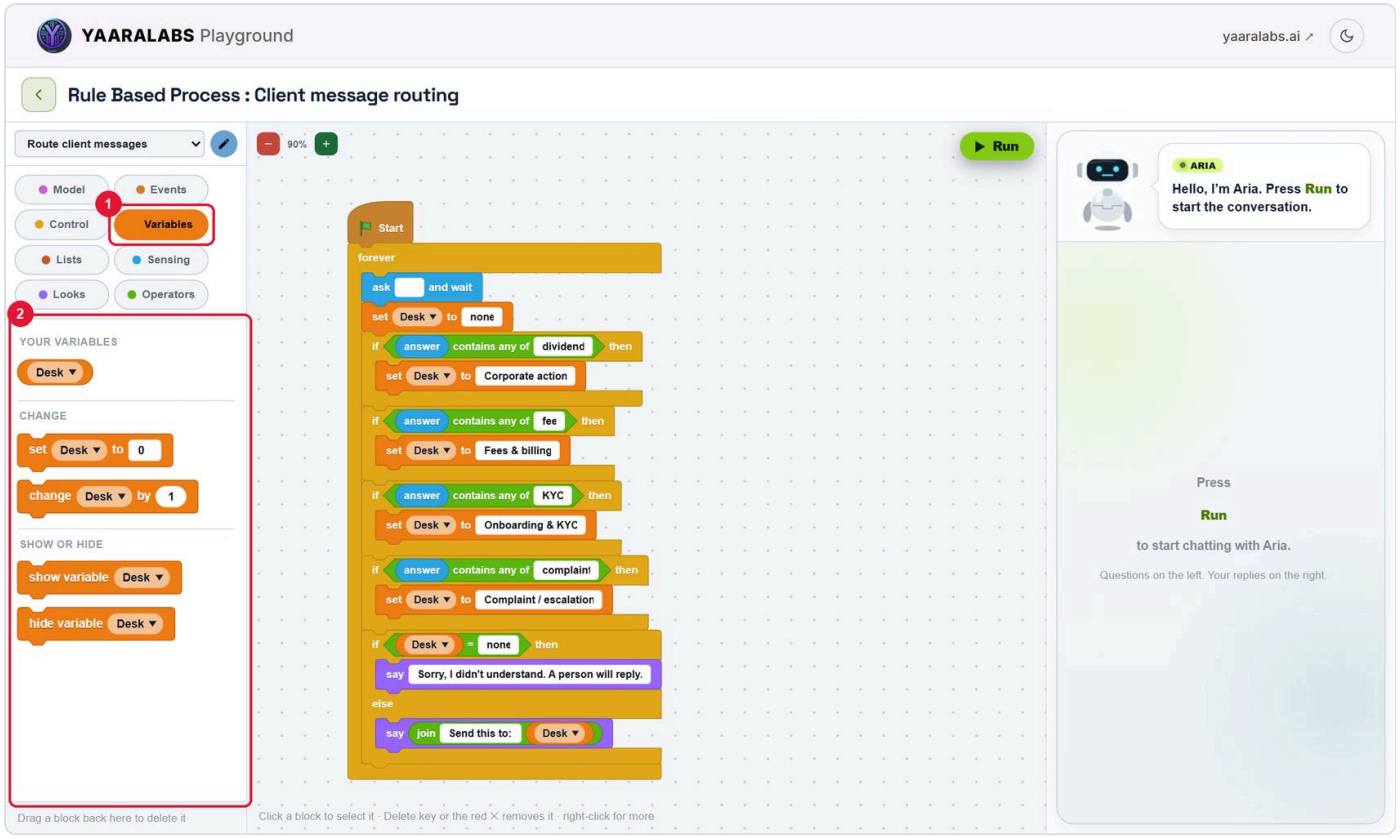
11 Blocks: Operators

Click Operators: contains any of, =, >, join and the other checks that go inside an if.



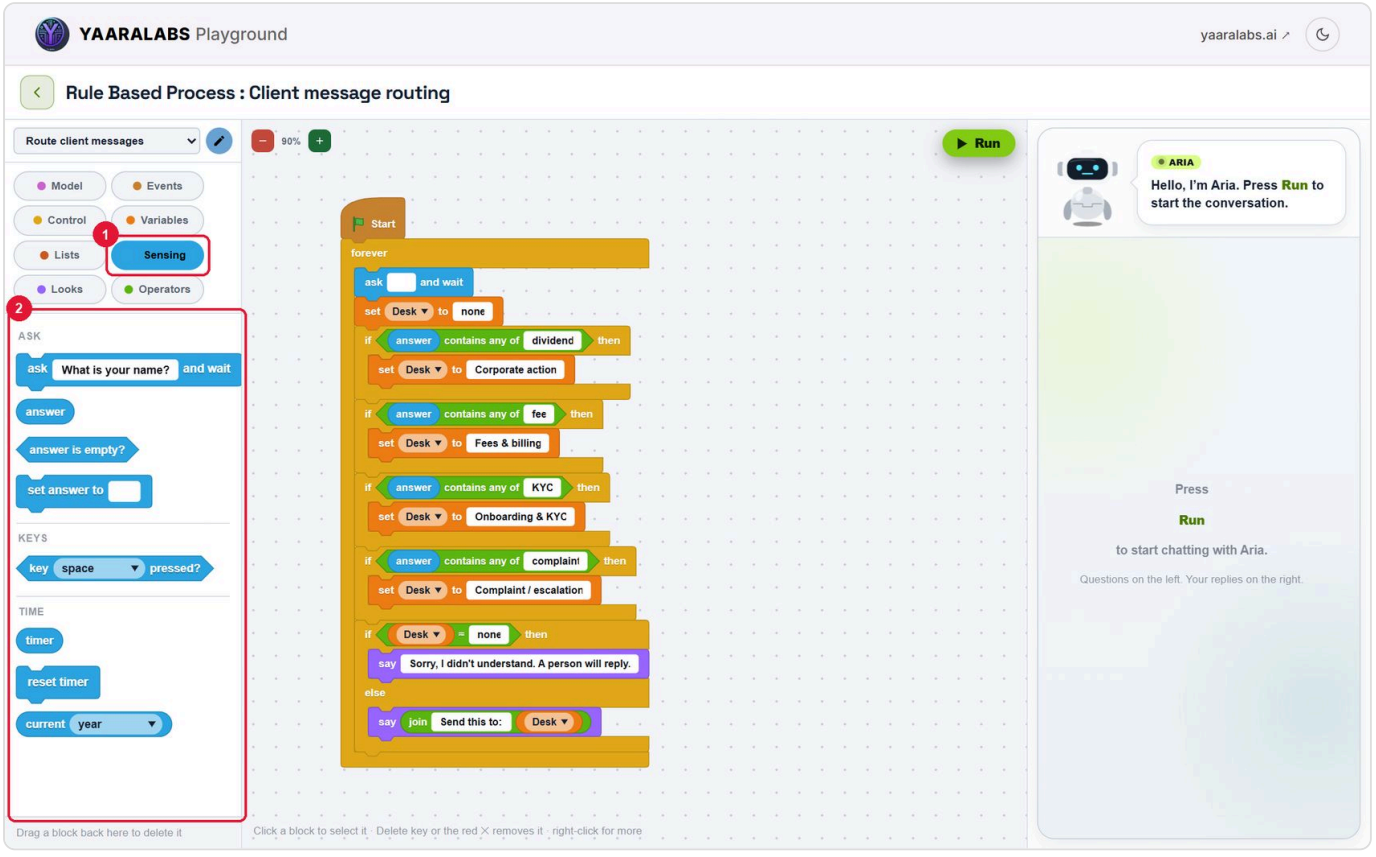
12 Blocks: Variables

Click Variables: set Desk to … and the Desk value: where the chosen desk is kept.



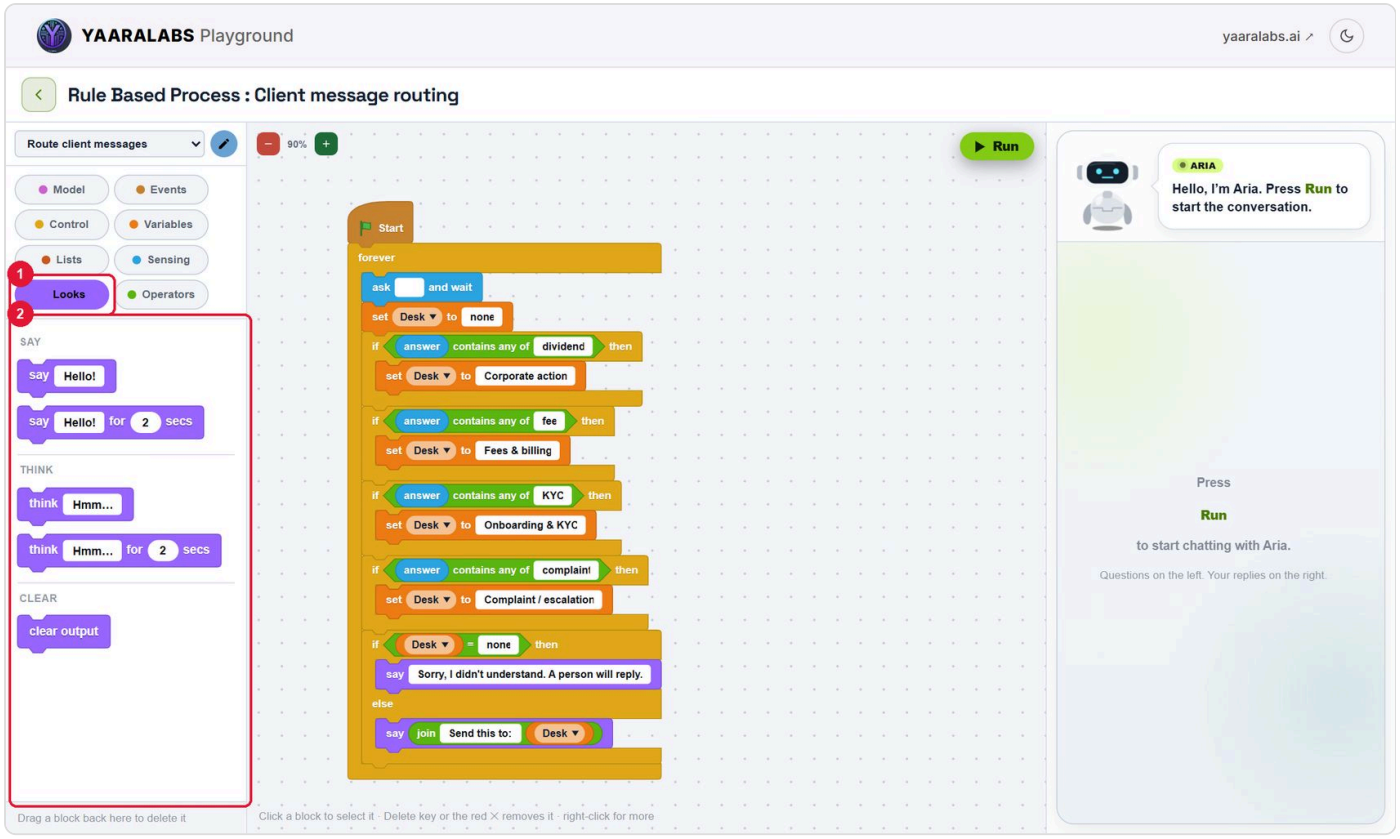
13 Blocks: Sensing

Click Sensing: ask … and wait and answer: how Aria reads the client message.



14 Blocks: Looks

Click Looks: say ...: how Aria replies in the chat.

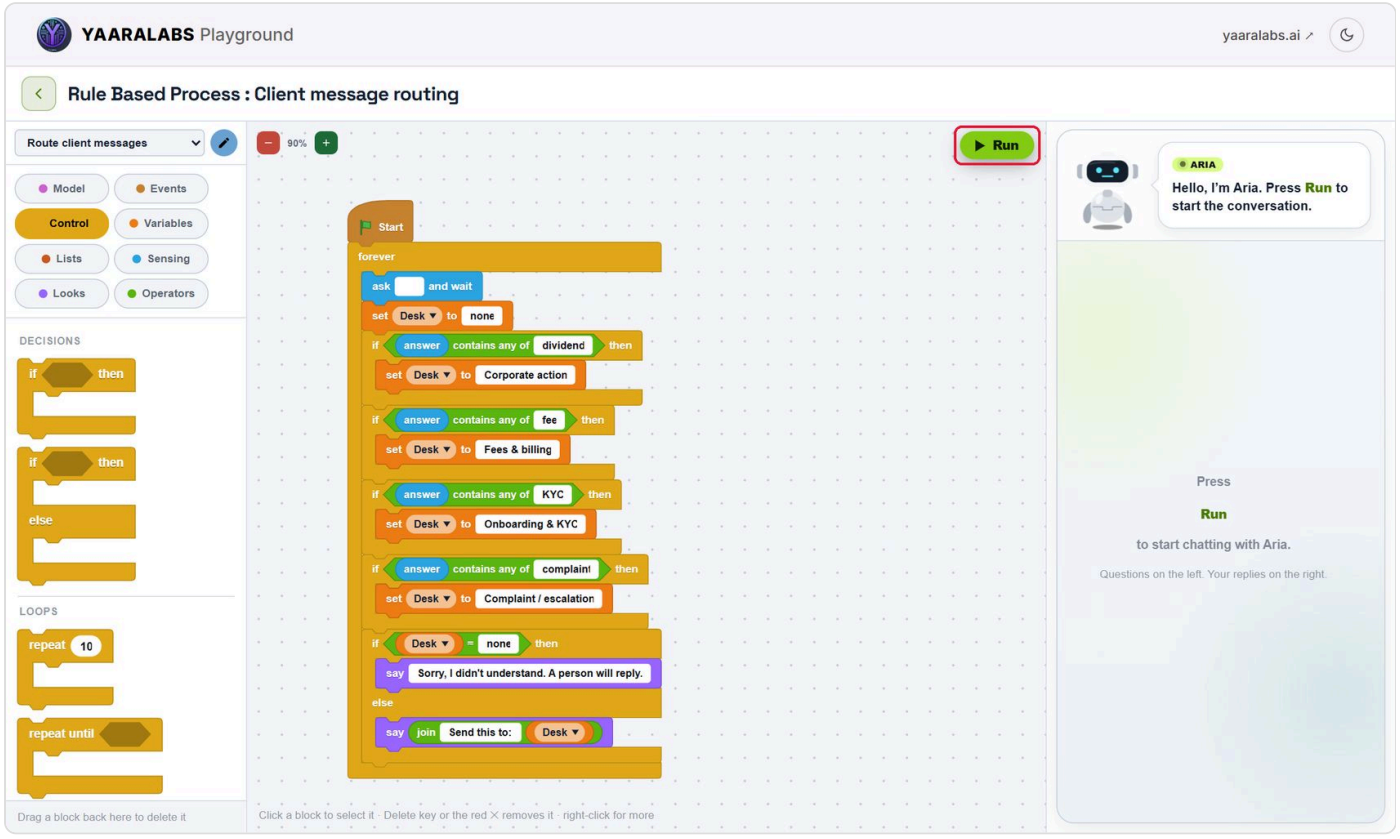


Run it

Press Run and type client messages to Aria: each one is routed by the keyword rules.

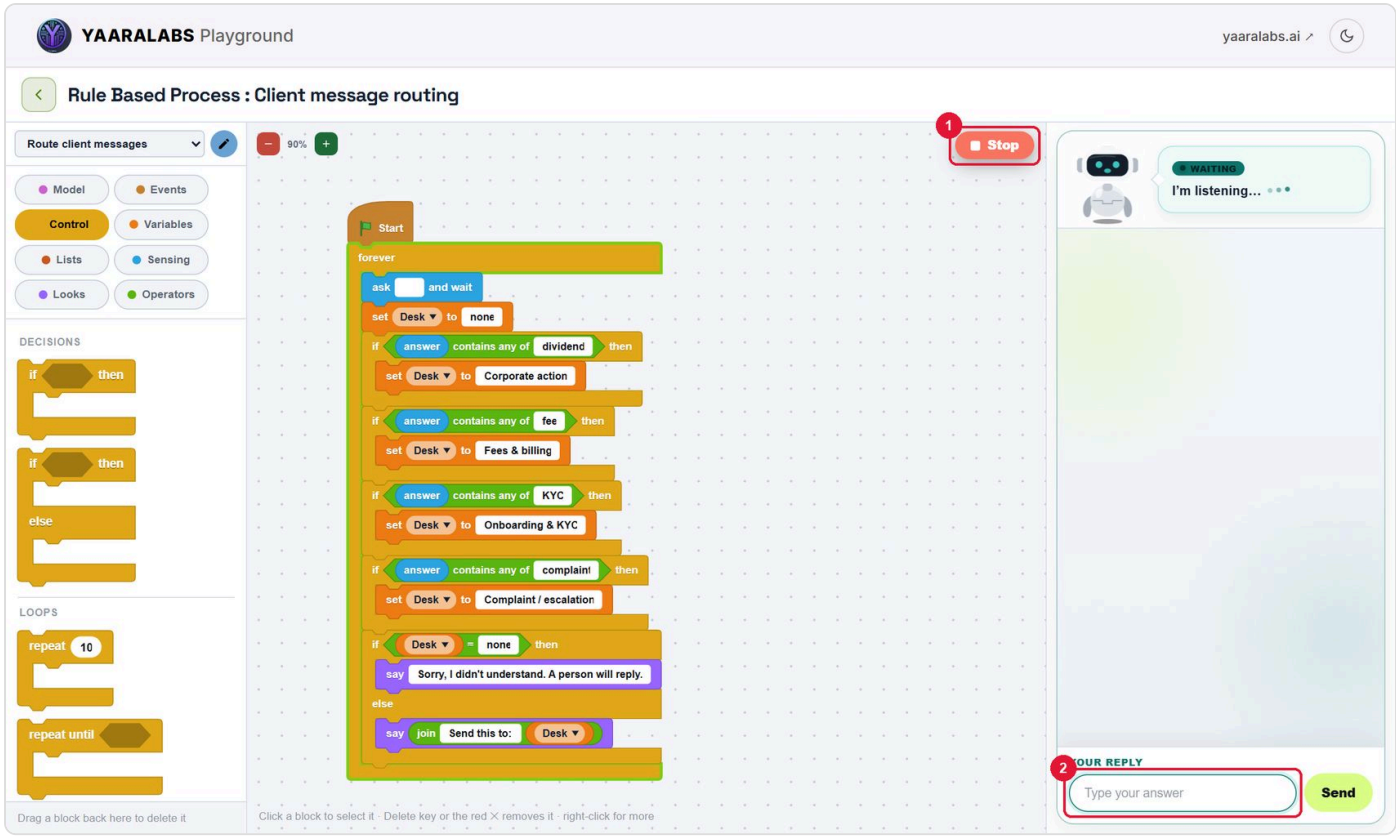
15 Click Run

Click Run on the canvas to start the script.



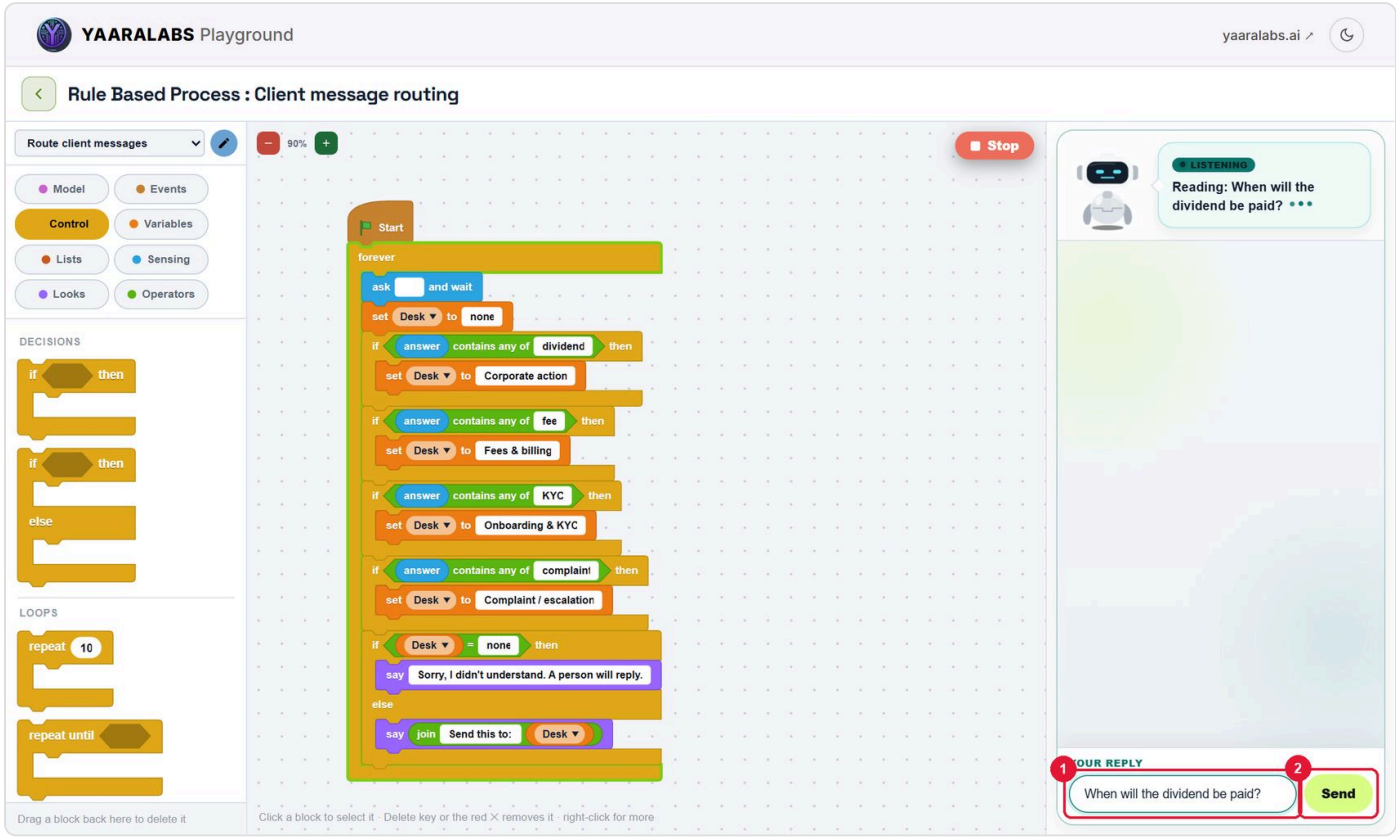
16 Aria waits for a message

The script is running: Run has turned into Stop, and the reply box at the bottom of the chat is ready.



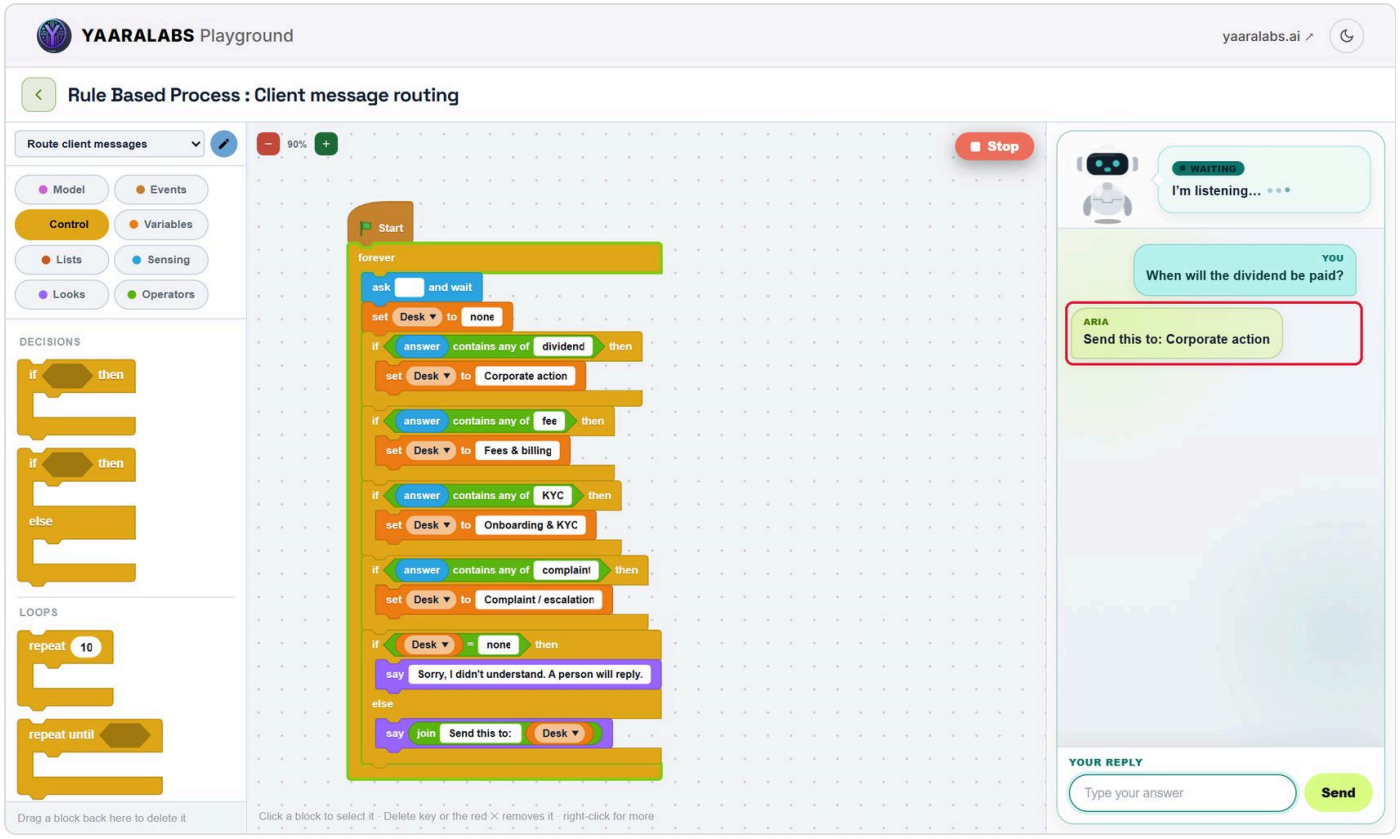
17 Type a client message

Type When will the dividend be paid? in the reply box and press Enter (or Send).



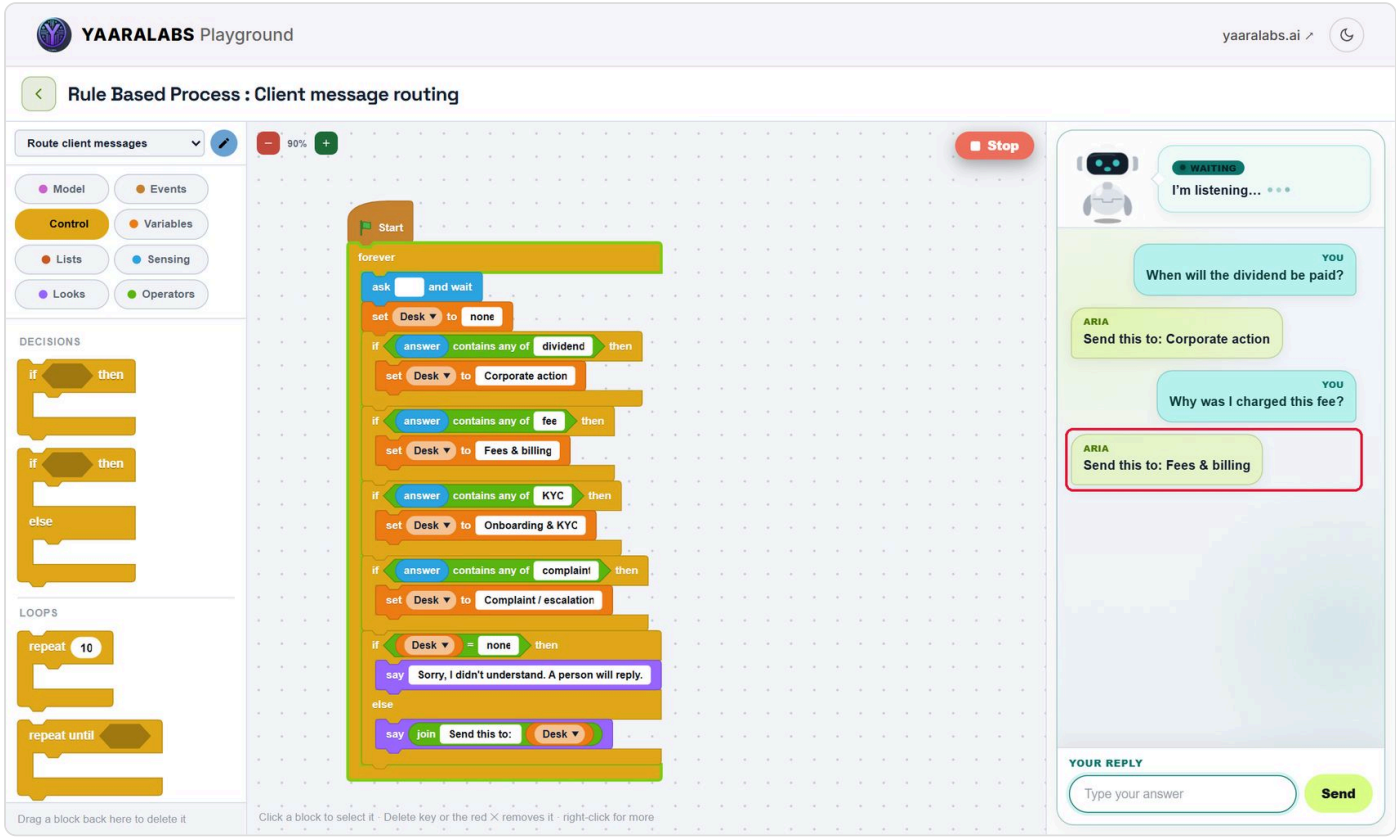
18 Dividend question

"dividend" matches the first rule: Aria says Send this to: Corporate action.



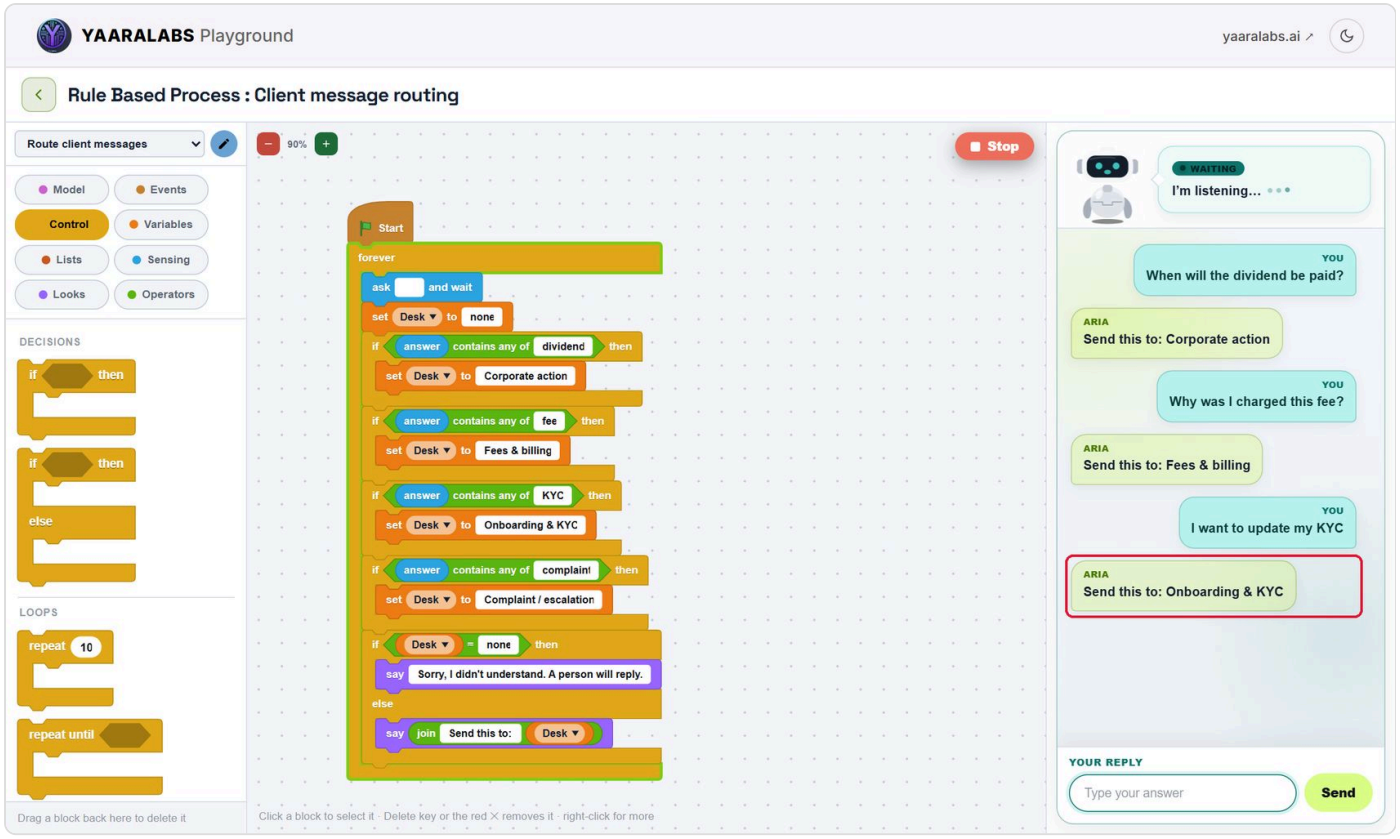
19 Fee question

Type Why was I charged this fee? and press Enter. "fee" matches: Send this to: Fees & billing.



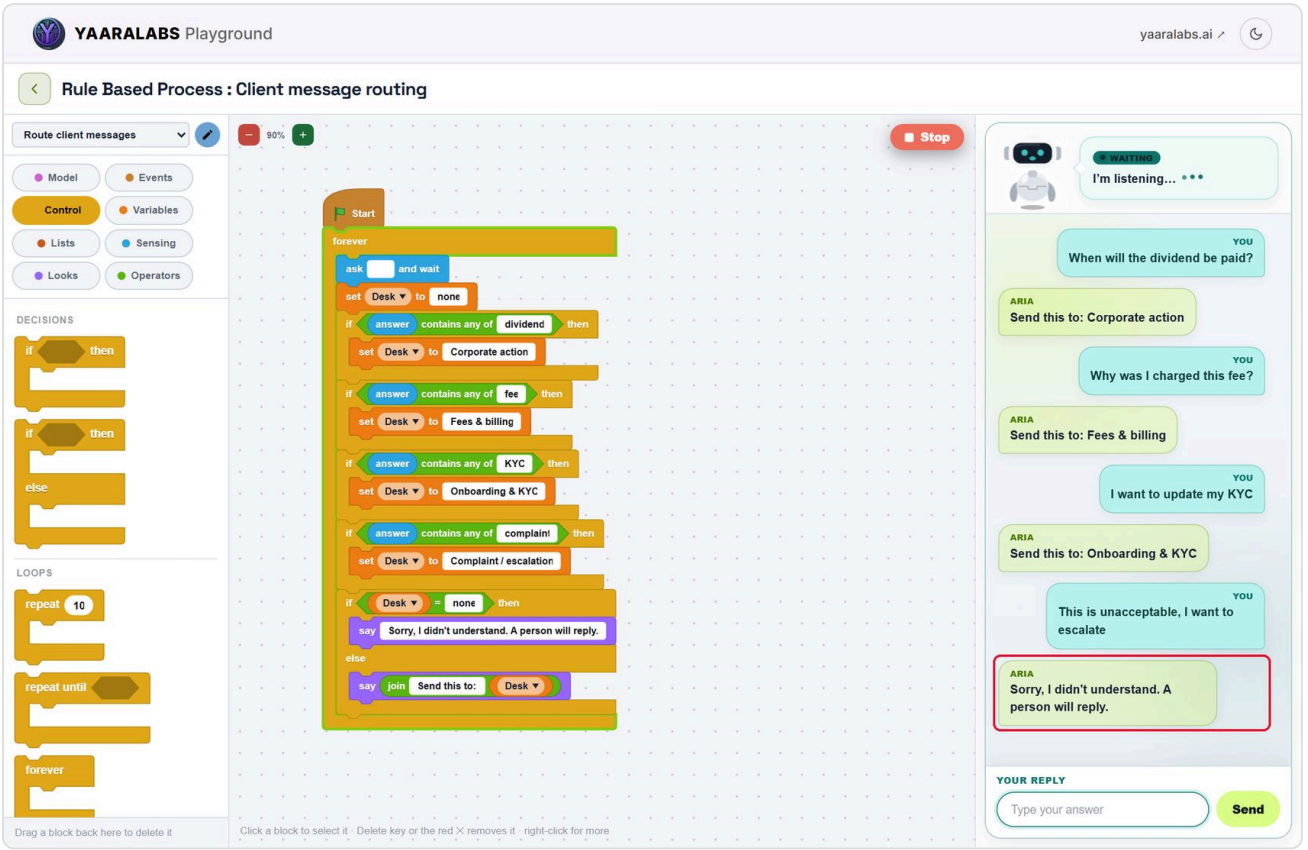
20 KYC request

Type I want to update my KYC and press Enter. “KYC” matches: Send this to: Onboarding & KYC.



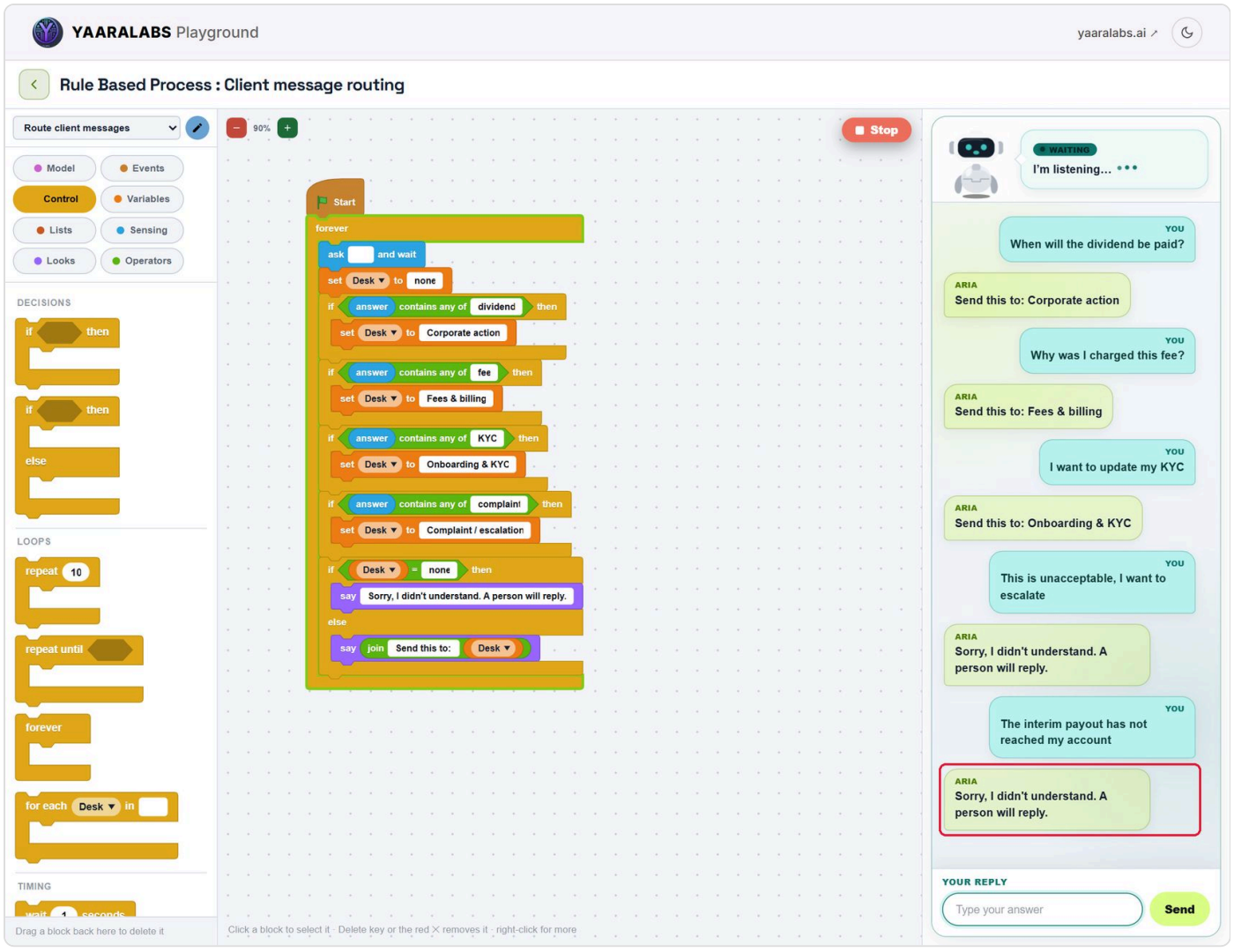
21 An escalation the rules miss

Type This is unacceptable, I want to escalate and press Enter. No keyword matches (the rule only knows “complaint”), so Aria says Sorry.



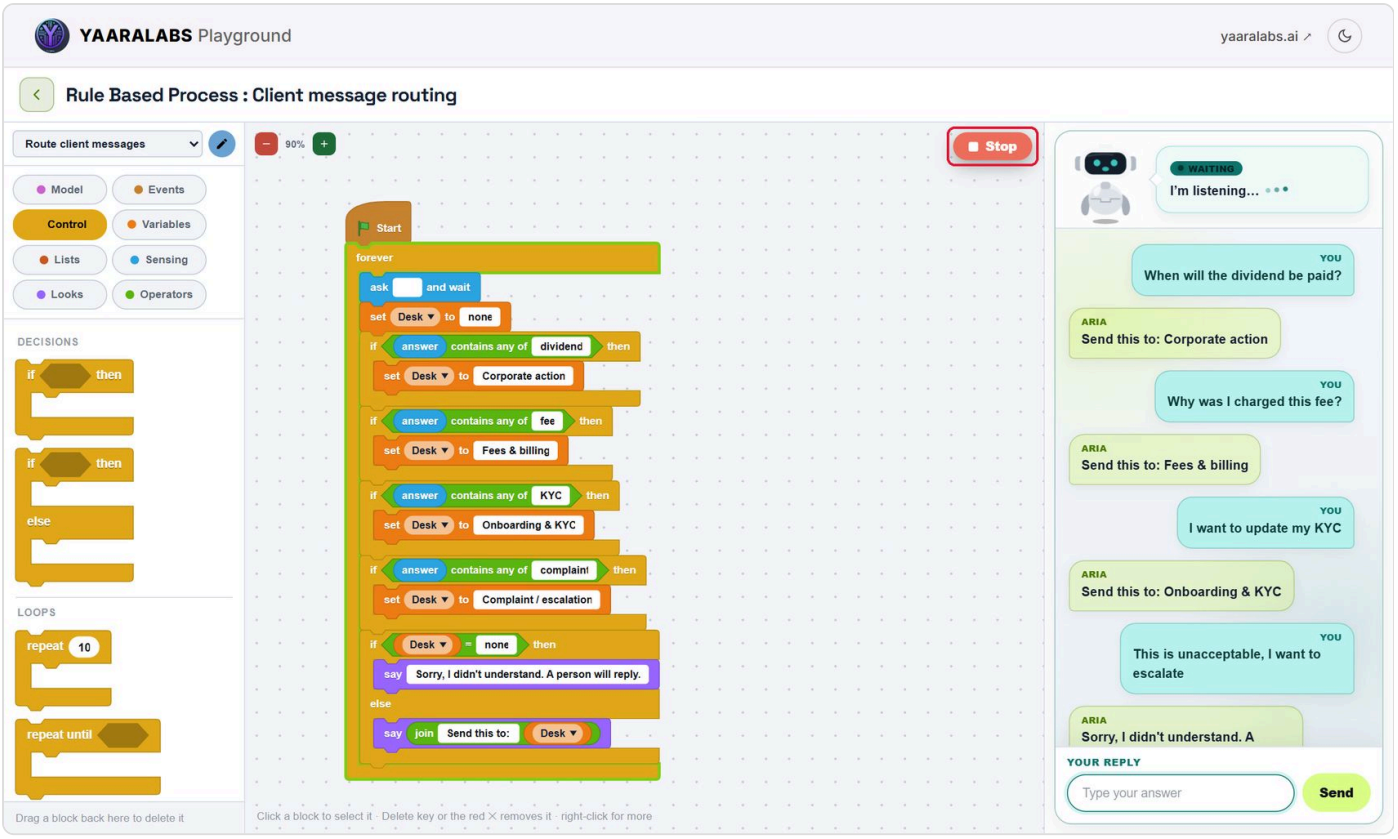
22 Another message the rules miss

Type The interim payout has not reached my account and press Enter. “payout” is not a keyword yet, so this one is not routed either: Sorry.



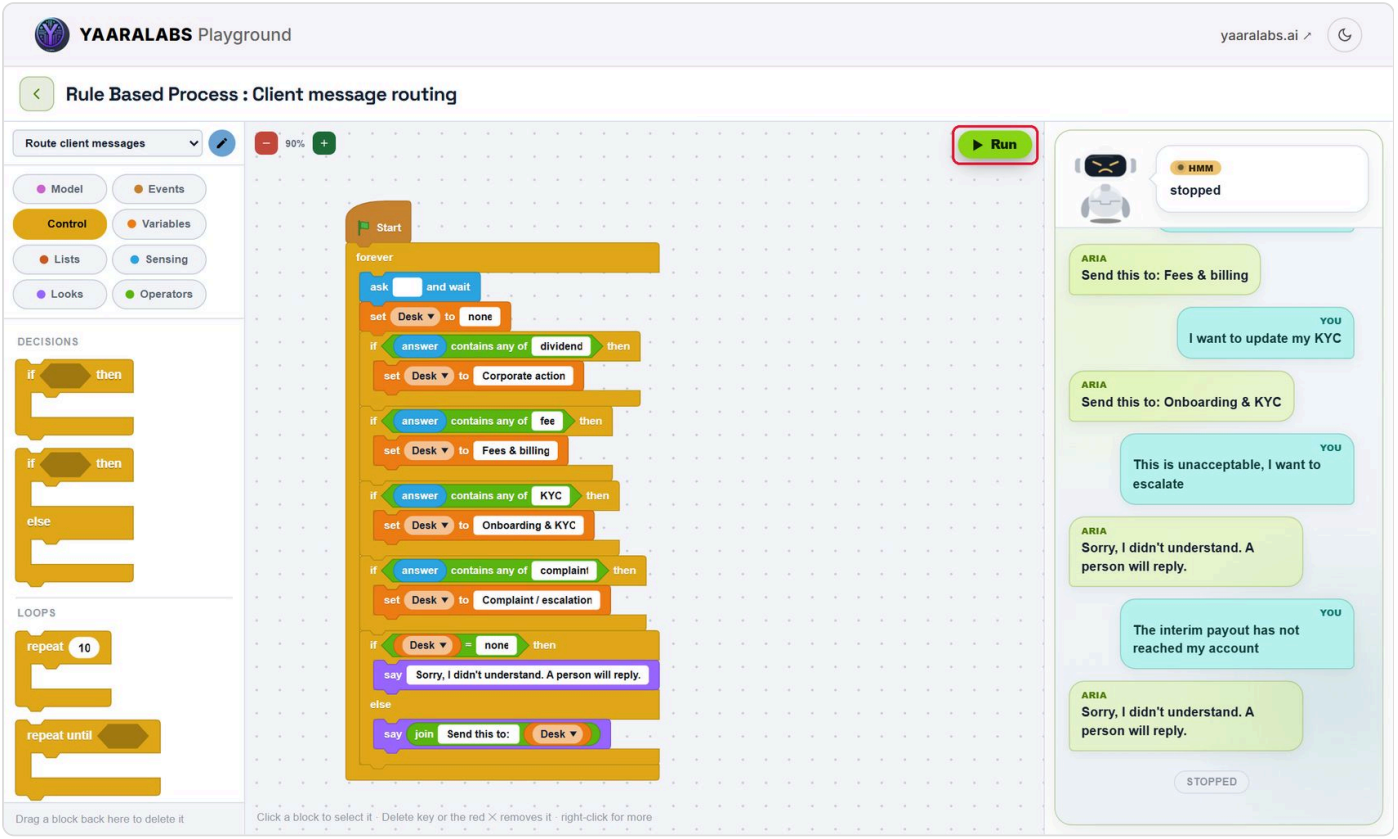
23 Stop the run

Click Stop (the Run button while the script runs) to end the chat.



24 Stopped

The script has stopped: the button shows Run again.

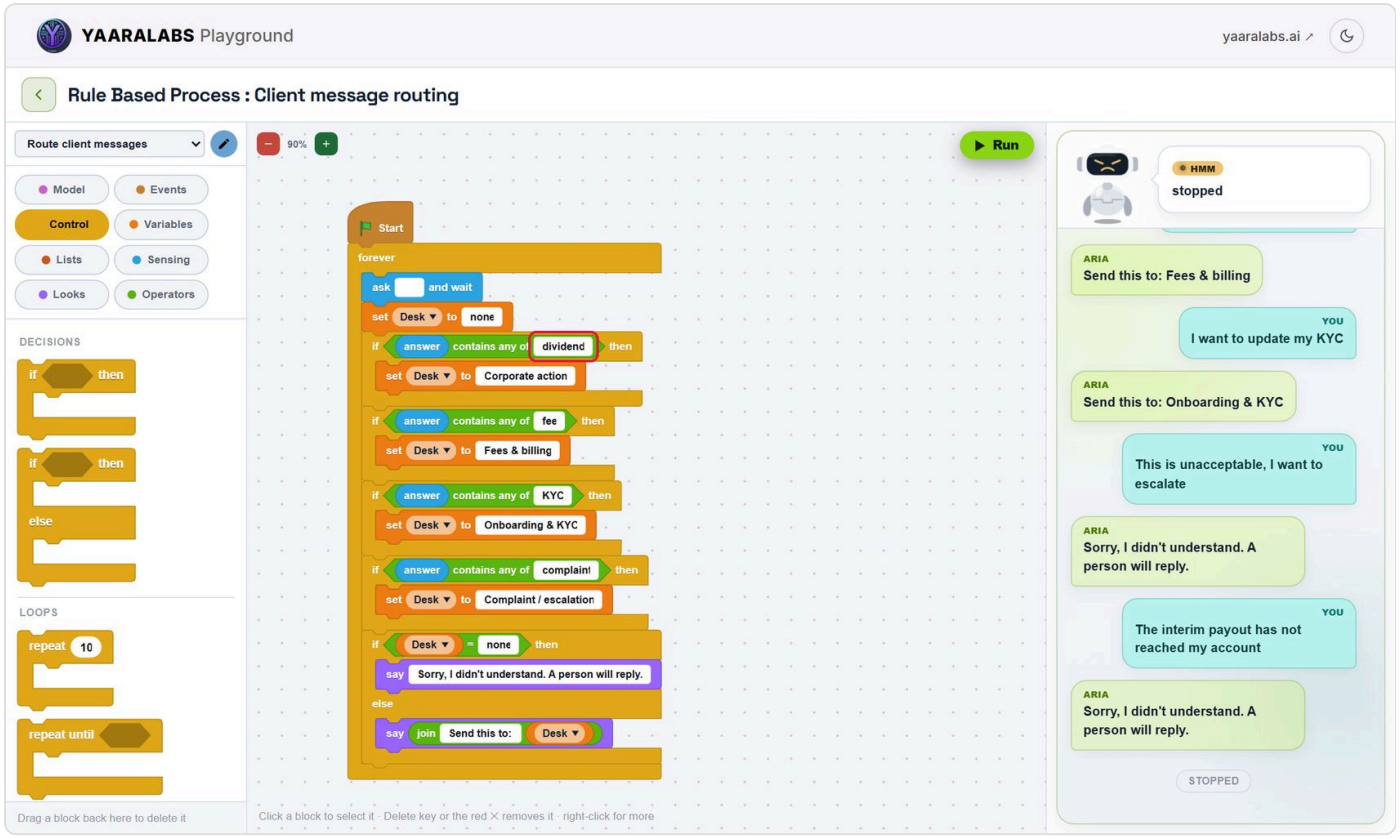


Make it smarter

Add more keywords to a rule, comma separated, and run the messages that failed again.

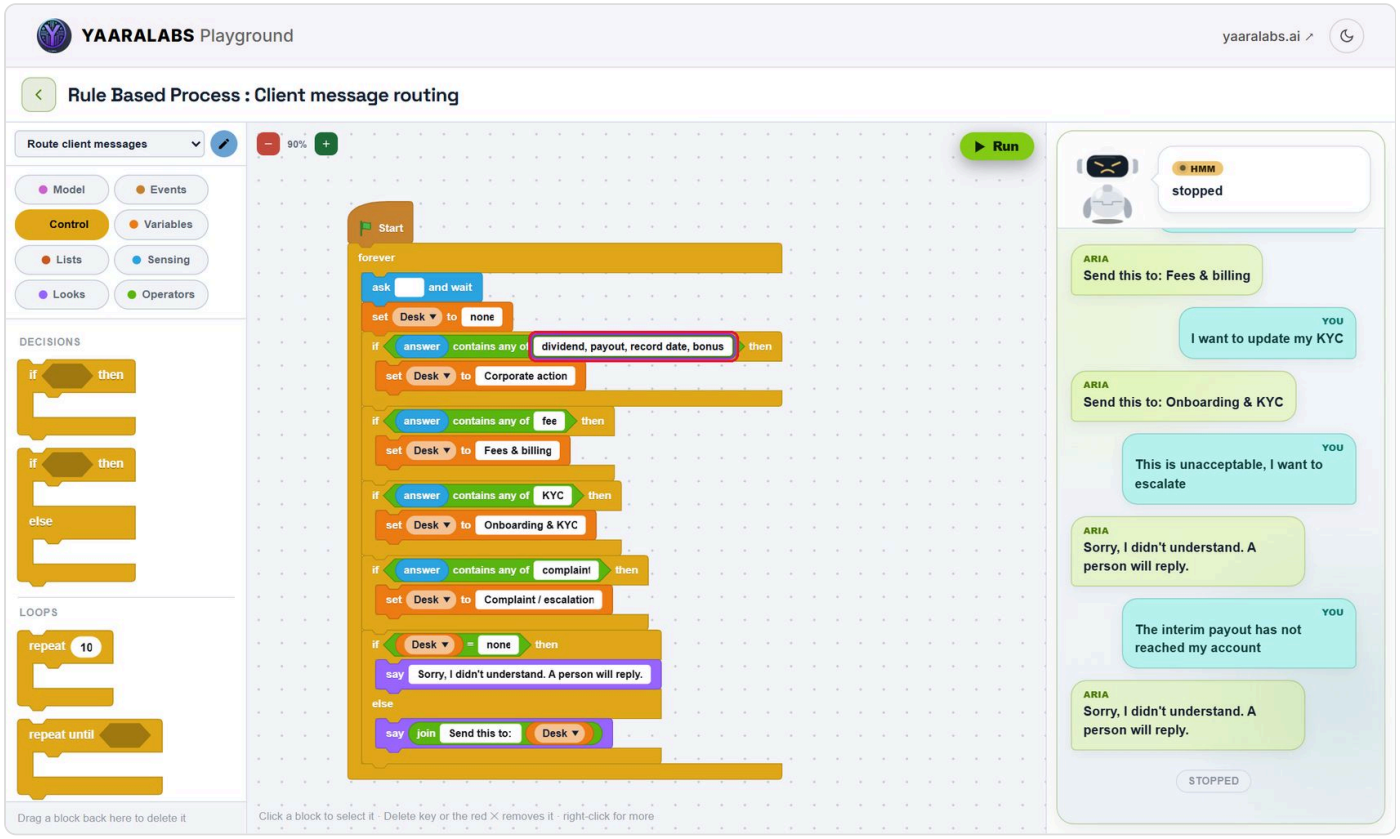
25 Find the Corporate action rule

Click the keyword box of the first rule (dividend).



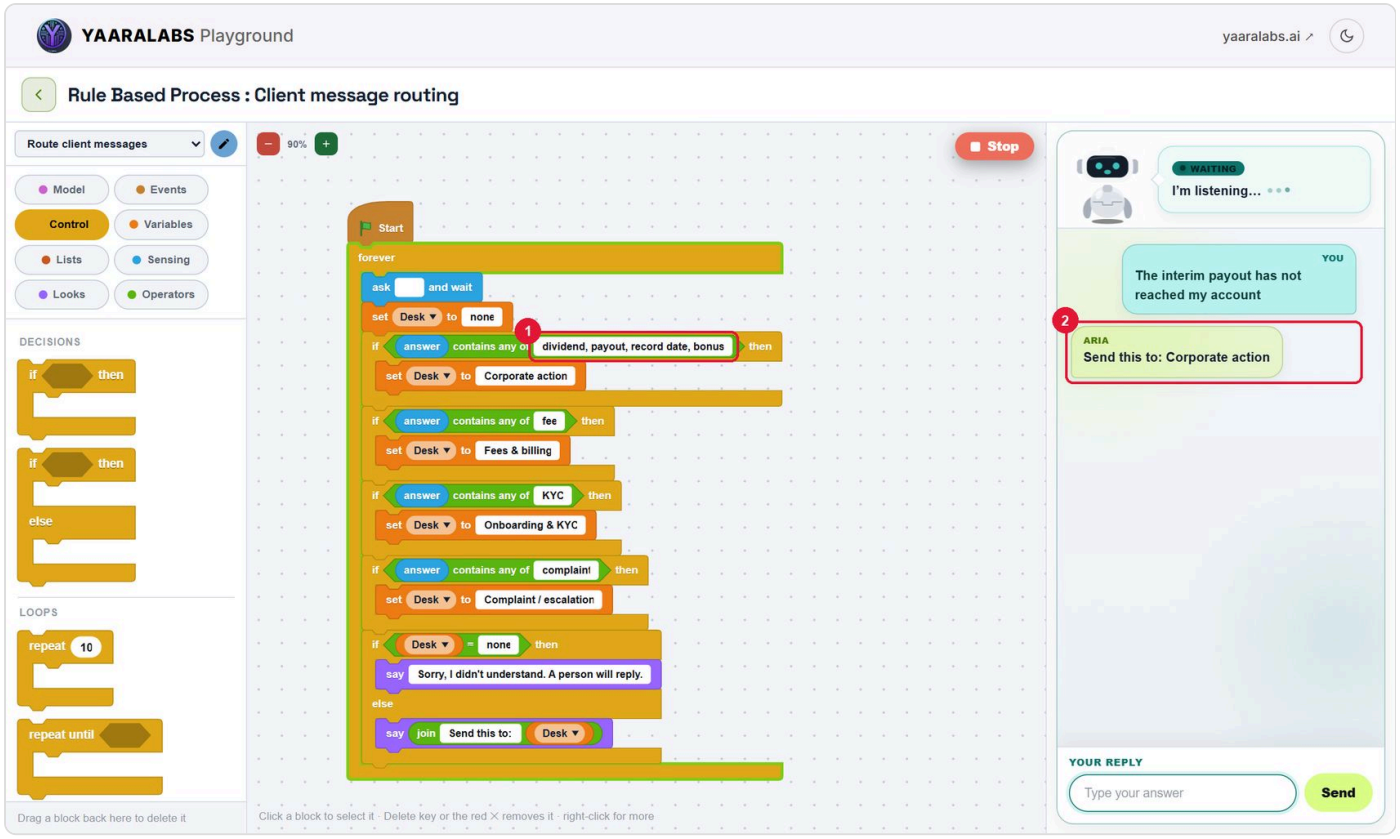
26 Add keywords, comma separated

Type more words after a comma: dividend, payout, record date, bonus.



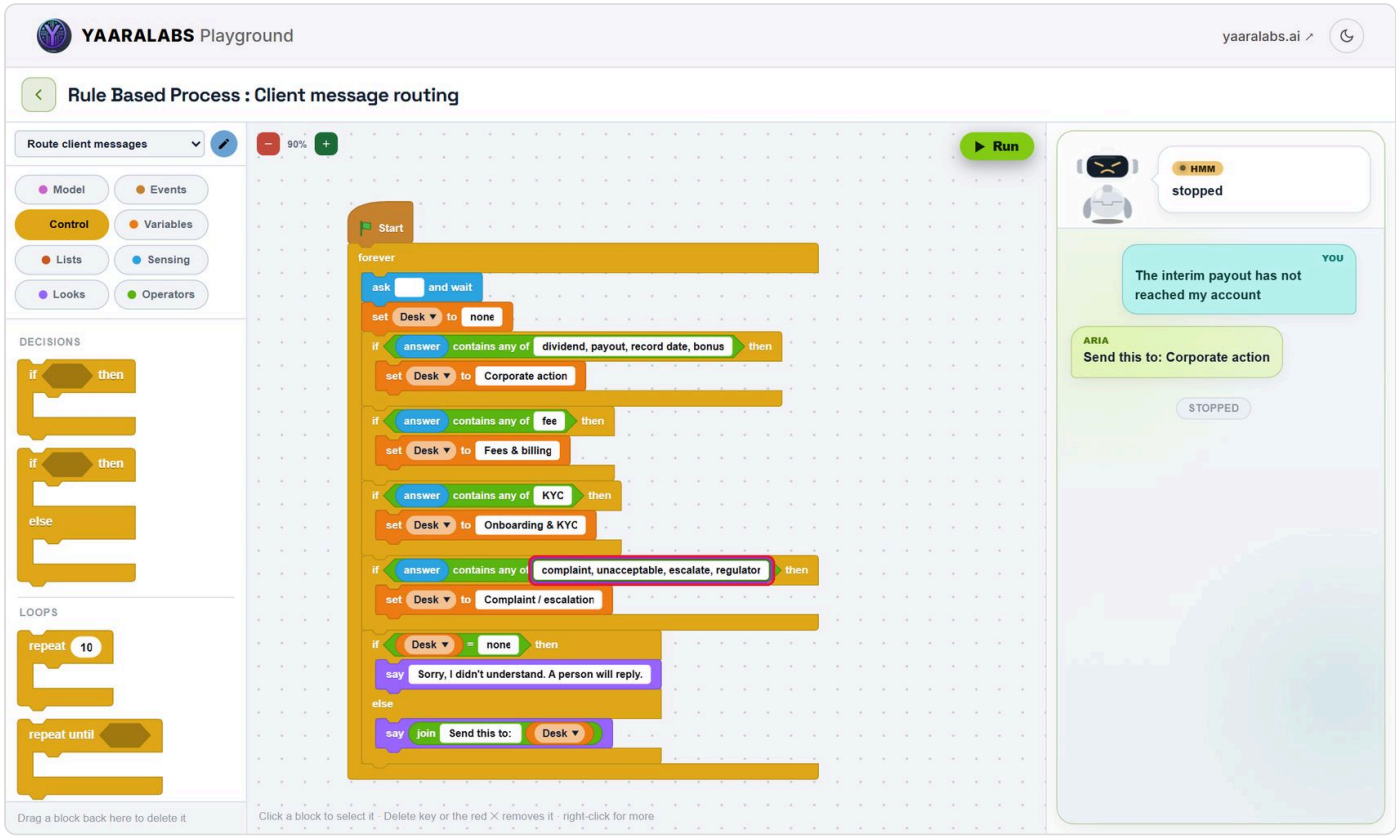
27 Run again: now it is routed

Click Run and type The interim payout has not reached my account: “payout” now matches, Send this to: Corporate action.



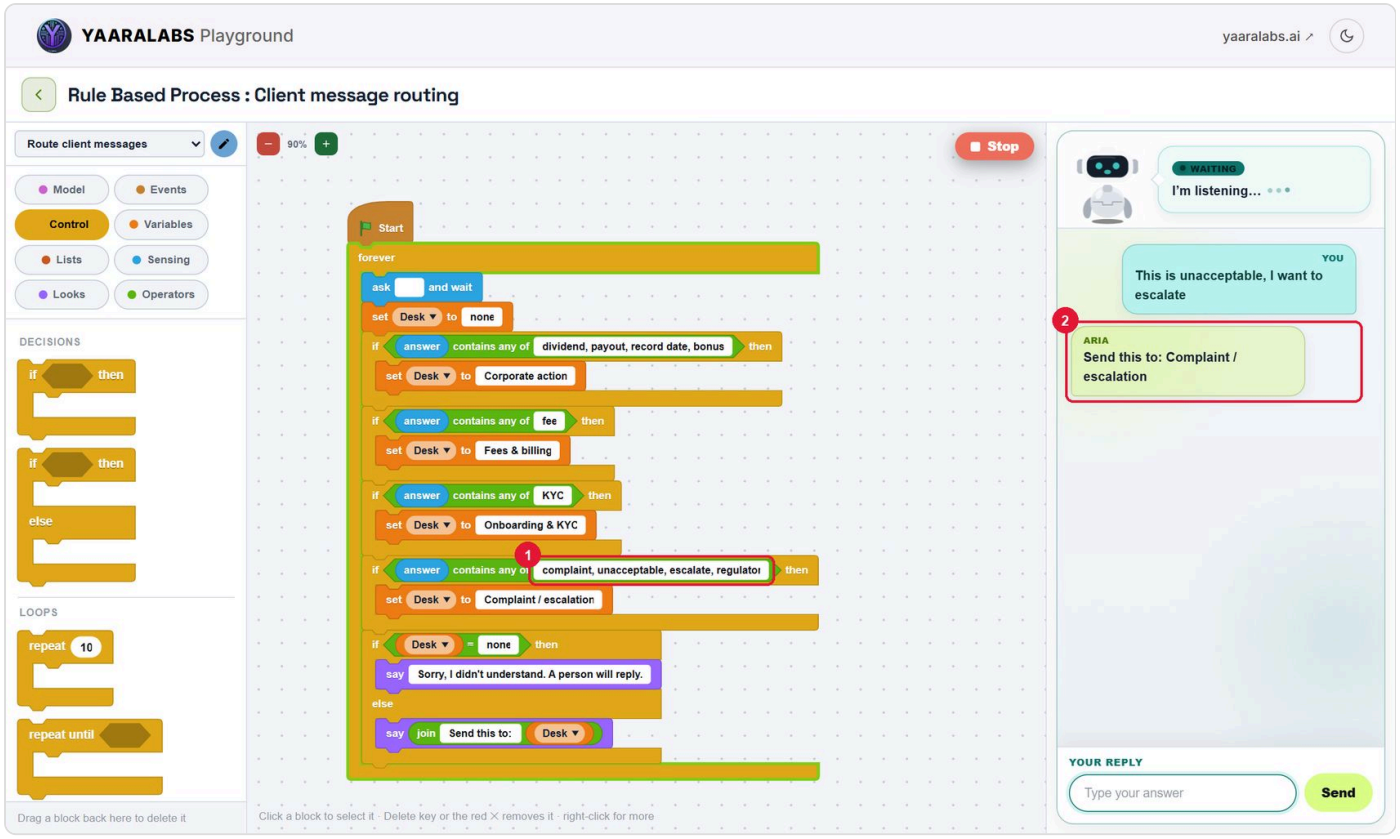
28 Do the same for Complaint / escalation

Stop the run, then in the complaint rule type: complaint, unacceptable, escalate, regulator.



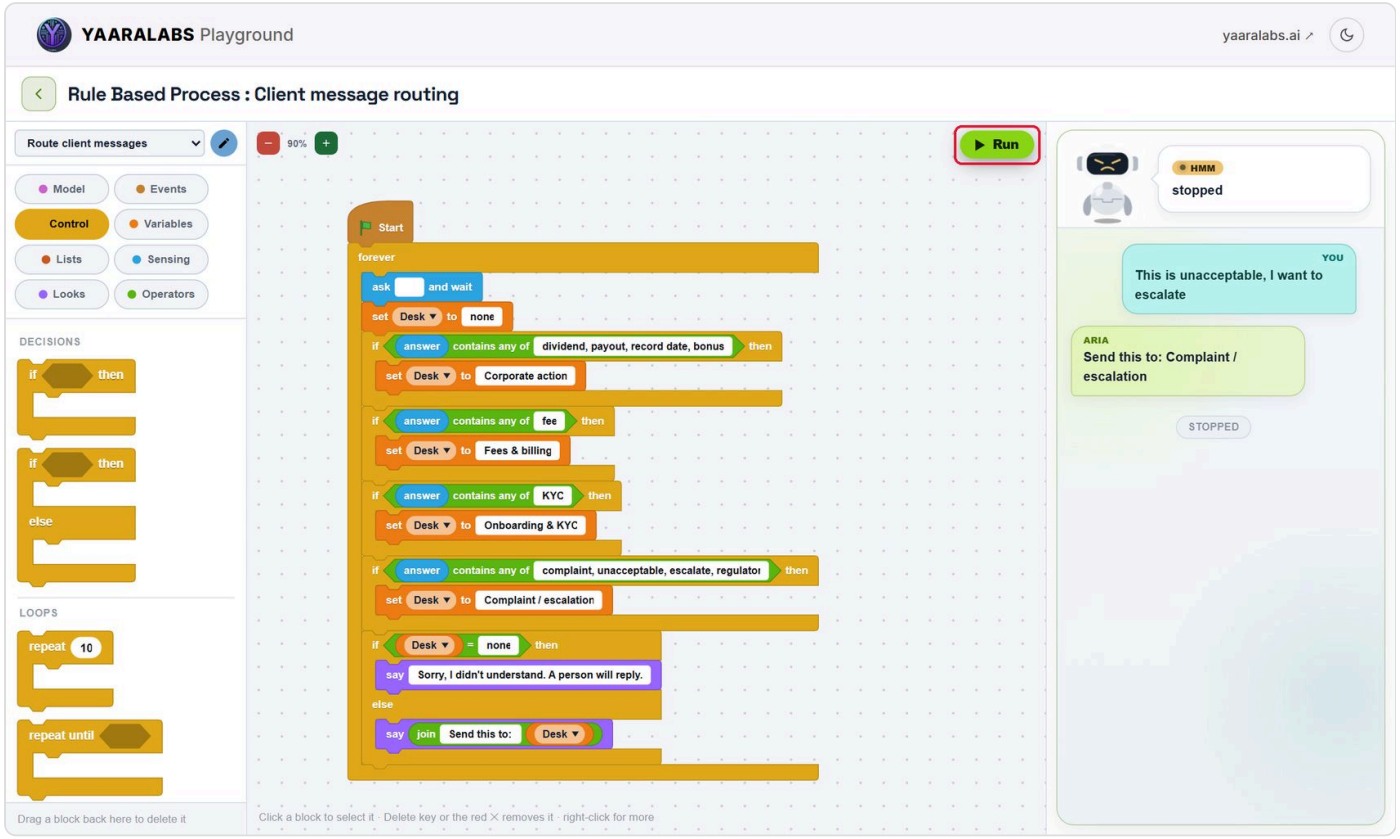
29 The escalation is routed

Run and type This is unacceptable, I want to escalate: now Send this to: Complaint / escalation.



30 Stop when you are done

Click Stop; the button turns back into Run. Your blocks are saved automatically.

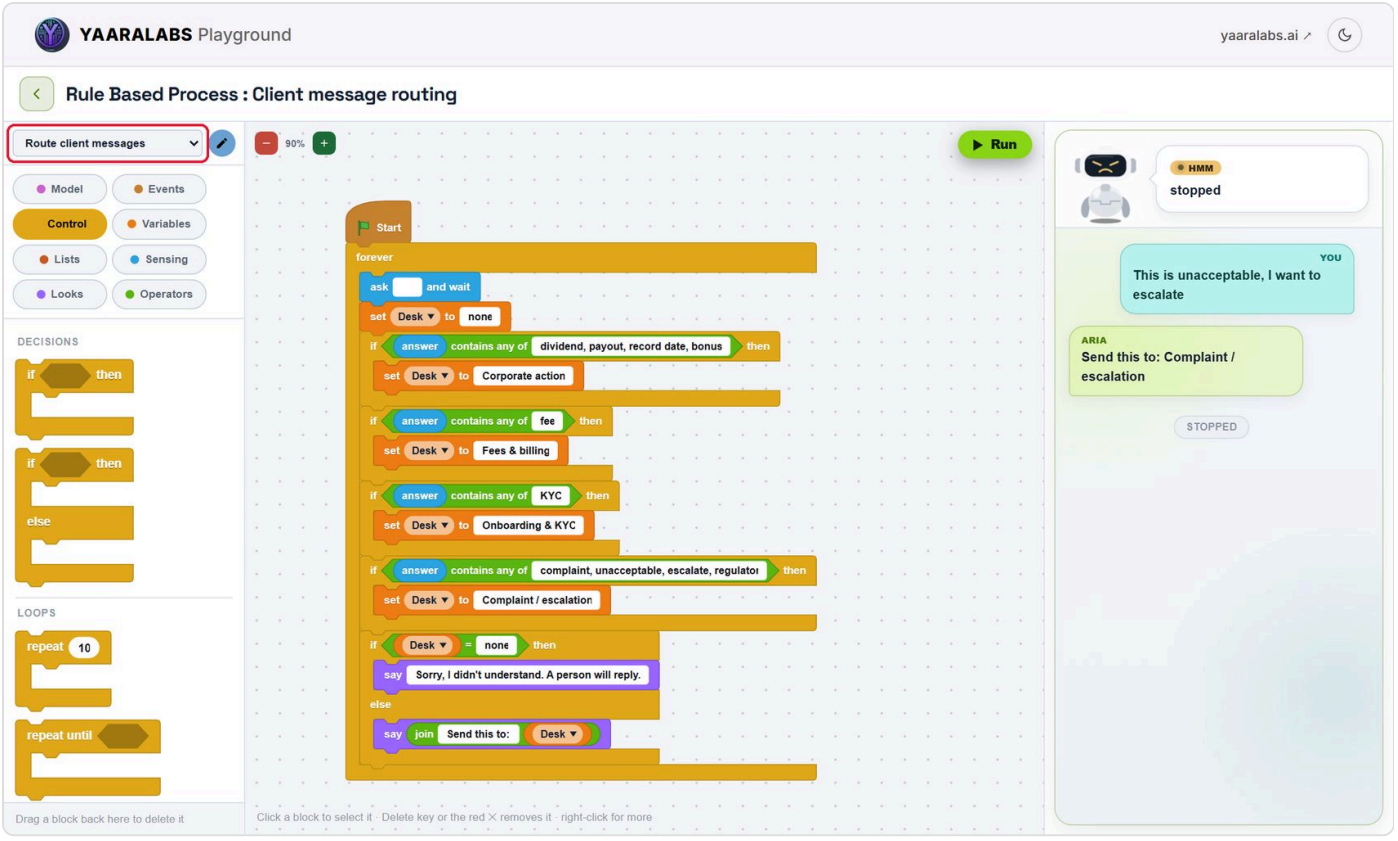


Other controls

The script list, the edit button, zoom and Back.

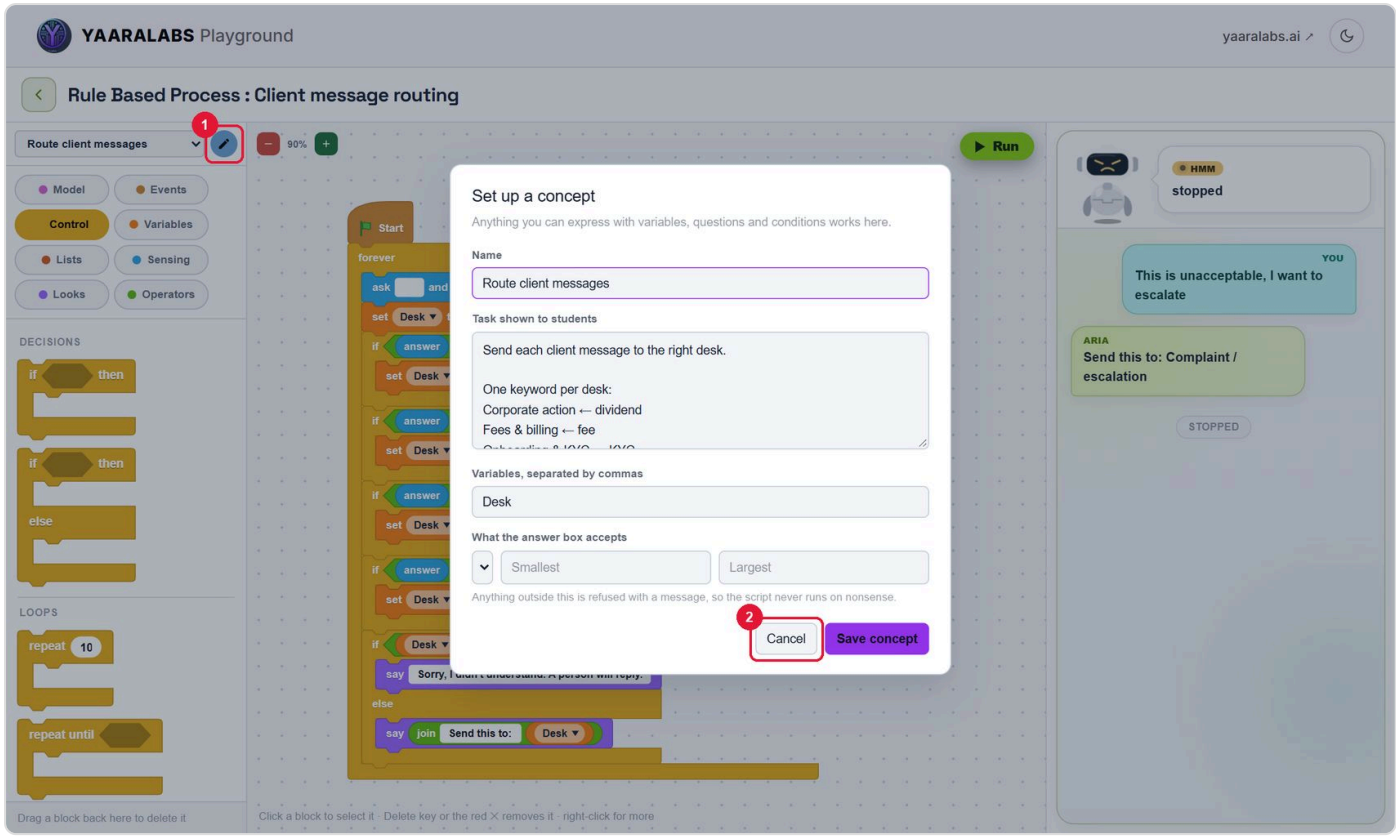
31 Choose a script

The dropdown at the top left lists other scripts to try (Marks to grade, Odd or even, Leap year…); Route client messages is this one.



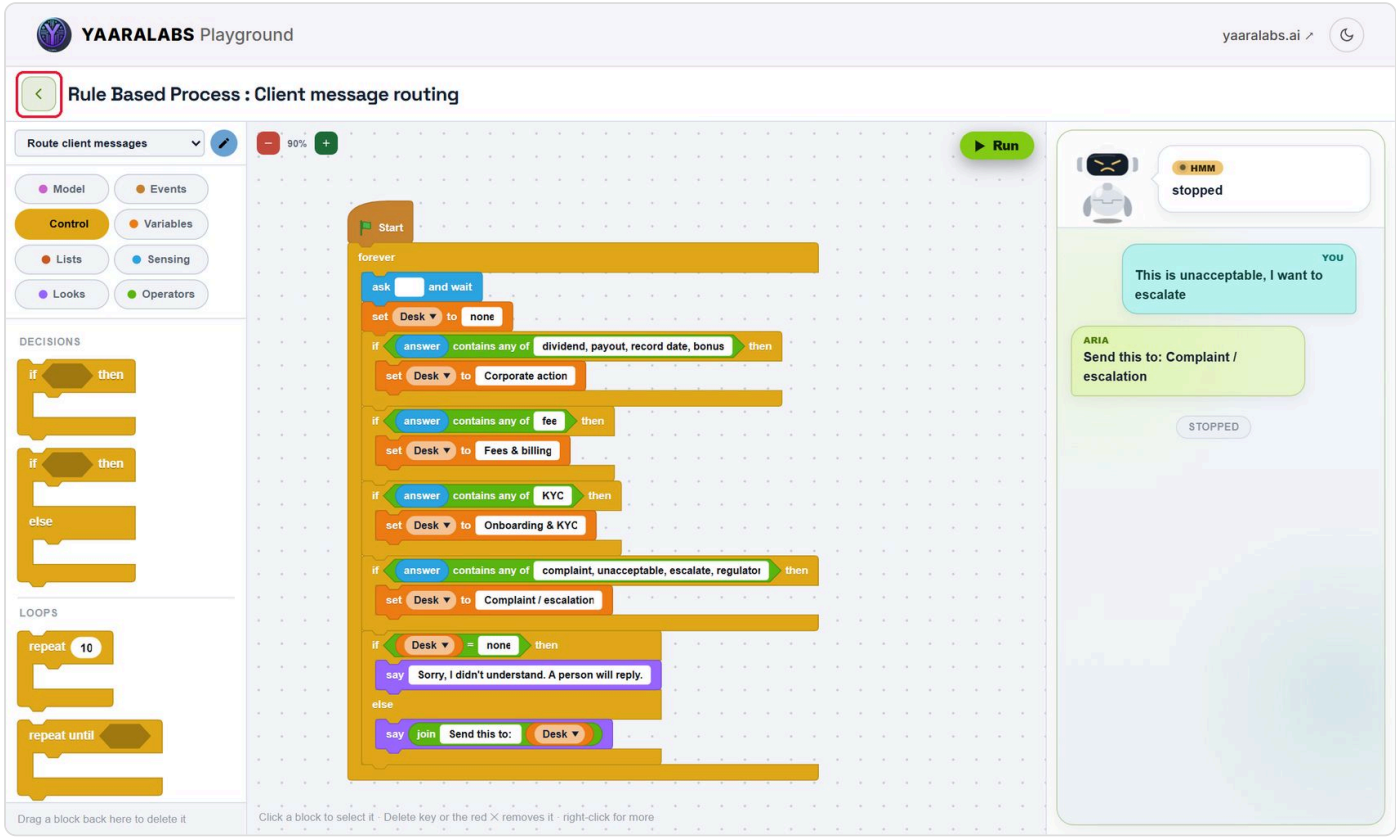
32 Edit the script details

The pencil opens Set up a concept: name, task text and variables. Click Cancel to close without changes.



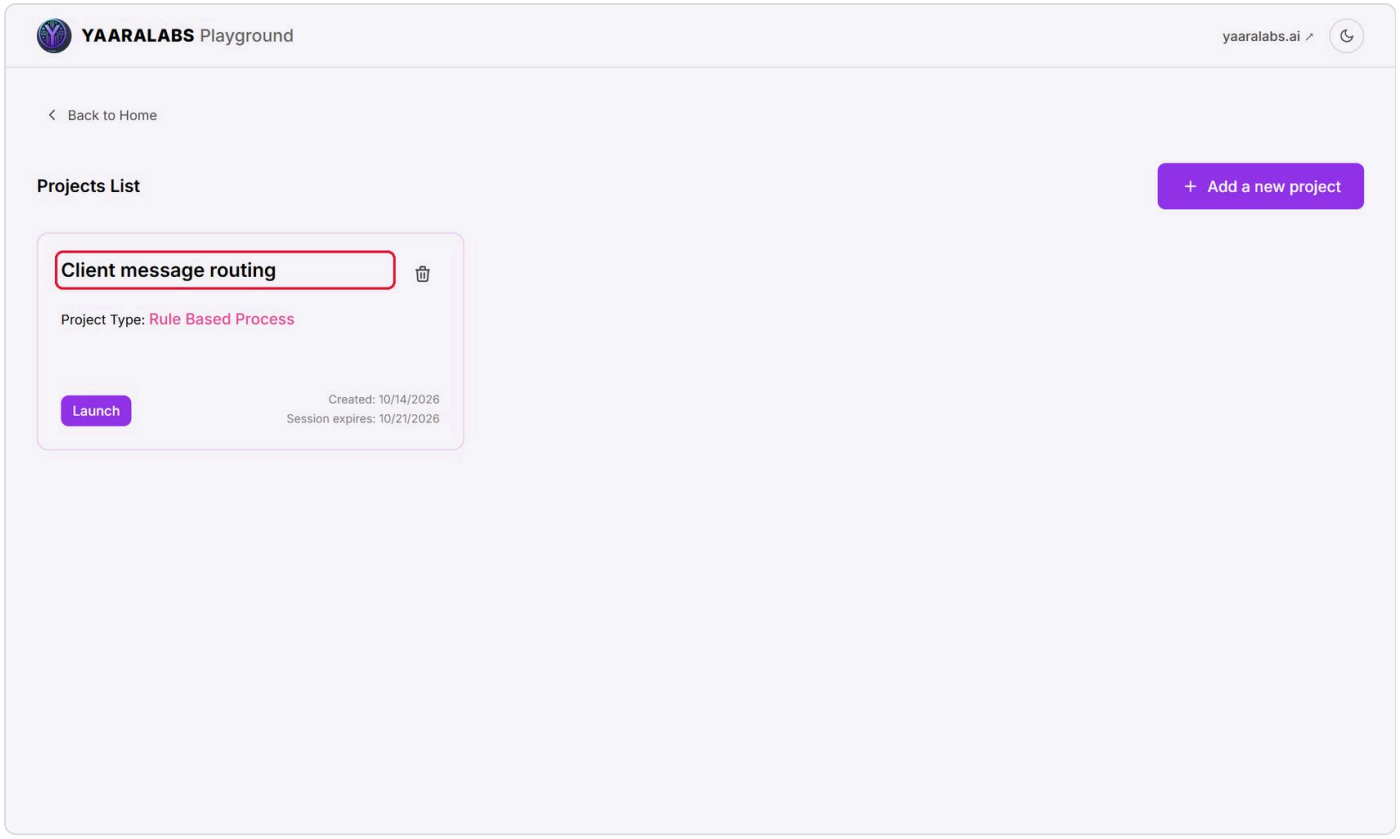
33 Back to your projects

Click the back arrow to return to your projects list.



34 Your project is saved

The project is in your list: click it any time to open the editor again.

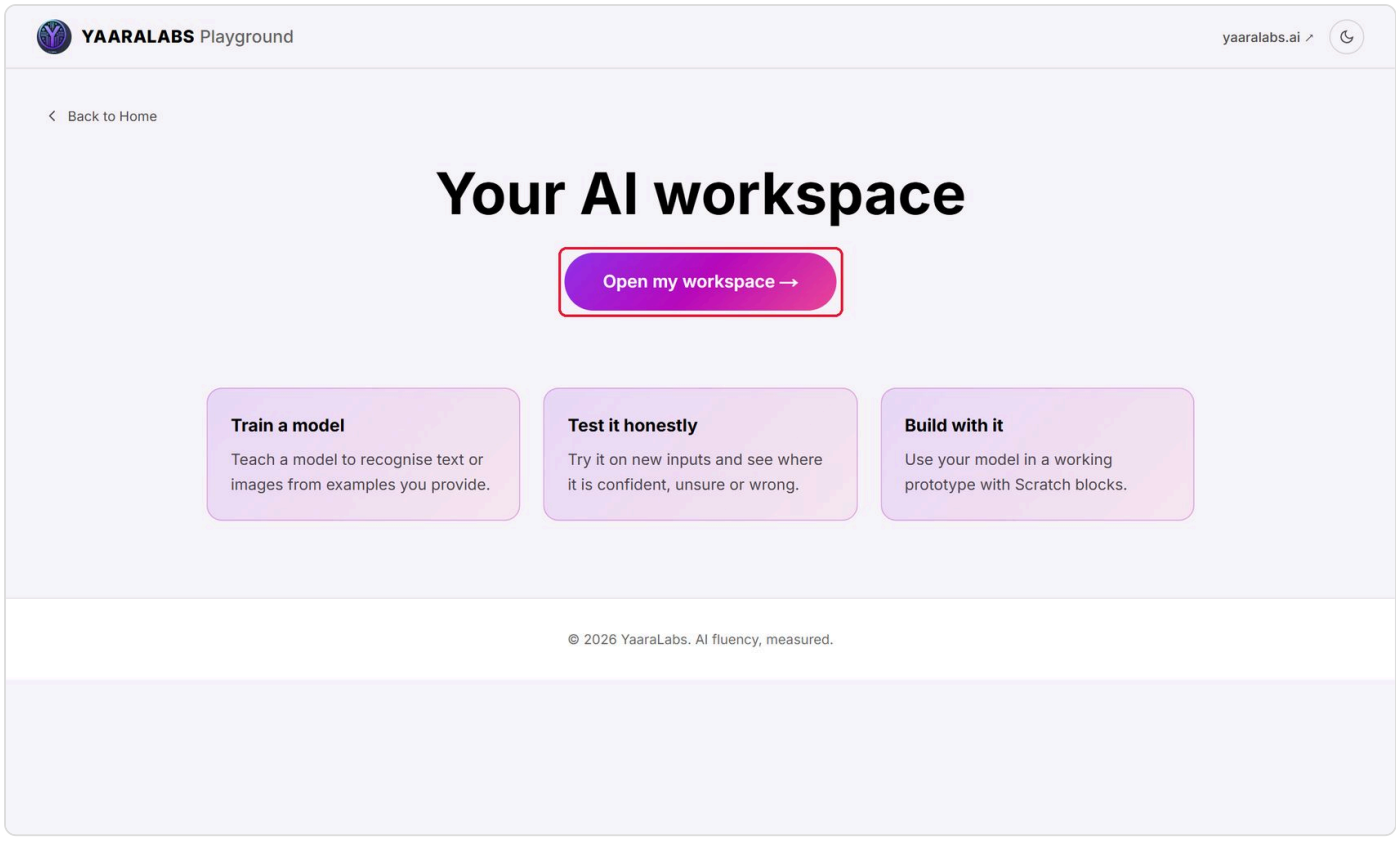


Experiment 2: Text Recognition

Teach an AI to sort Nuvama custody client queries into four desks, test it, then use it in a chatbot.

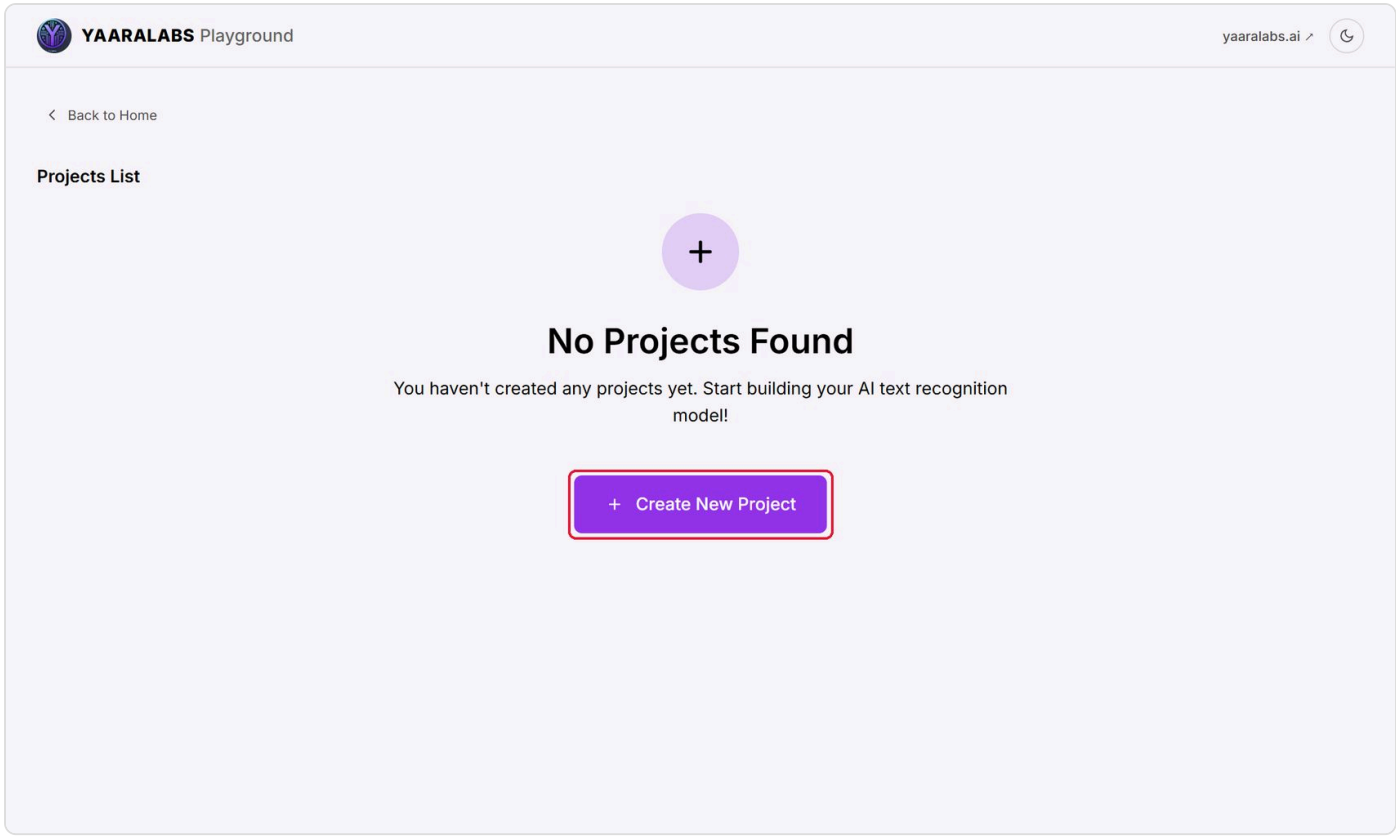
35 Open the workspace

Open the site and click Open my workspace.



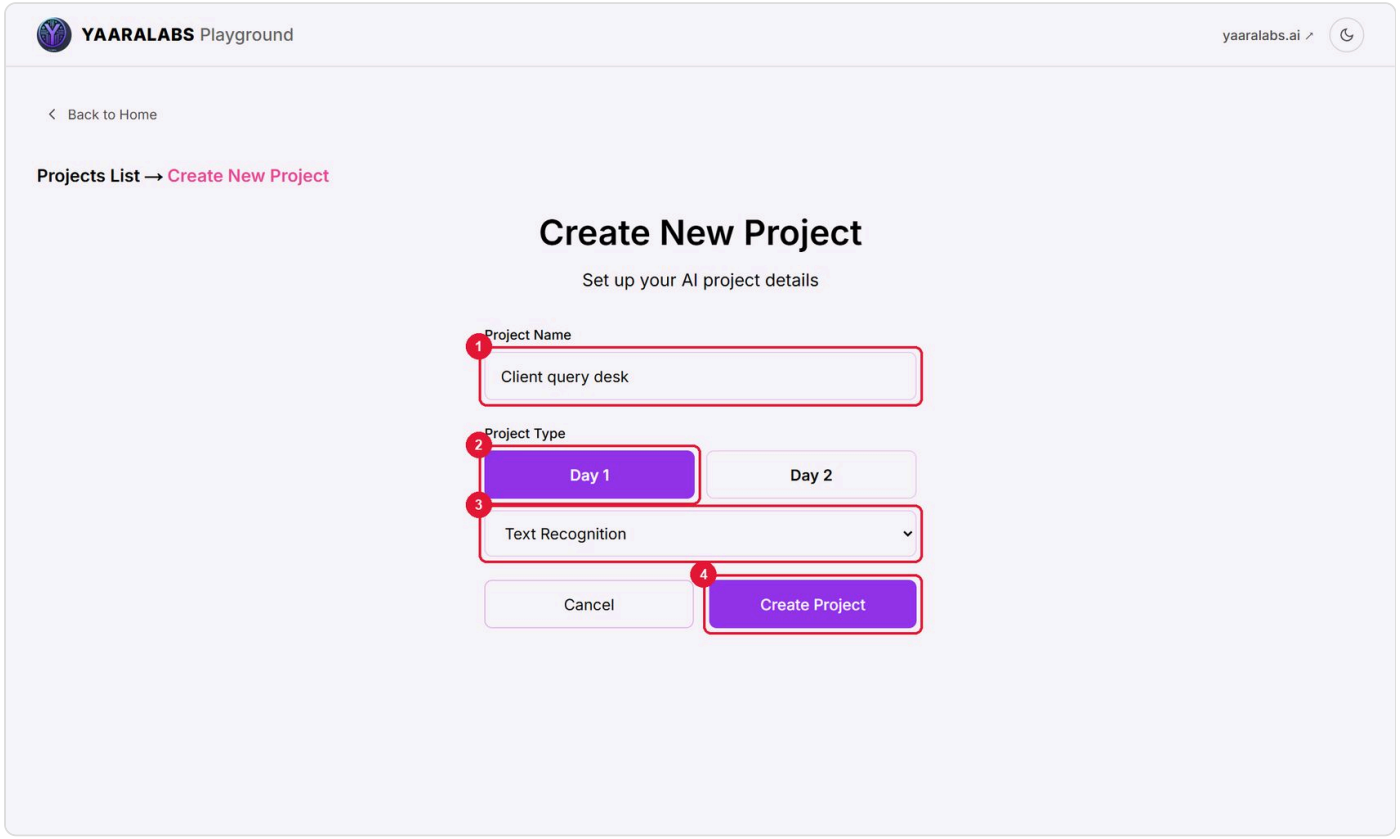
36 Create a new project

Click Create New Project.



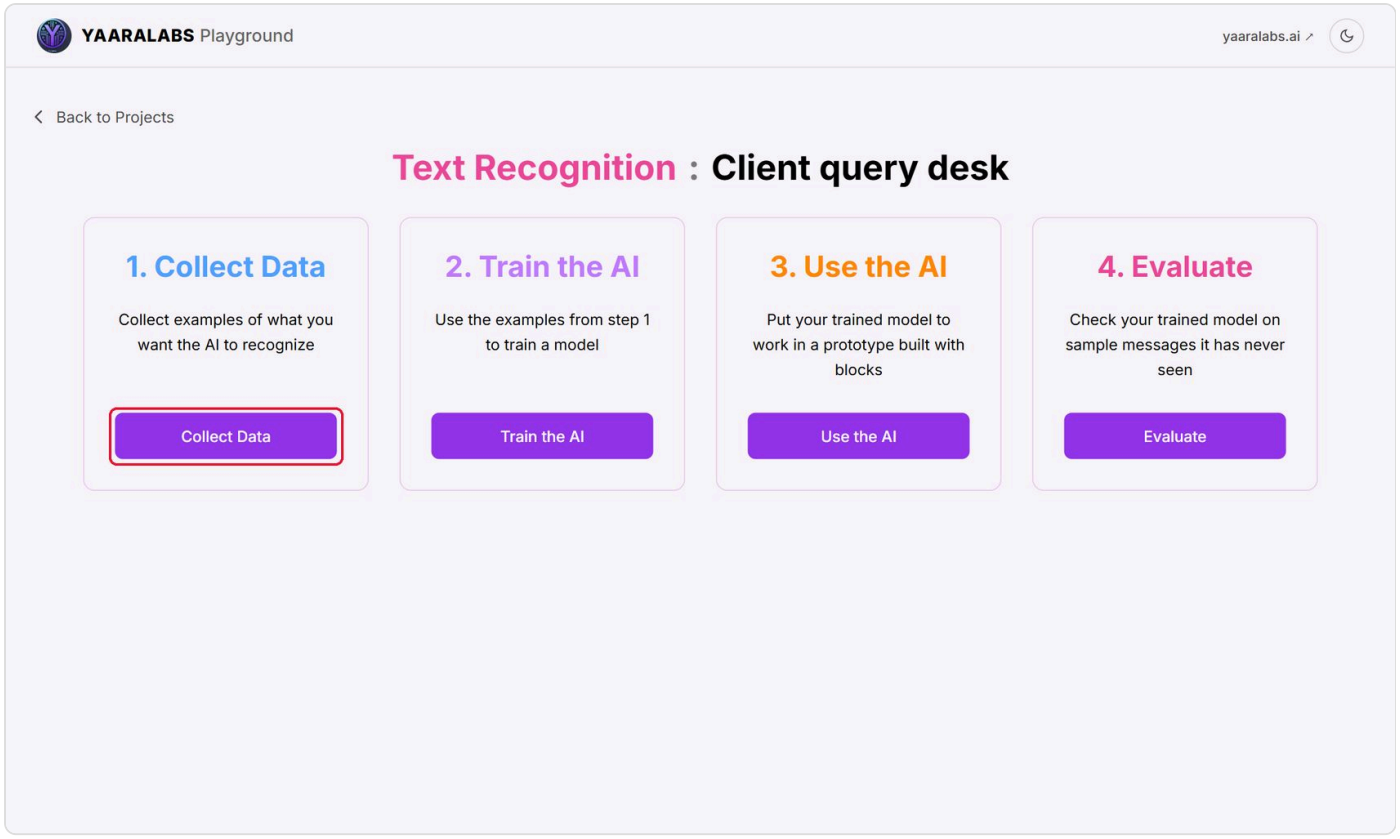
37 Name it and choose Text Recognition

Type a name (Client query desk), pick Day 1, choose Text Recognition, then click Create Project.



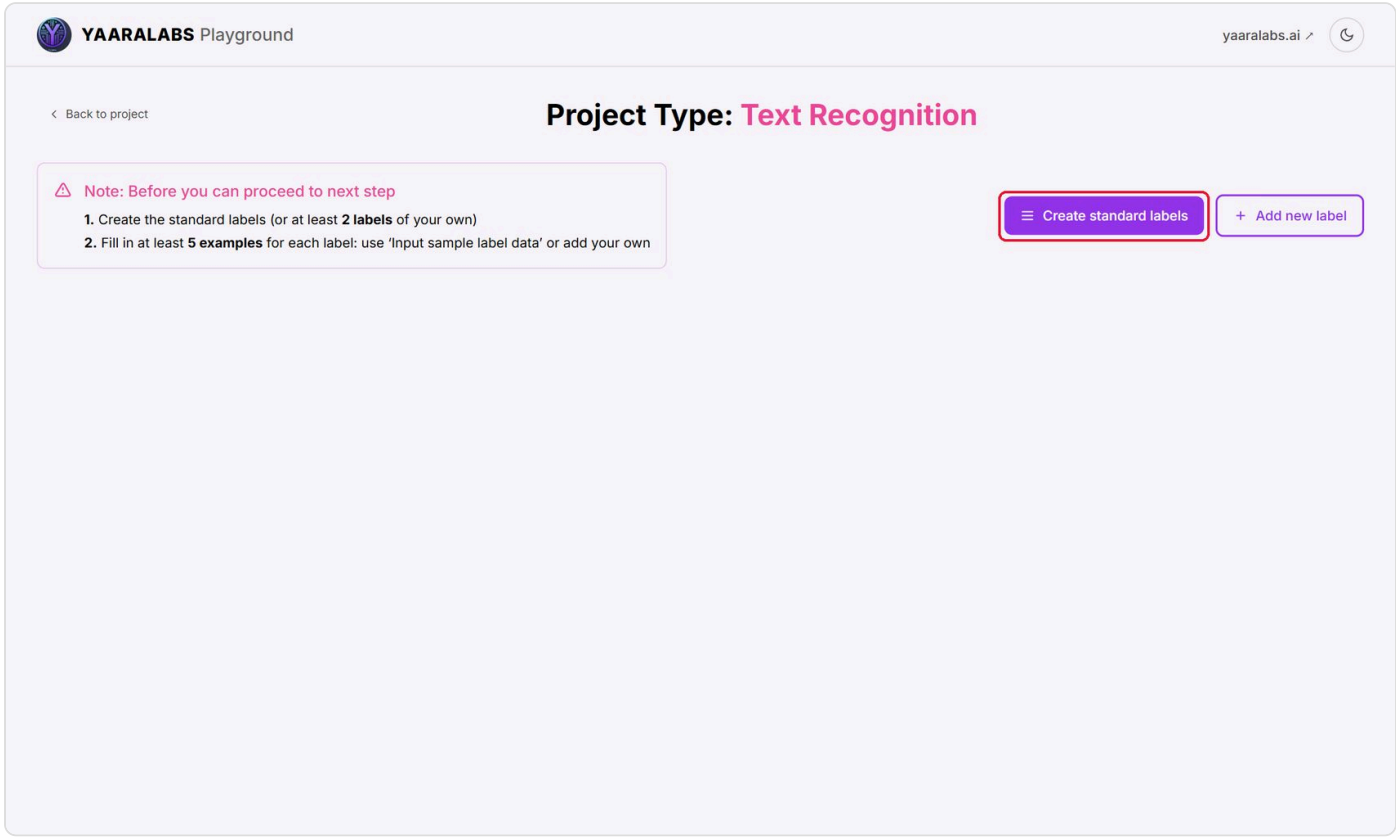
38 The project page

The project has four steps: Collect Data, Train the AI, Use the AI, Evaluate. Click Collect Data.



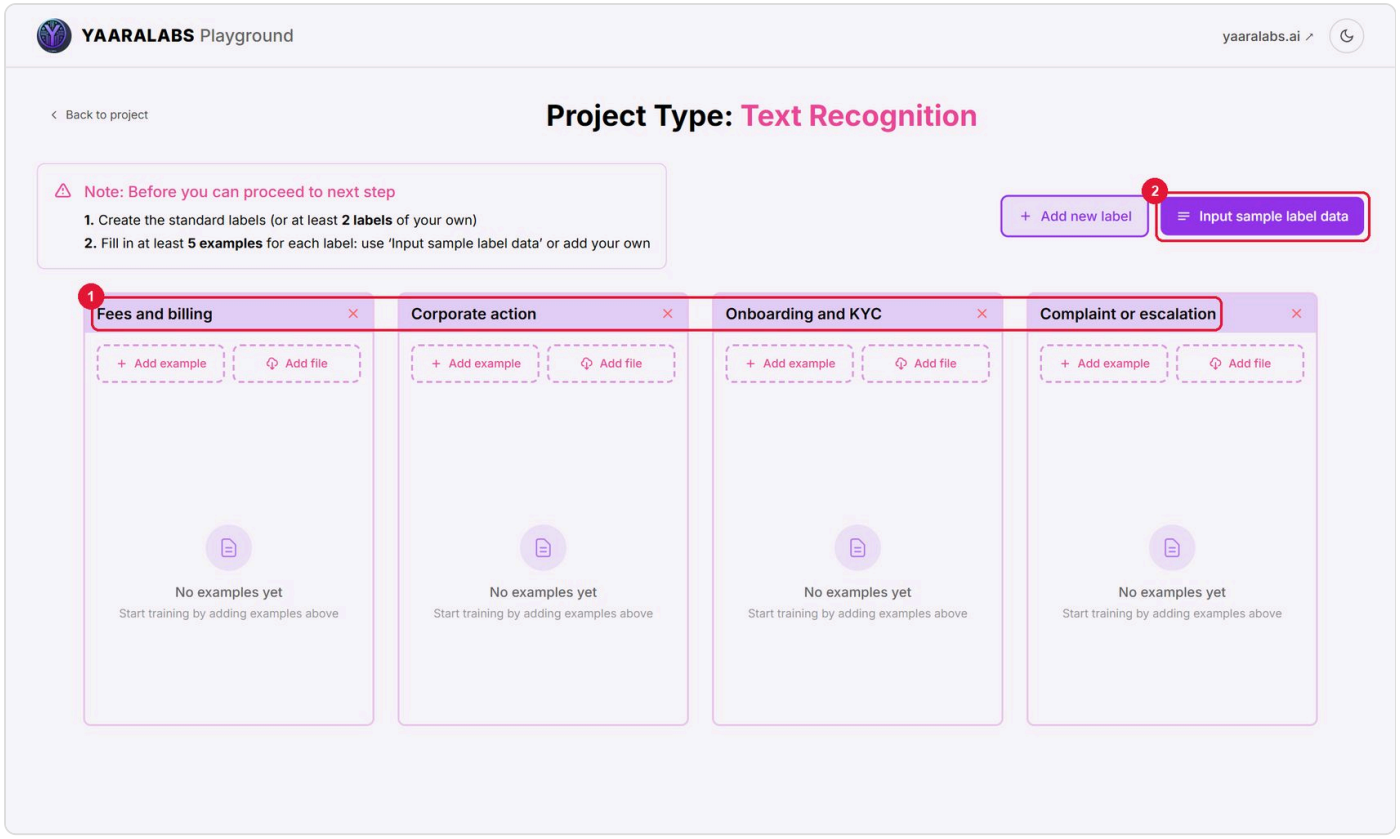
39 The data page

Read the note at the top, then click Create standard labels.



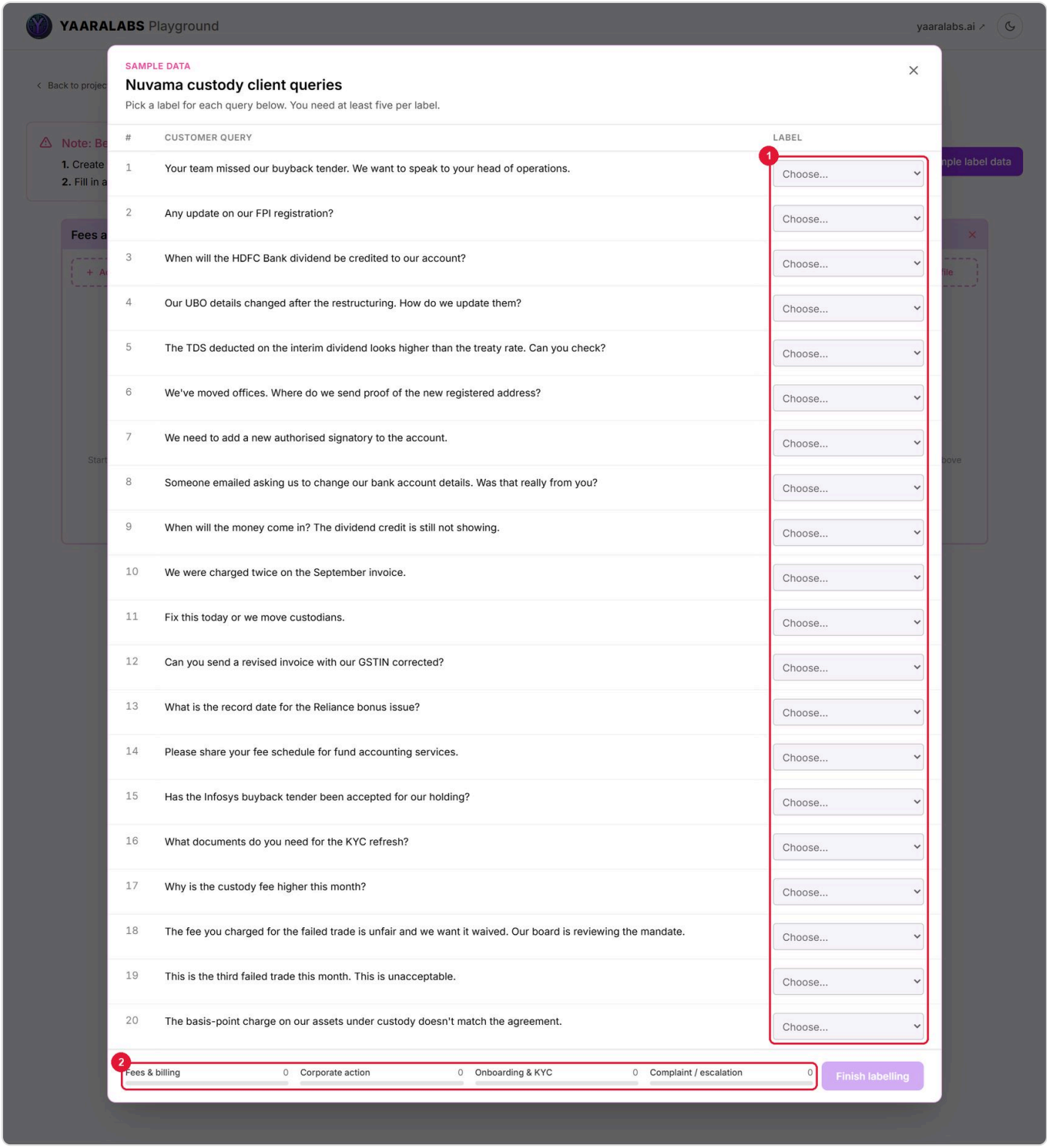
40 Four labels, no examples yet

Four desks appear as labels, each still empty. Click Input sample label data.



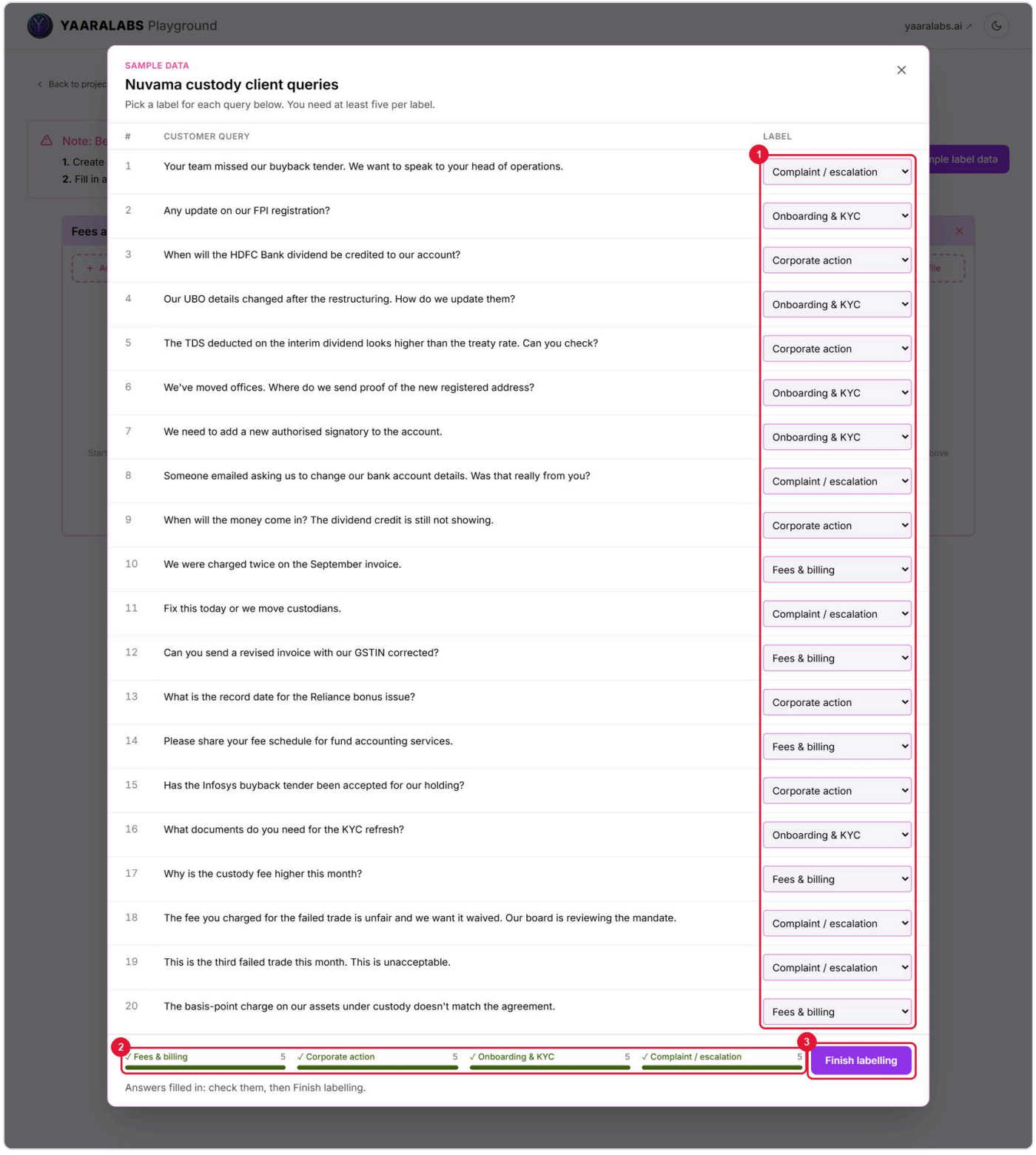
41 Sample data: the client queries

Each client query needs a label from its menu. Press Ctrl + Shift + Z to fill the answers.



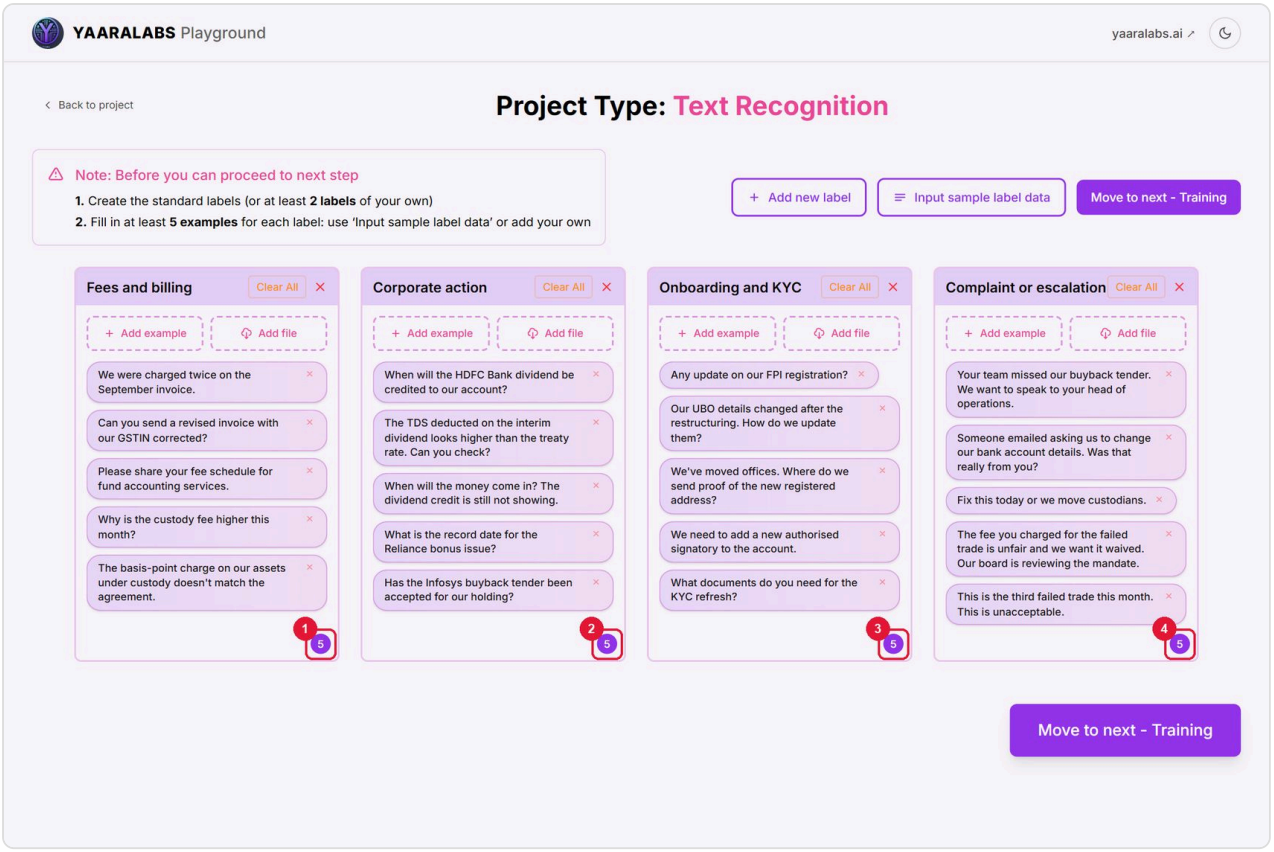
42 Every query labelled

Ctrl + Shift + Z picks the correct label for every query: each label now has 5. Click Finish labelling.



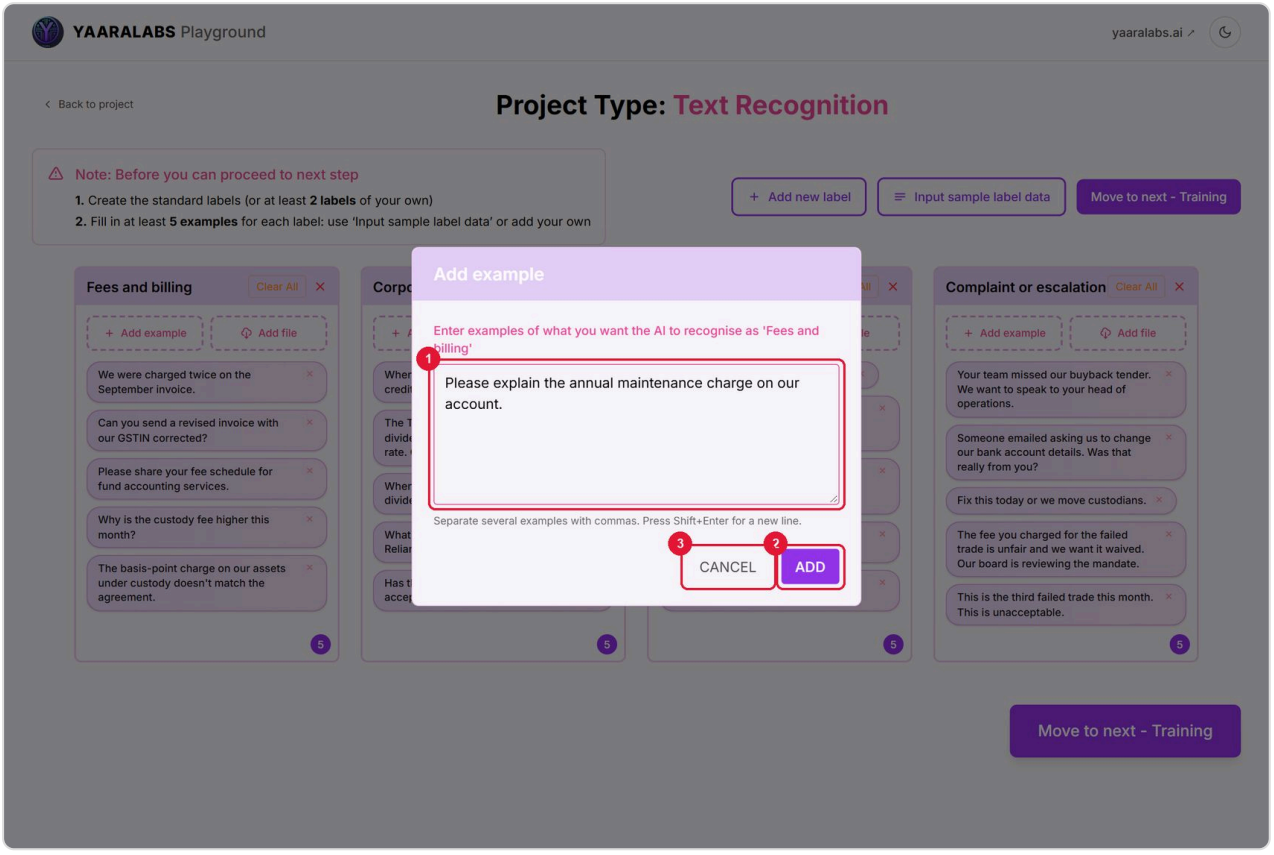
43 Labels with their examples

Each label now holds its 5 example queries (the number in the corner).



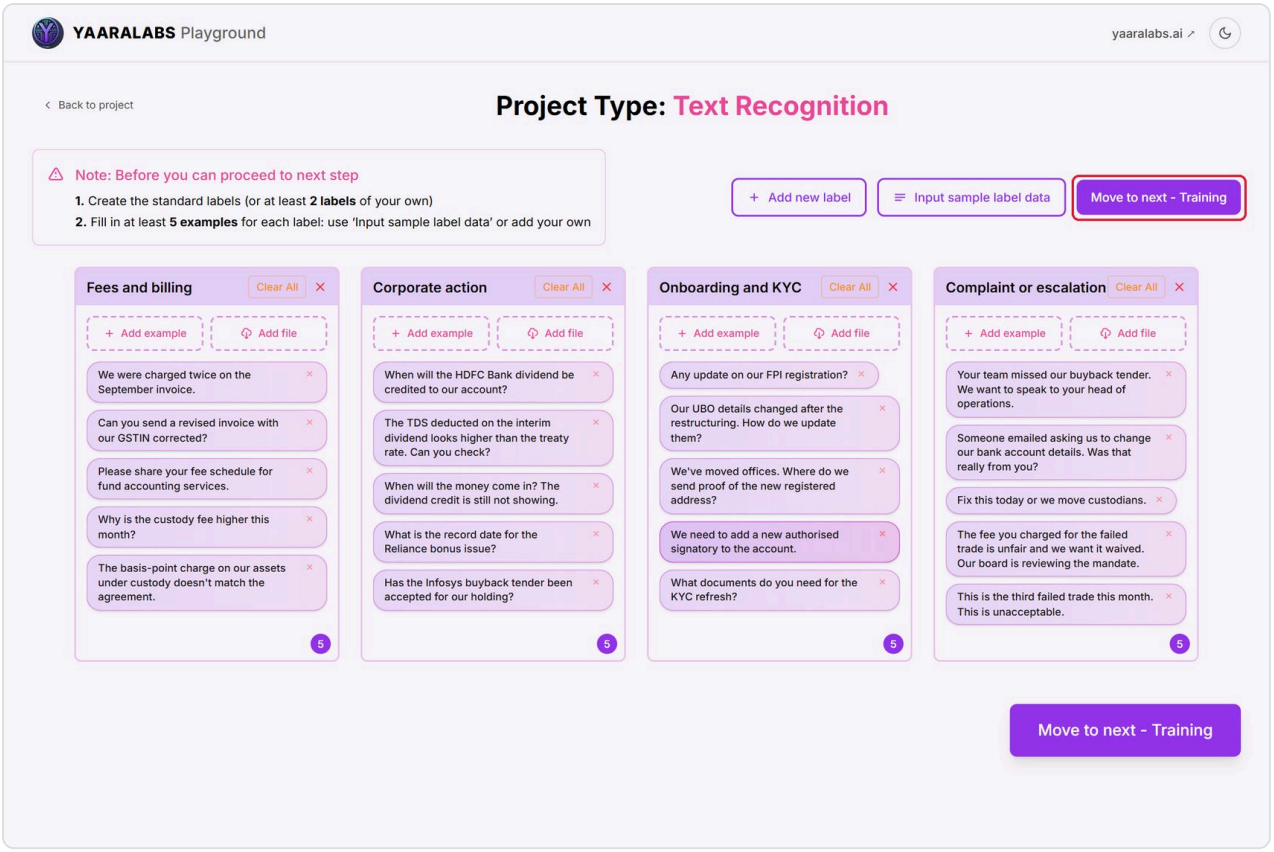
44 Add an example by hand

To add your own, click + Add example under a label, type the query and click ADD (CANCEL closes it).



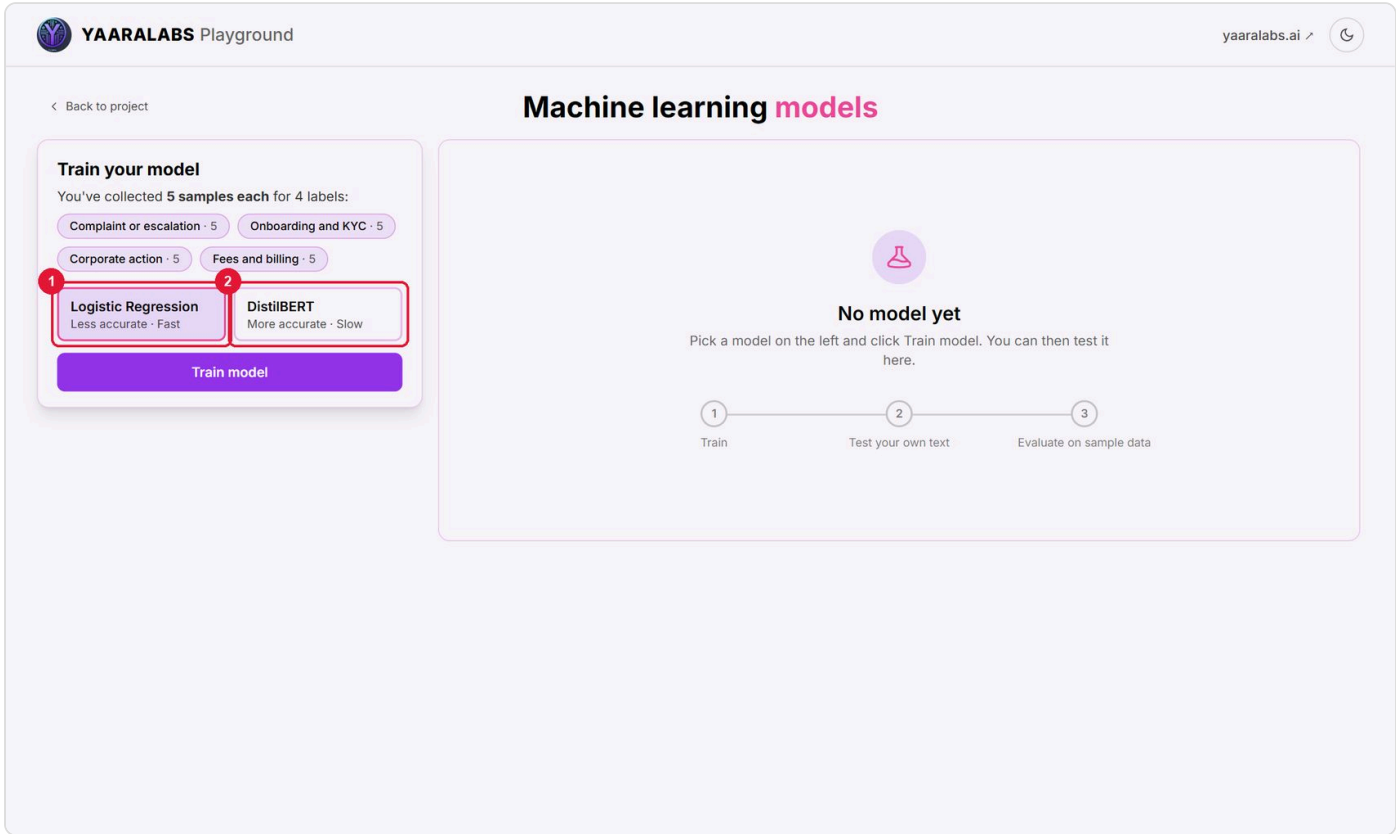
45 Go to training

The data is ready. Click Move to next - Training.



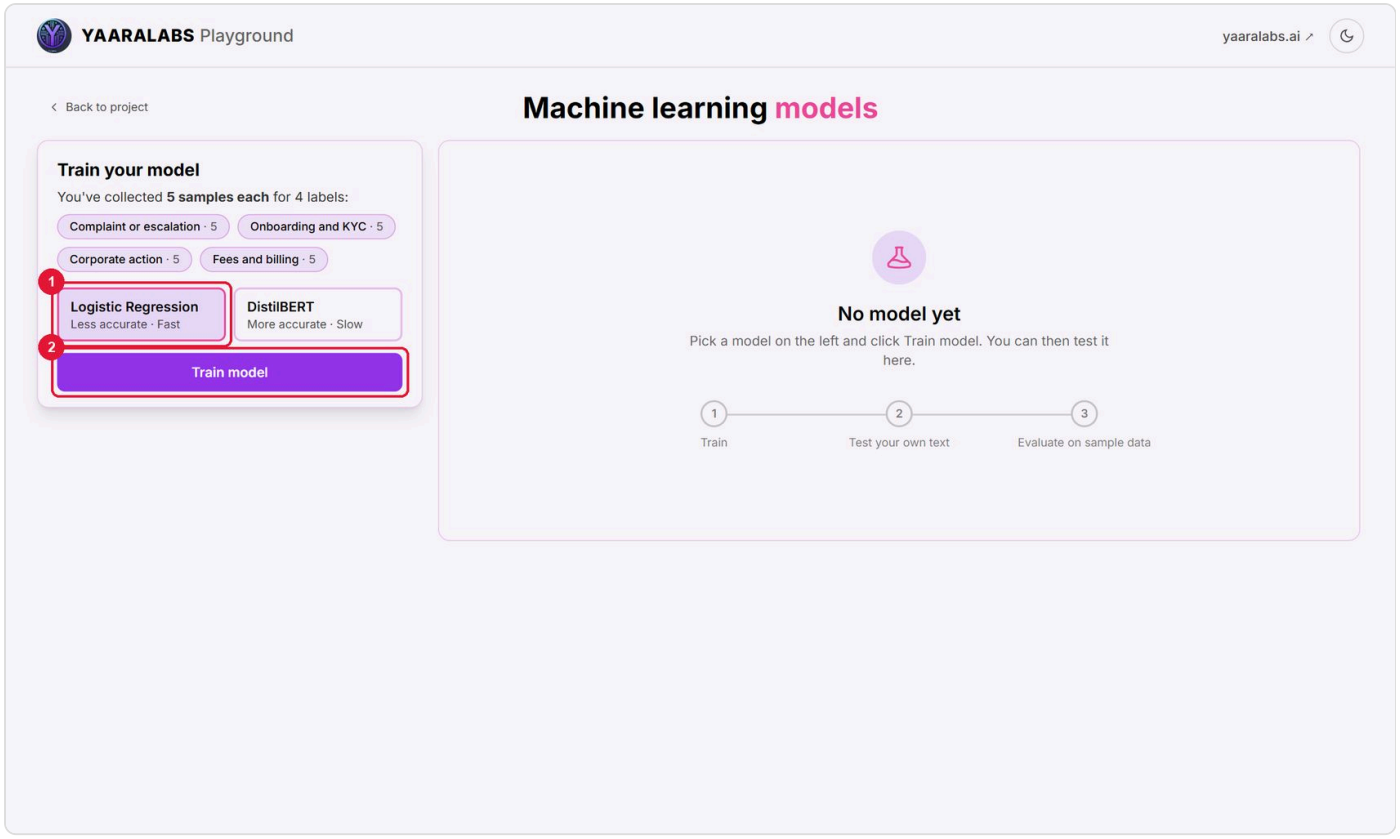
46 Train your model

Two models: Logistic Regression (less accurate, fast) and DistilBERT (more accurate, slow).



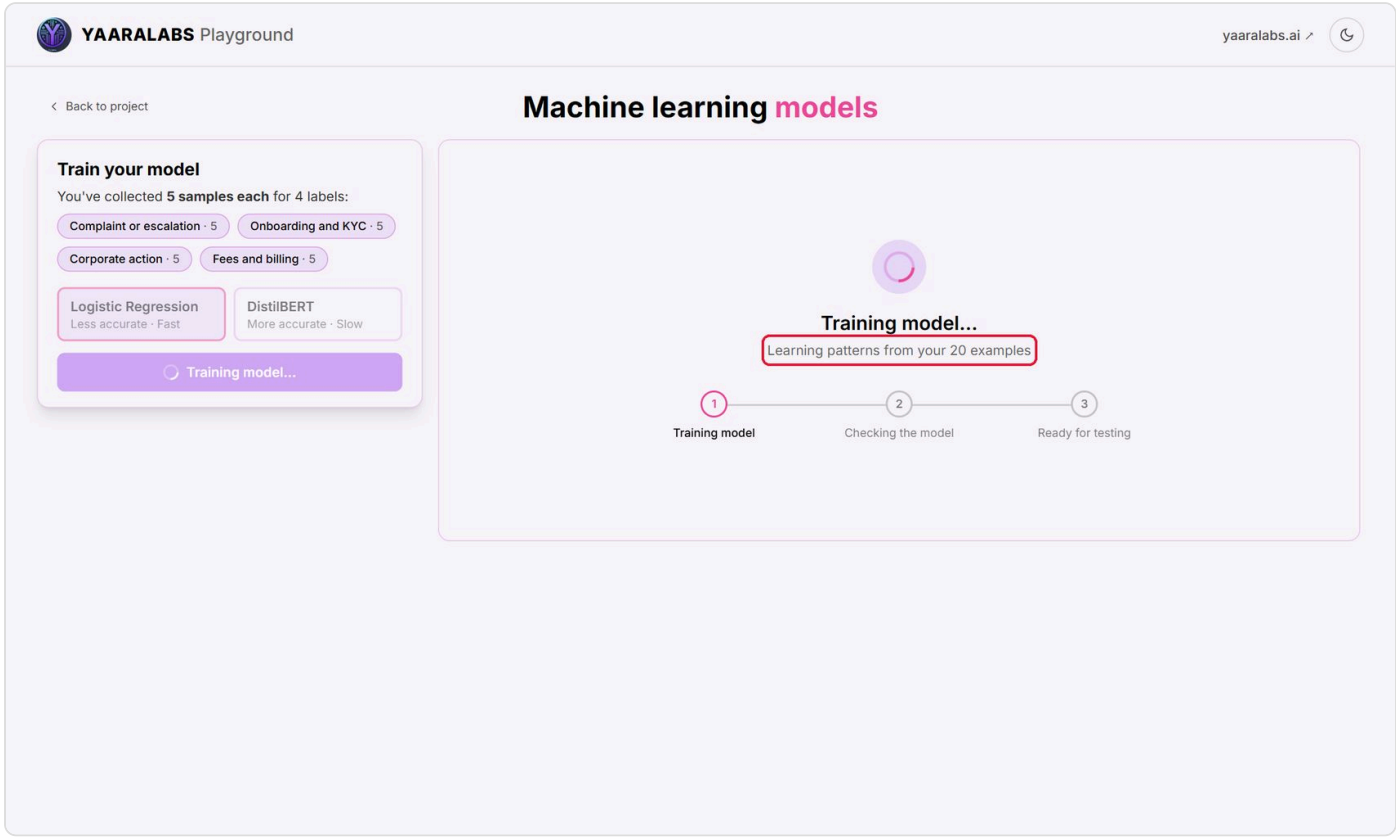
47 Pick Logistic Regression

Click Logistic Regression, then click Train model.



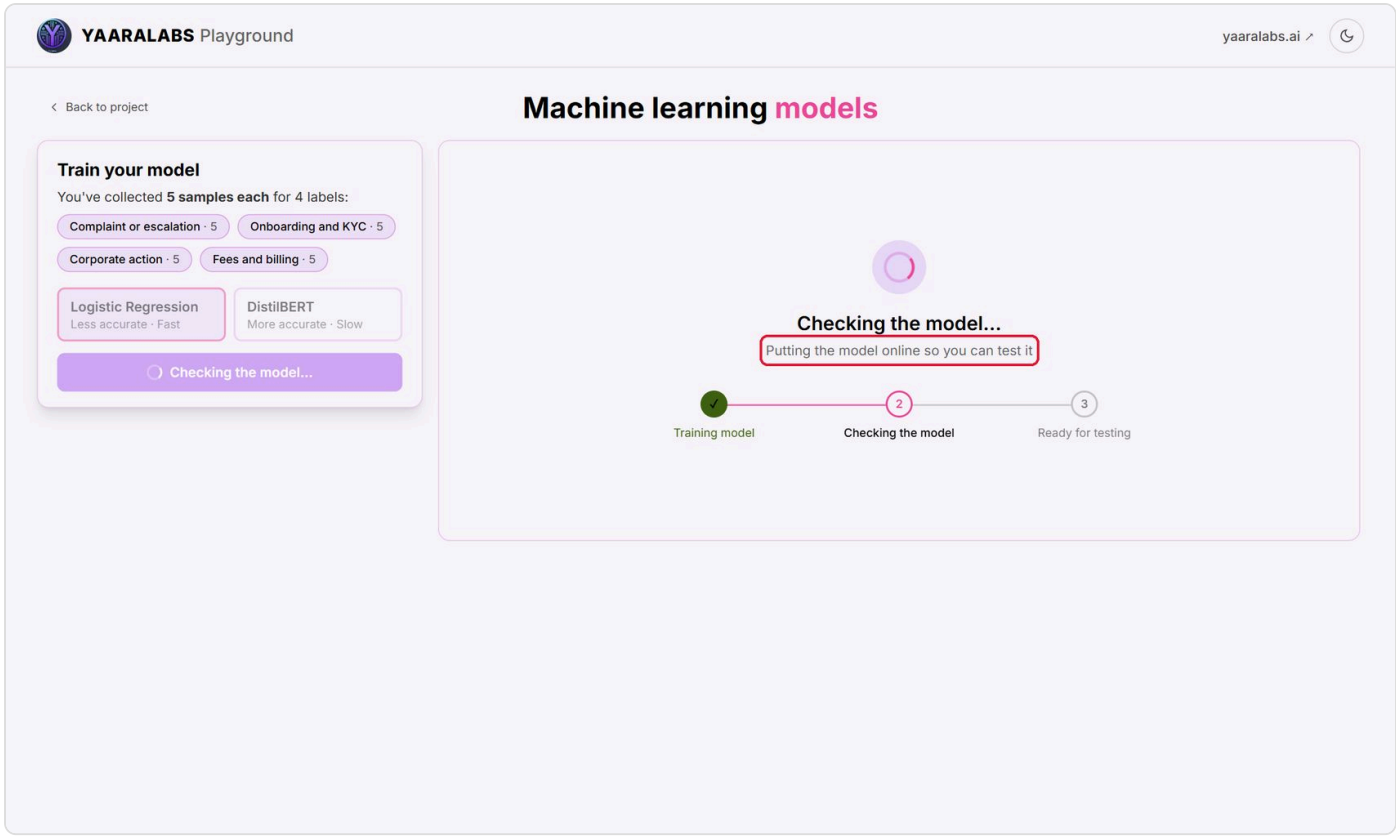
48 Training model

Step 1: the model learns patterns from your 20 examples.



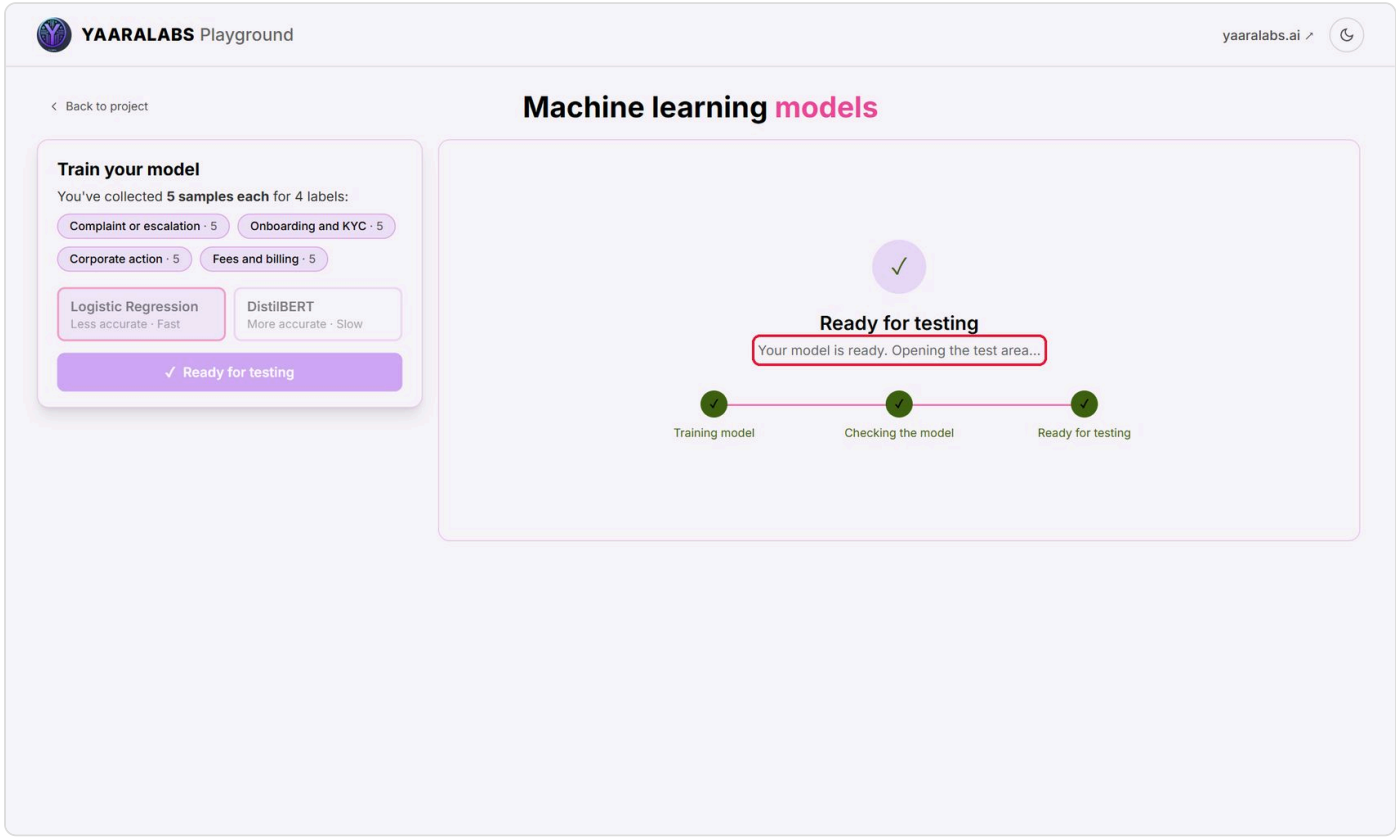
49 Checking the model

Step 2: the model is checked and put online.



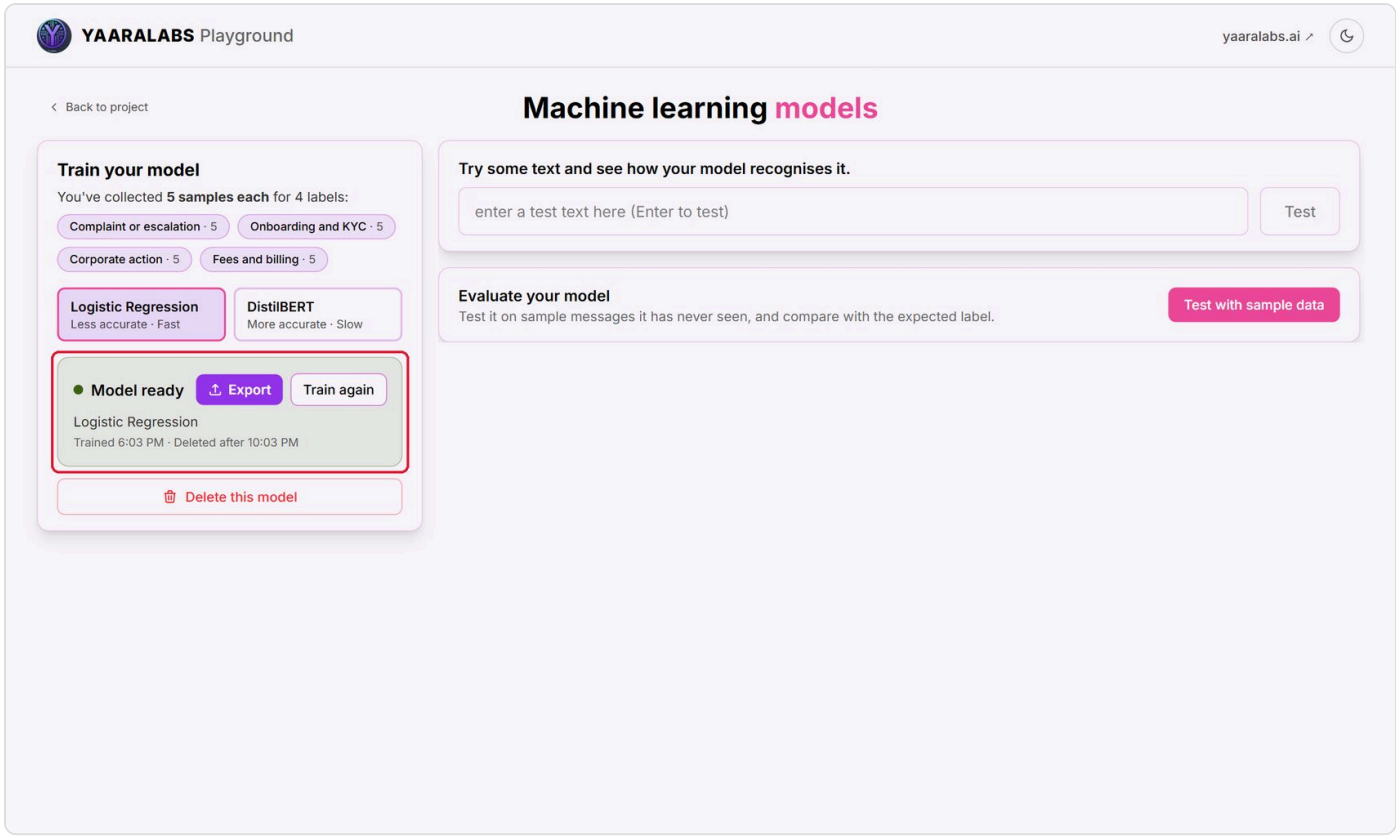
50 Ready for testing

Step 3: the model is ready; the test area opens next.



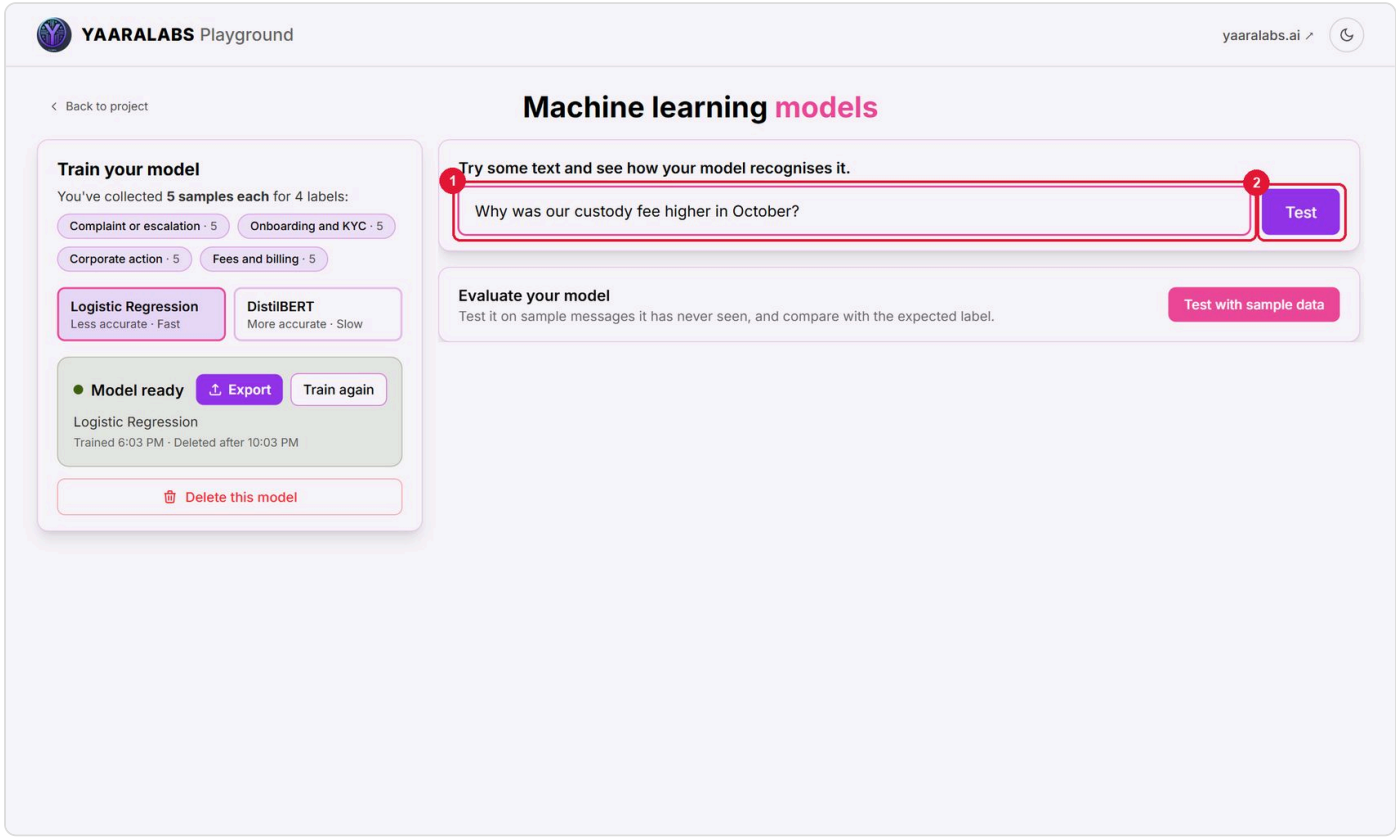
51 Model ready

The Model ready card shows the trained model, Export and Train again.



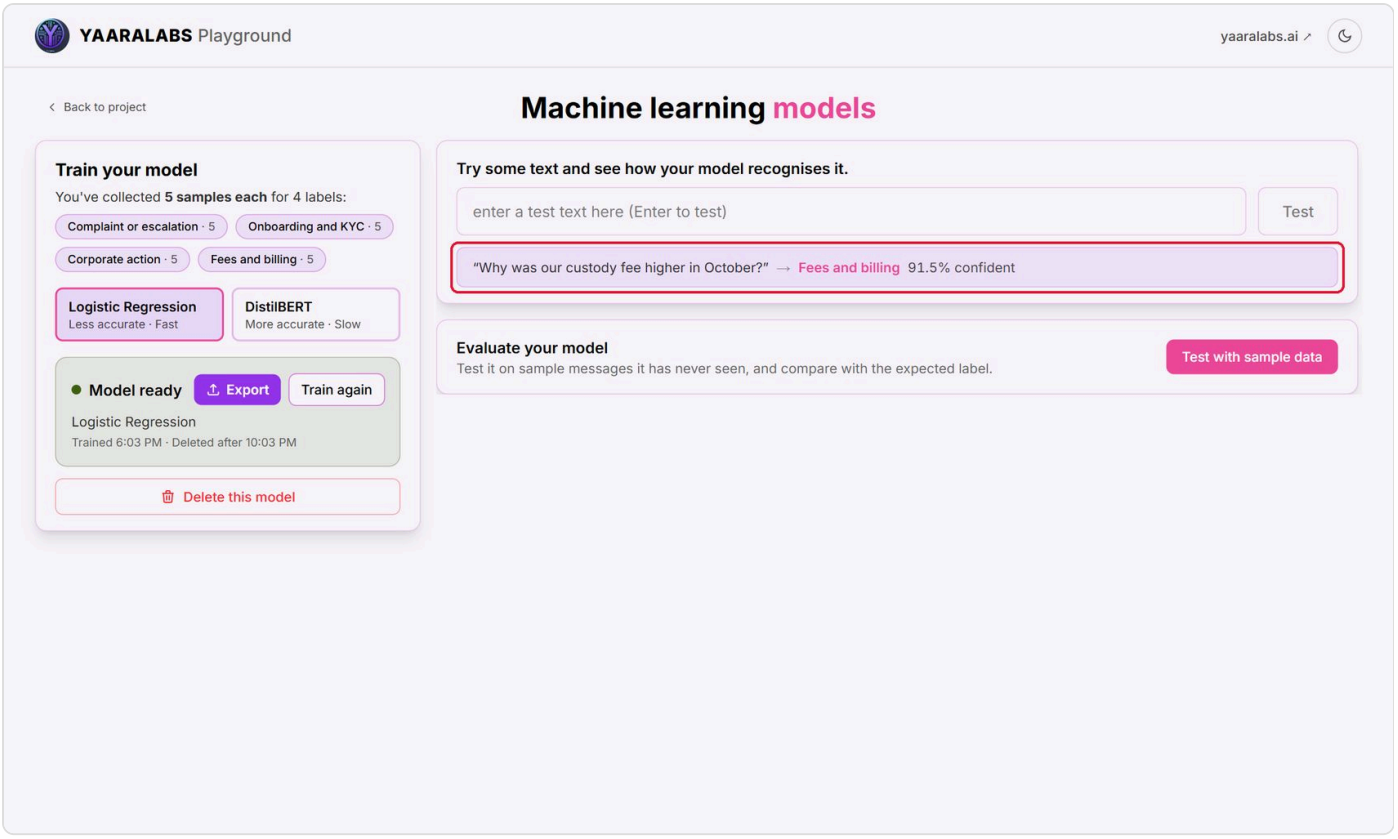
52 Try some text

Type a client message in Try some text and click Test (or press Enter).



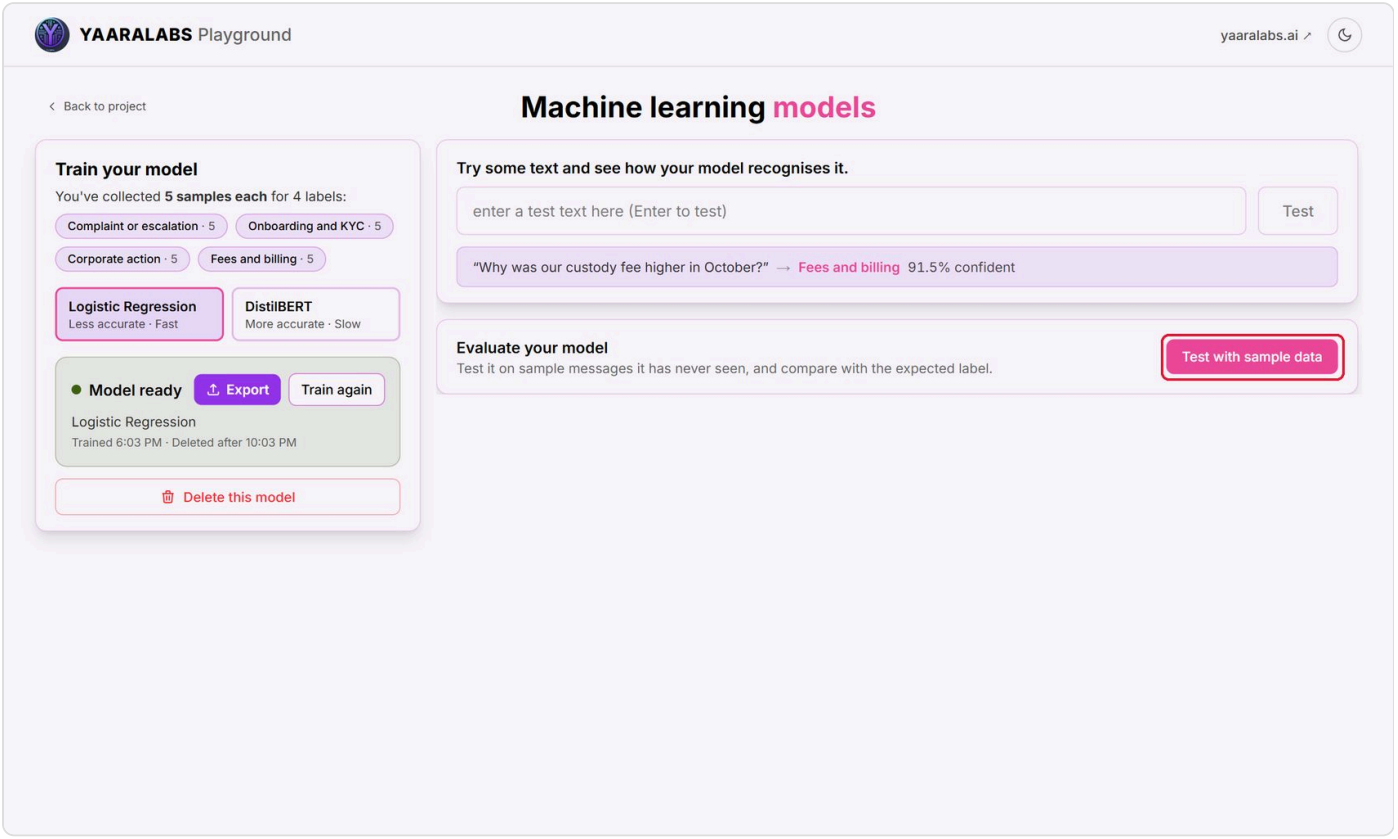
53 The model's answer

The model picks a label (Fees and billing) and shows how confident it is.



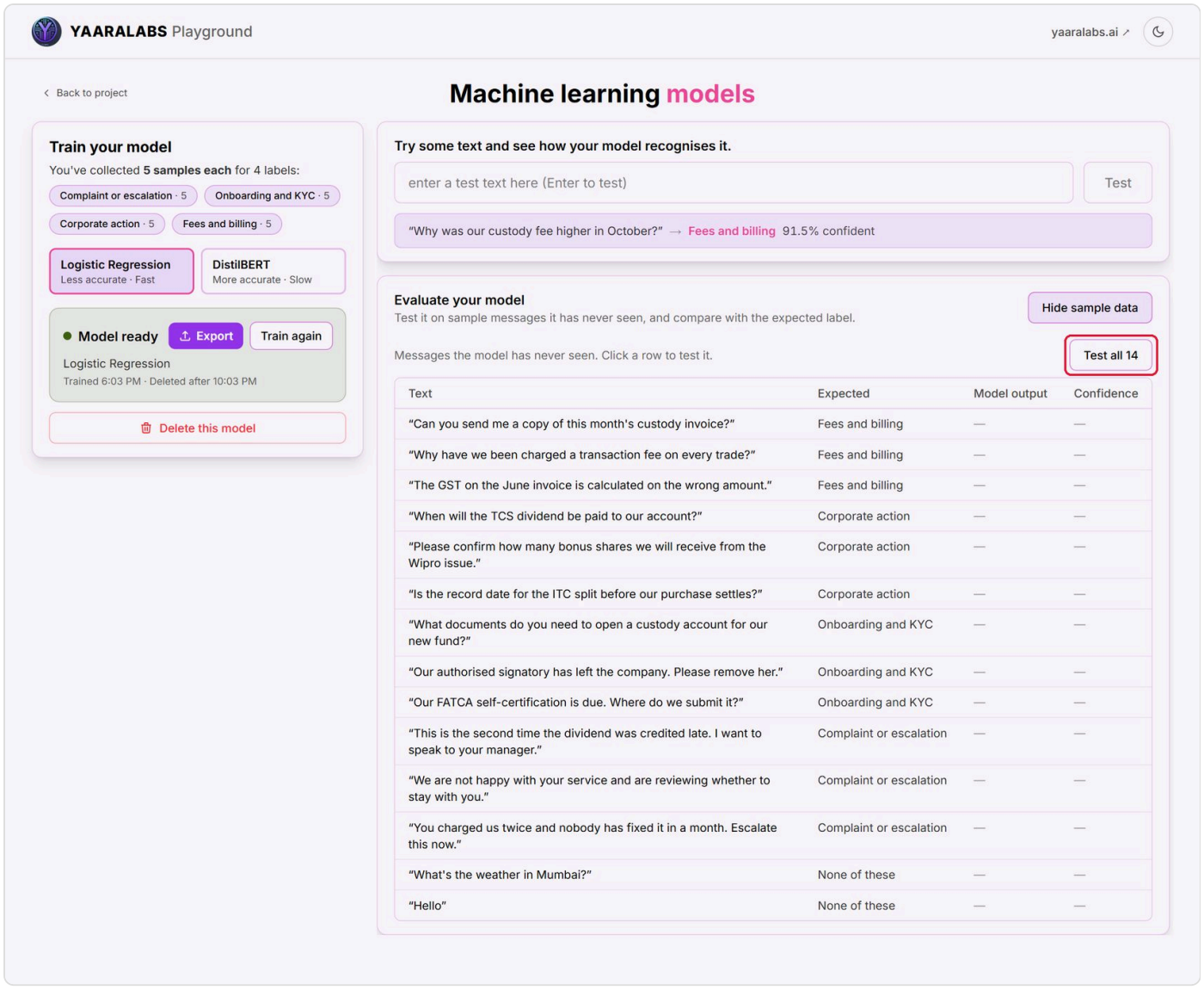
54 Test with sample data

Under Evaluate your model, click Test with sample data.



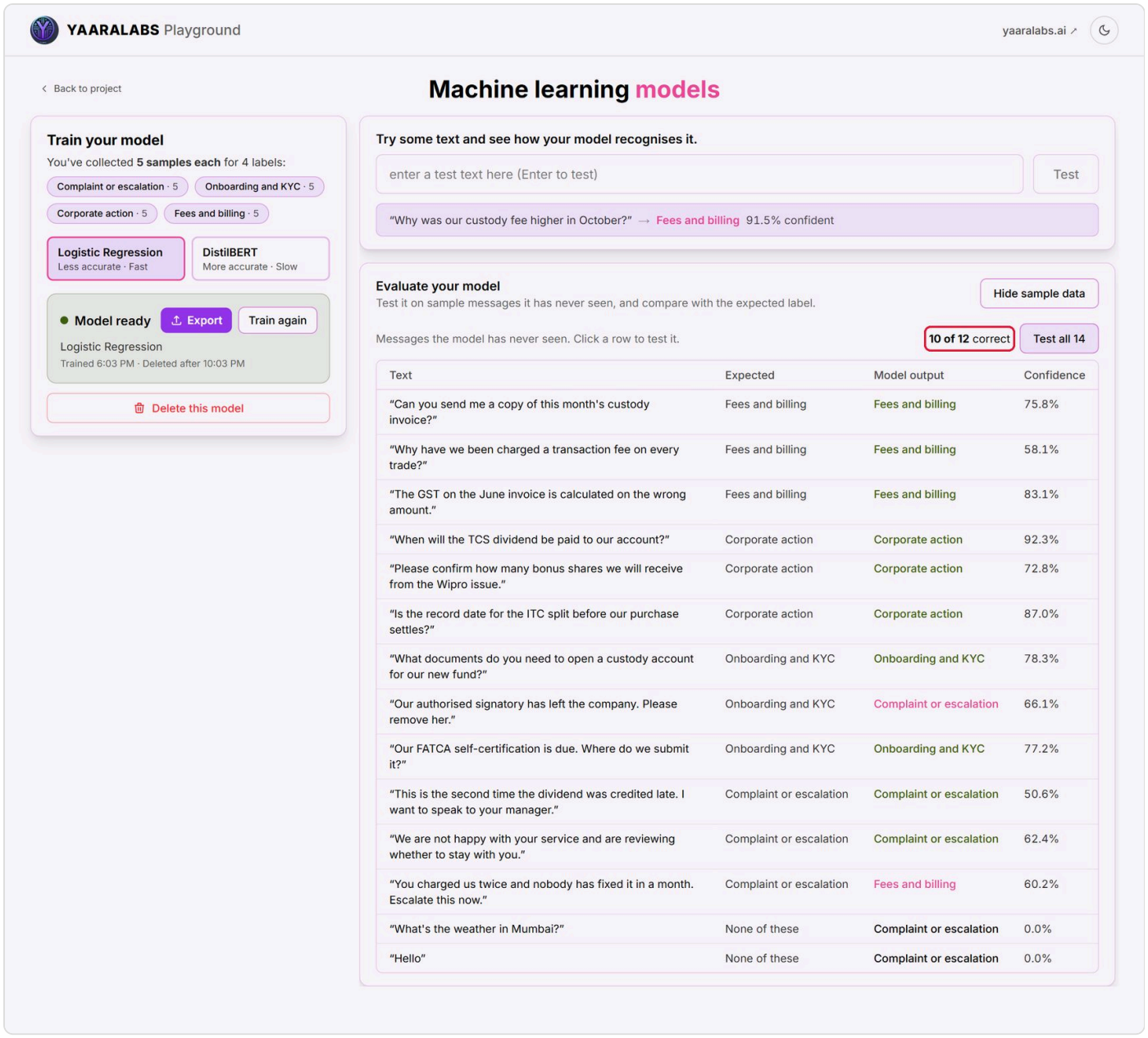
55 Messages it has never seen

14 new messages with the expected label. Click Test all 14 (or click one row to test it).



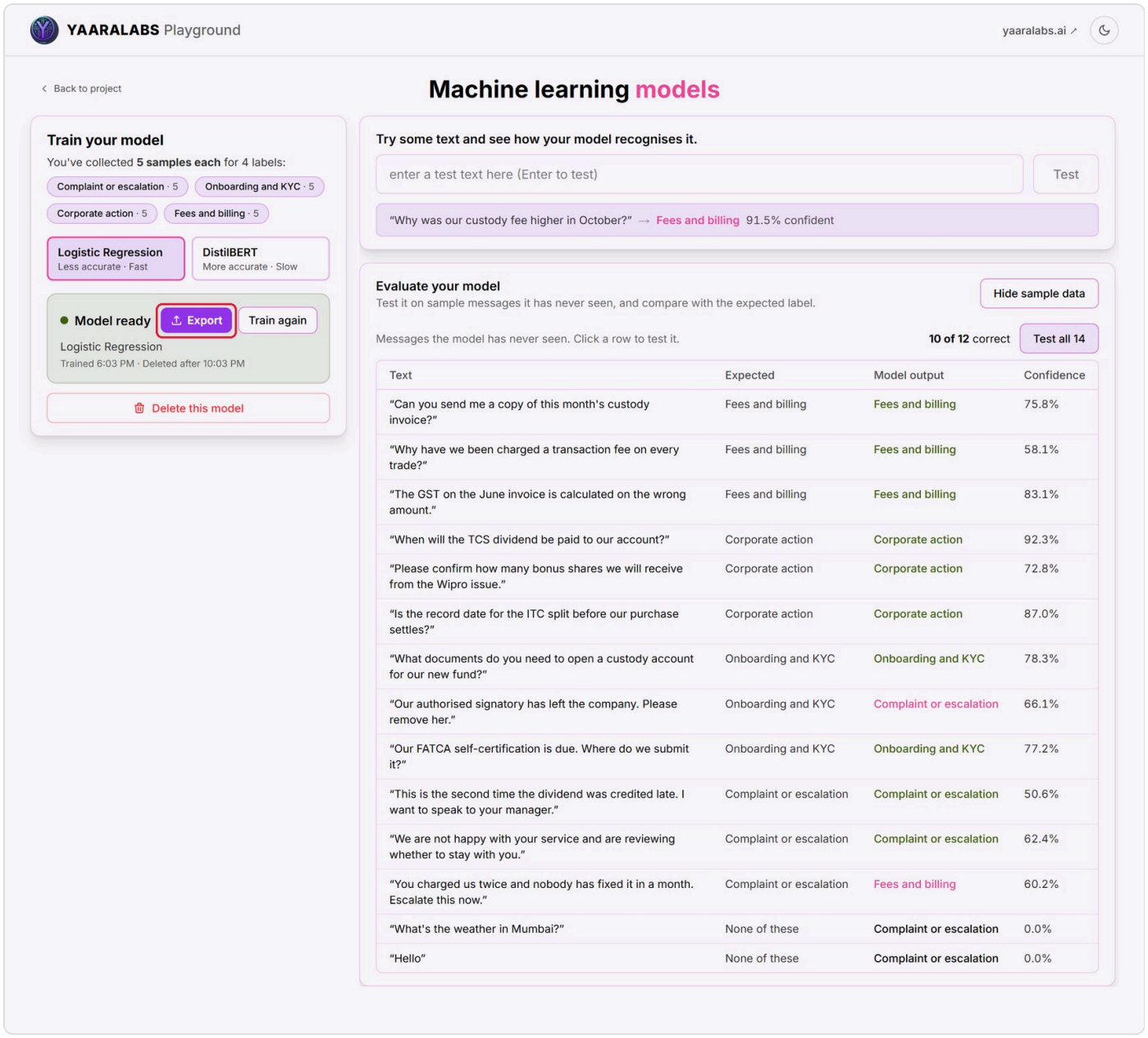
56 Sample data results

Each row shows the model's label and confidence; the count shows how many it got right.



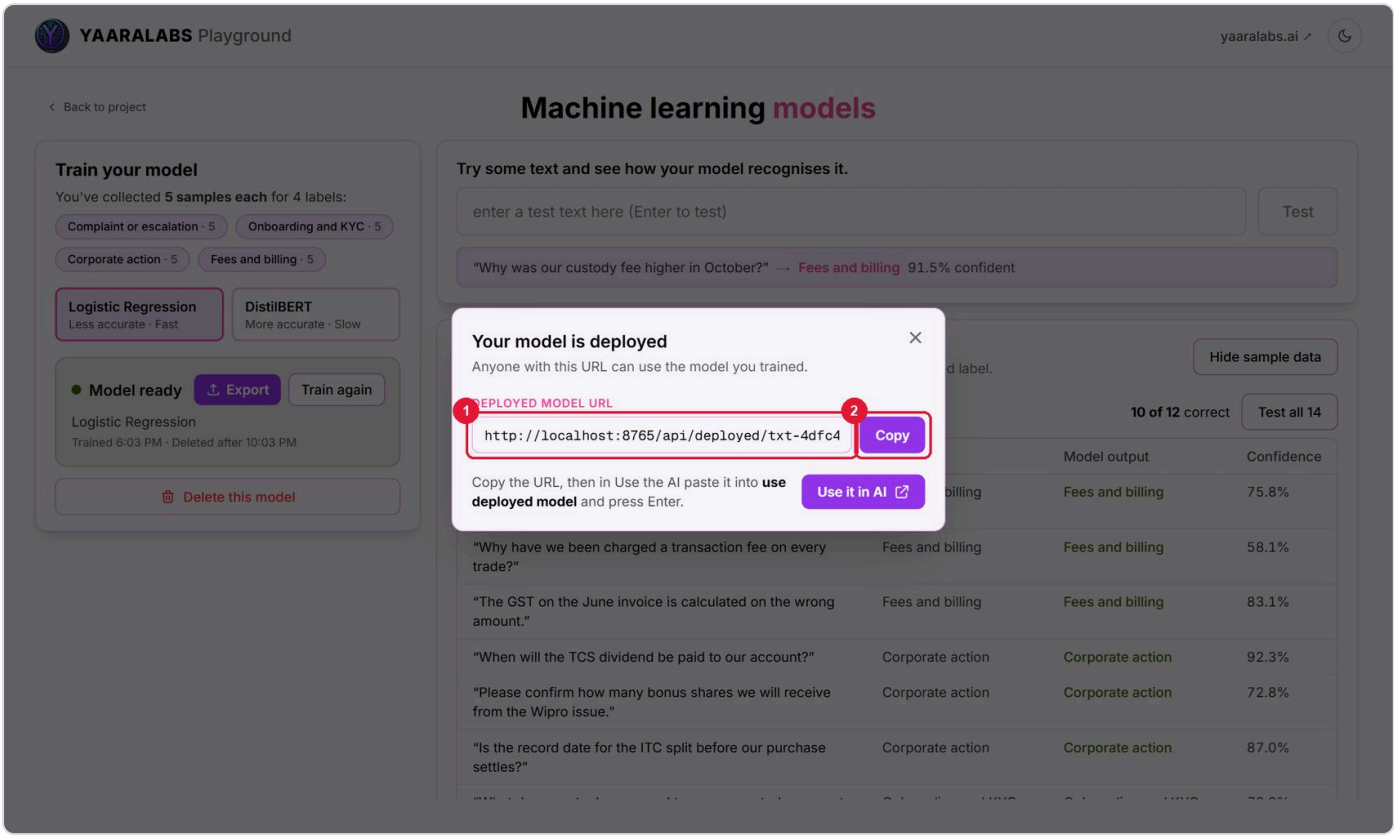
57 Export the model

Click Export on the Model ready card.



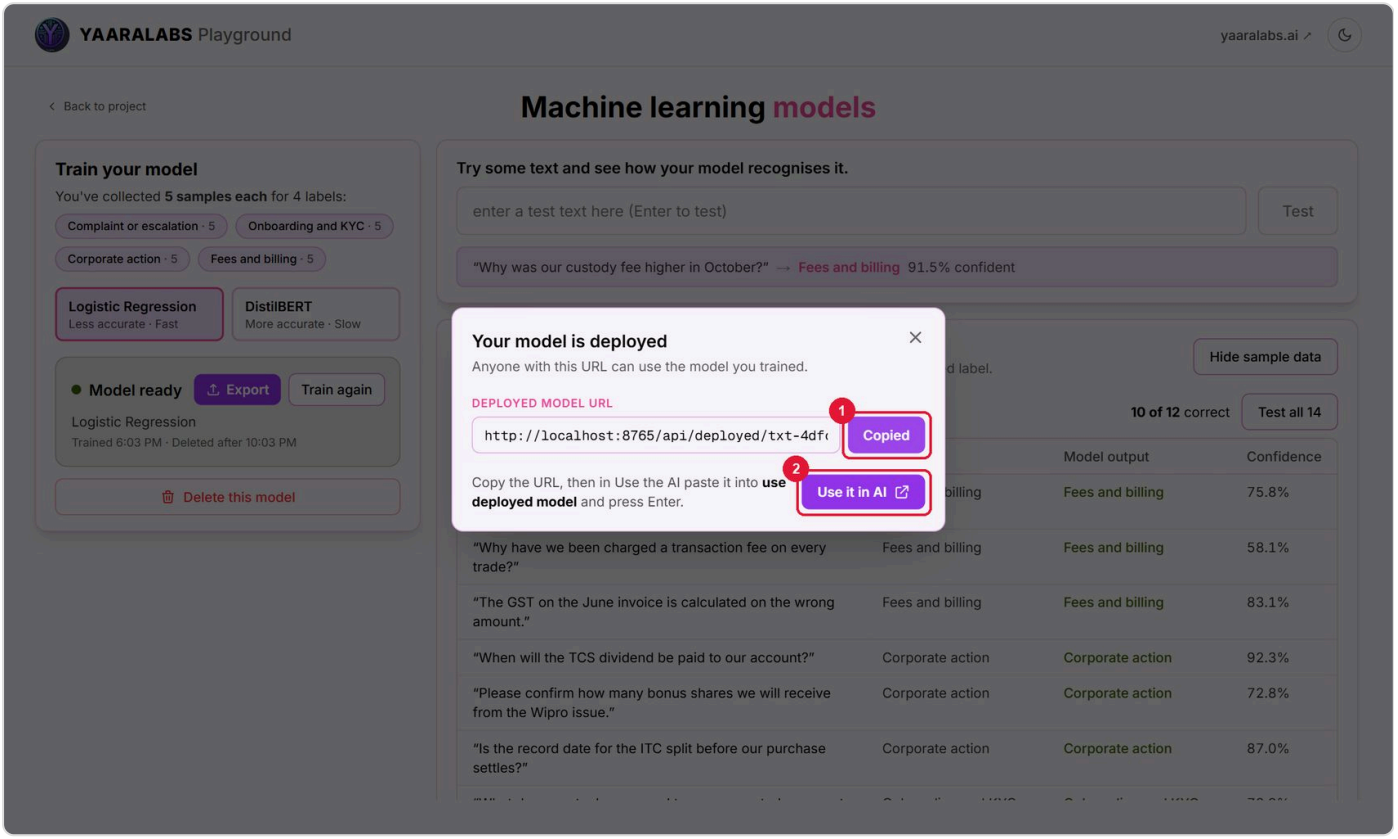
58 Your model is deployed

The dialog shows the deployed model URL. Click Copy.



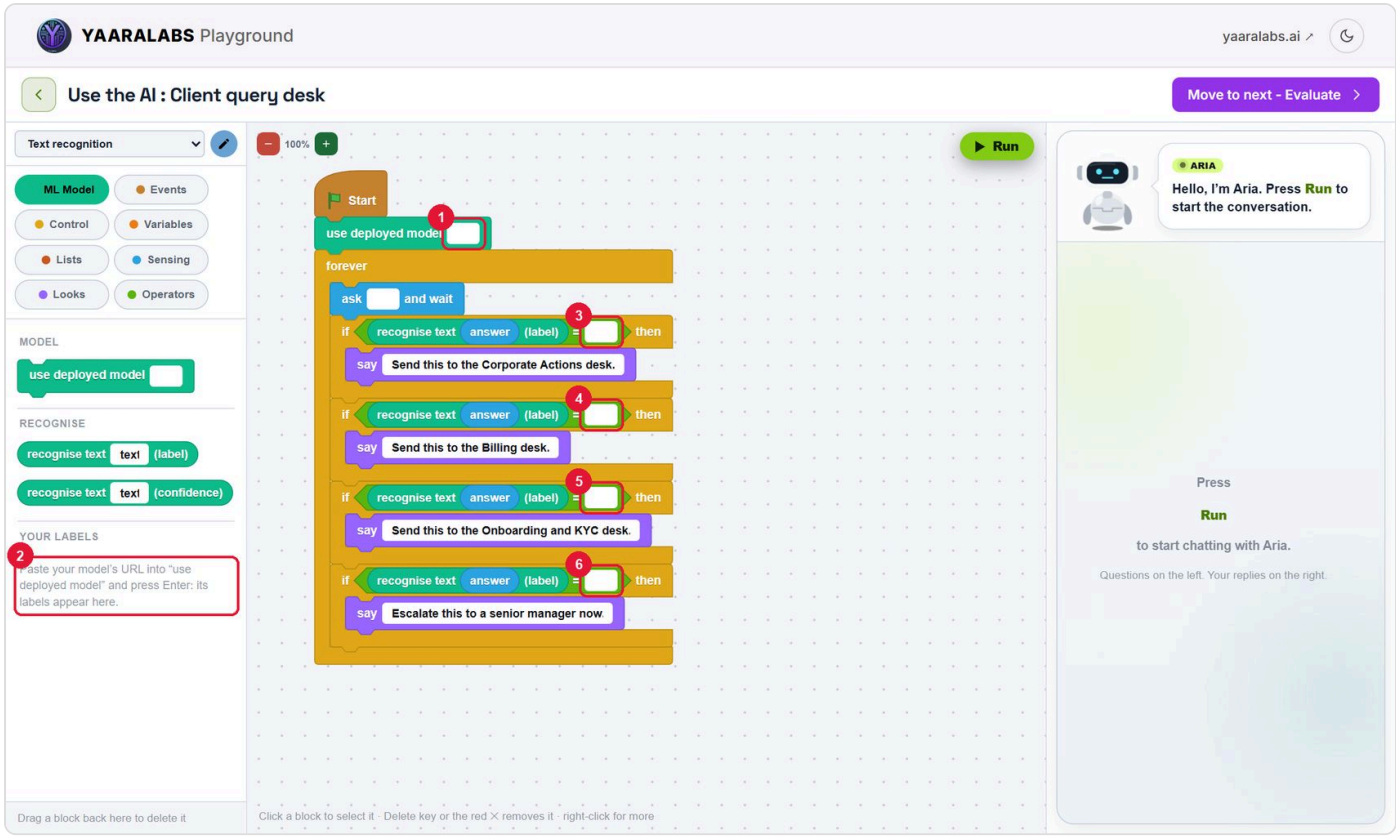
59 Copied

The URL is copied. Click Use it in AI: it opens in a new tab.



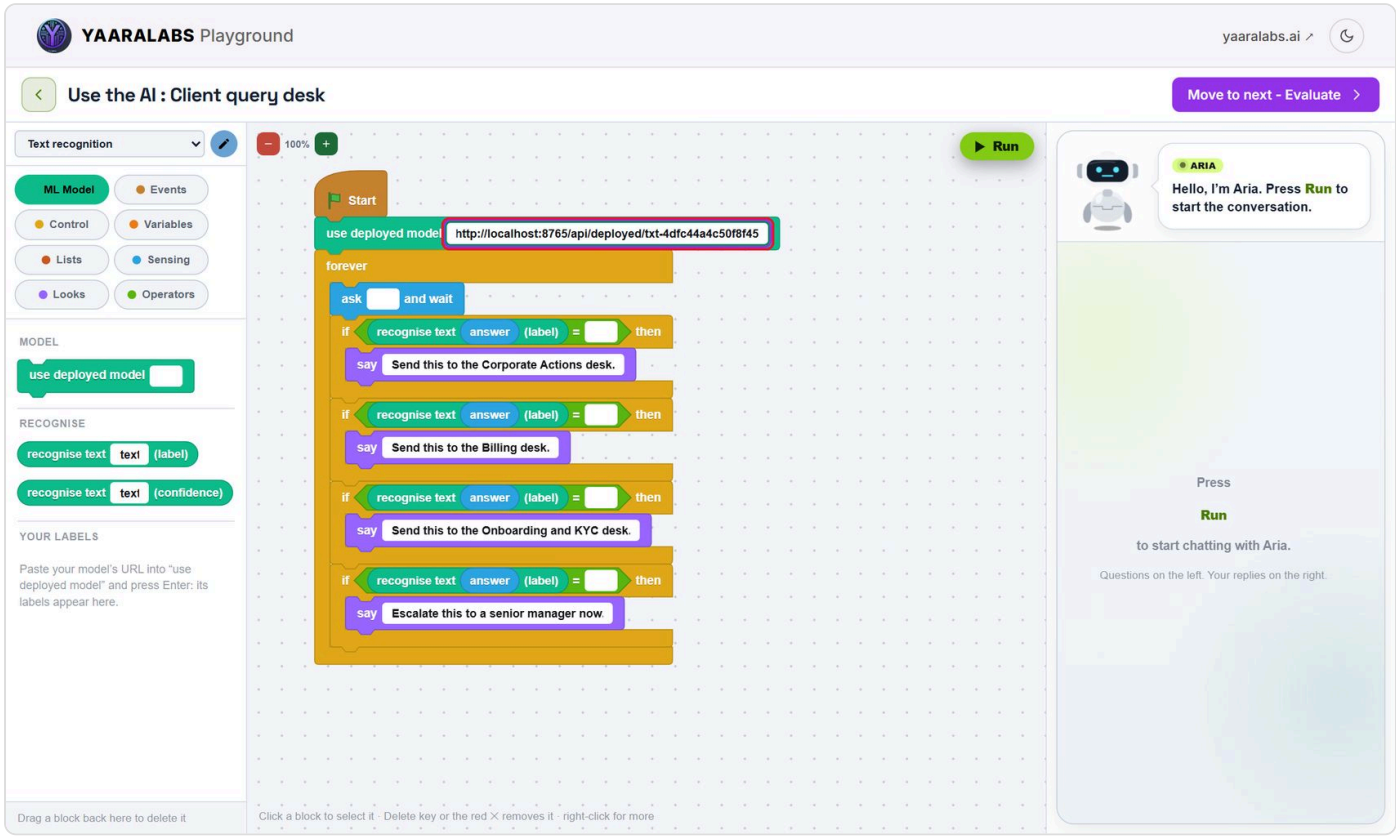
60 Use the AI opens in a new tab

The starter blocks are ready: use deployed model (1) and the four label slots (3-6) are empty; the left panel says to paste the URL.



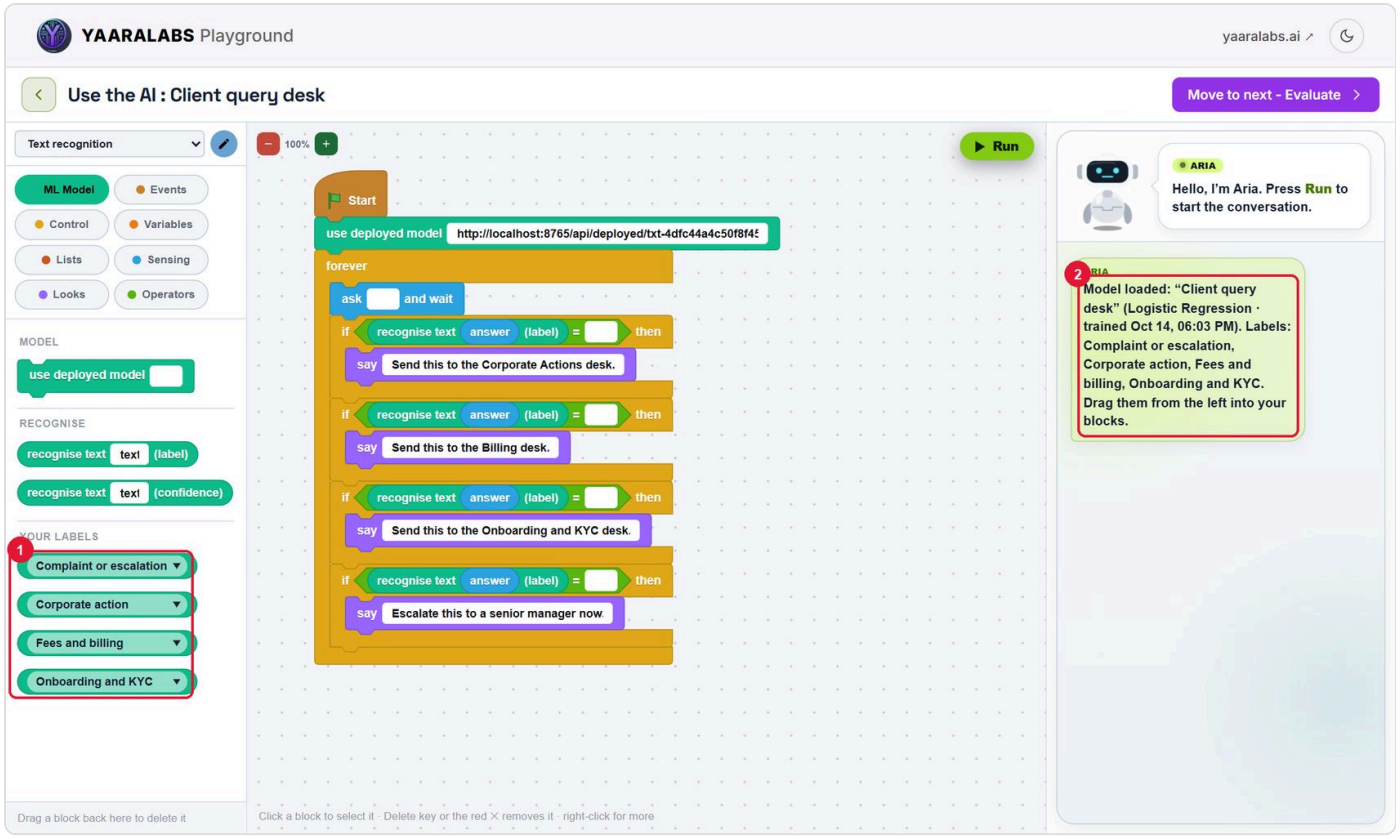
61 Paste the model URL

Click the white box in use deployed model, paste the URL (Ctrl + V) and press Enter.



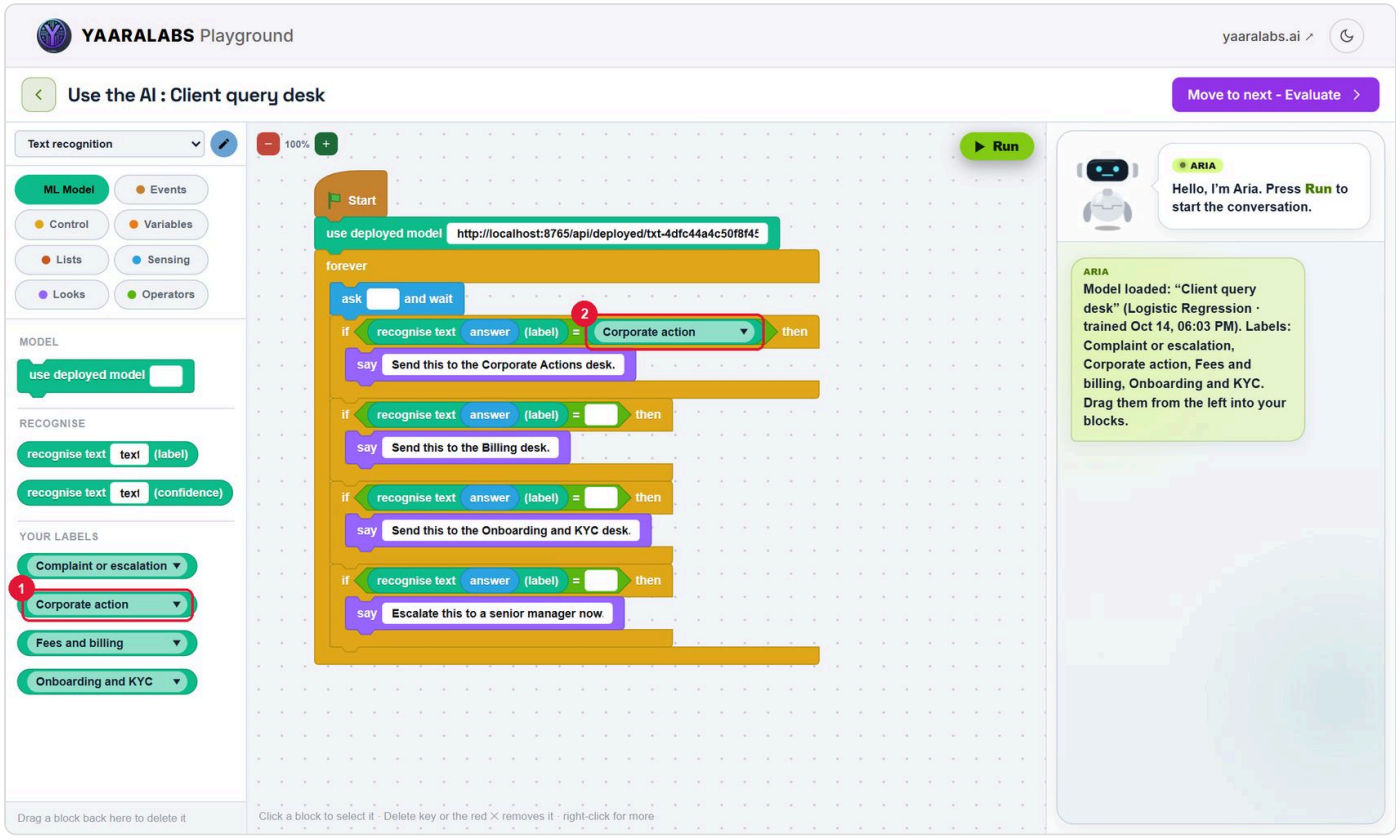
62 The model is loaded

Your labels appear on the left under YOUR LABELS, and Aria says Model loaded.



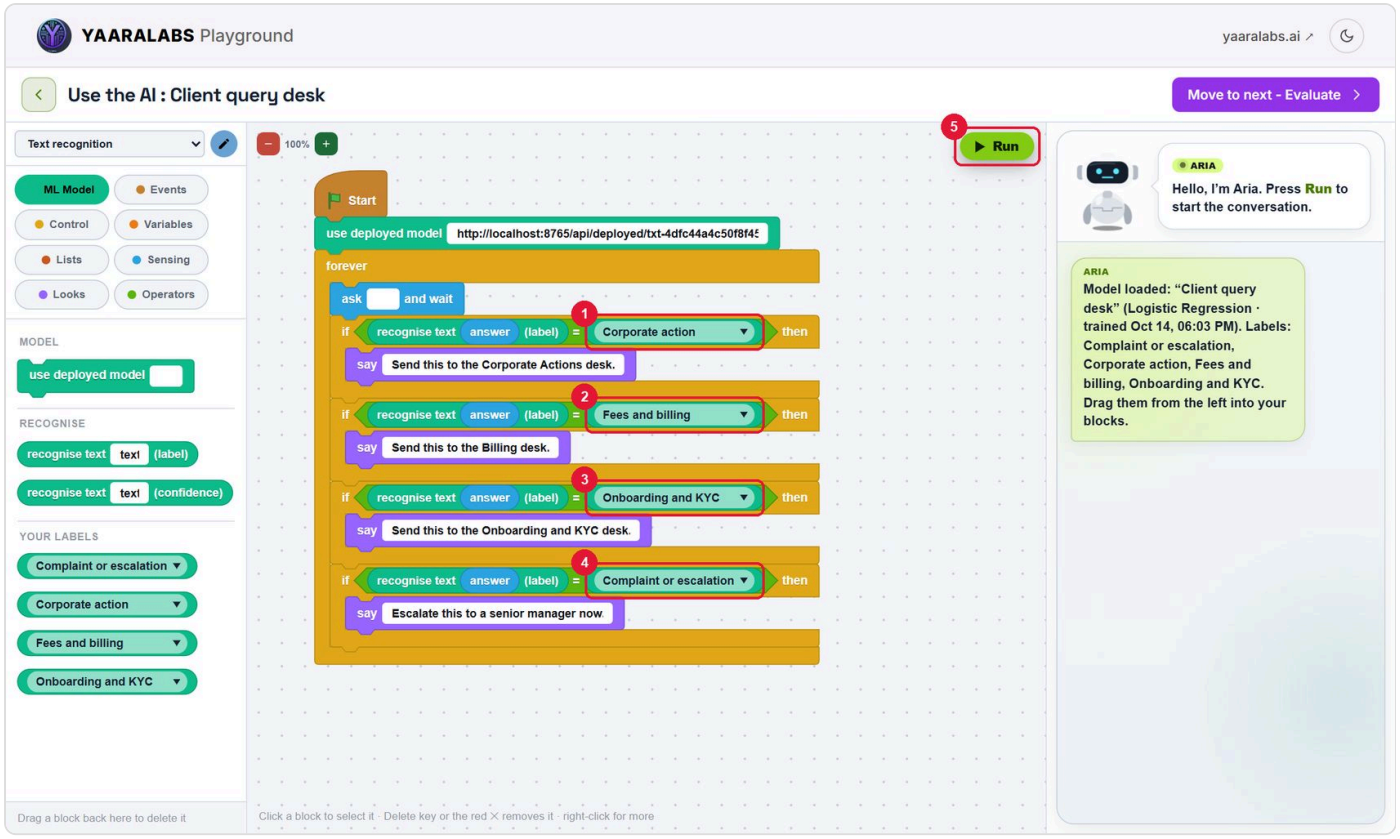
63 Drag a label into its slot

Drag Corporate action from the left into the empty slot of the if block that says Send this to the Corporate Actions desk.



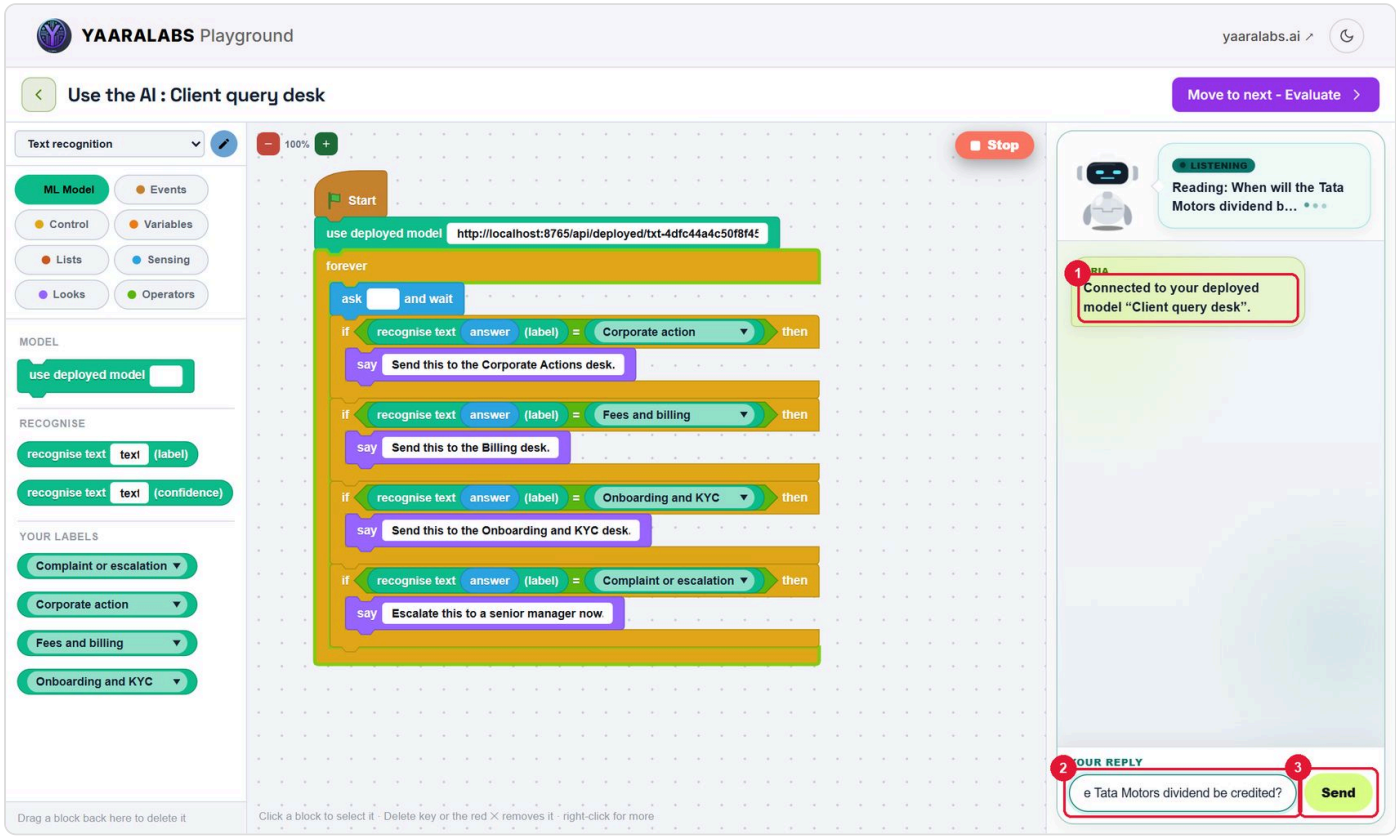
64 All four labels in place

Fees and billing goes to the Billing if, Onboarding and KYC to the KYC desk, Complaint or escalation to escalate. Then click Run.



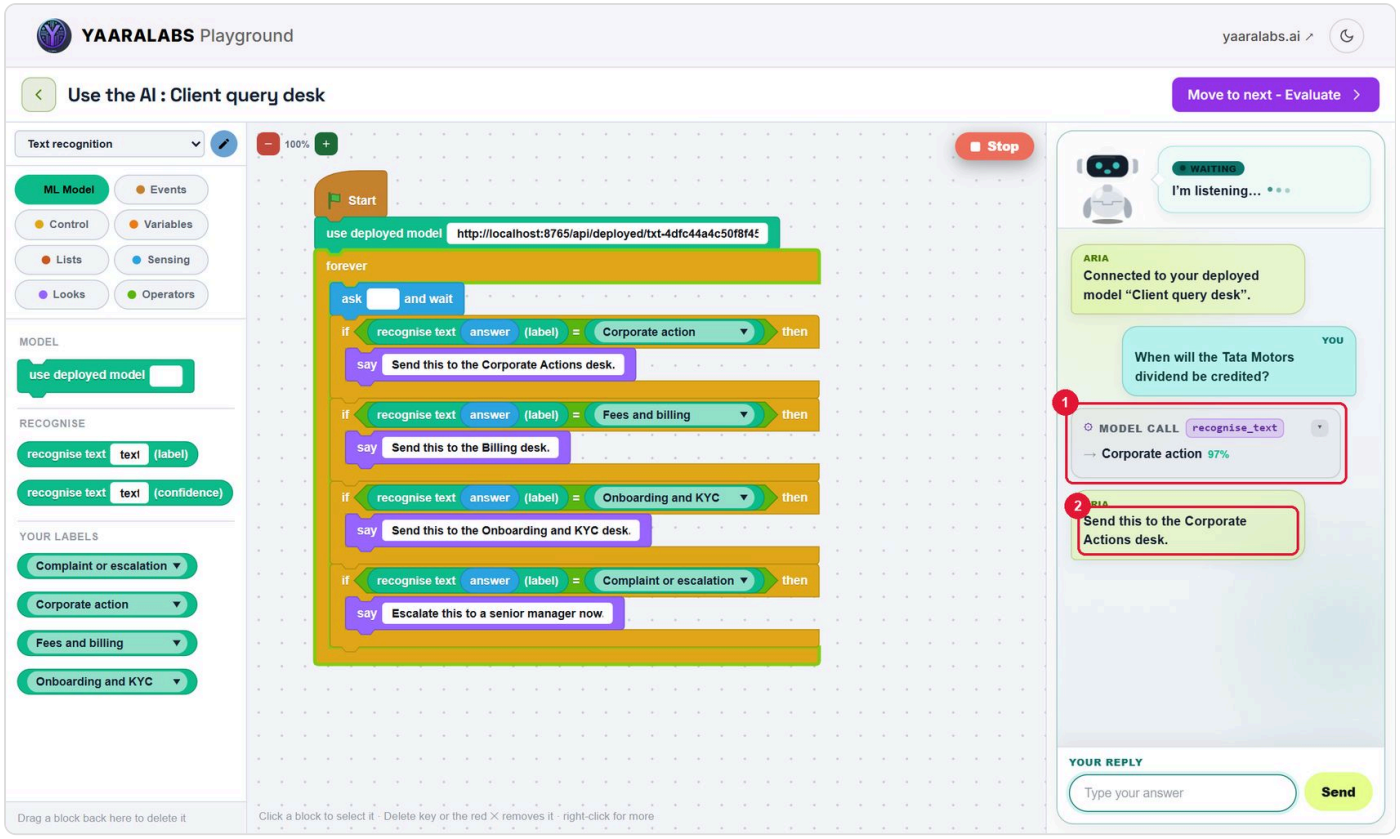
65 Chat with Aria

Aria is listening. Type a client message in Your reply and click Send.



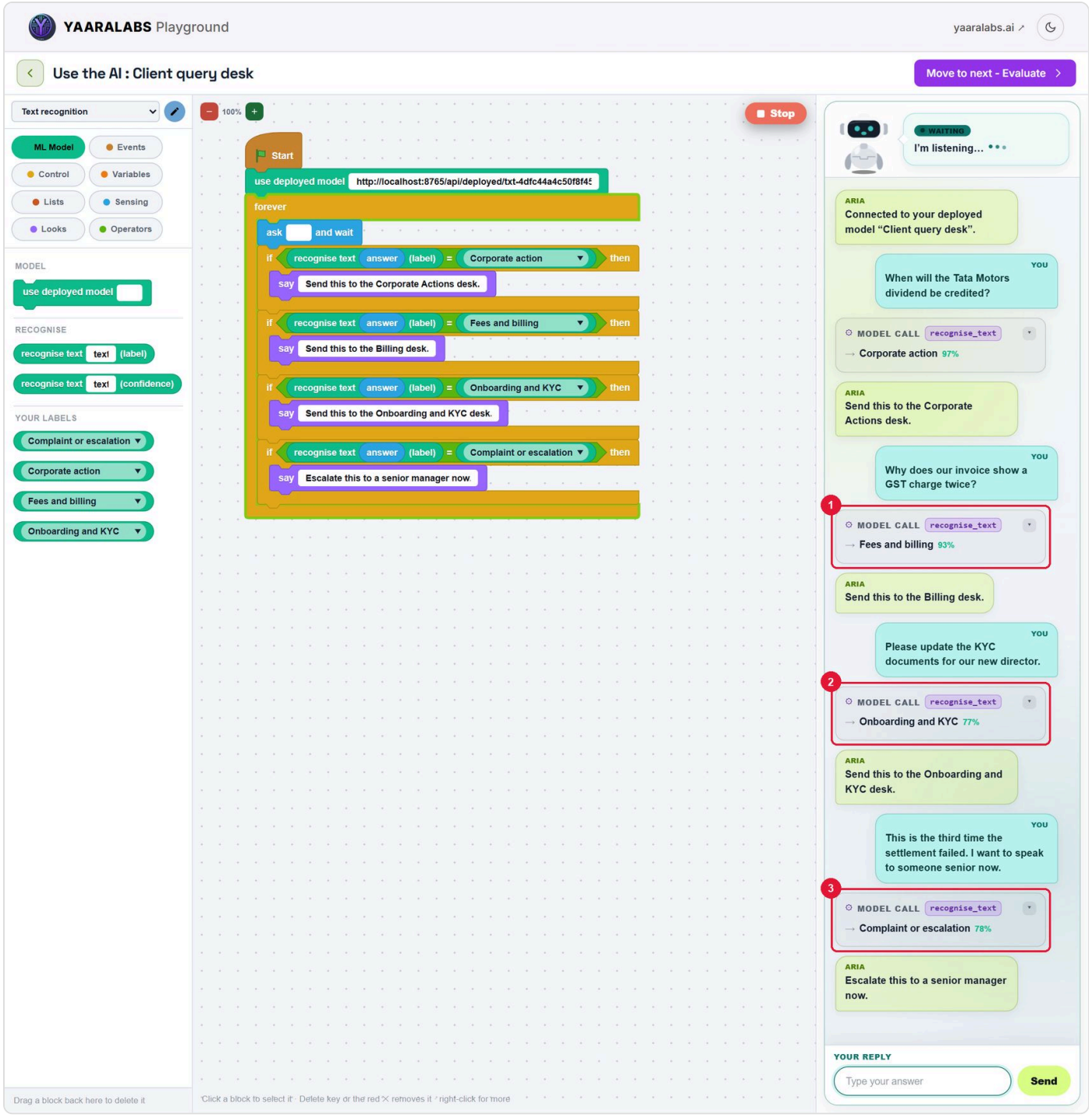
66 Aria answers

The model call card shows the label the model chose (Corporate action) and how sure it is; Aria routes it to that desk.



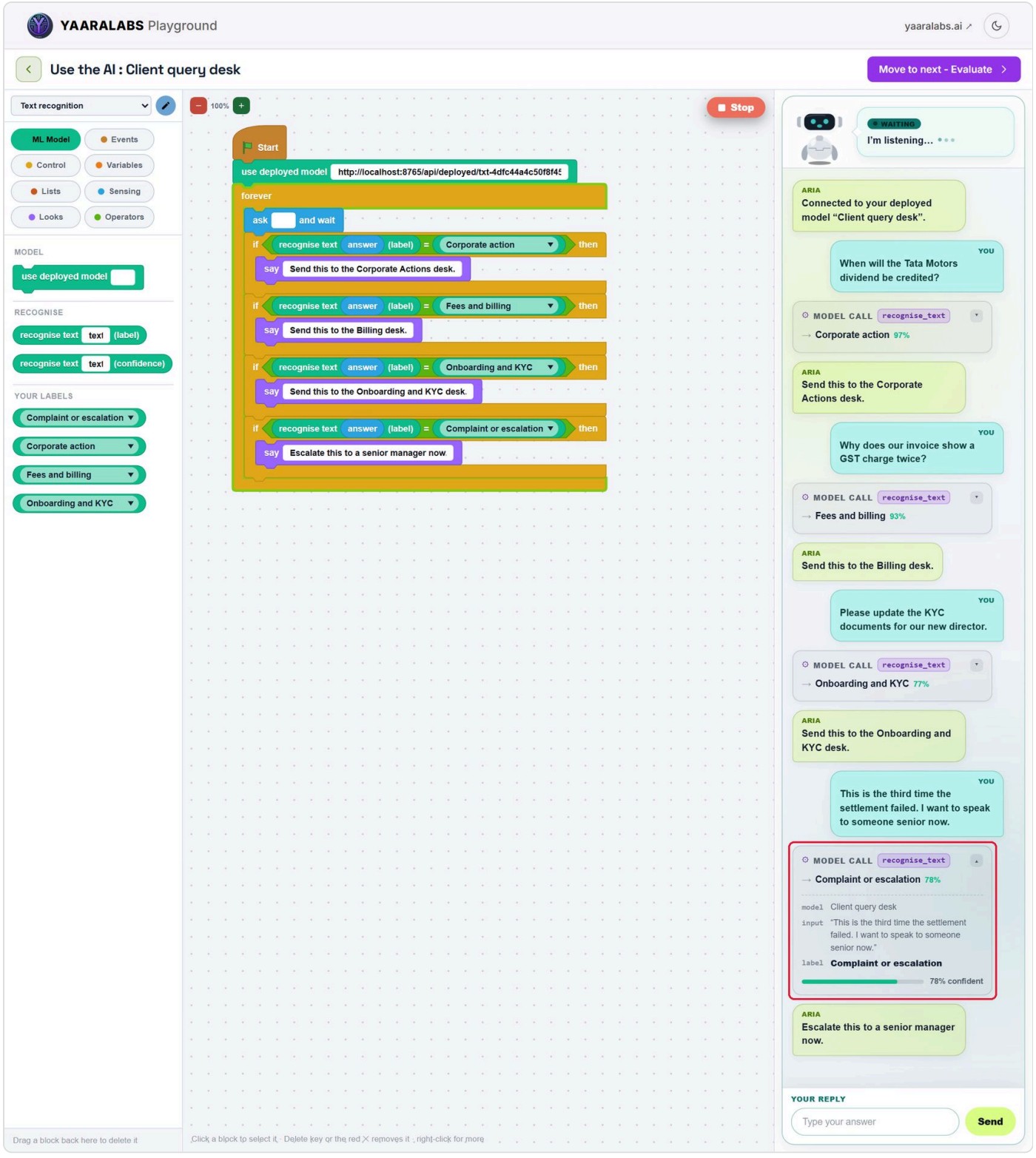
67 More client messages

Send a few more: each message gets a model call and the matching desk answer.



68 Open a model call

Click a model call card to see the model, the input text, the label and the confidence.

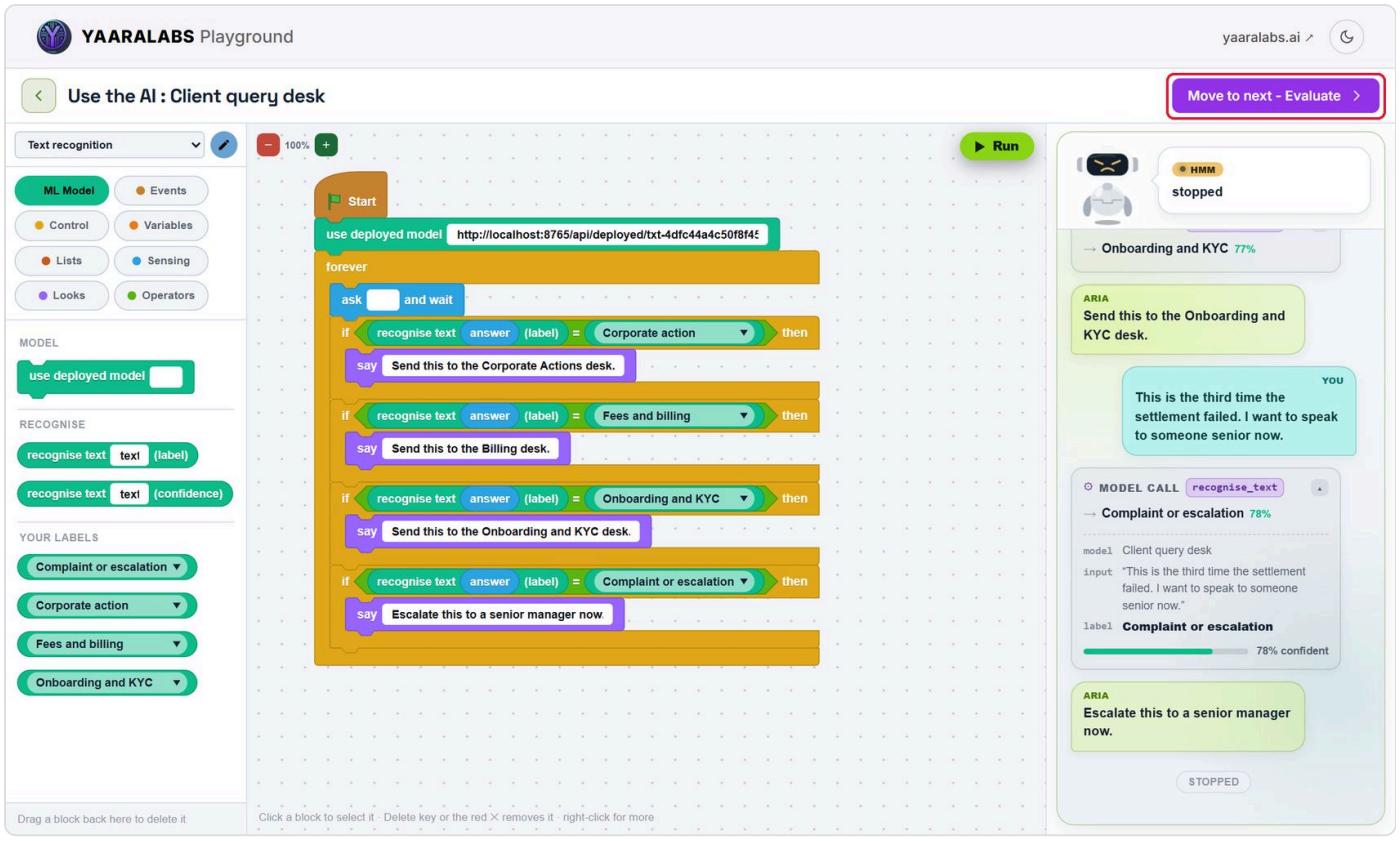


Experiment 3: Evaluate the model

Give the trained model an exam on messages it has never seen and read its accuracy, precision and recall.

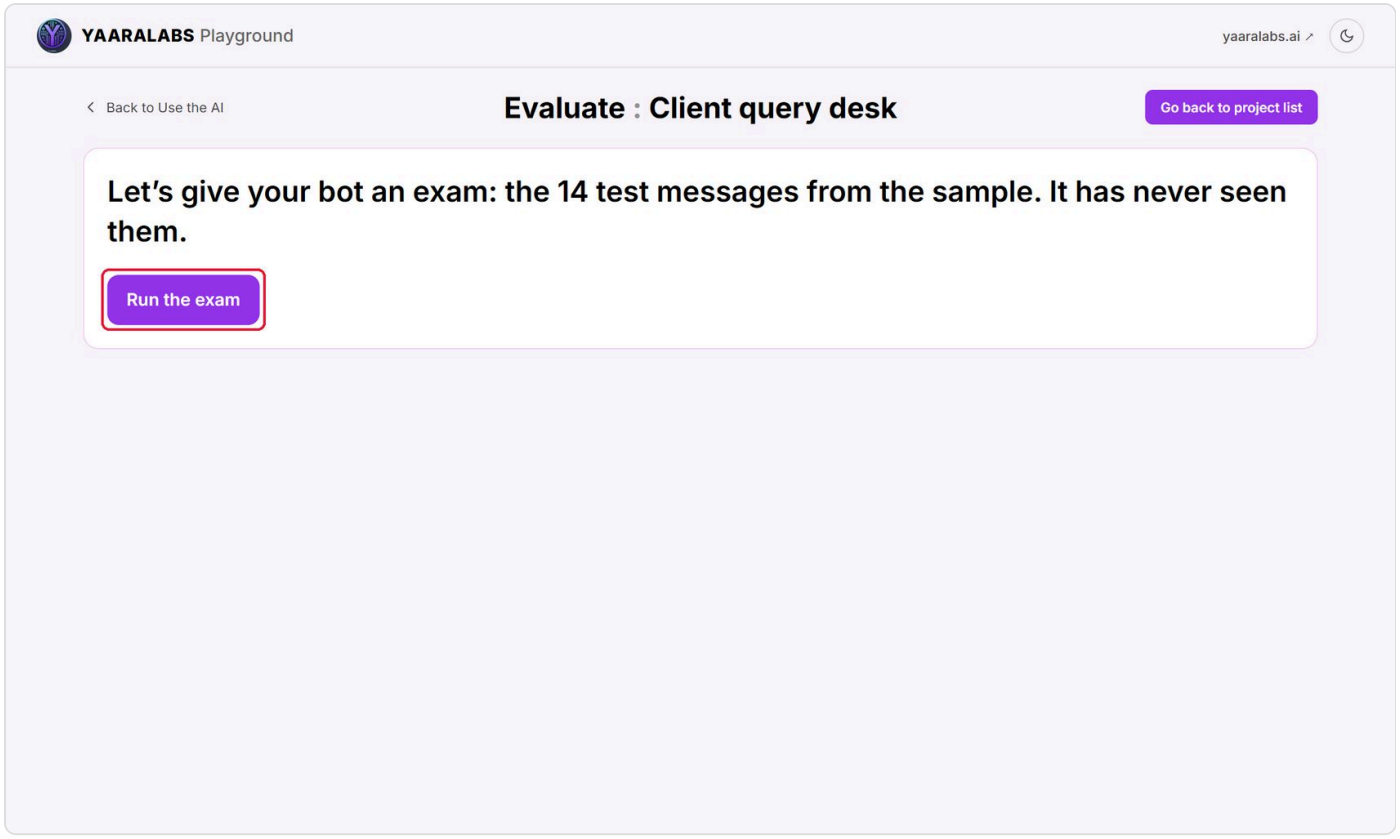
69 Move to Evaluate

Click Move to next - Evaluate at the top right.



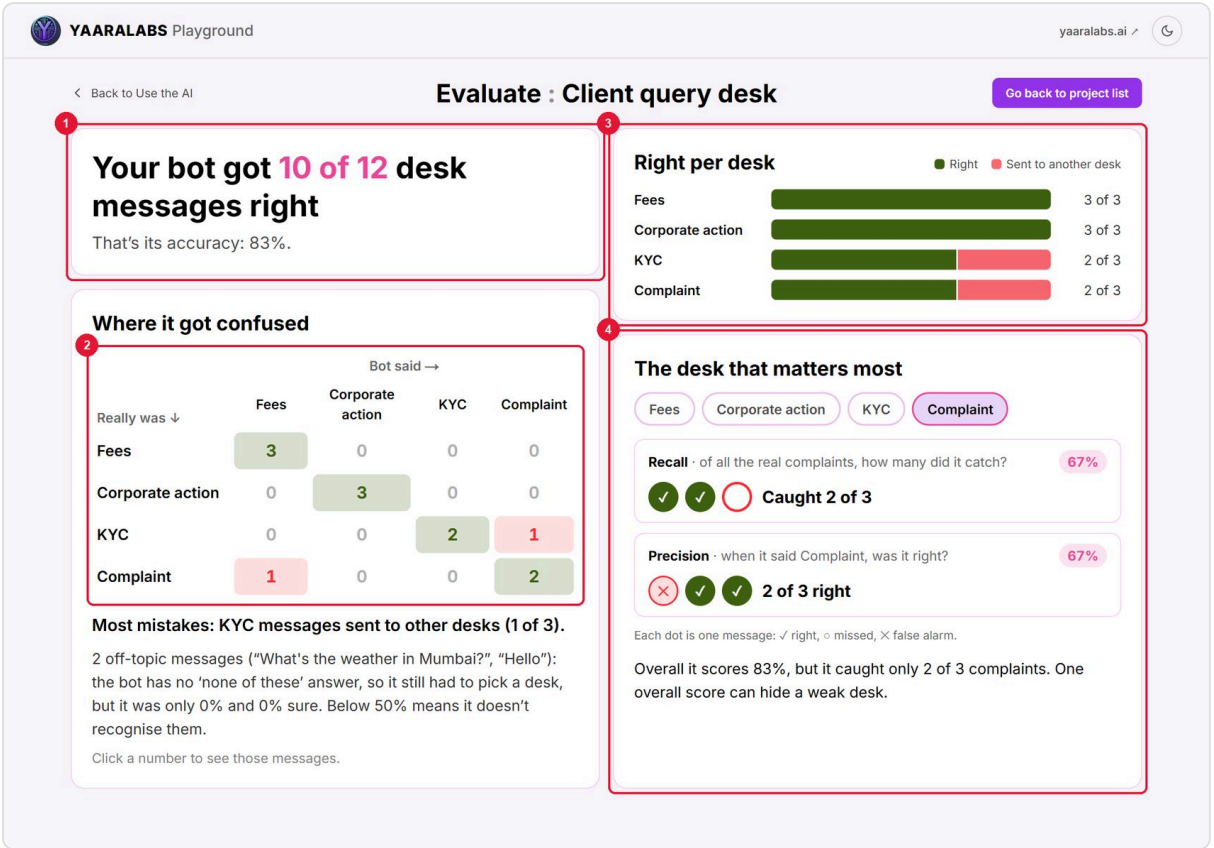
70 The exam

The bot is tested on the 14 sample messages it has never seen. Click Run the exam.



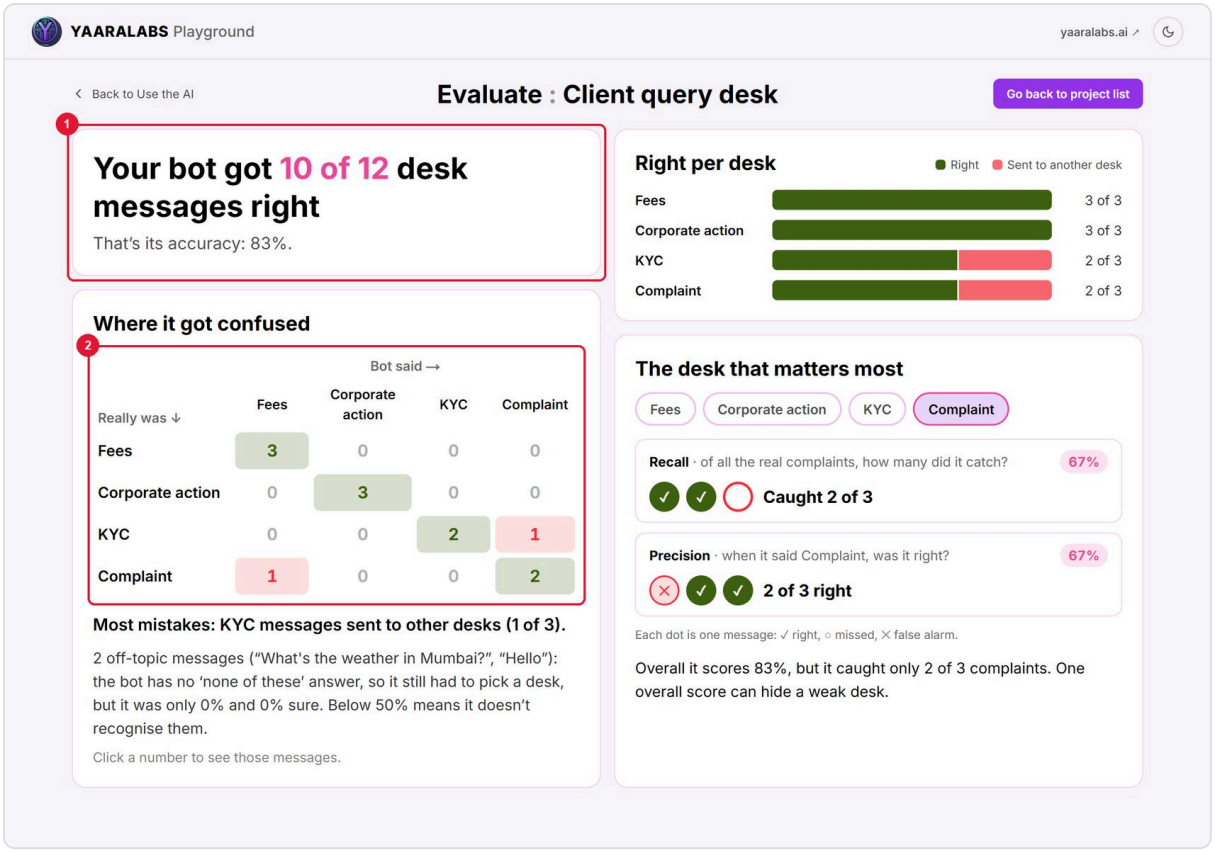
71 The results

Read the overall score (accuracy), Where it got confused, Right per desk and The desk that matters most.



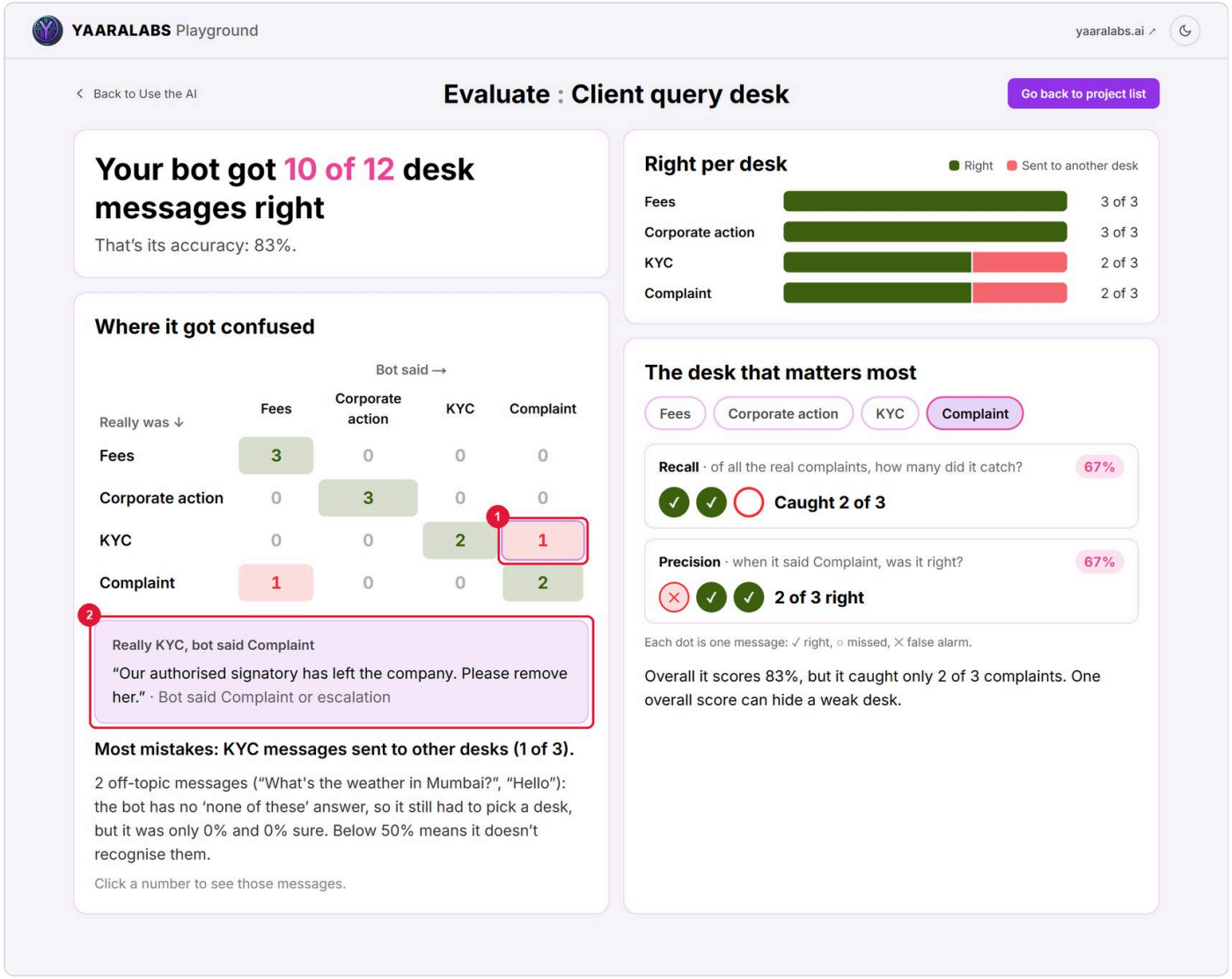
72 Accuracy and confusion

Accuracy = right answers out of all desk messages. In the grid, green is right; red numbers are mix-ups.



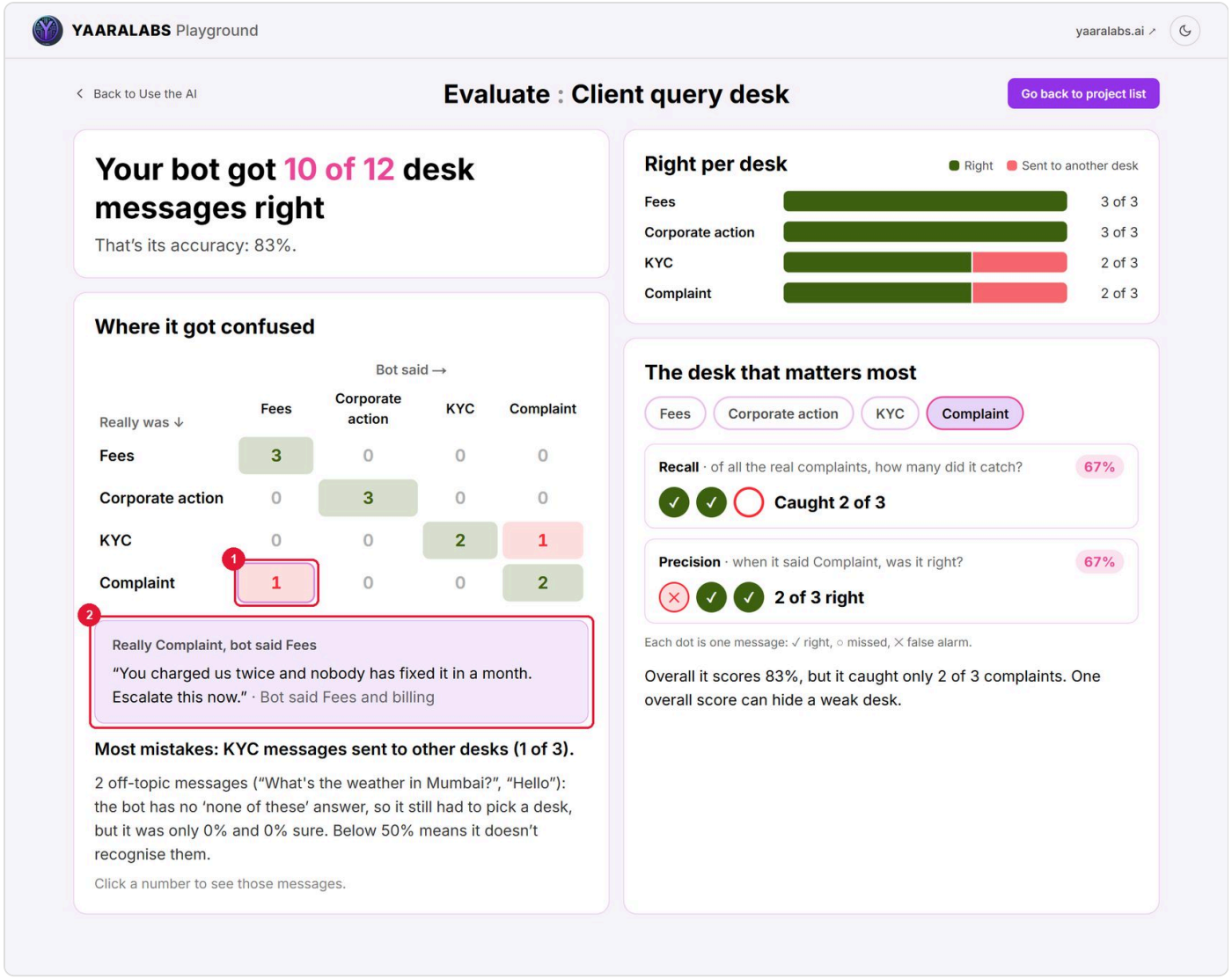
73 Really KYC, bot said Complaint

Click a red number to see the message behind it: a KYC message the bot sent to Complaint.



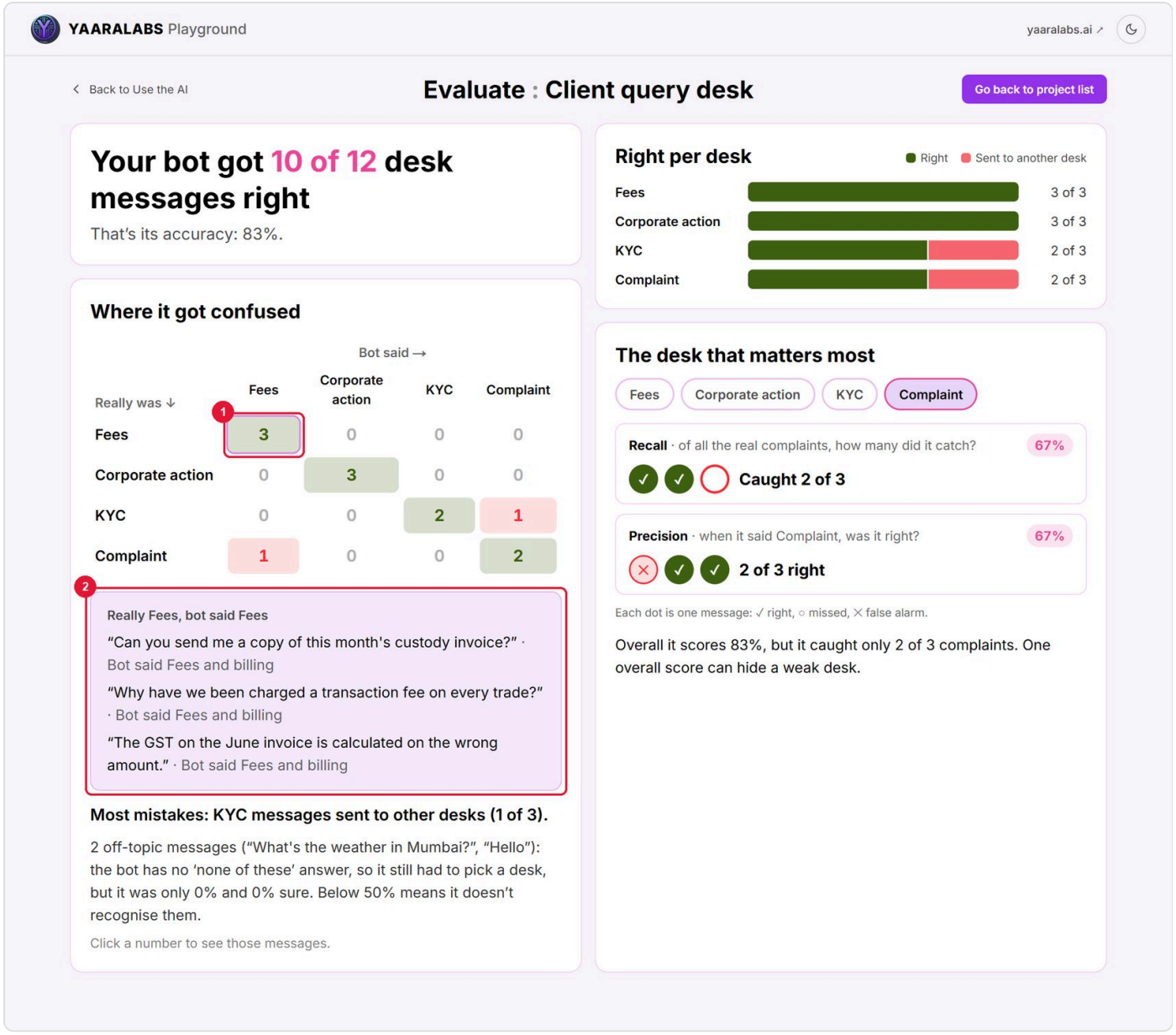
74 Really Complaint, bot said Fees

Click a red number to see the message behind it: a Complaint message the bot sent to Fees.



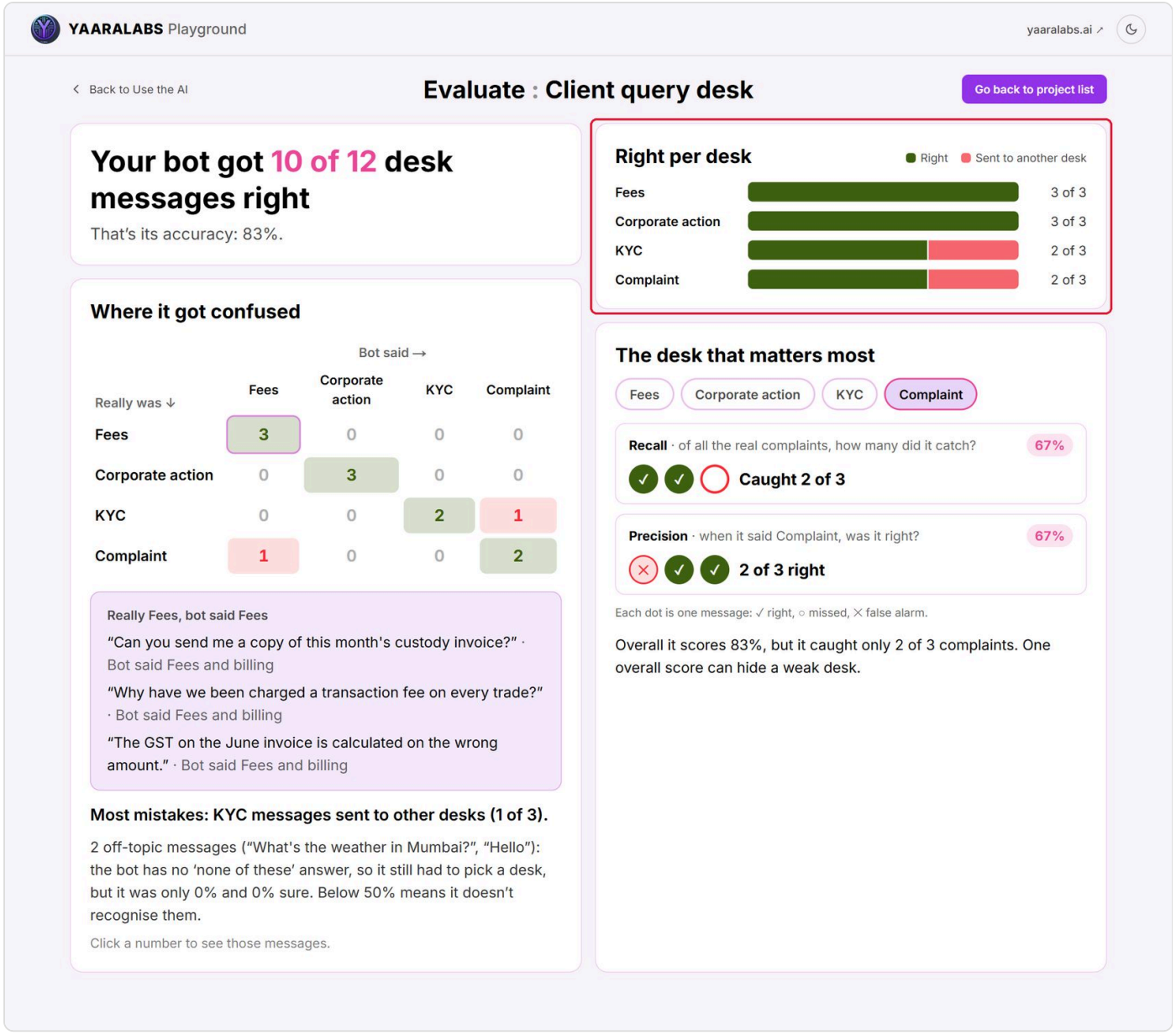
75 A green number

Green numbers are the messages the bot got right; click one to list them.



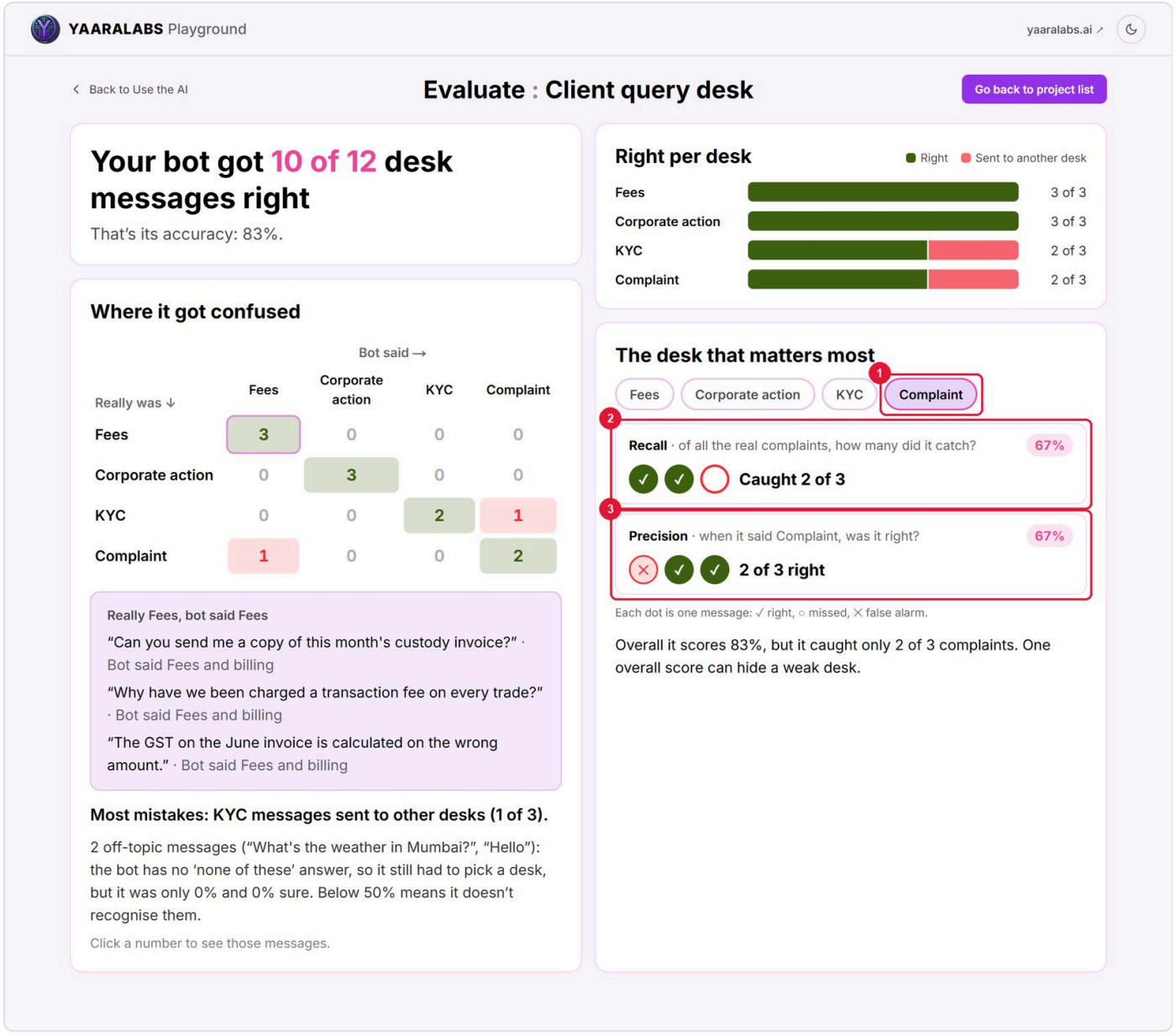
76 Right per desk

Each bar shows, per desk, how many messages went to the right desk (green) or another desk (red).



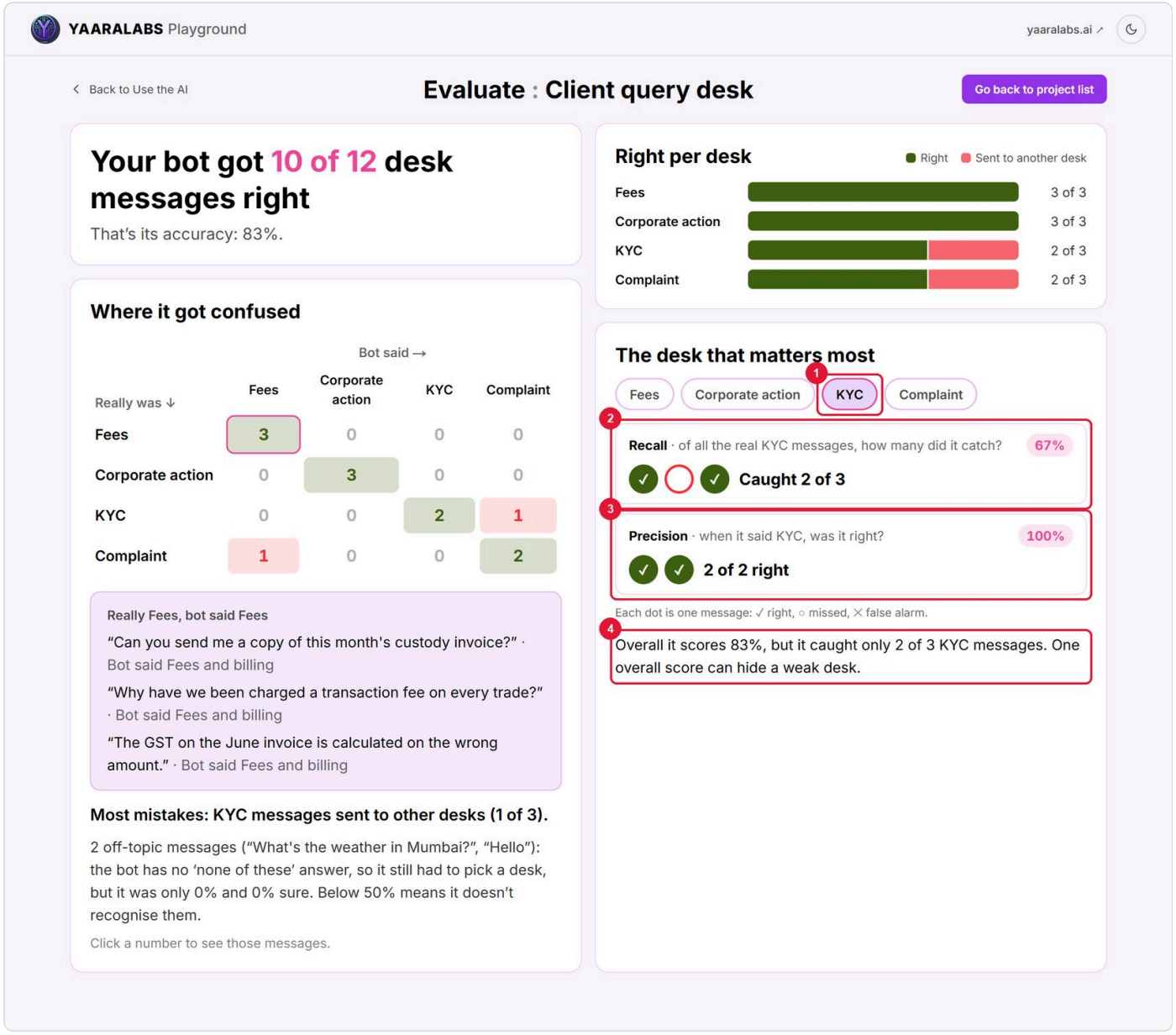
77 The desk that matters most: Complaint

Recall: of all real complaints, how many it caught. Precision: when it said Complaint, how often it was right.



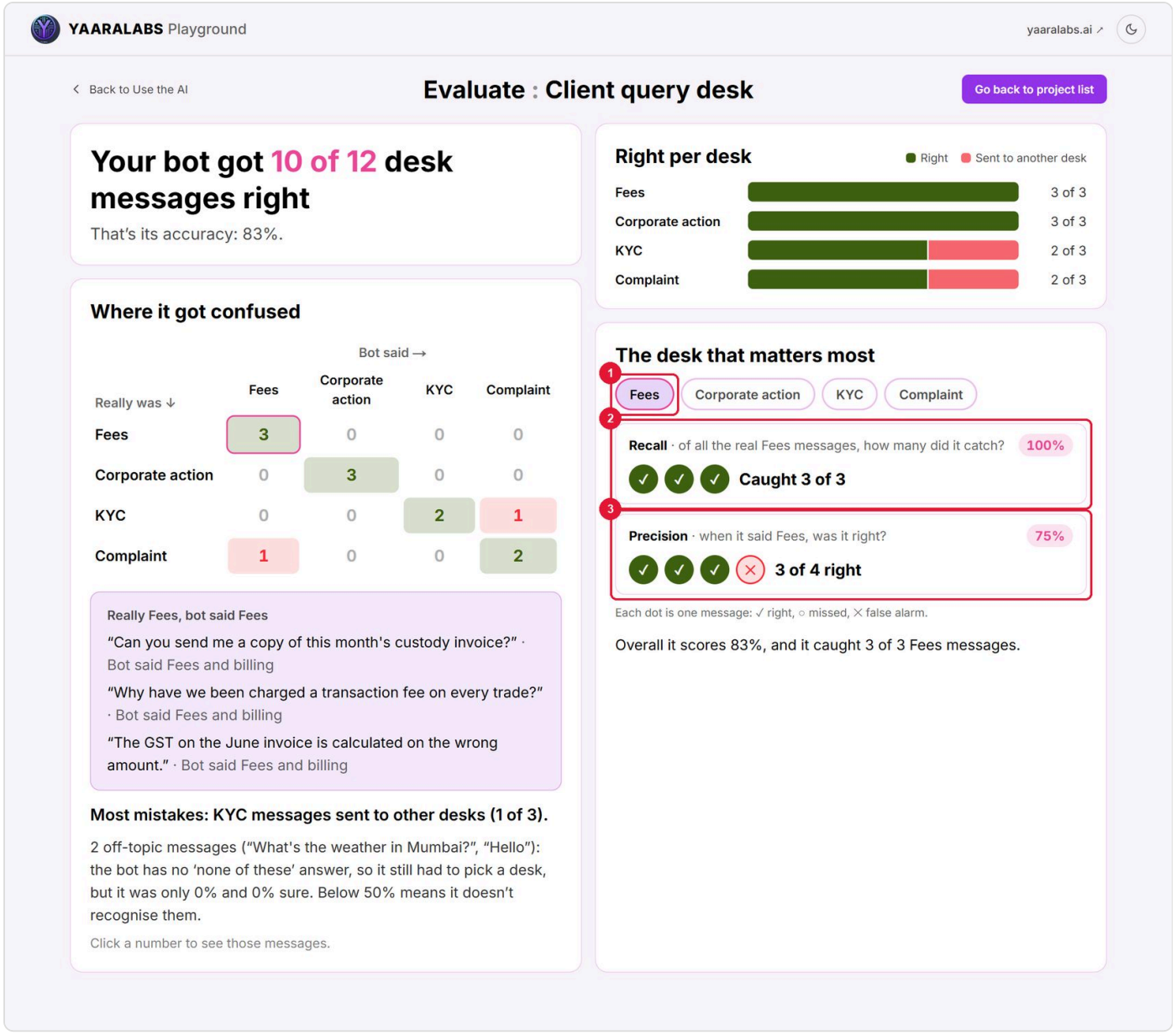
78 Pick another desk: KYC

Click KYC: recall is 67% and precision 100% for this desk; each desk tells a different story.



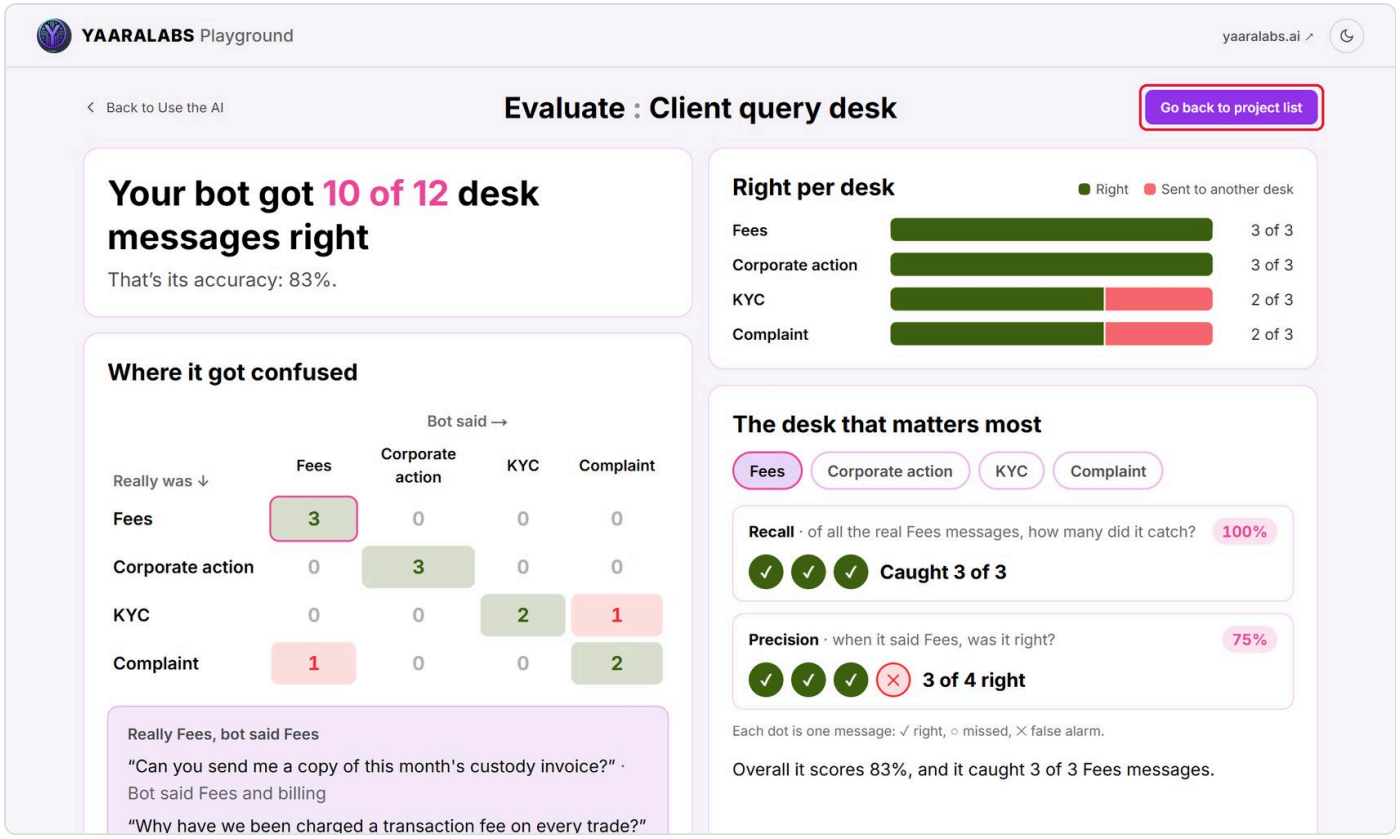
79 Pick another desk: Fees

Click Fees: recall is 100% and precision 75% for this desk; each desk tells a different story.



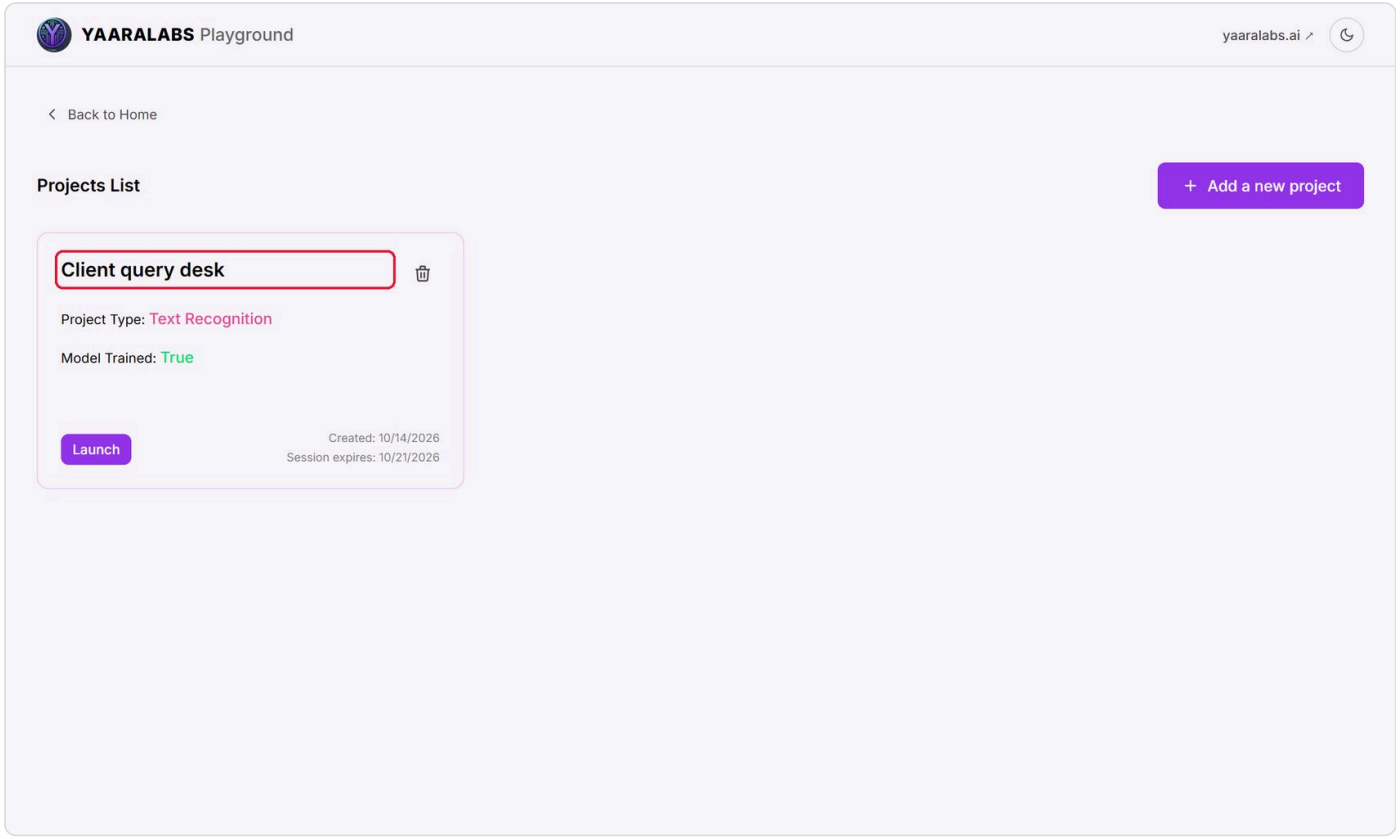
80 Back to the project list

When you are done, click Go back to project list at the top right.



81 The project list

You are back on the projects page; Client query desk is in your list.

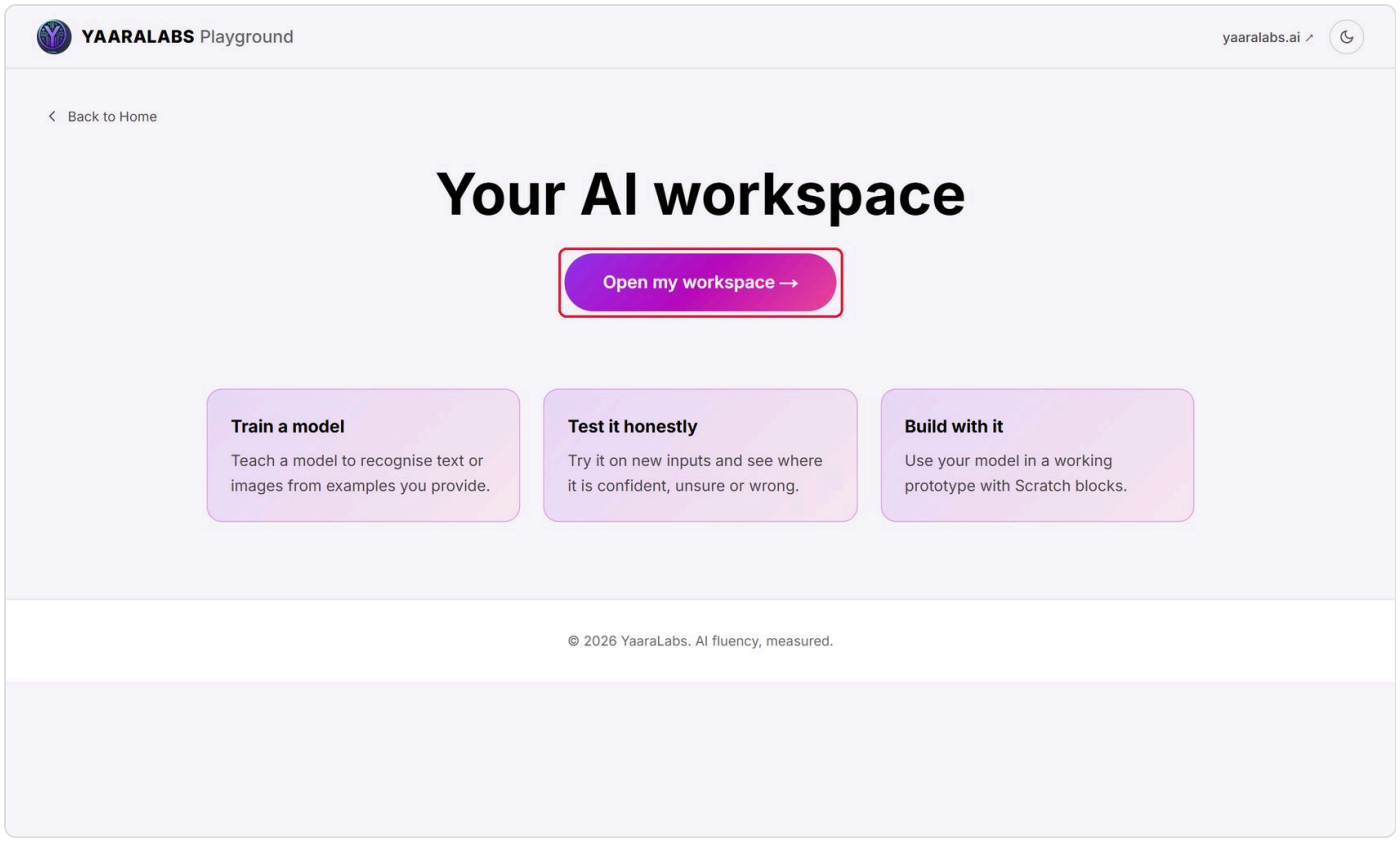


Experiment 4: Image Recognition

Teach a model to sort client documents (KYC, corporate action, invoice, cheque) from images, then use it in blocks.

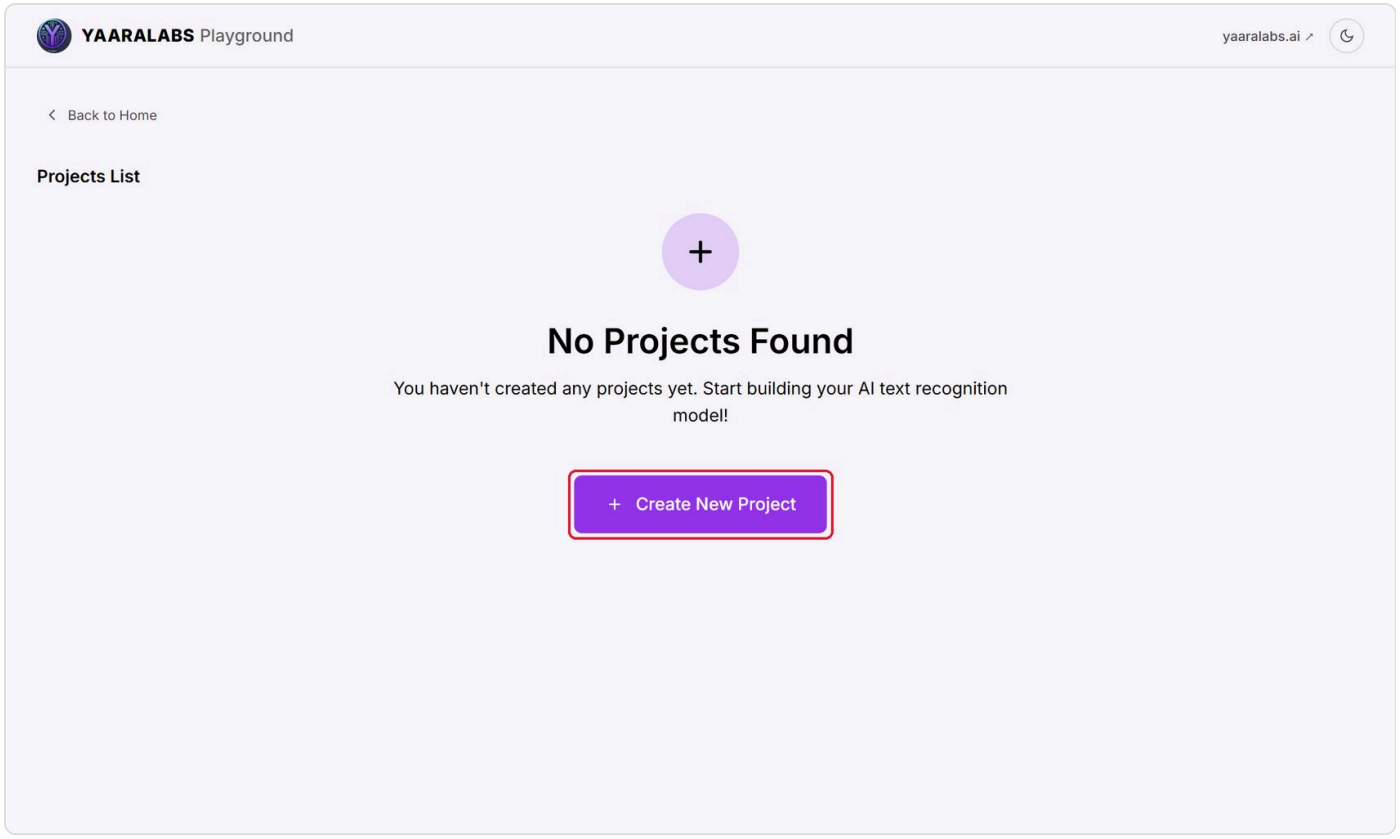
82 Open the workspace

Open the site and click Open my workspace.



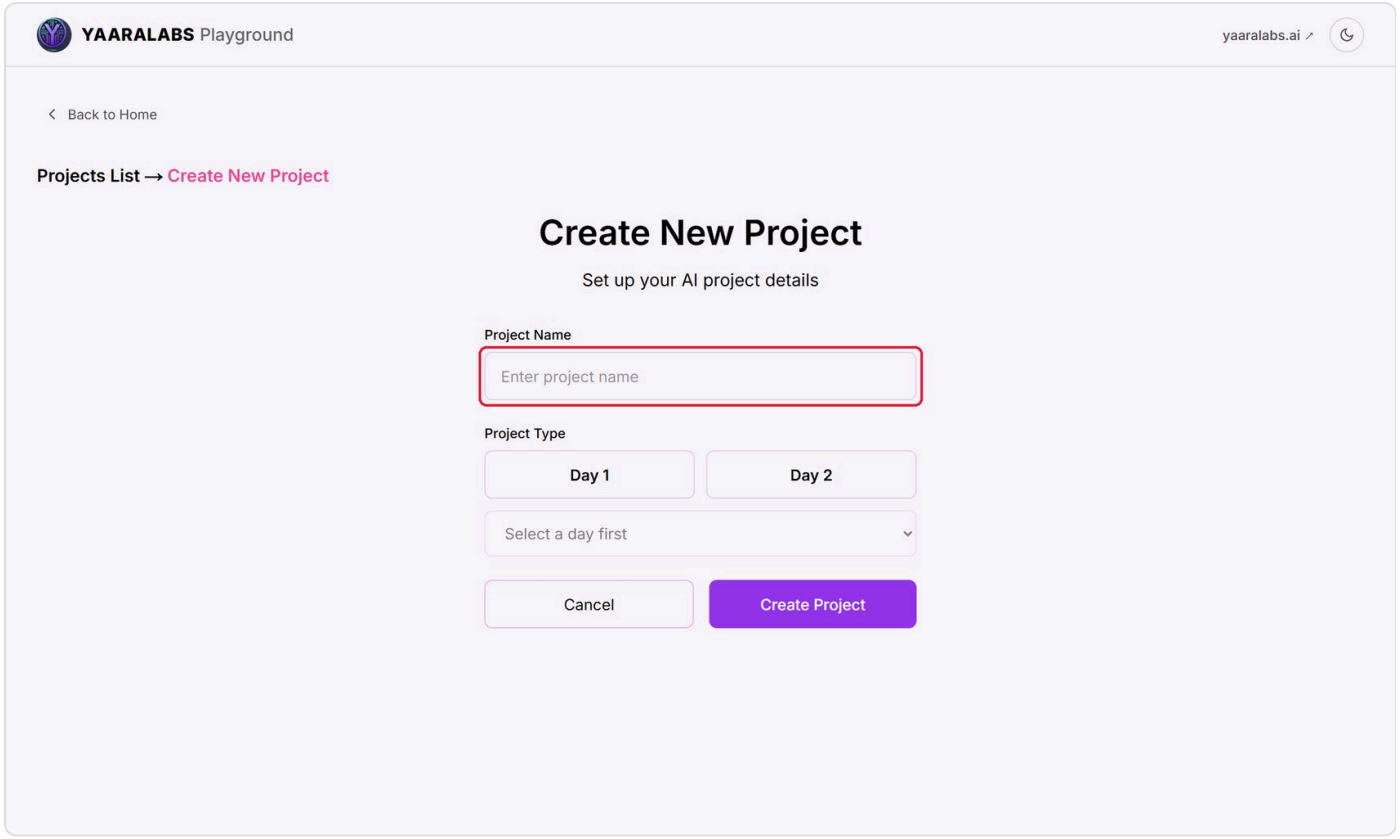
83 Create a new project

Click Create New Project.



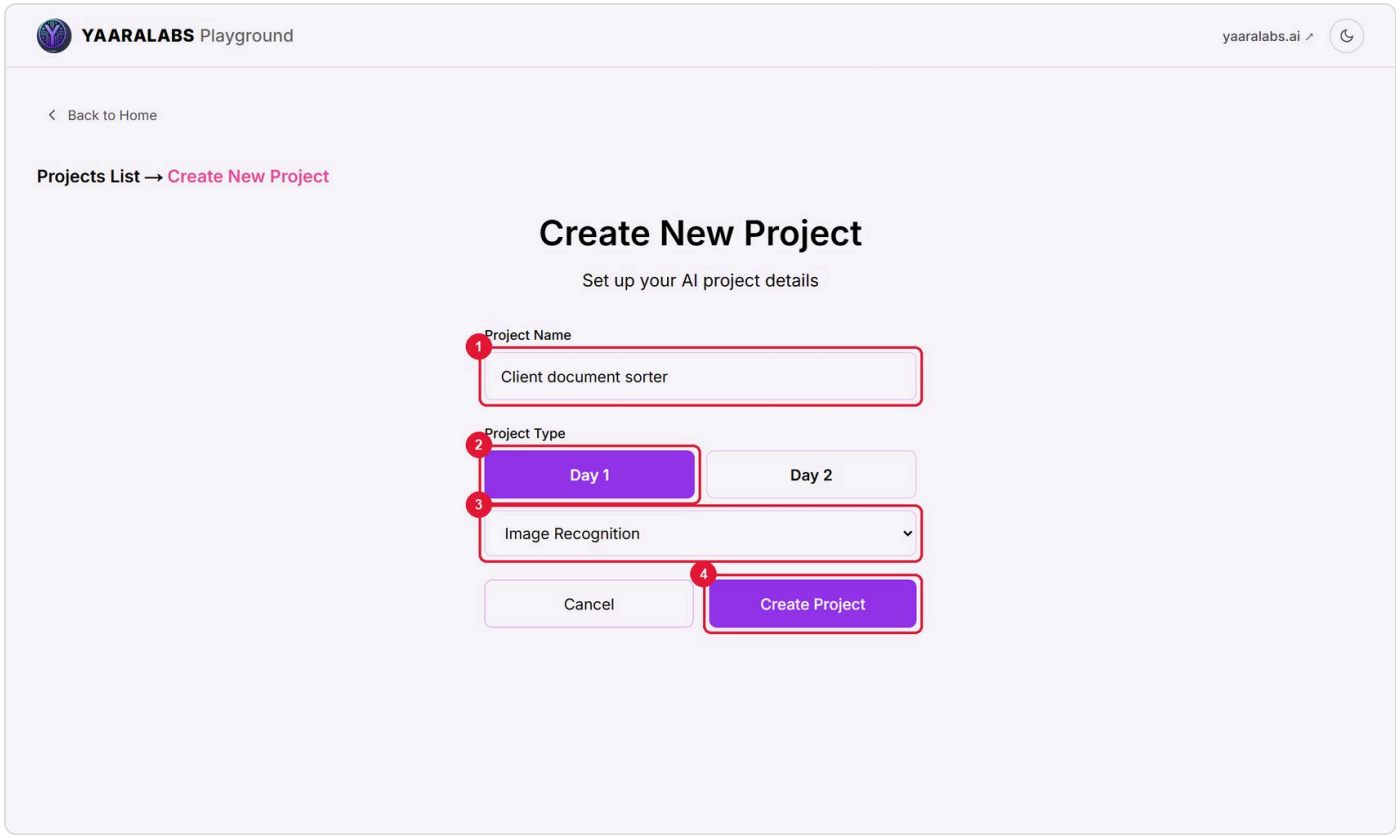
84 The new project form

The form asks for a name, the day and the project type.



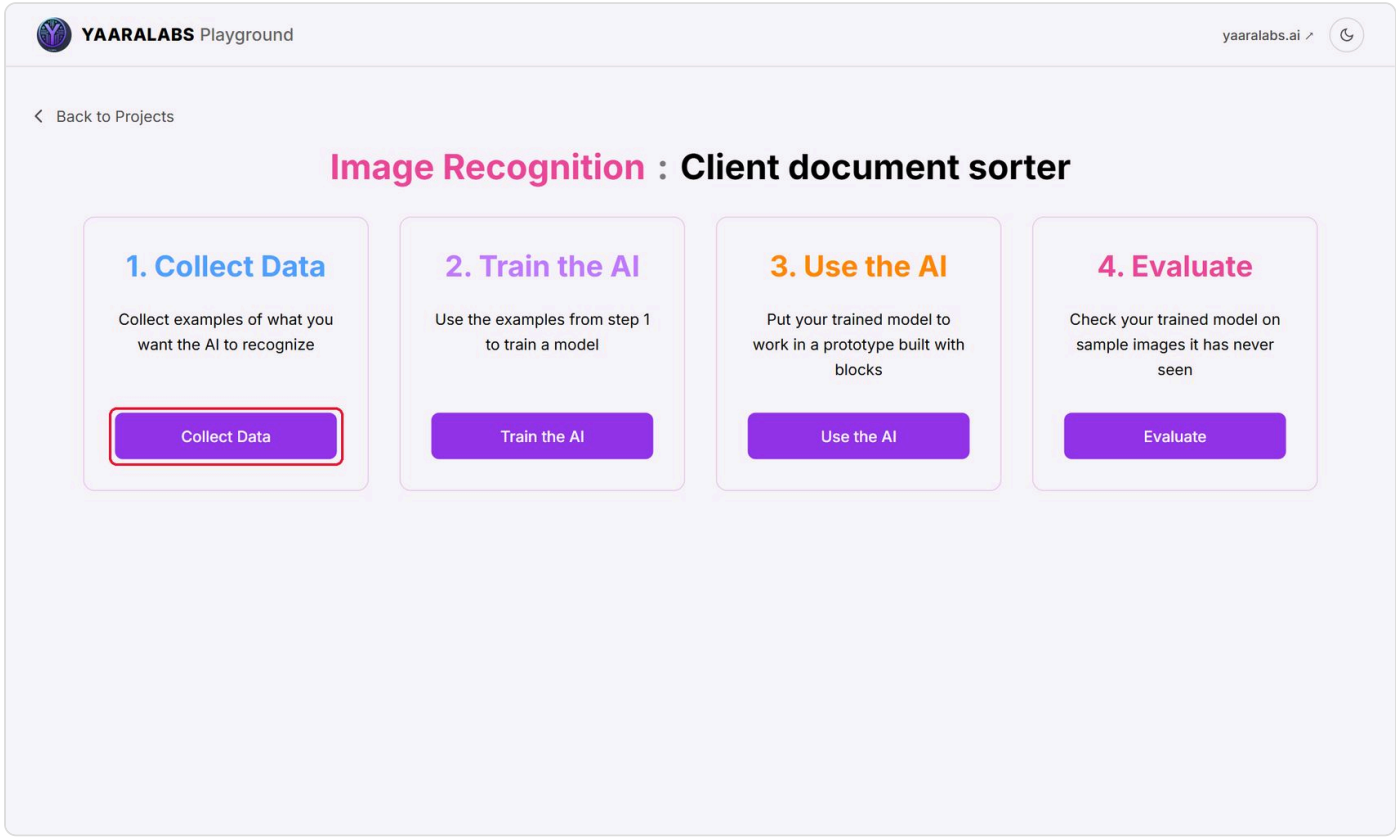
85 Name it and choose Image Recognition

Type a name (e.g. Client document sorter), click Day 1, choose Image Recognition, then Create Project.



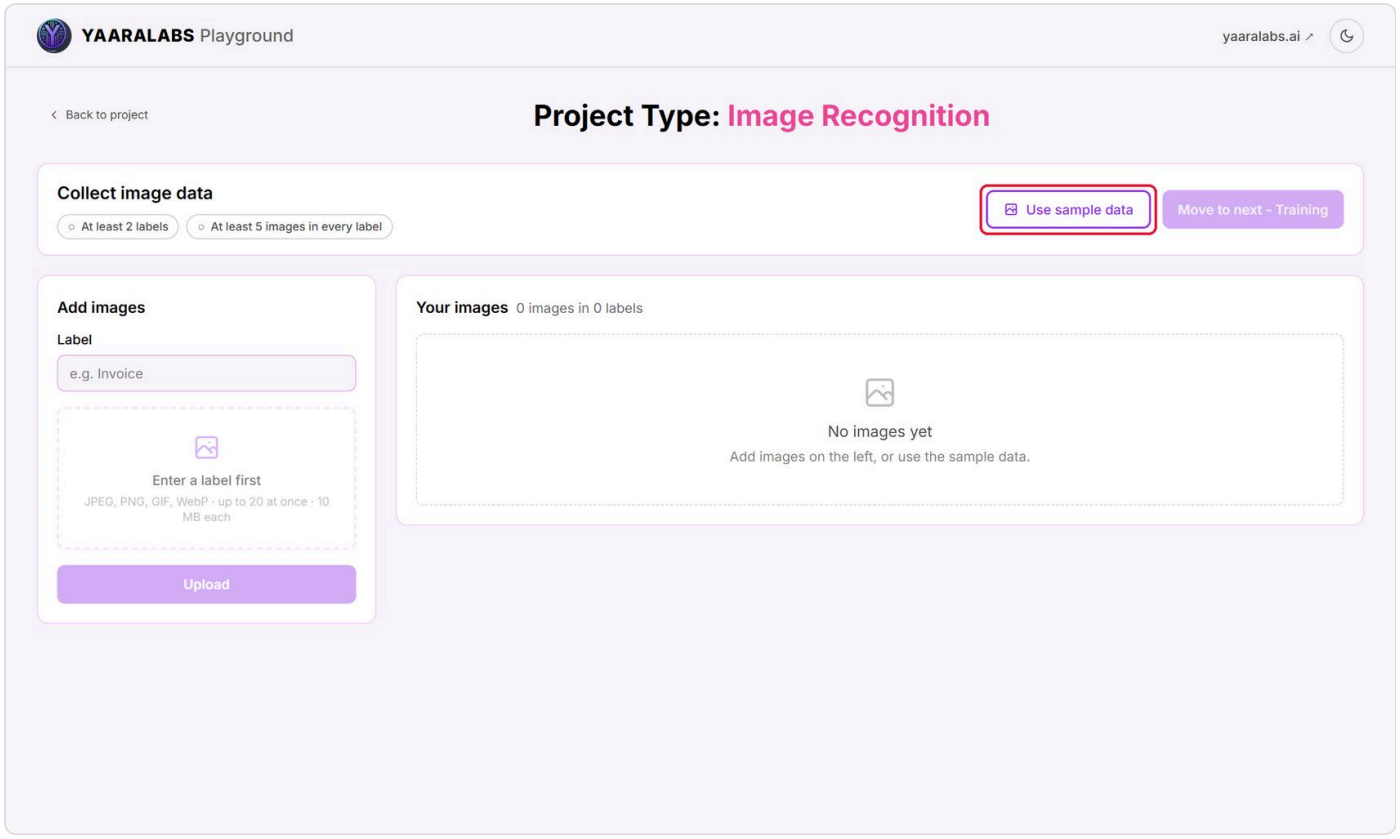
86 The project page

The project has four steps: Collect Data, Train the AI, Use the AI, Evaluate. Click Collect Data.



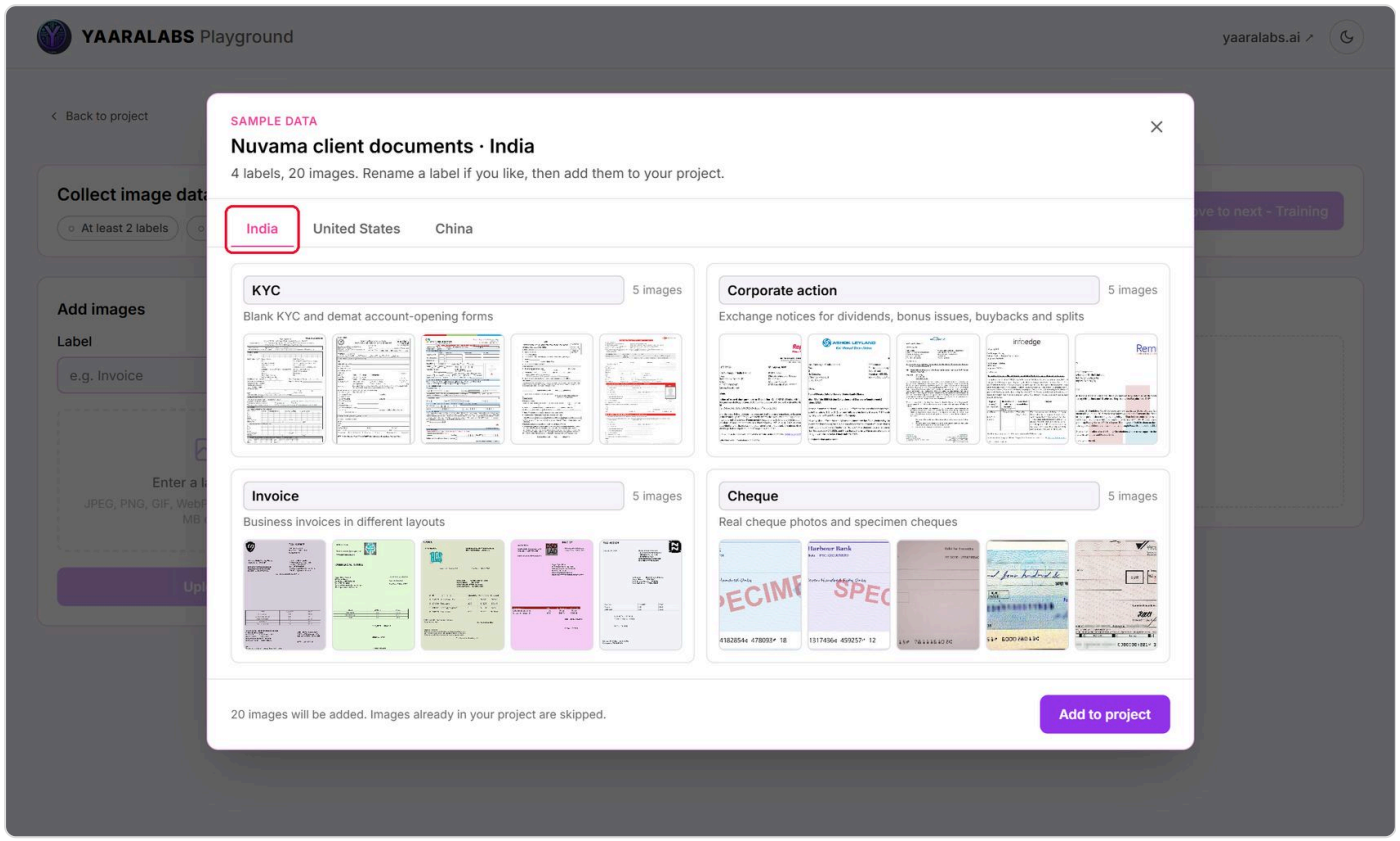
87 Collect image data

The page is empty: add images on the left or click Use sample data.



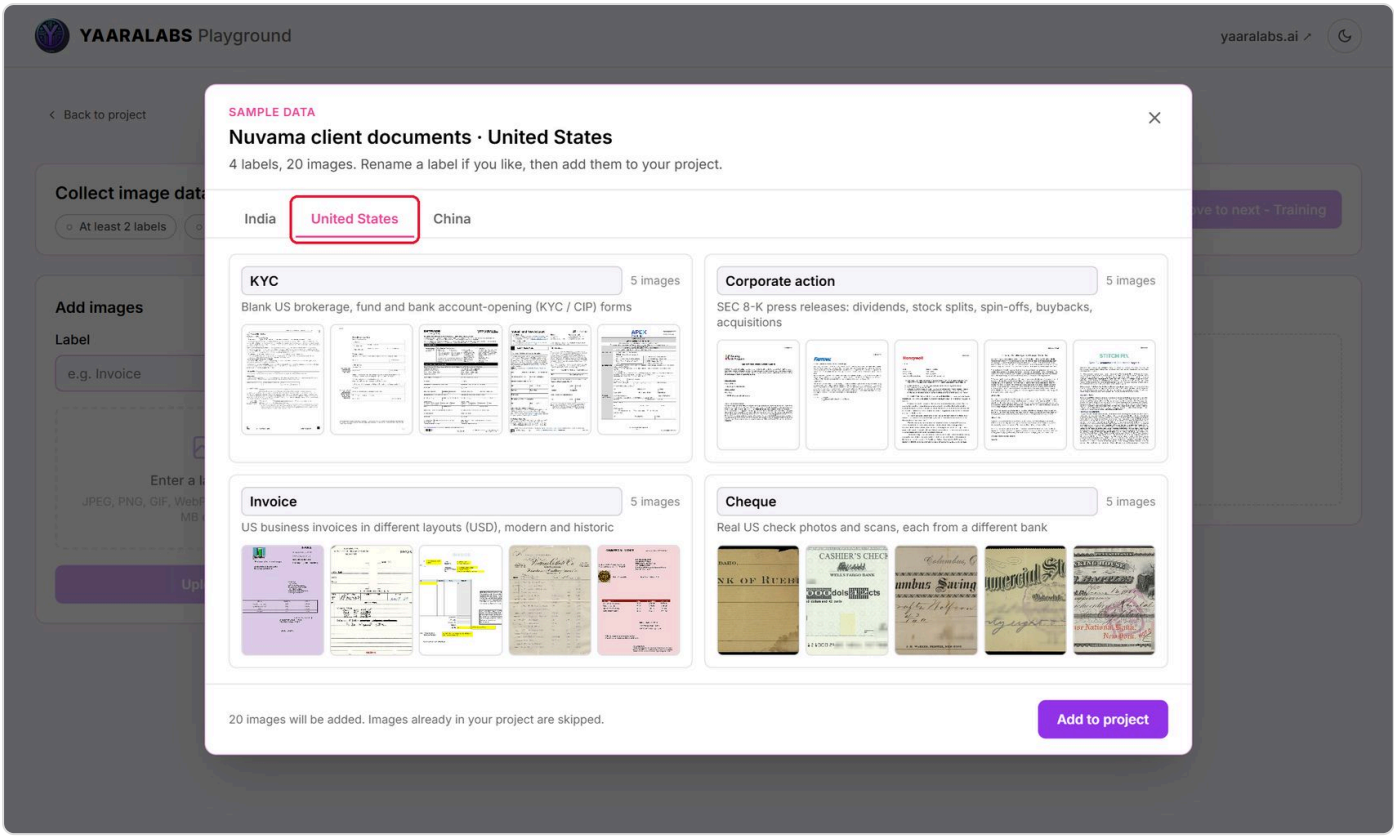
88 Sample data: India

The dialog Nuvama client documents opens on India: 4 labels with 5 images each.



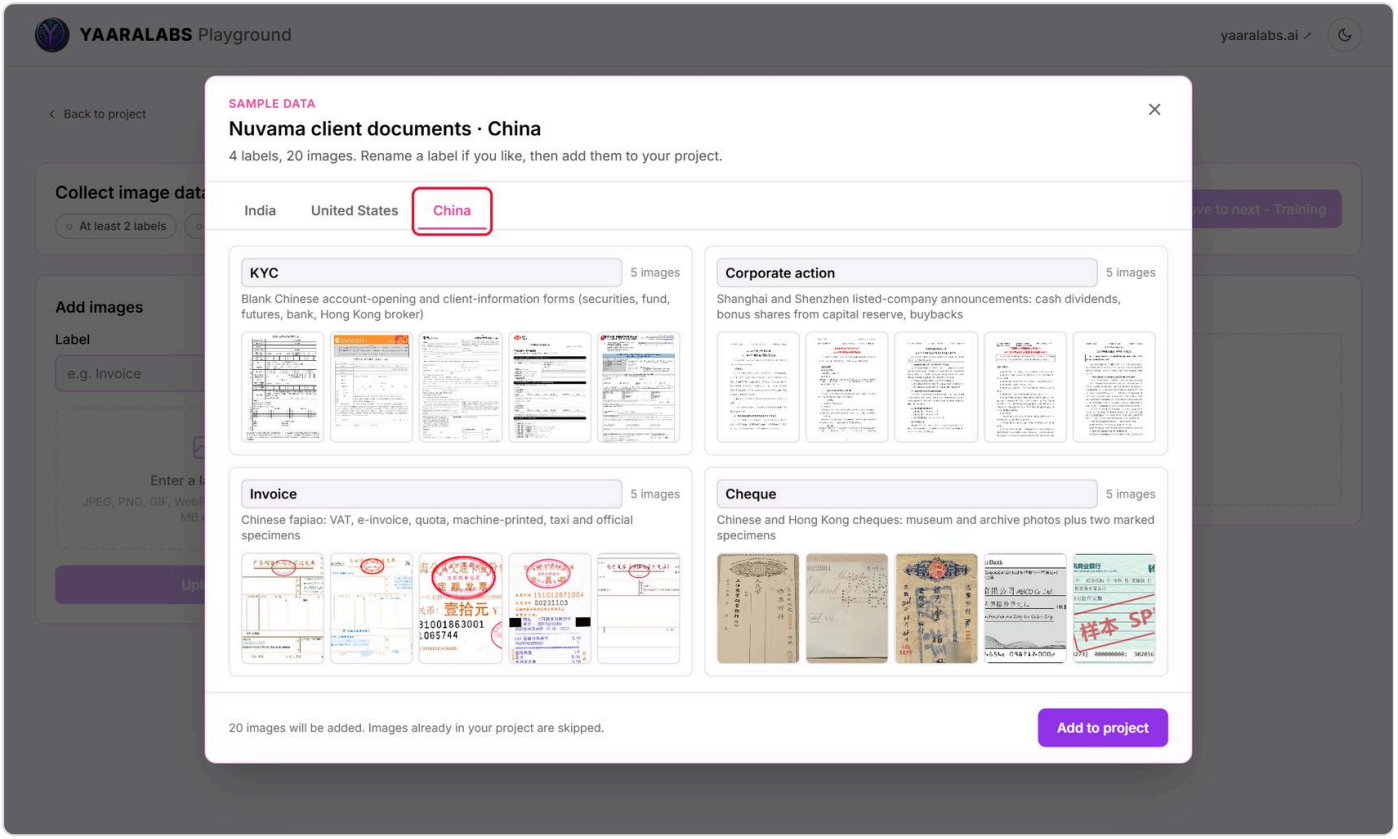
89 Sample data: United States

Click United States: the same 4 labels with US documents.



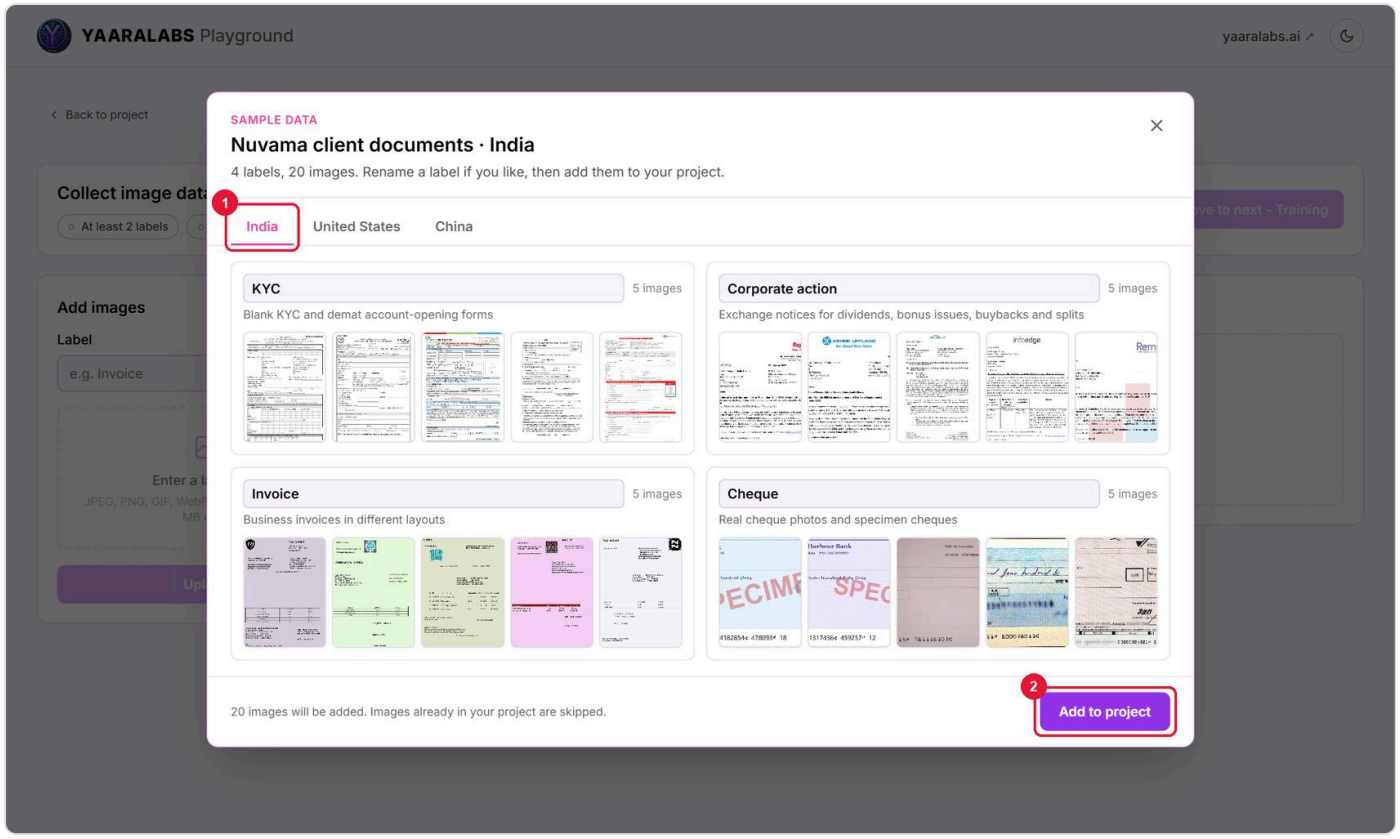
90 Sample data: China

Click China: the same 4 labels with Chinese documents.



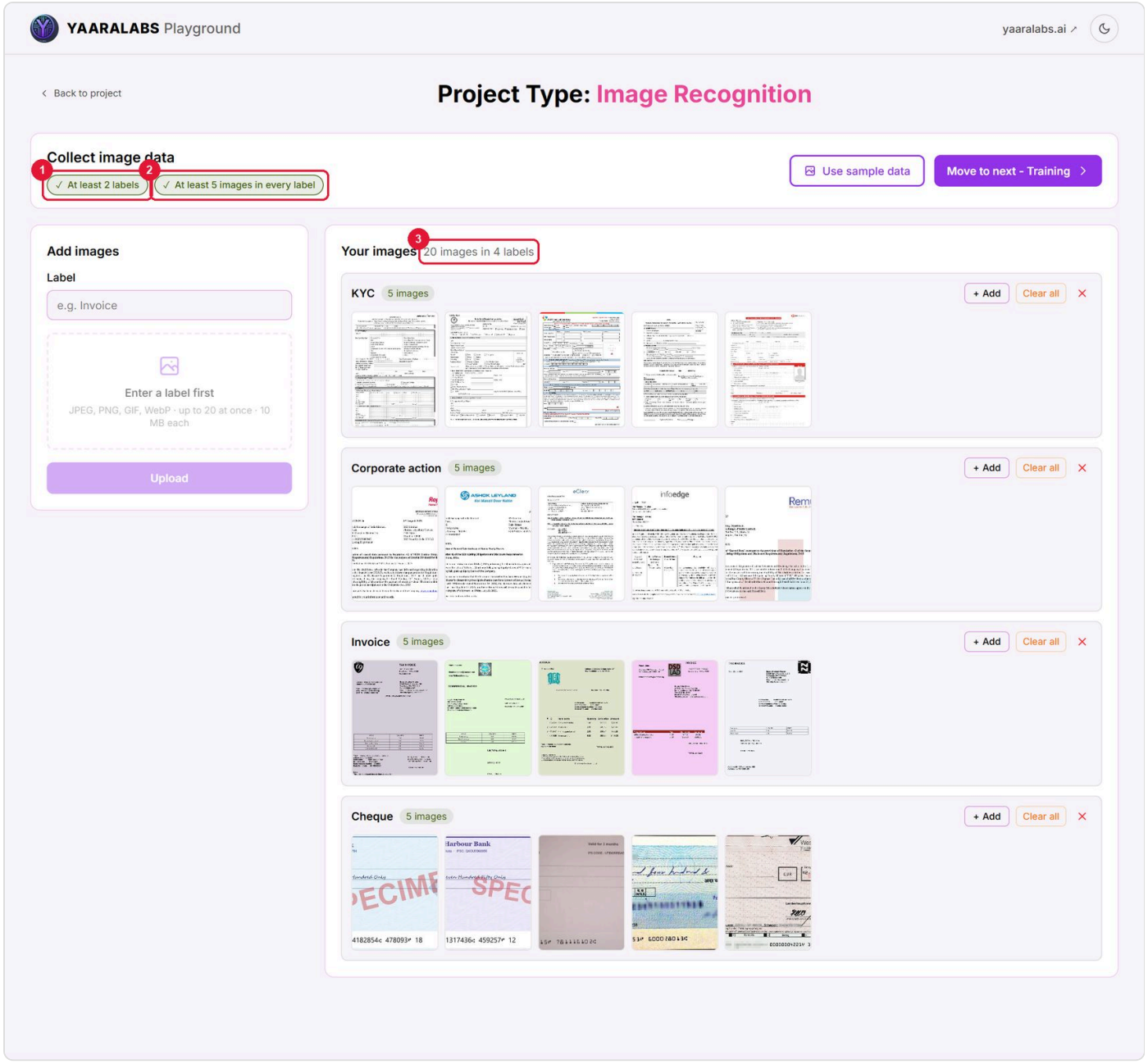
91 Add the India set

Go back to India and click Add to project.



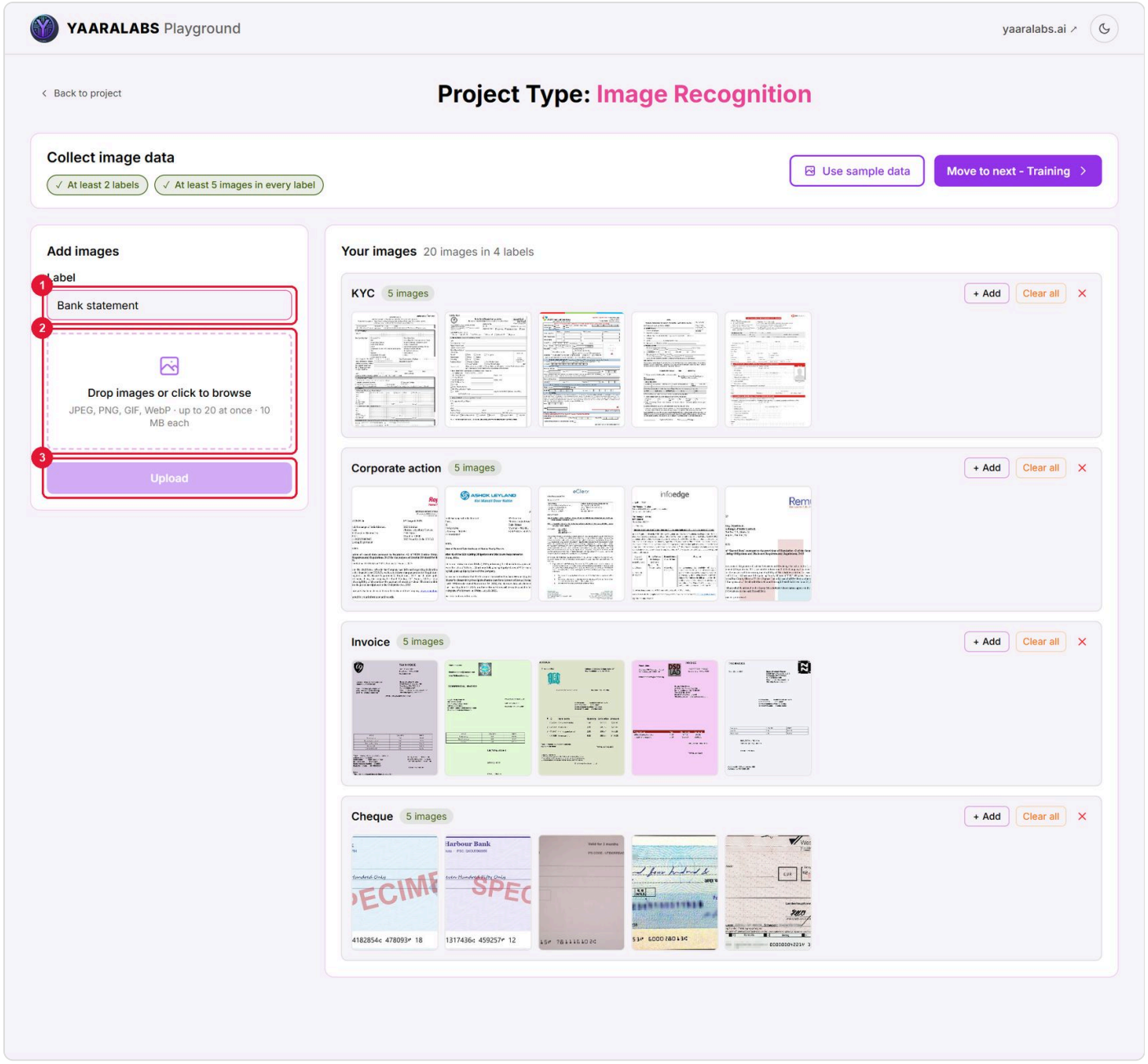
92 4 labels, 5 images each

The images are in your project: KYC, Corporate action, Invoice and Cheque, 5 each. Both checks turn green.



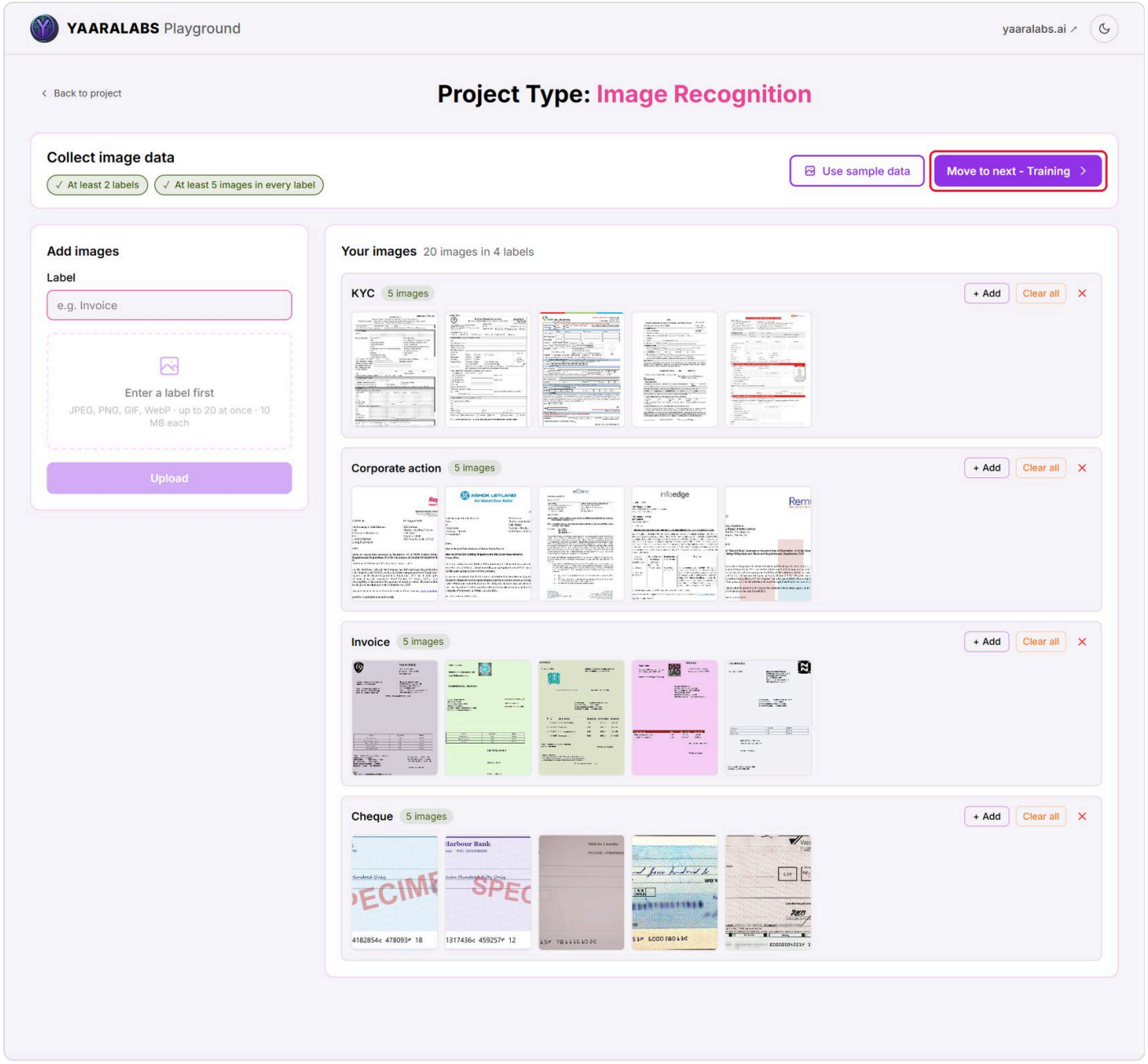
93 Adding your own images (optional)

To add images by hand: type a label, drop or choose images in the box, then click Upload.



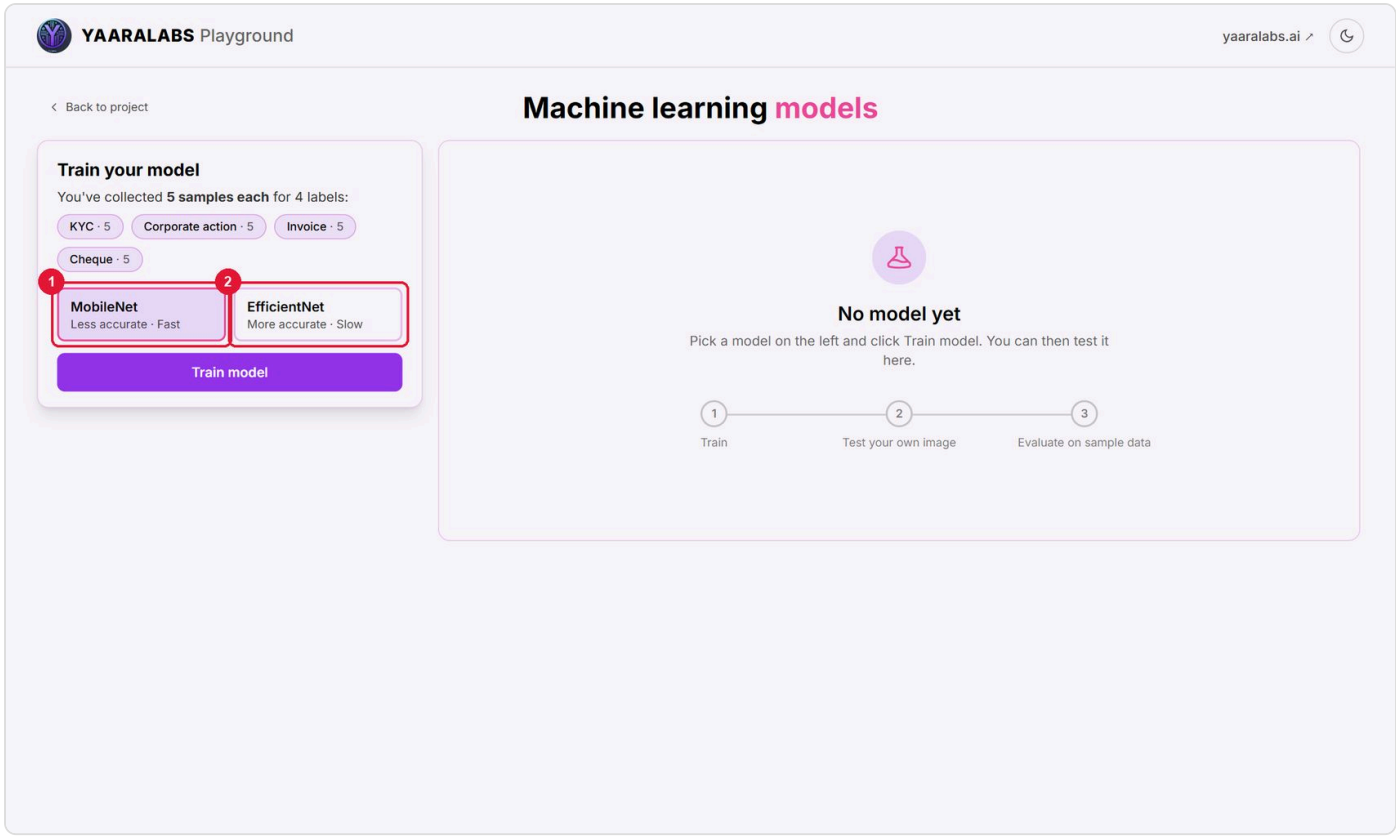
94 Move to training

Click Move to next - Training.



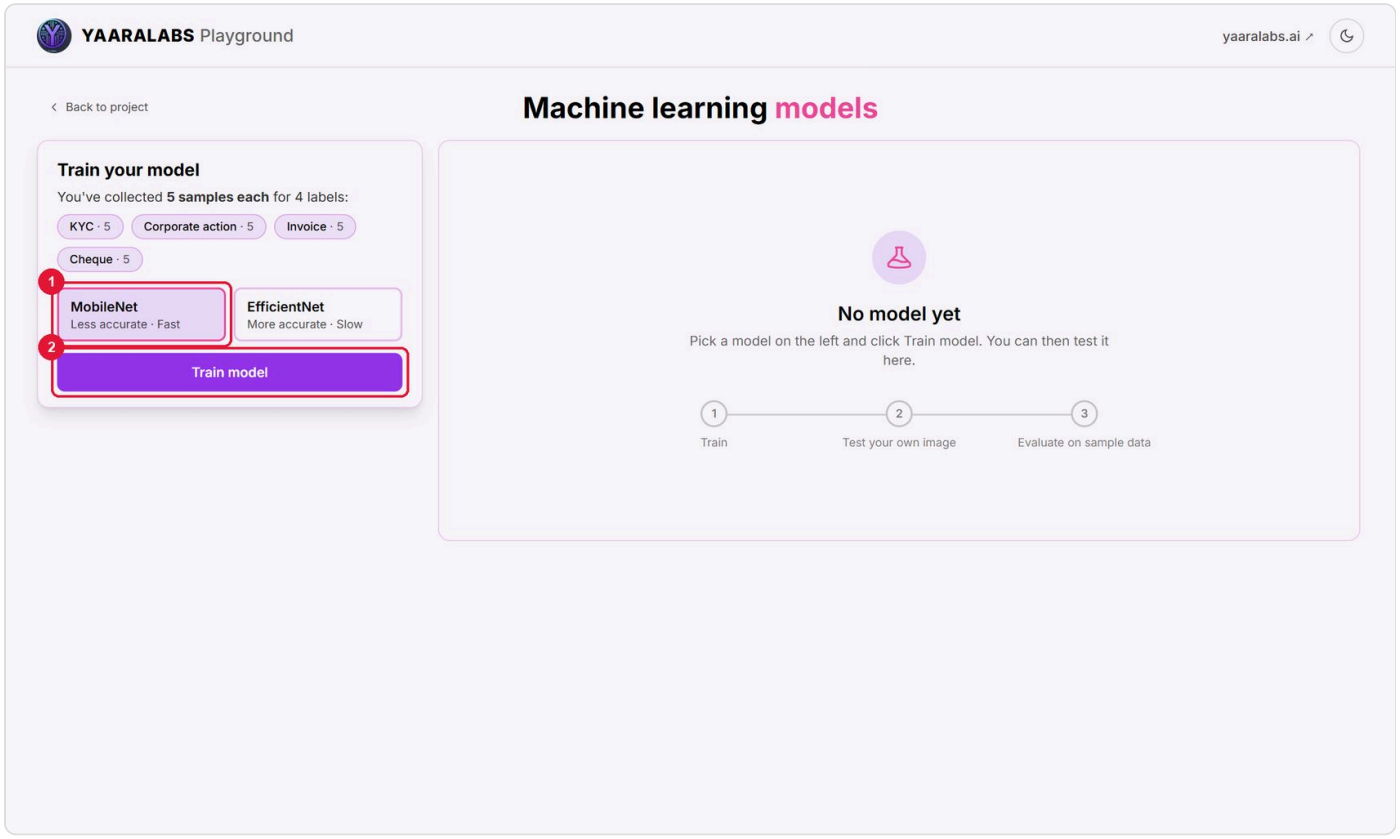
95 Train your model

The card lists your labels and two models: MobileNet (Less accurate · Fast) and EfficientNet (More accurate · Slow).



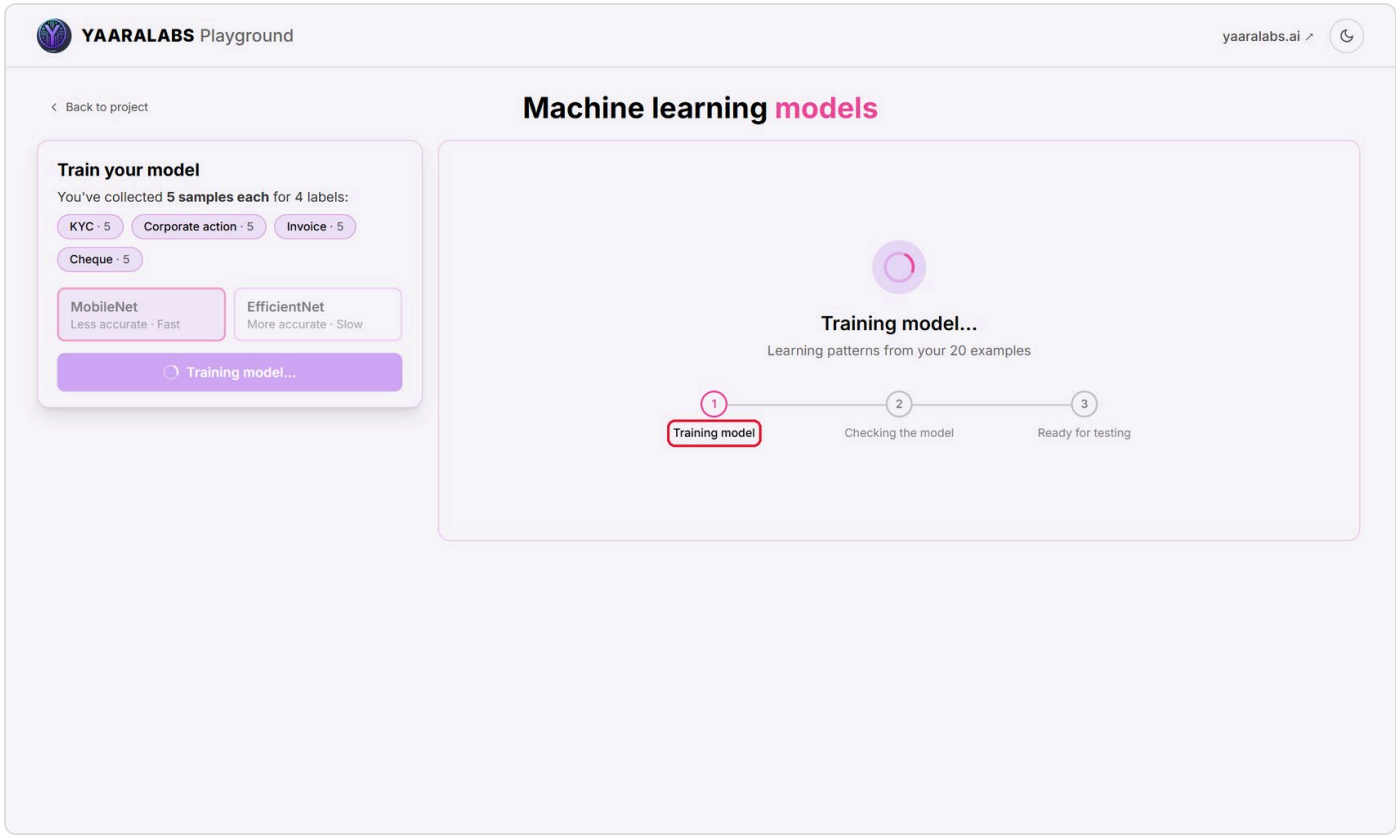
96 Pick MobileNet and train

Click MobileNet, then Train model.



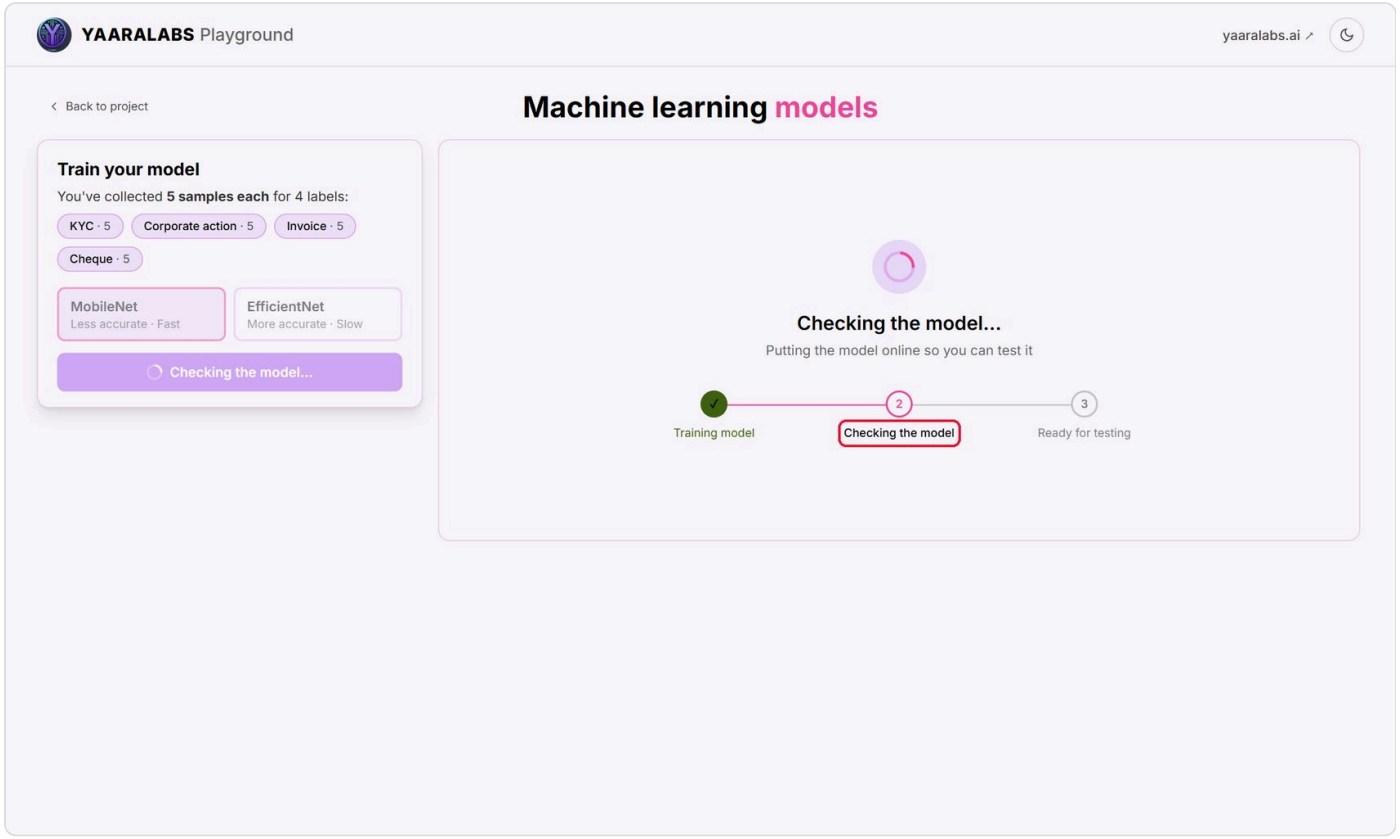
97 Step 1: Training model

The model learns from your 20 images (about a minute).



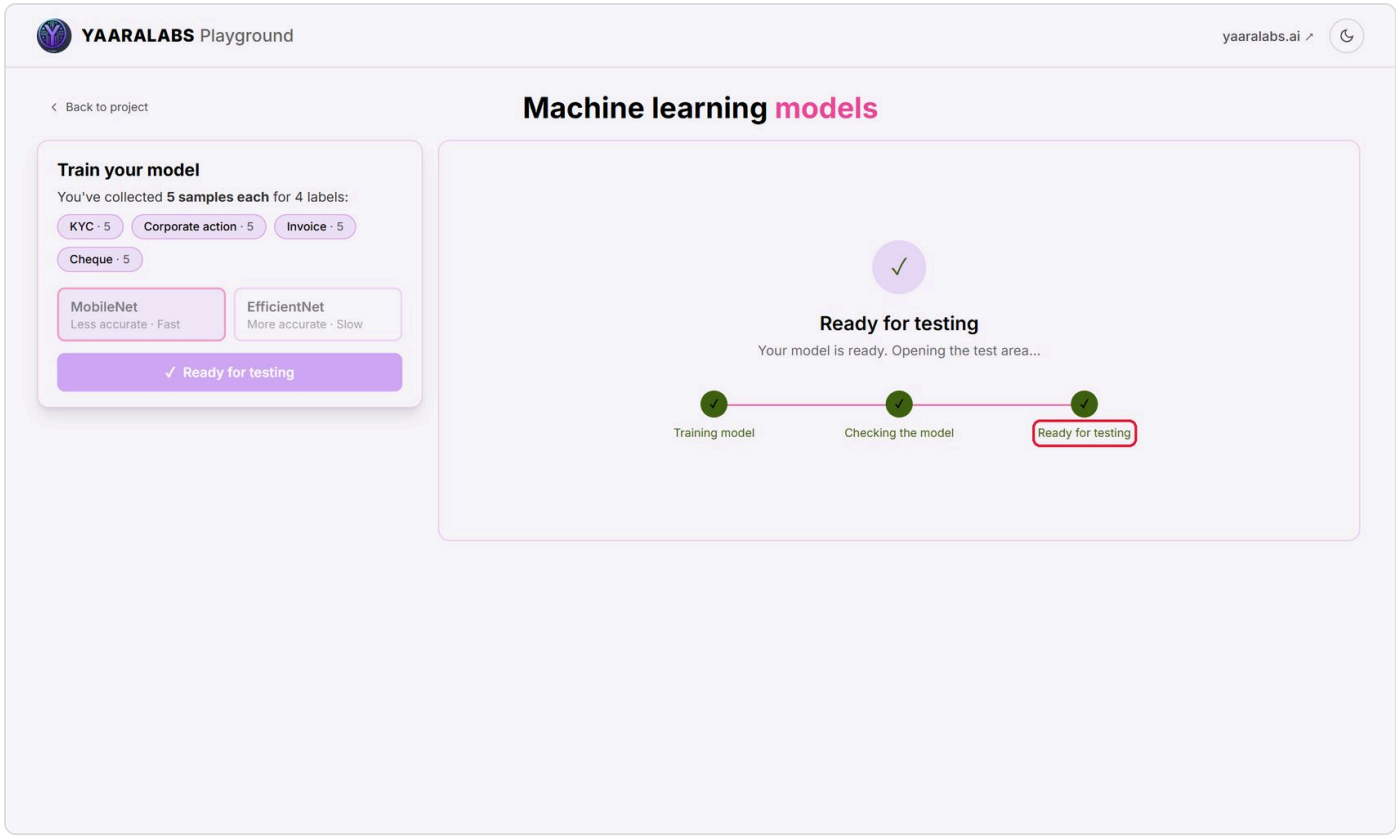
98 Step 2: Checking the model

Training is done; the model is checked and put online.



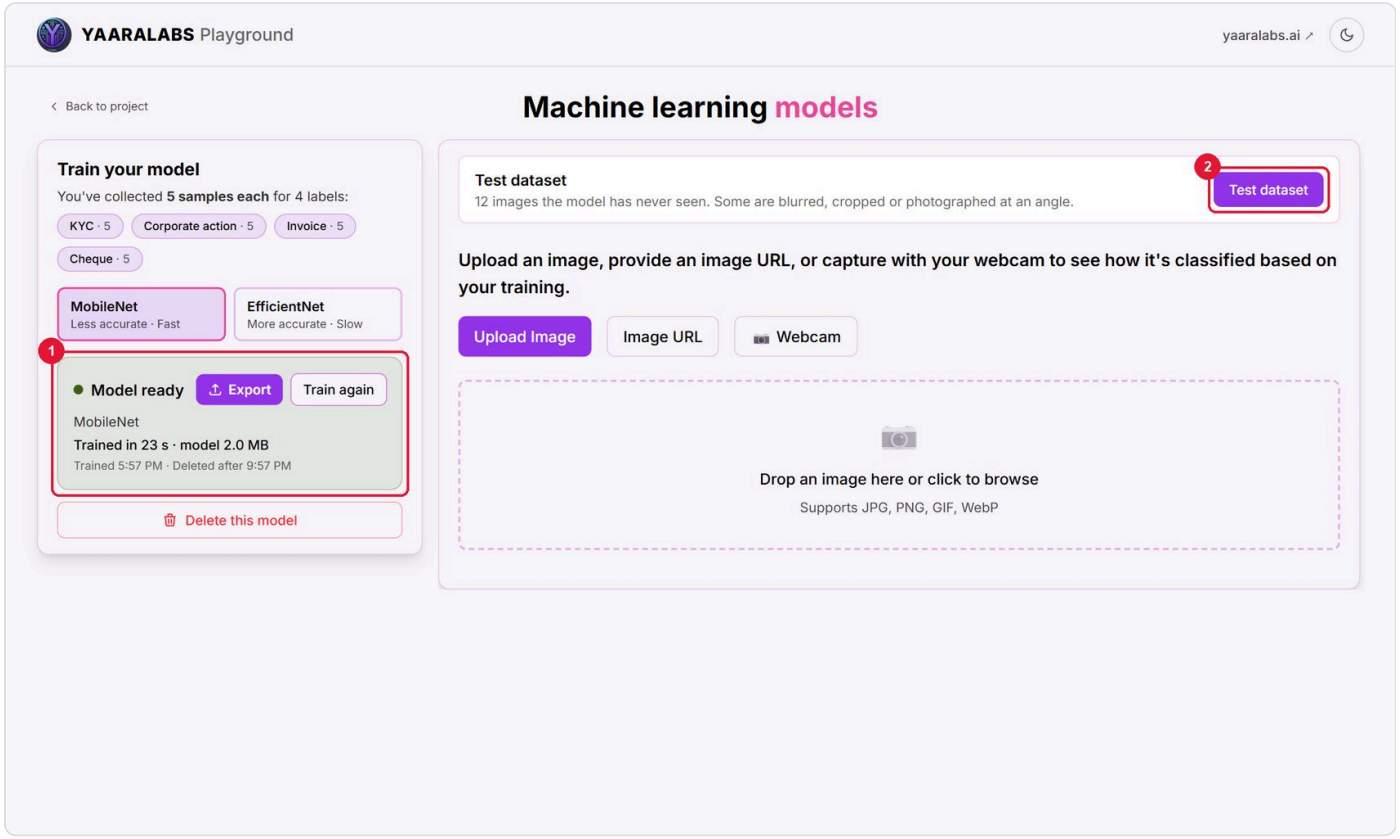
99 Step 3: Ready for testing

All three steps are ticked: the test area opens next.



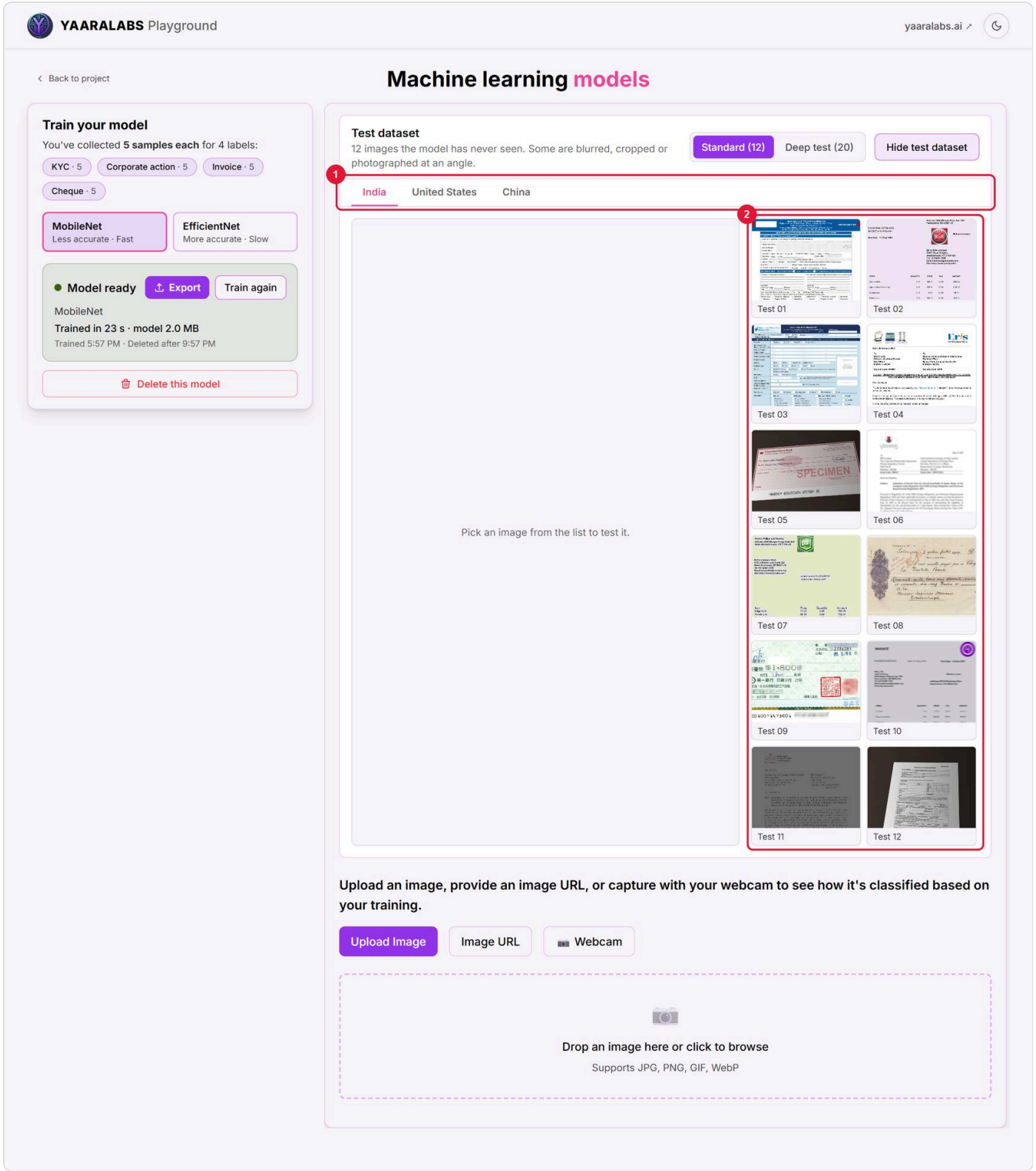
100 Model ready

Model ready shows the model and its training time. Click Test dataset to try it.



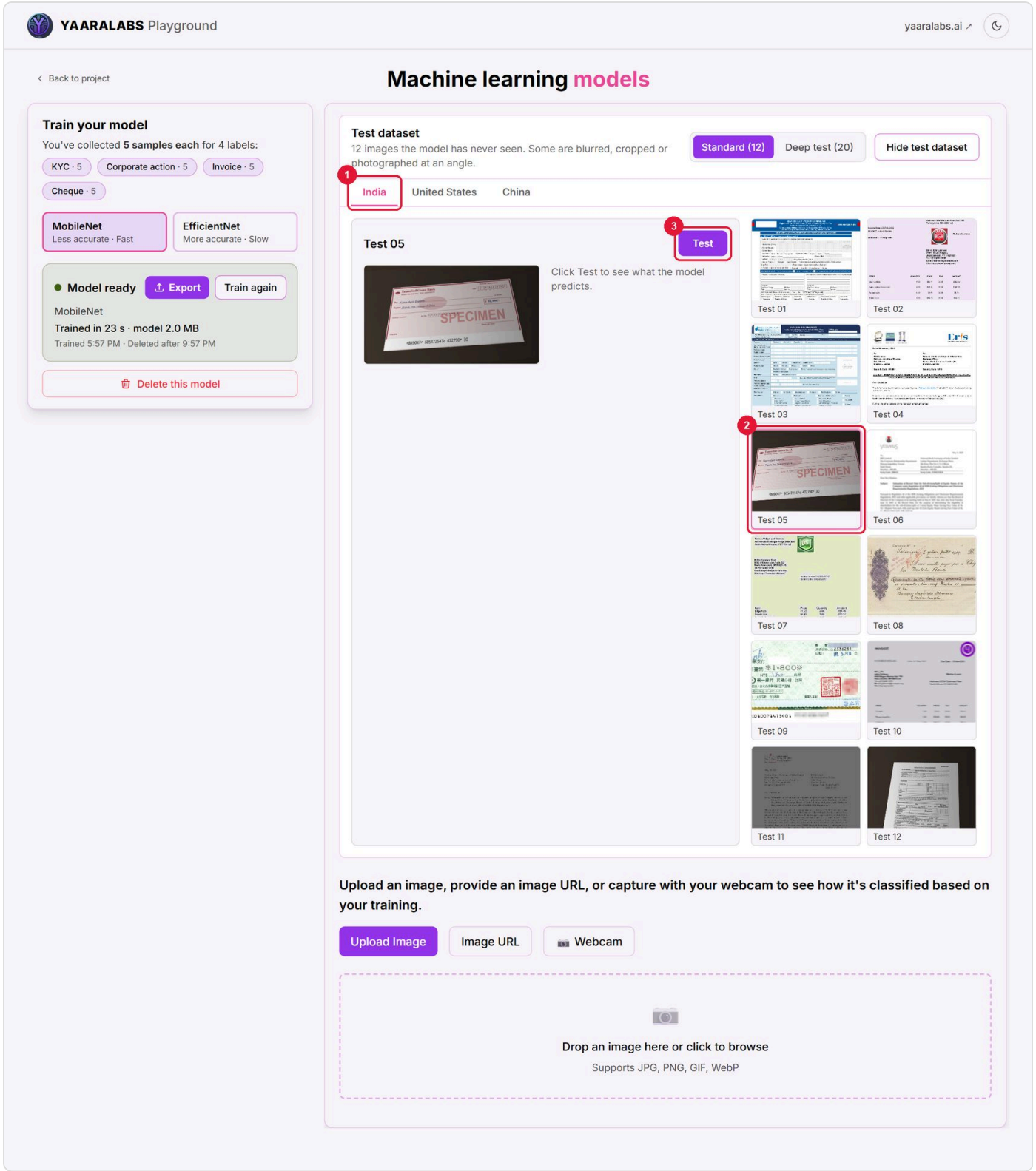
101 The test dataset

The panel has a tab per region and images the model has never seen. India is open.



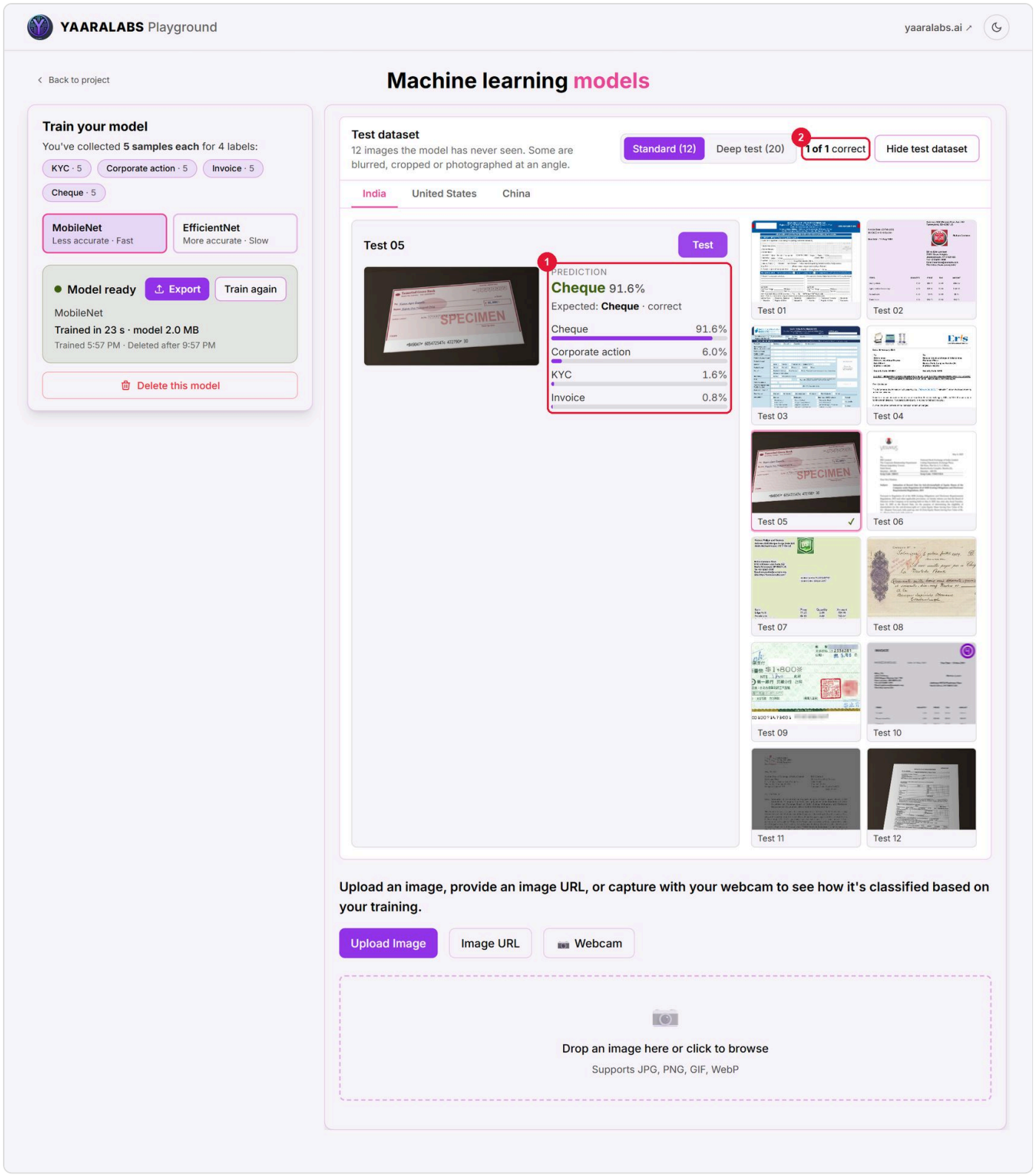
102 India: pick an image

Click Test 05, then Test.



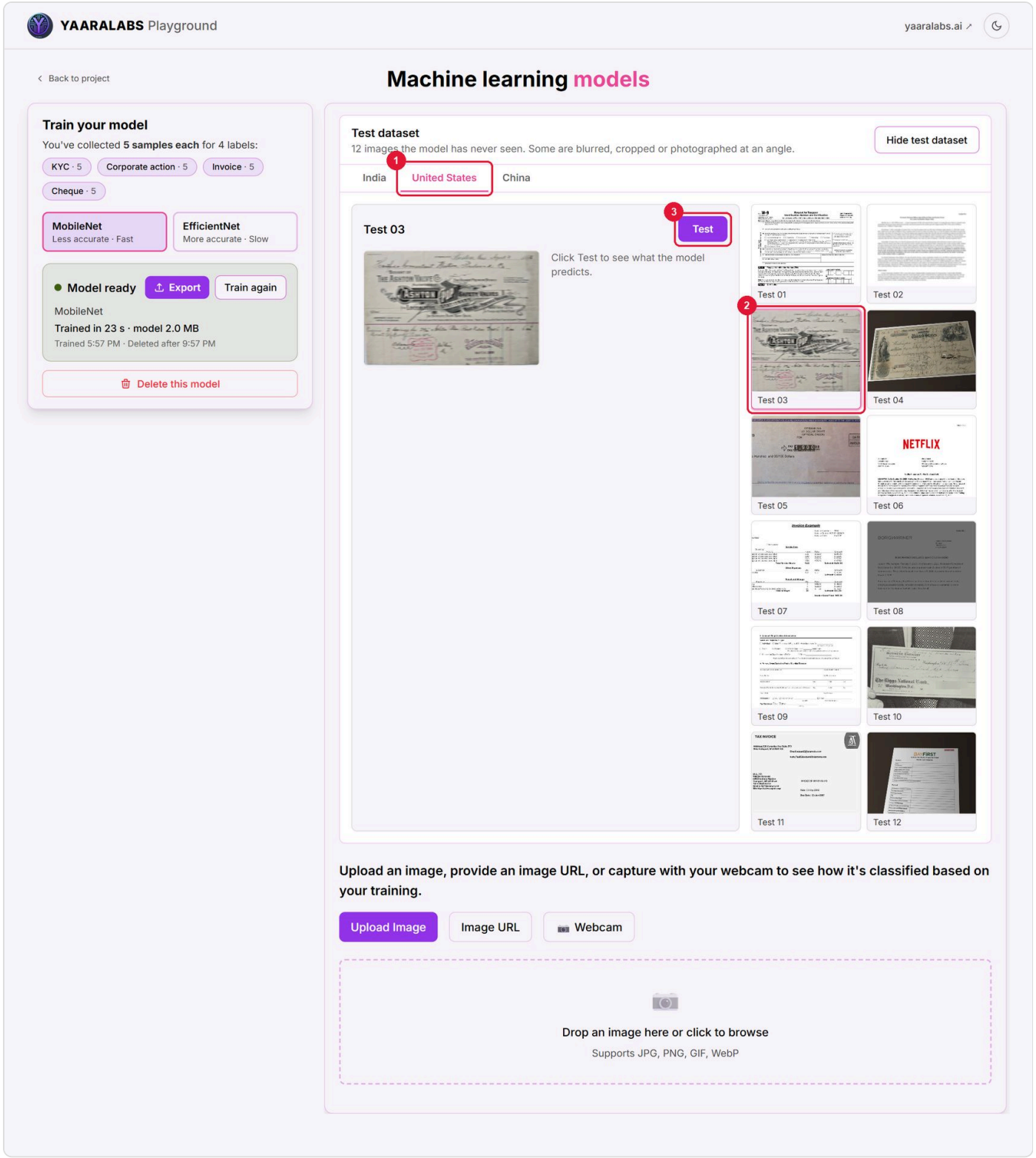
103 India: the prediction

The model says Cheque (91.6%) and it was Cheque: correct. Every label gets a confidence; the counter at the top shows x of y correct.



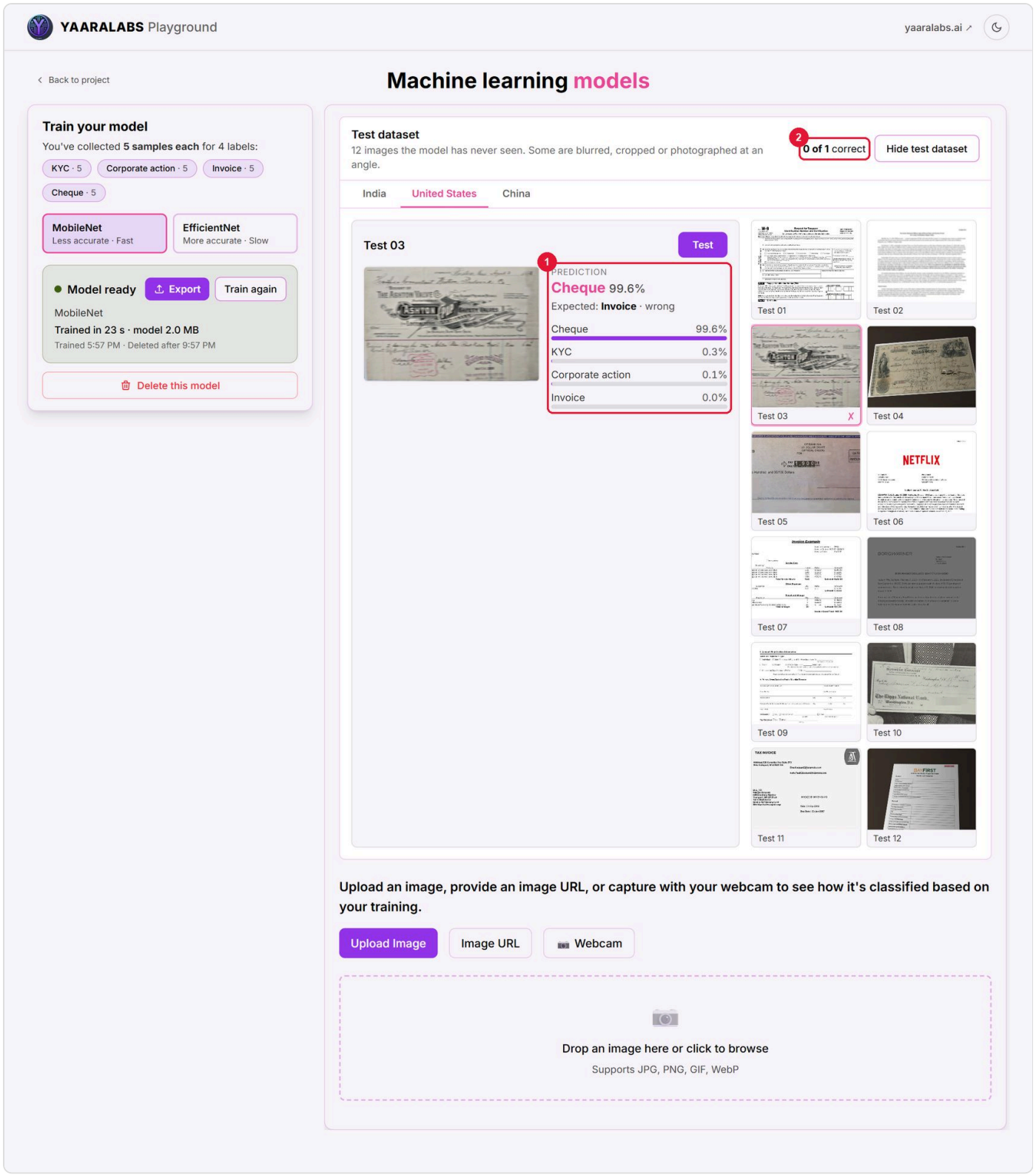
104 United States: pick an image

Click United States, pick Test 03, then Test.



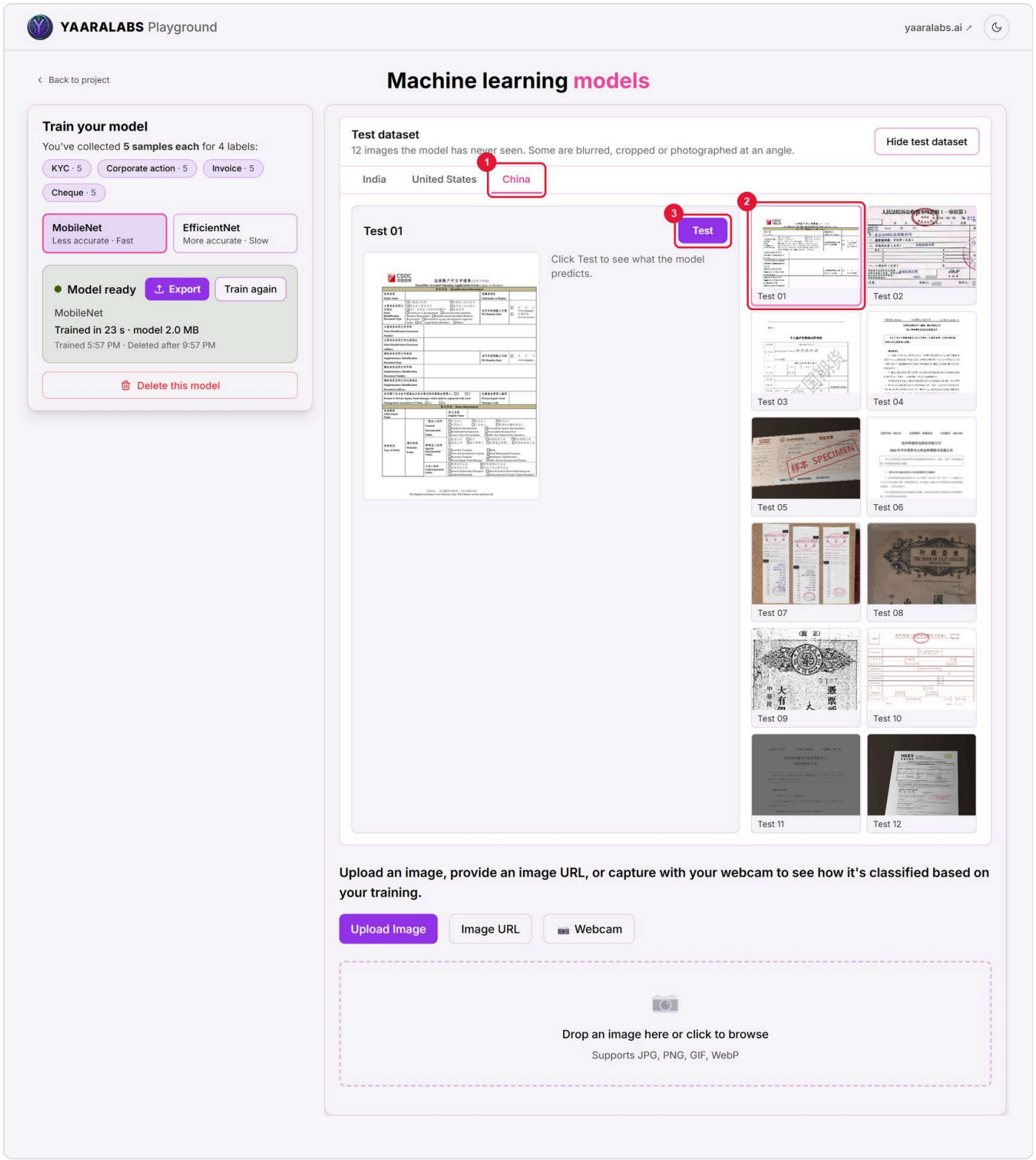
105 United States: the prediction

The model says Cheque (99.6%) but it was Invoice: wrong. The model learned from Indian documents; US layouts can trip it. The counter keeps score.



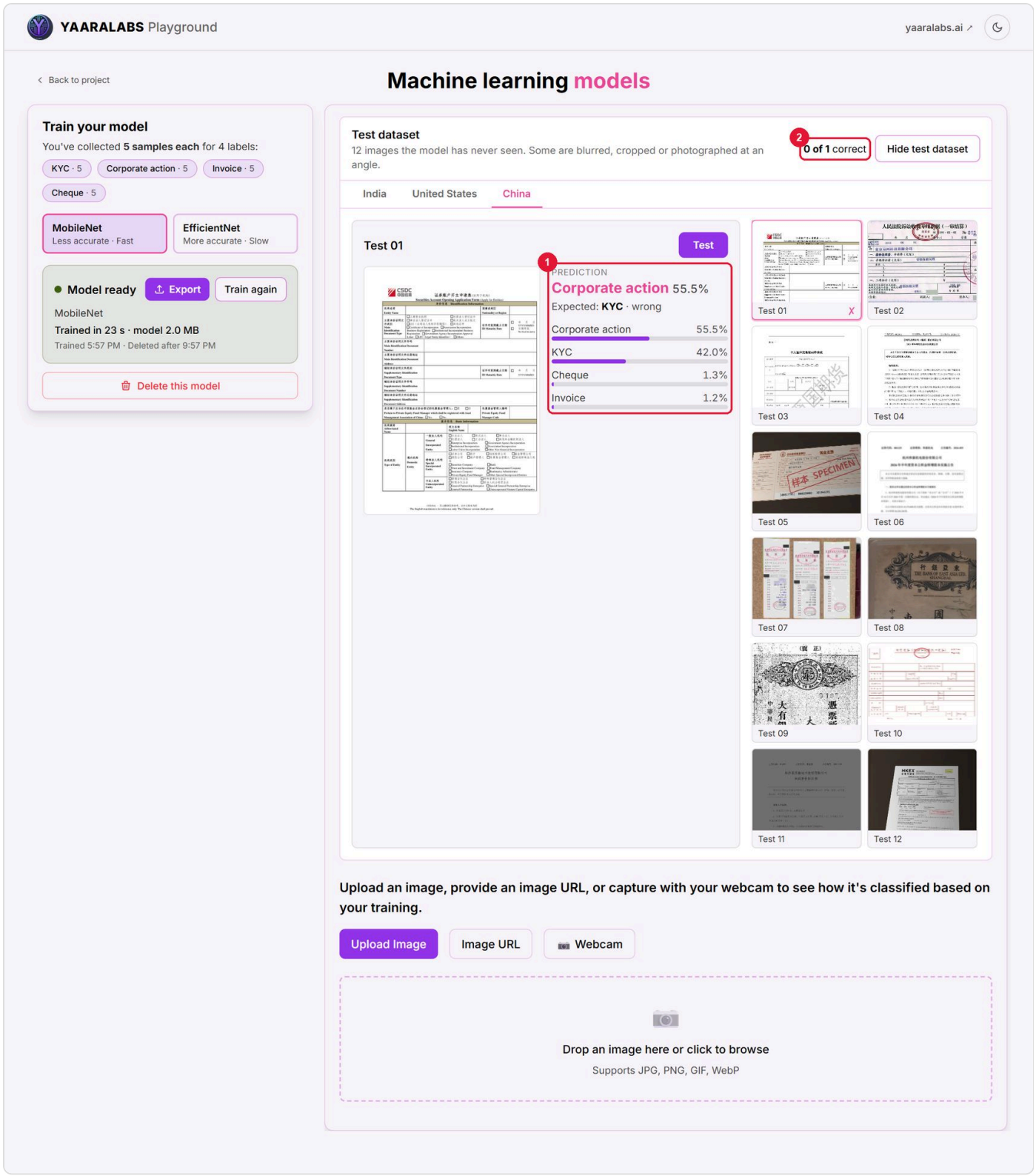
106 China: pick an image

Click China, pick Test 01, then Test.



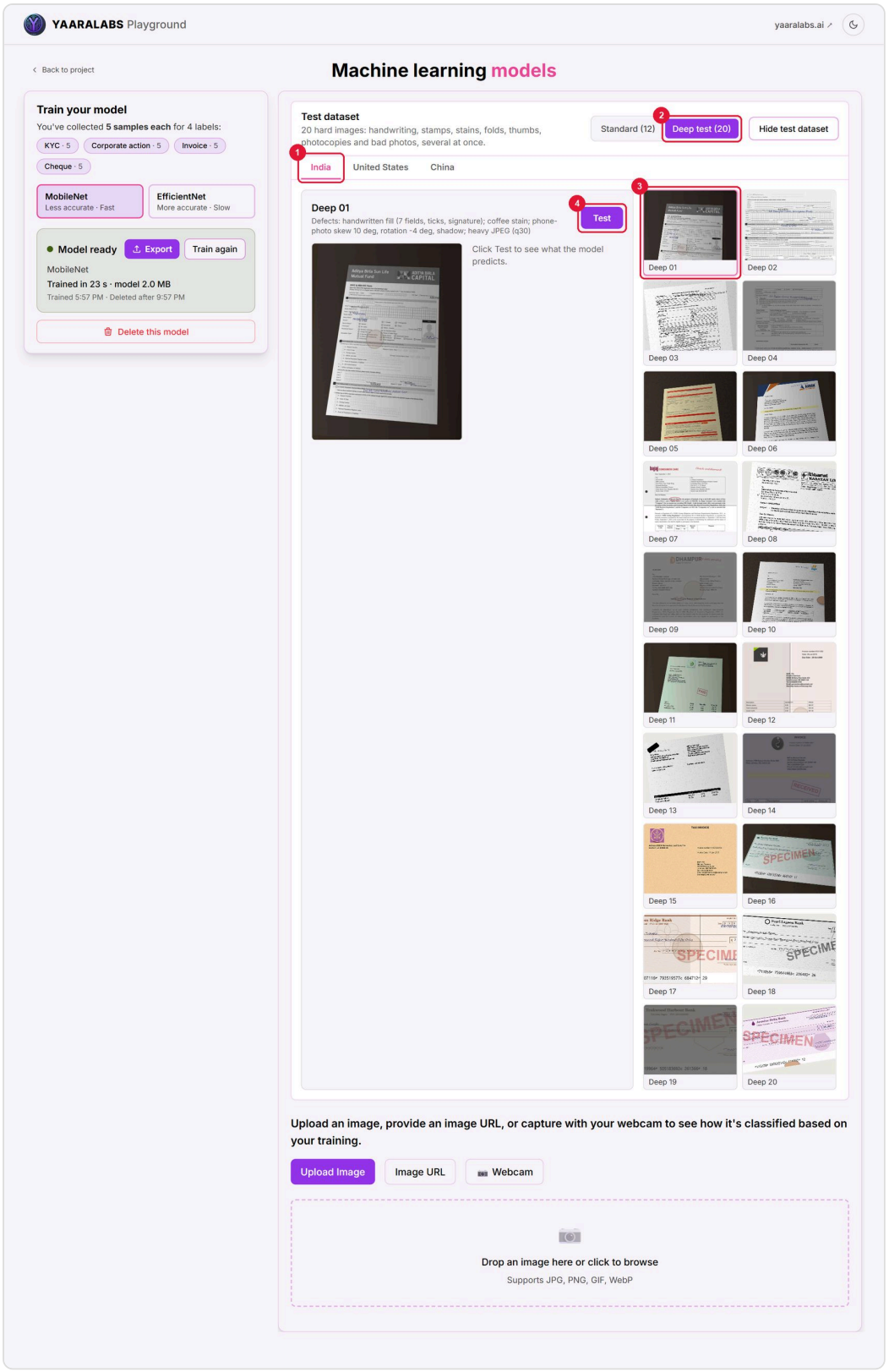
107 China: the prediction

The model says Corporate action (55.5%) but it was KYC: wrong. Chinese documents look different again; check the confidences and the counter.



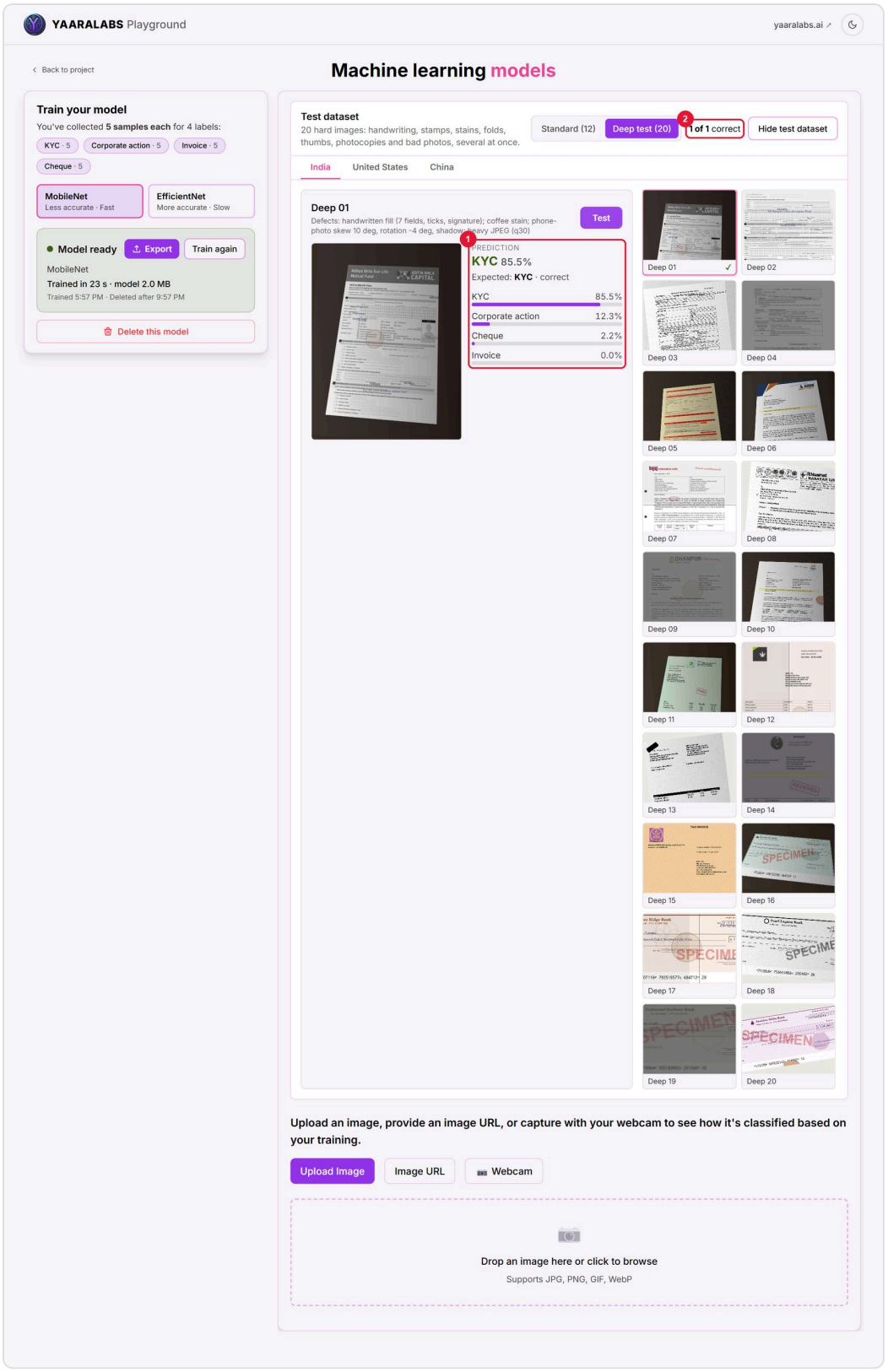
108 India deep test: pick an image

Back on India, open Deep test (hard images: handwriting, stains, bad photos), pick Deep 01, then Test.



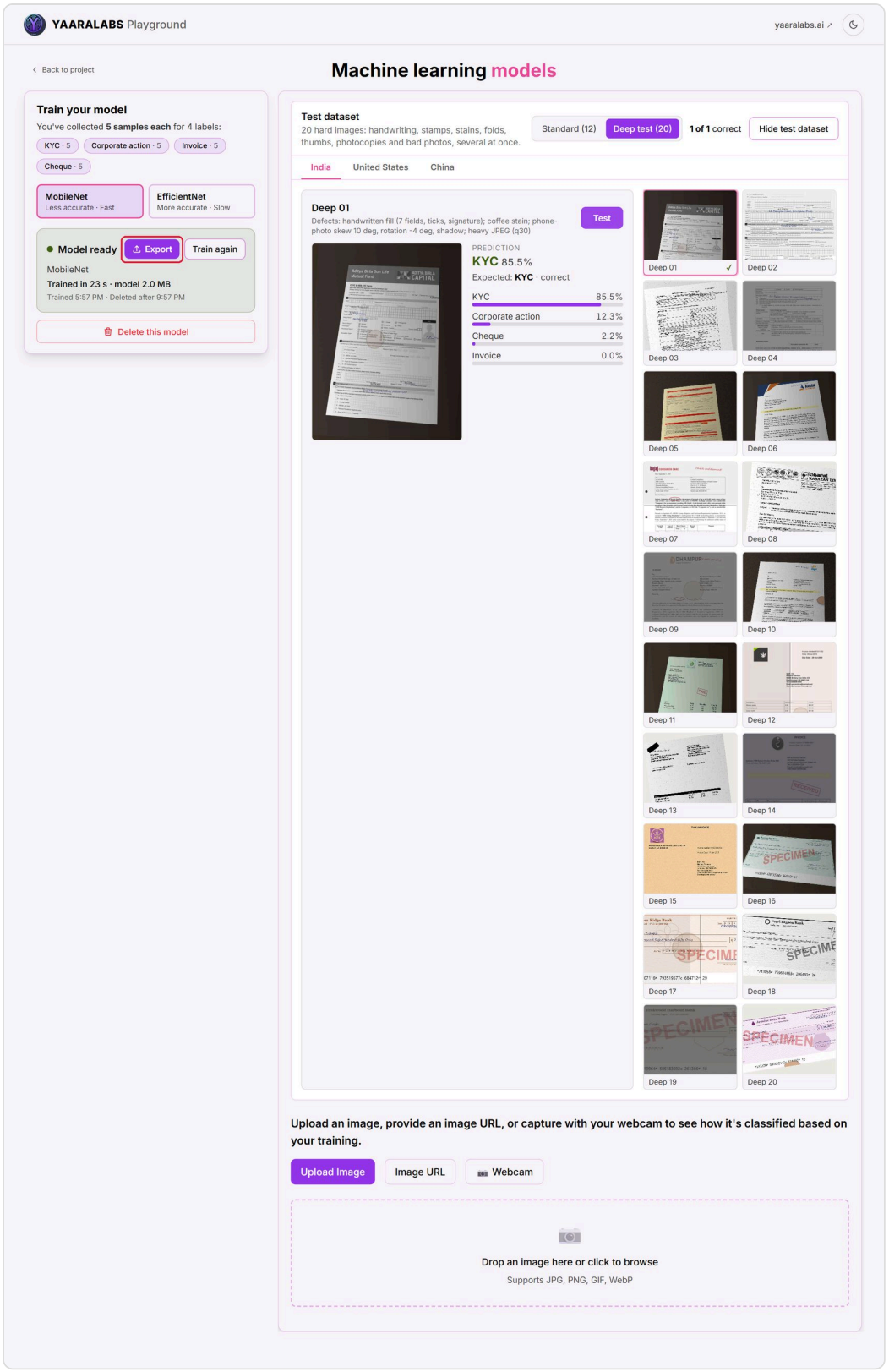
109 India deep test: the prediction

The model says KYC (85.5%) and it was KYC: correct. Hard images trip the model more often; the counter keeps score.



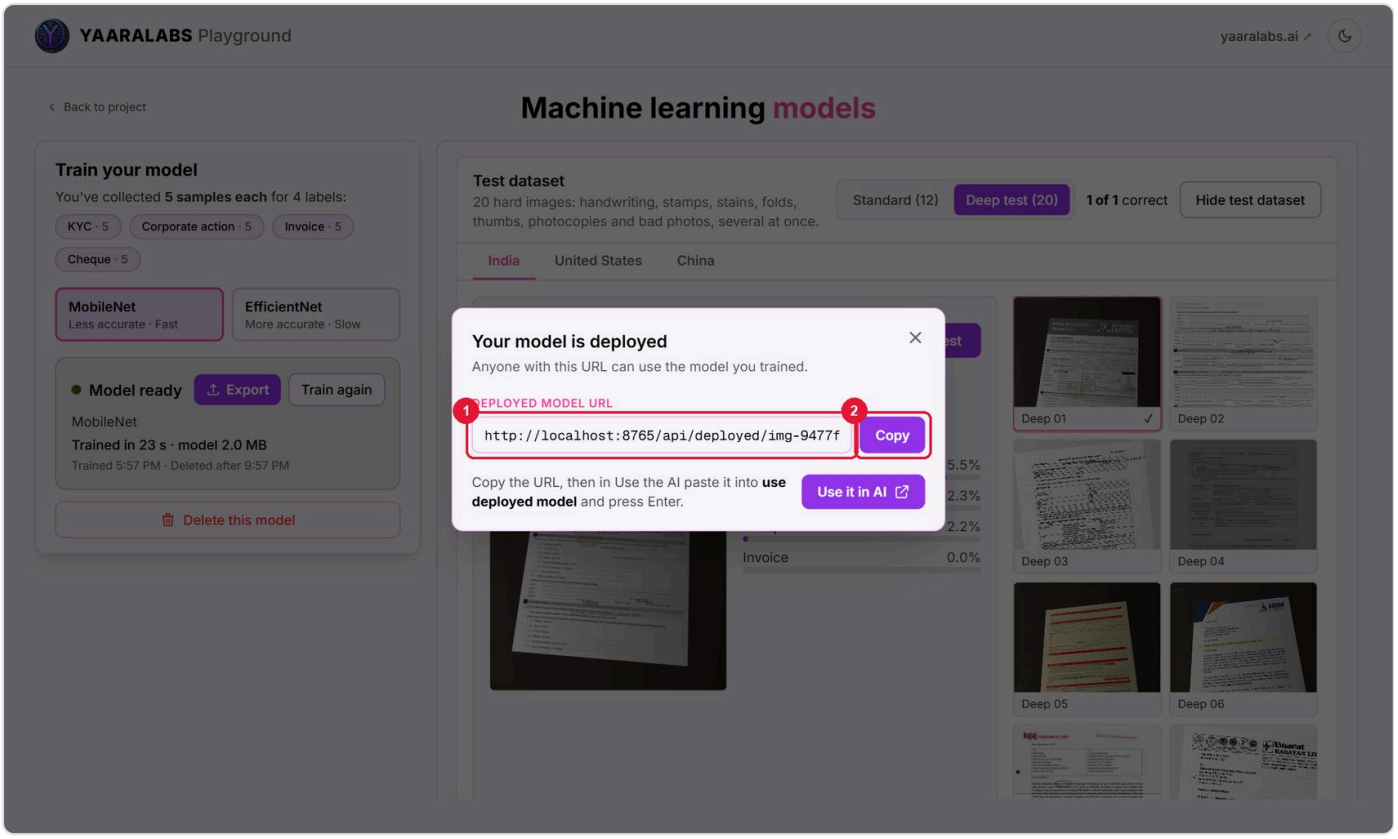
110 Export the model

Click Export (next to Model ready).



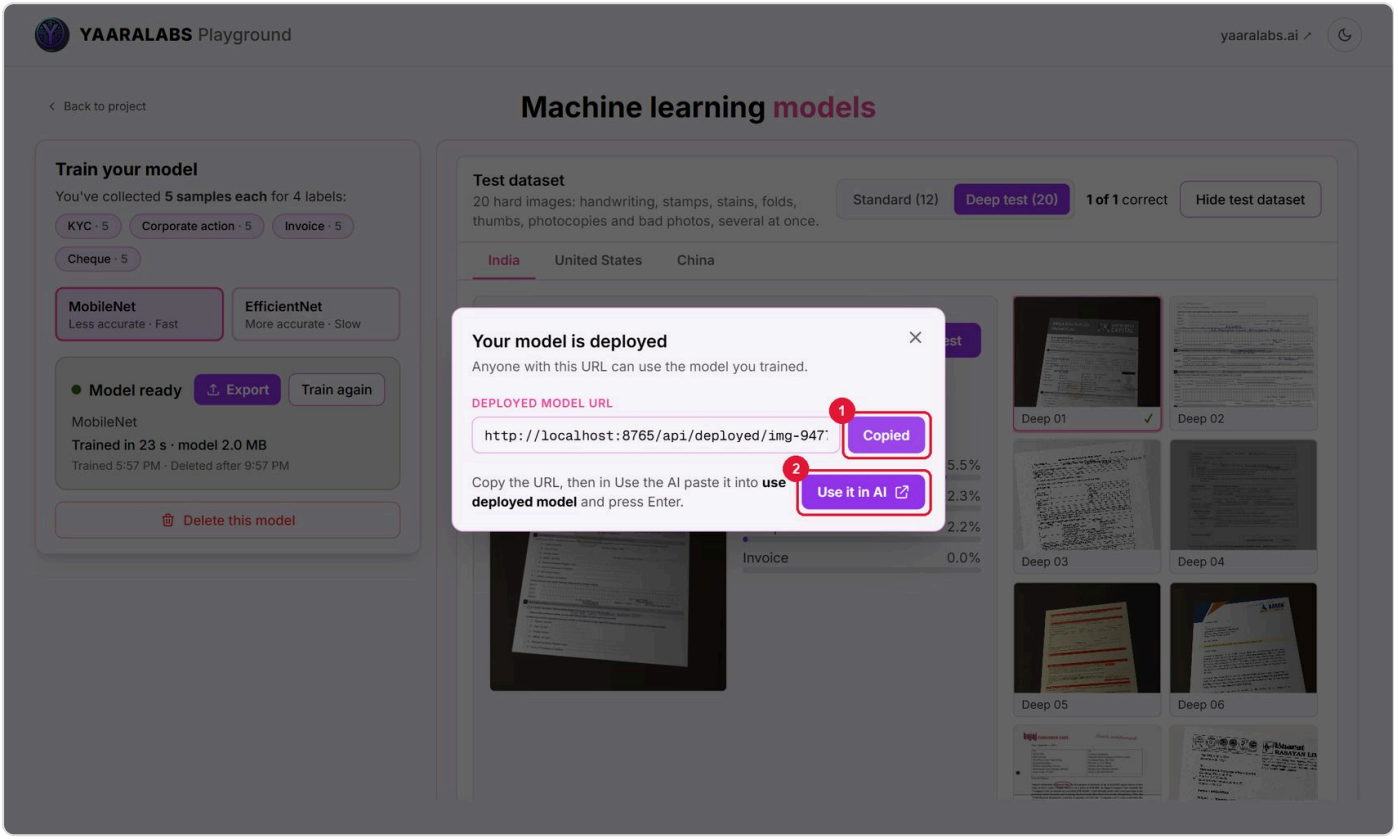
111 Your model is deployed

The dialog shows the deployed model URL. Click Copy.



112 Copied

Copy turns into Copied. Then click Use it in AI: it opens in a new tab.

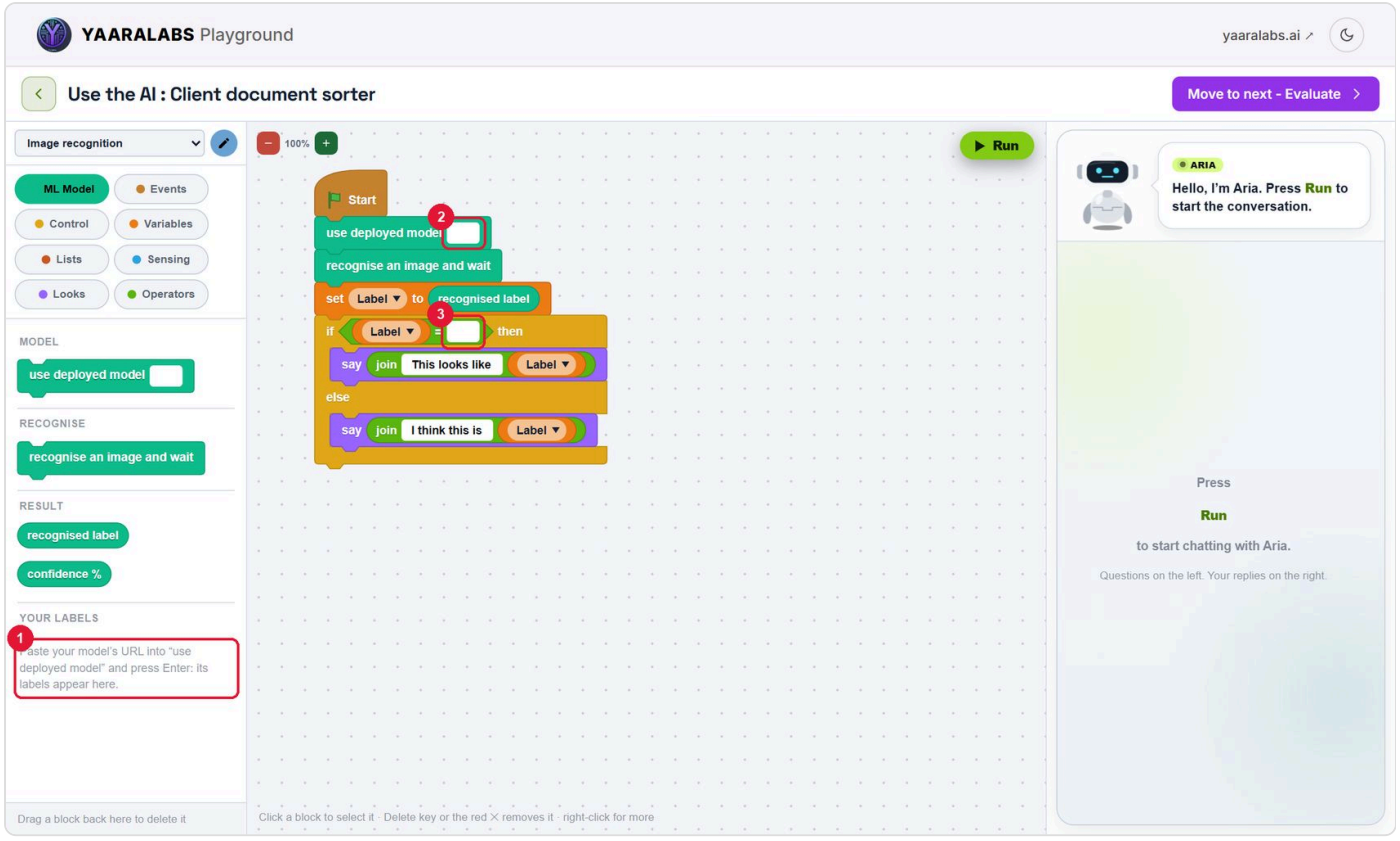


Use the AI

Use the deployed model in blocks: it looks at an image and Aria says what it is.

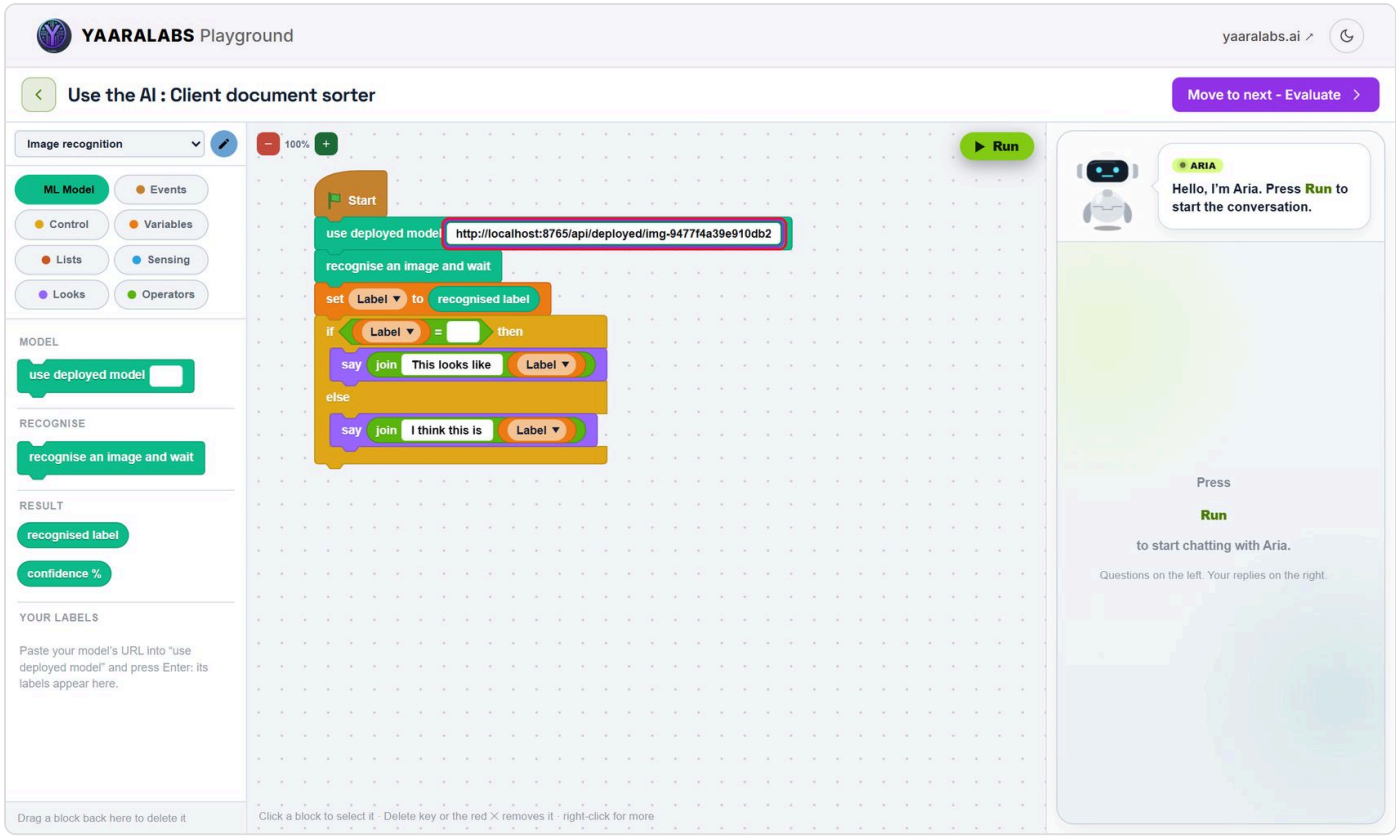
113 The Use the AI page

Starter blocks: use deployed model (empty), recognise an image and wait, set Label, if Label = (empty) say … else say …. The left side says to paste the URL.



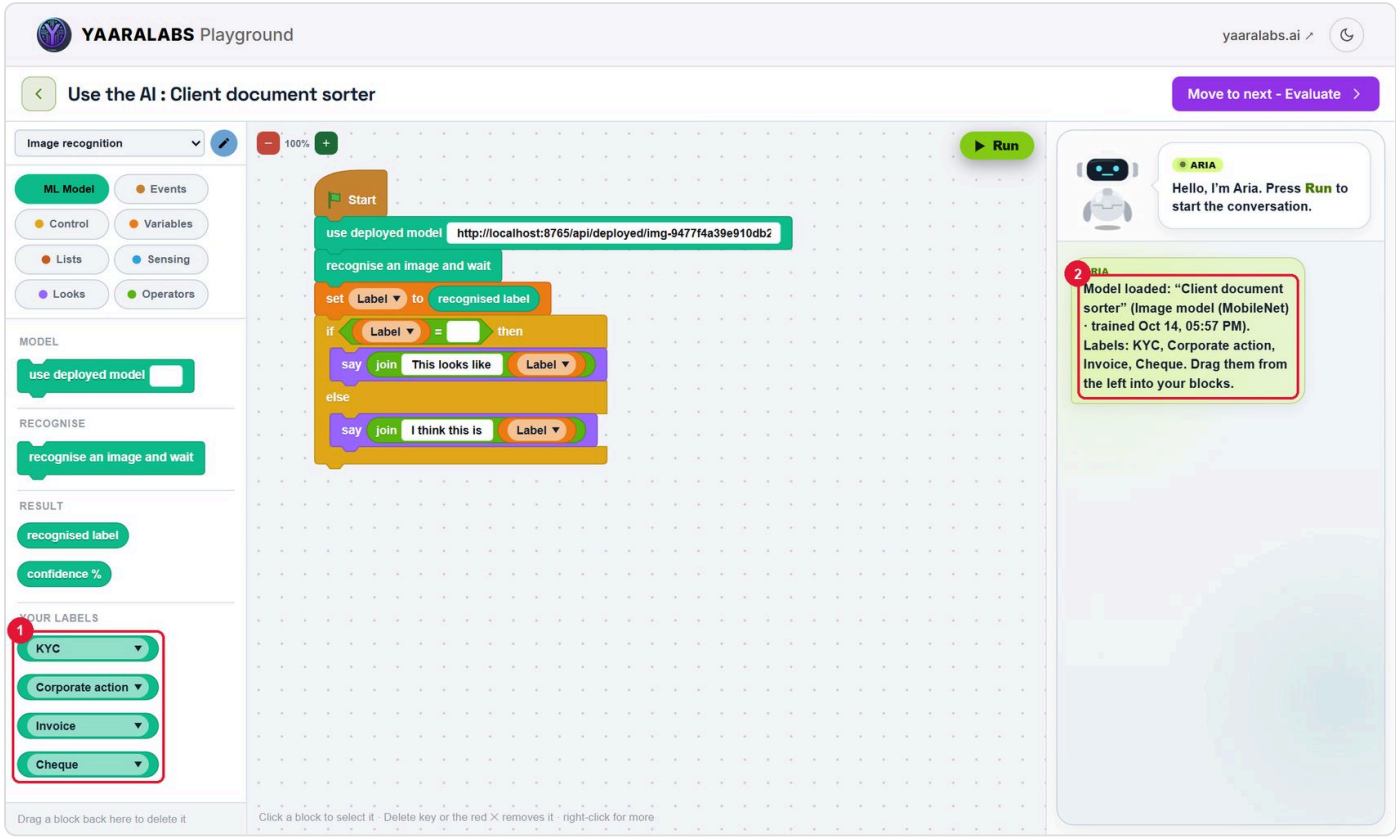
114 Paste the URL

Click the empty box in use deployed model, paste the URL (Ctrl + V), then press Enter.



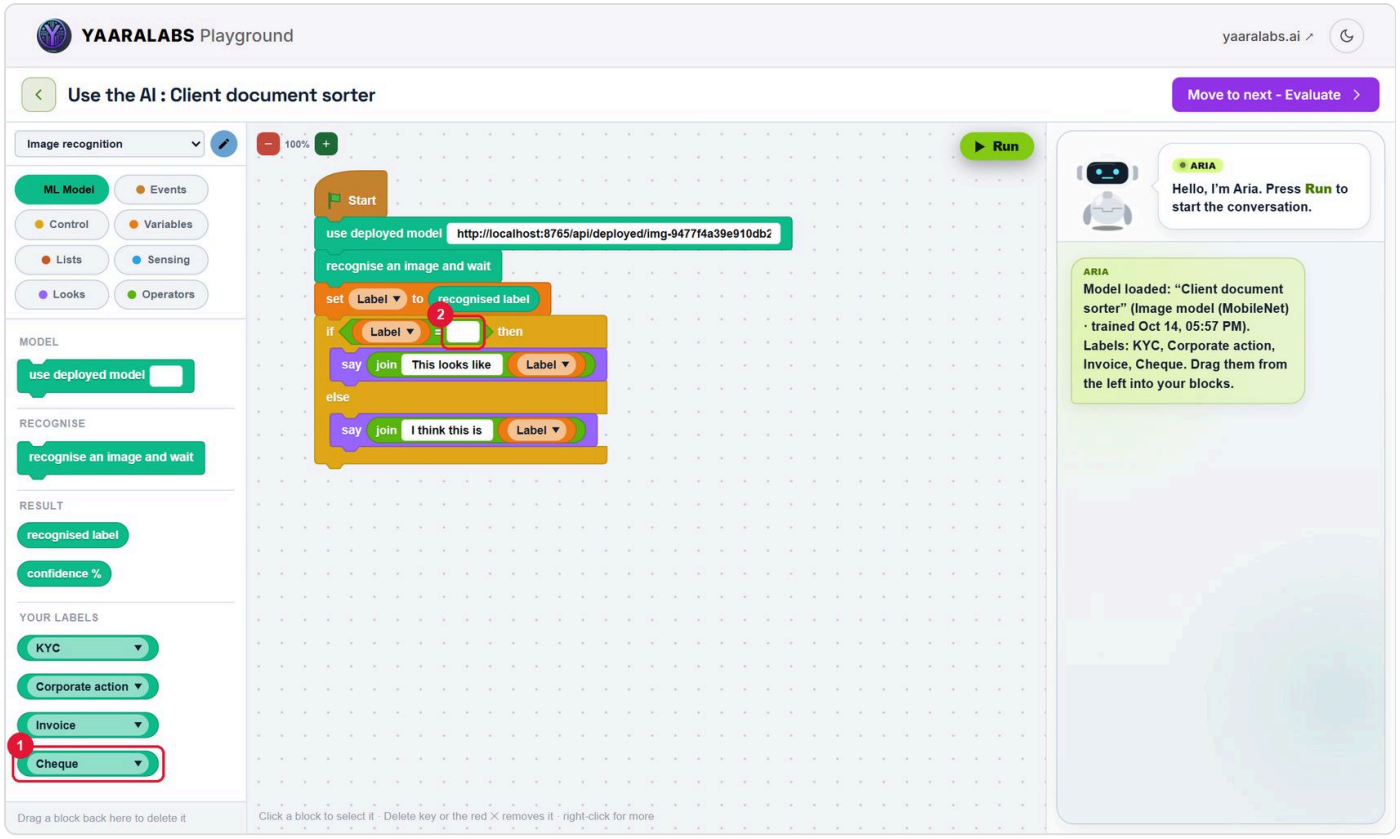
115 Model loaded

Your labels appear on the left under Your labels, and Aria says Model loaded with the four labels.



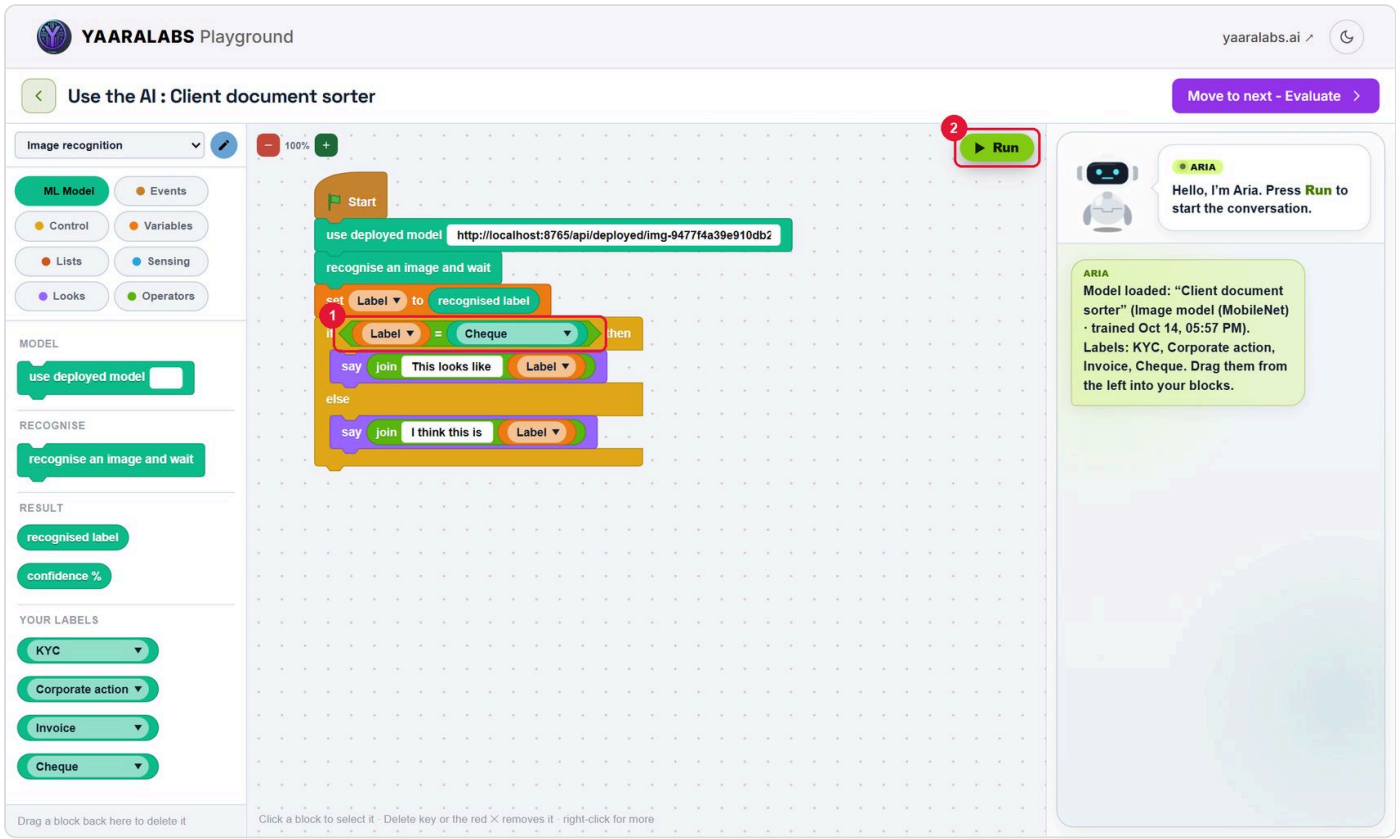
116 Drag a label into the empty slot

Drag the Cheque label from the left into the empty slot after if Label =.



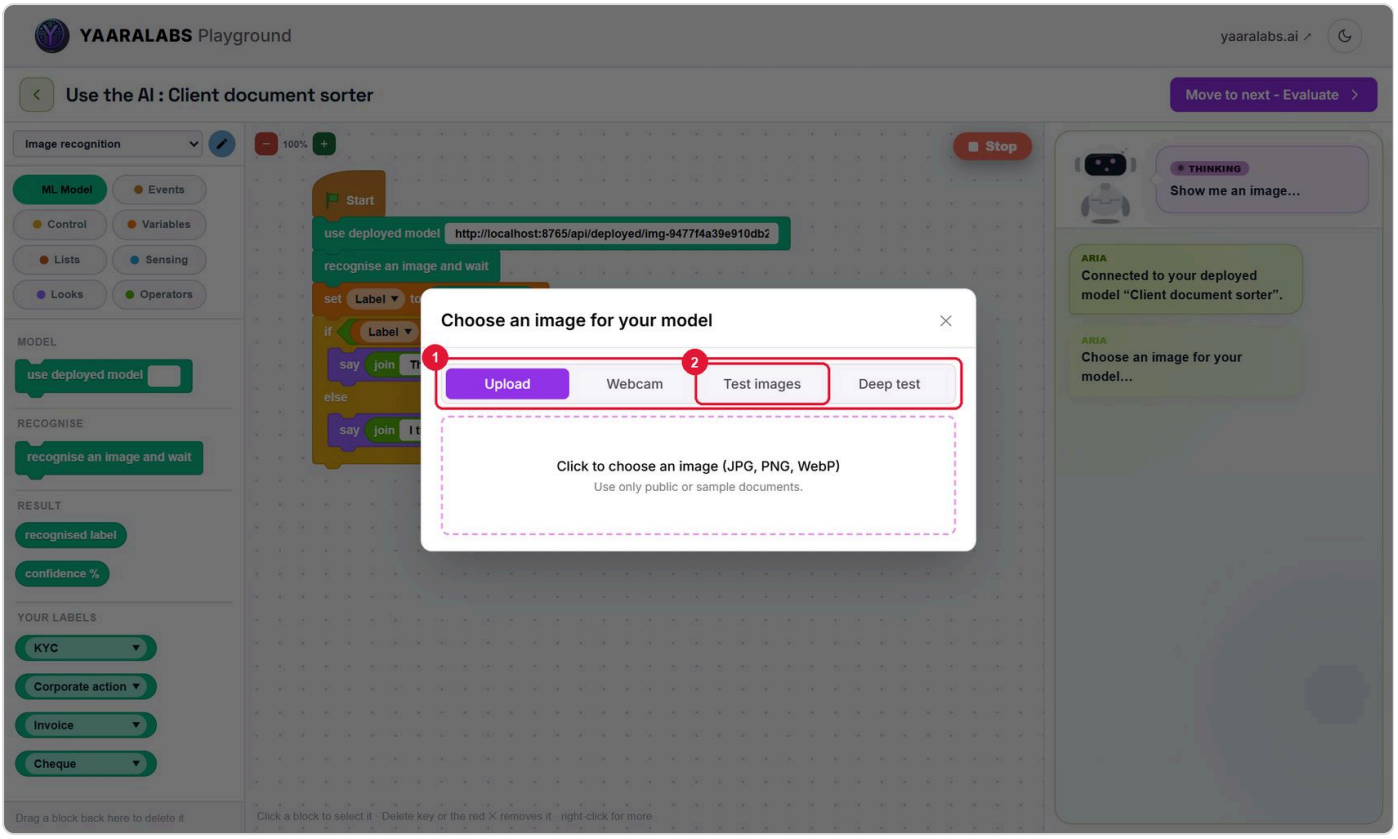
117 The program is ready

The if now reads: if Label = Cheque, say This looks like …, else say I think this is …. Click Run.



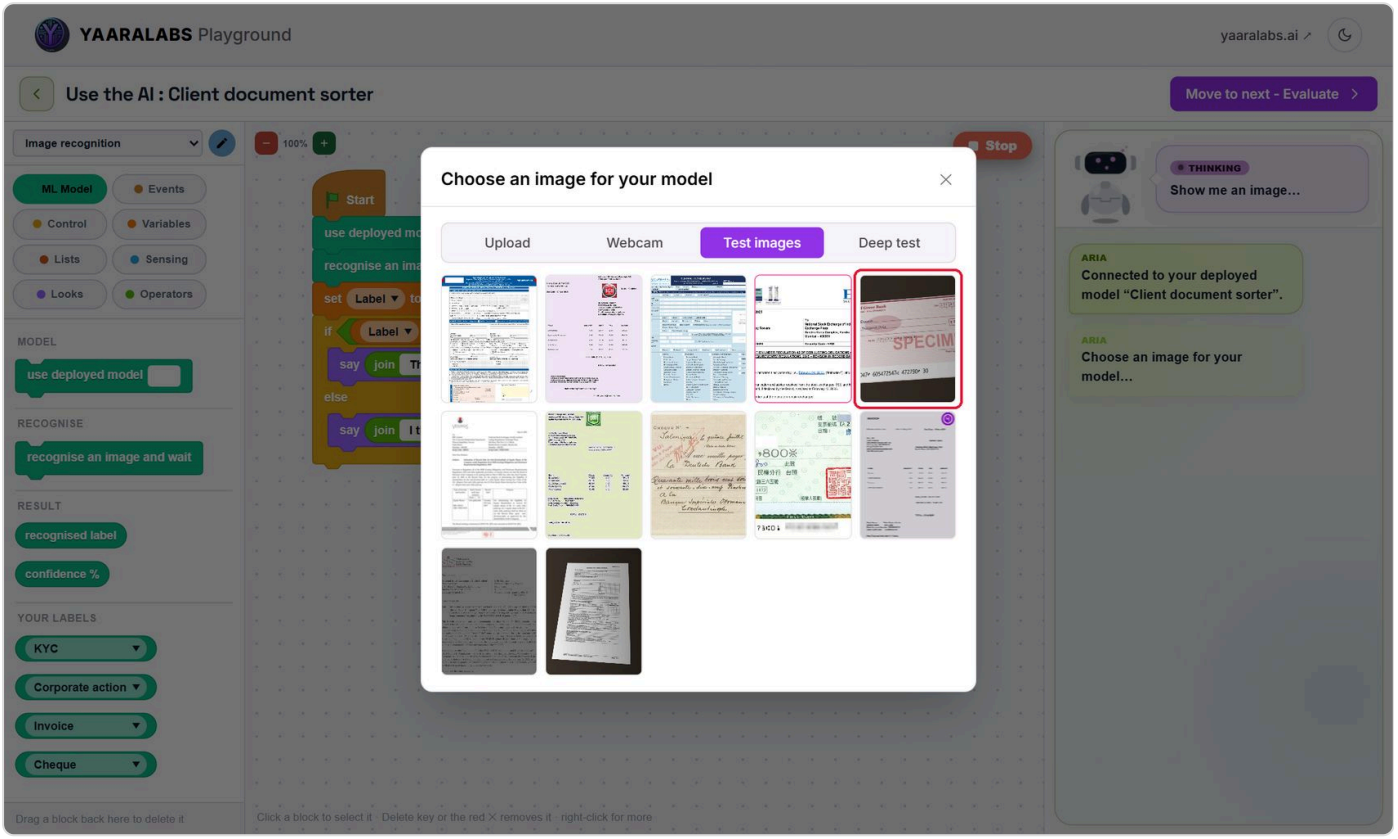
118 Choose an image

recognise an image and wait opens a picker: Upload, Webcam or the Test images. Click Test images.



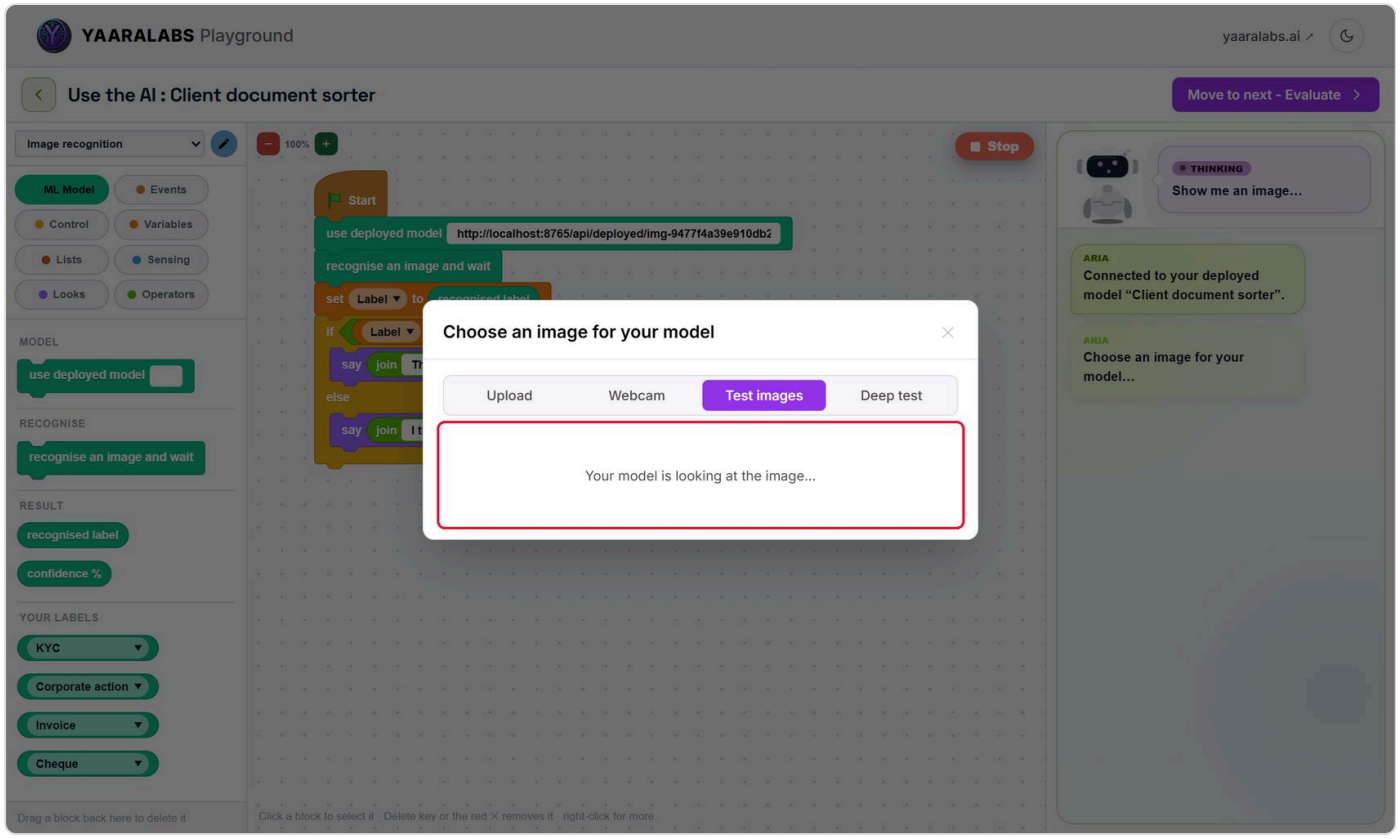
119 Pick a cheque

Pick a test image: here a cheque (the 5th one).



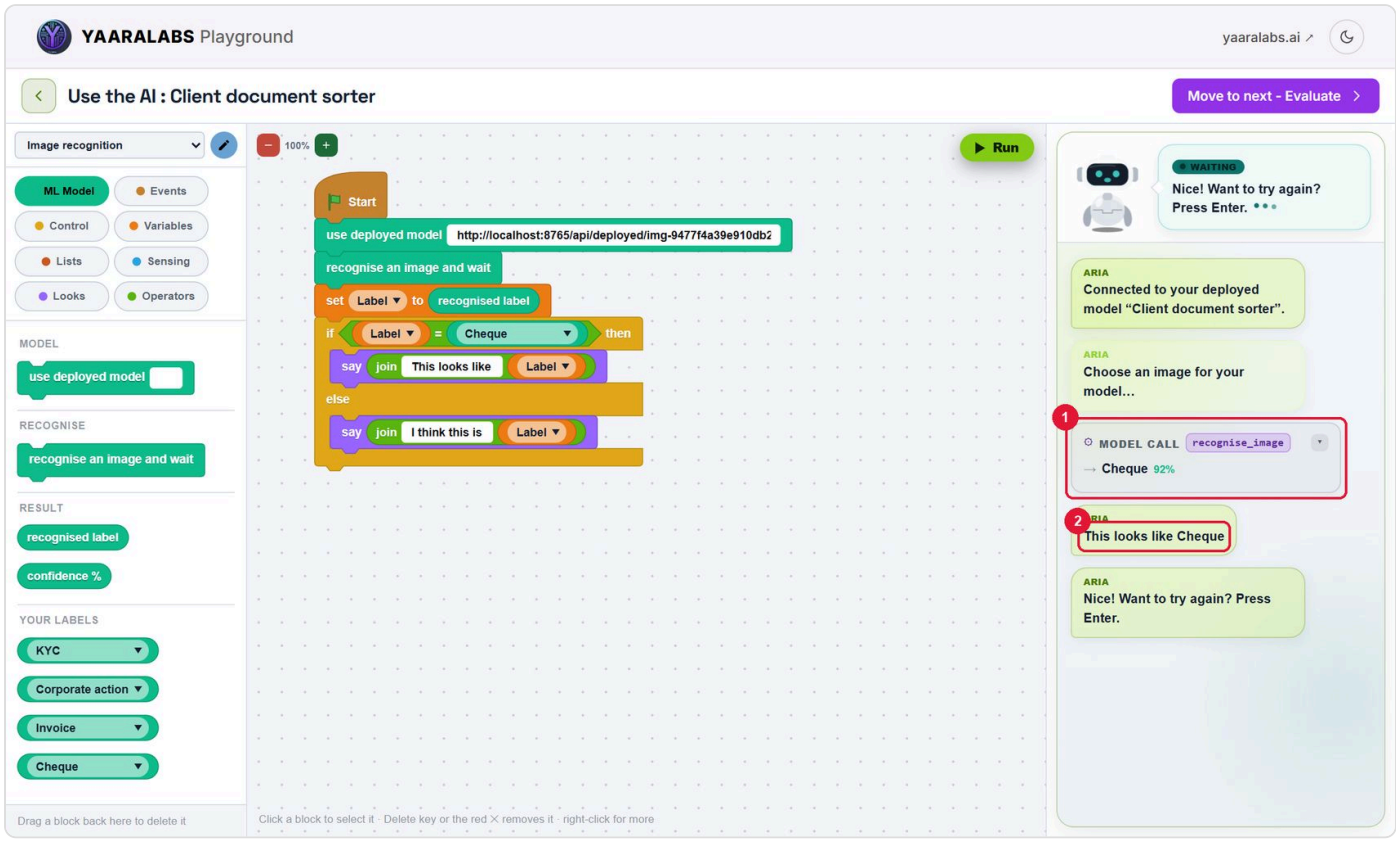
120 The model looks at the image

The model is called with the image; it takes a moment.



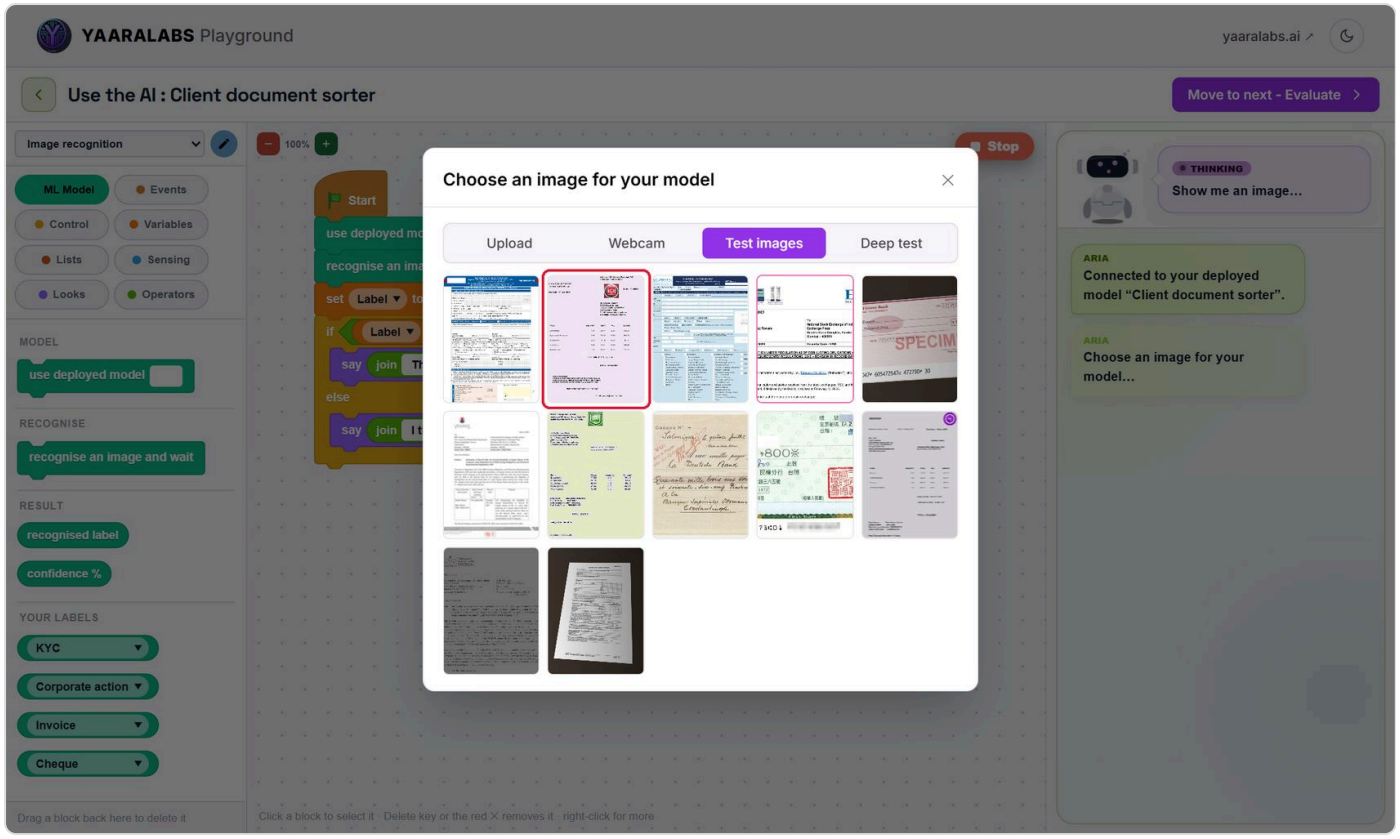
121 The answer: a cheque

The model call shows Cheque and its confidence. Label = Cheque, so Aria says This looks like Cheque.



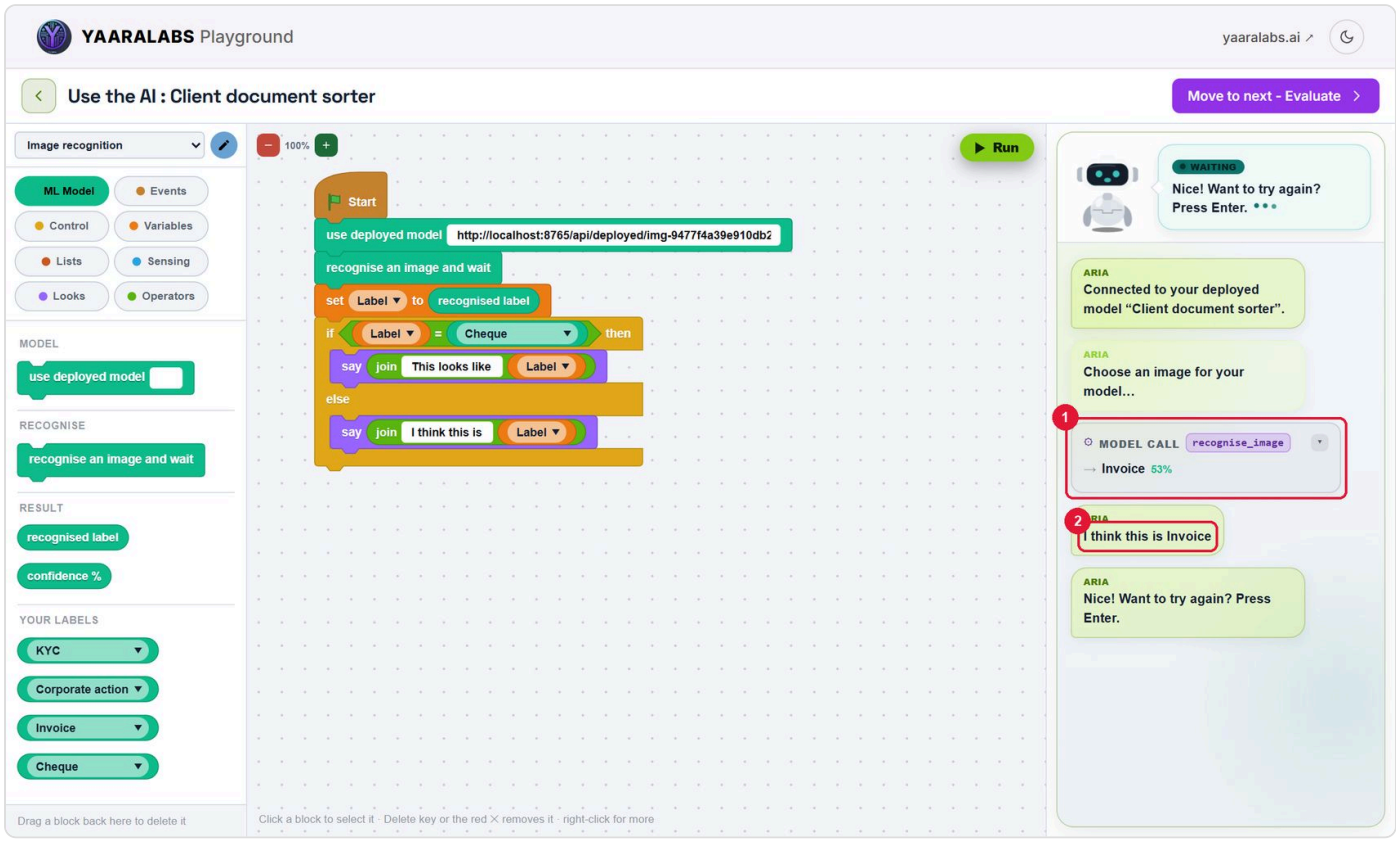
122 Run again with another image

Press Enter (or Run) to try again, open Test images and pick a different document, e.g. the 2nd (an invoice).



123 The answer: not a cheque

Not a cheque, so the else branch runs: Aria says I think this is ... with the label the model found.

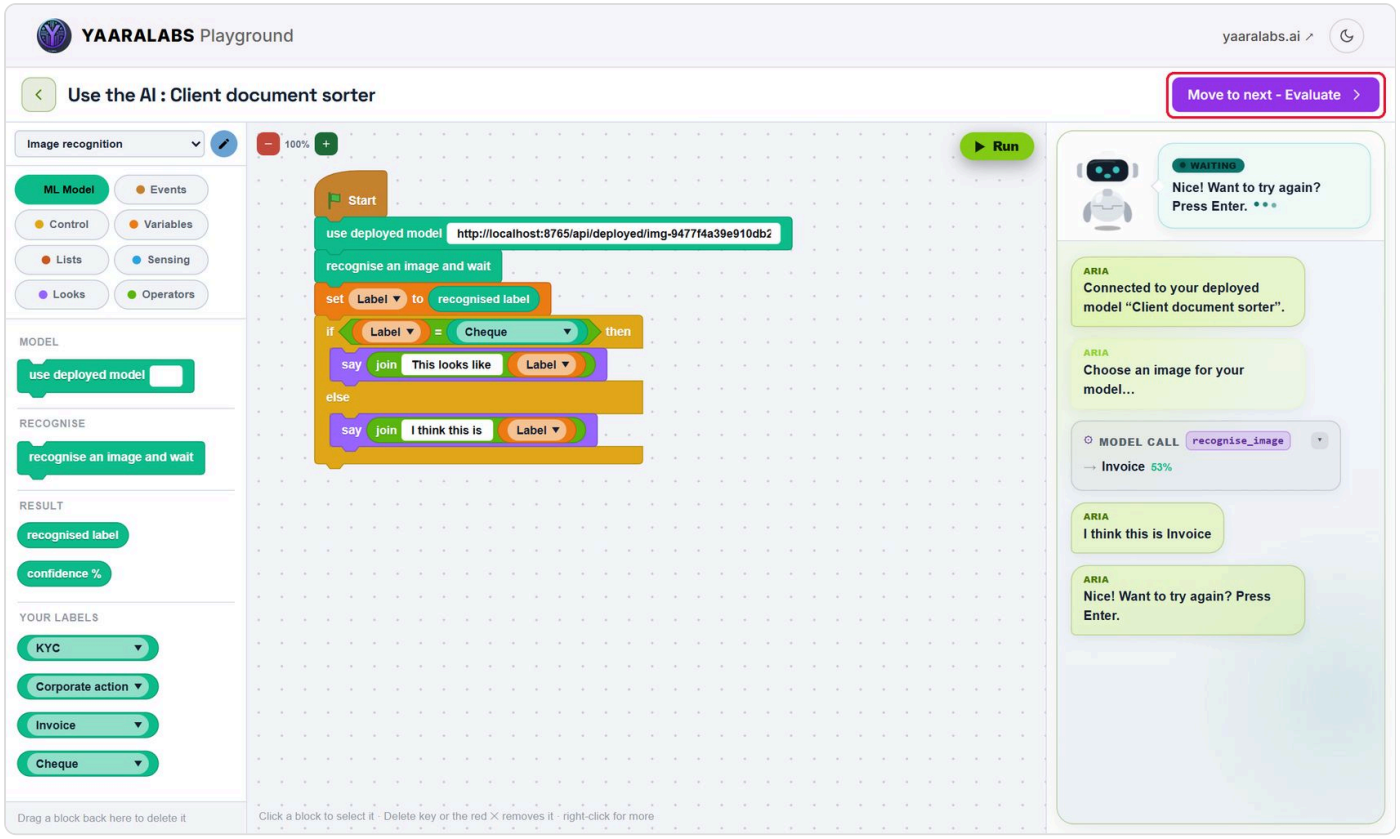


Evaluate

Give the model an exam on sample images it has never seen.

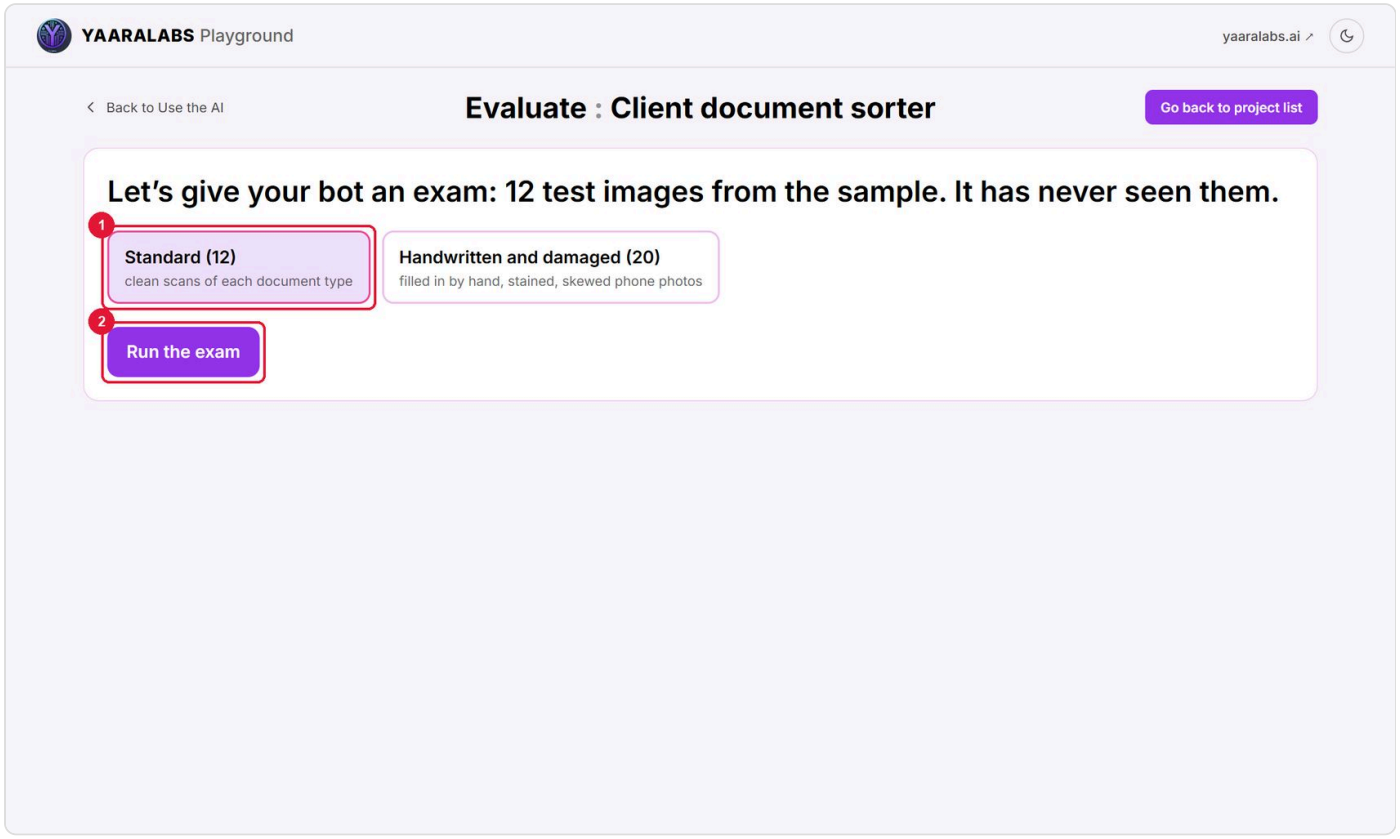
124 Move to Evaluate

Click Move to next - Evaluate.



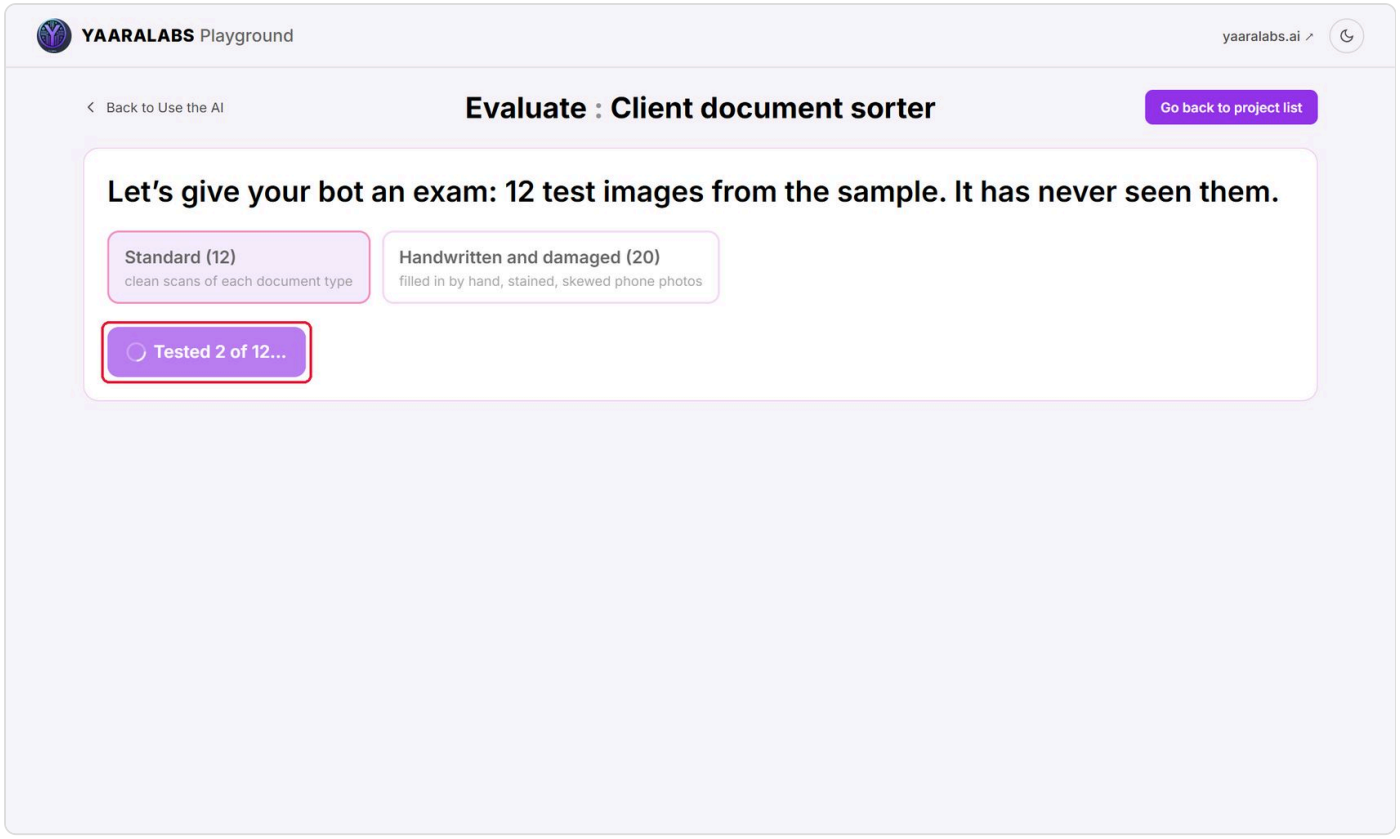
125 The exam

Standard (12) clean scans is picked; Handwritten and damaged (20) is the harder set. Click Run the exam.



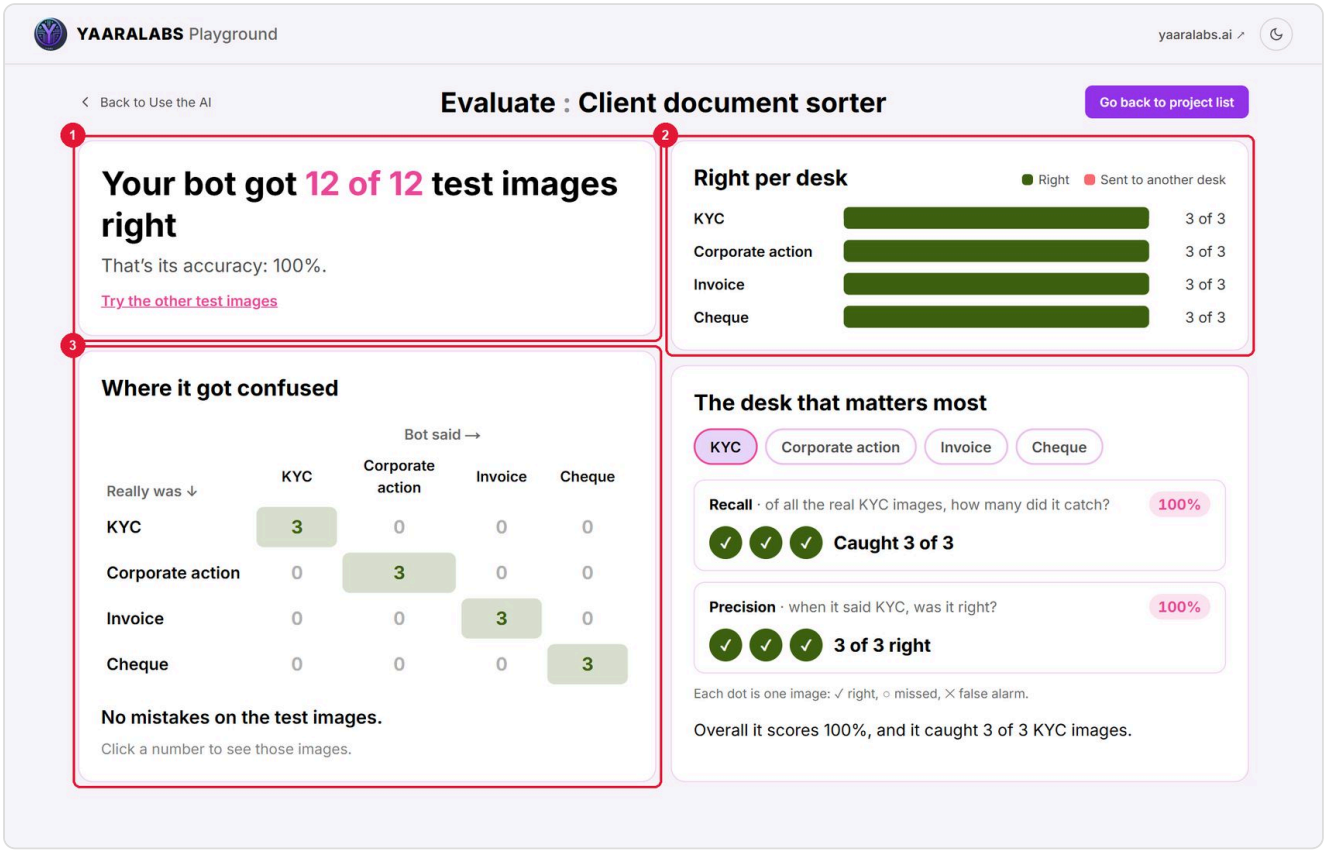
126 The exam runs

Each image is sent to your model; the button counts Tested x of 12.



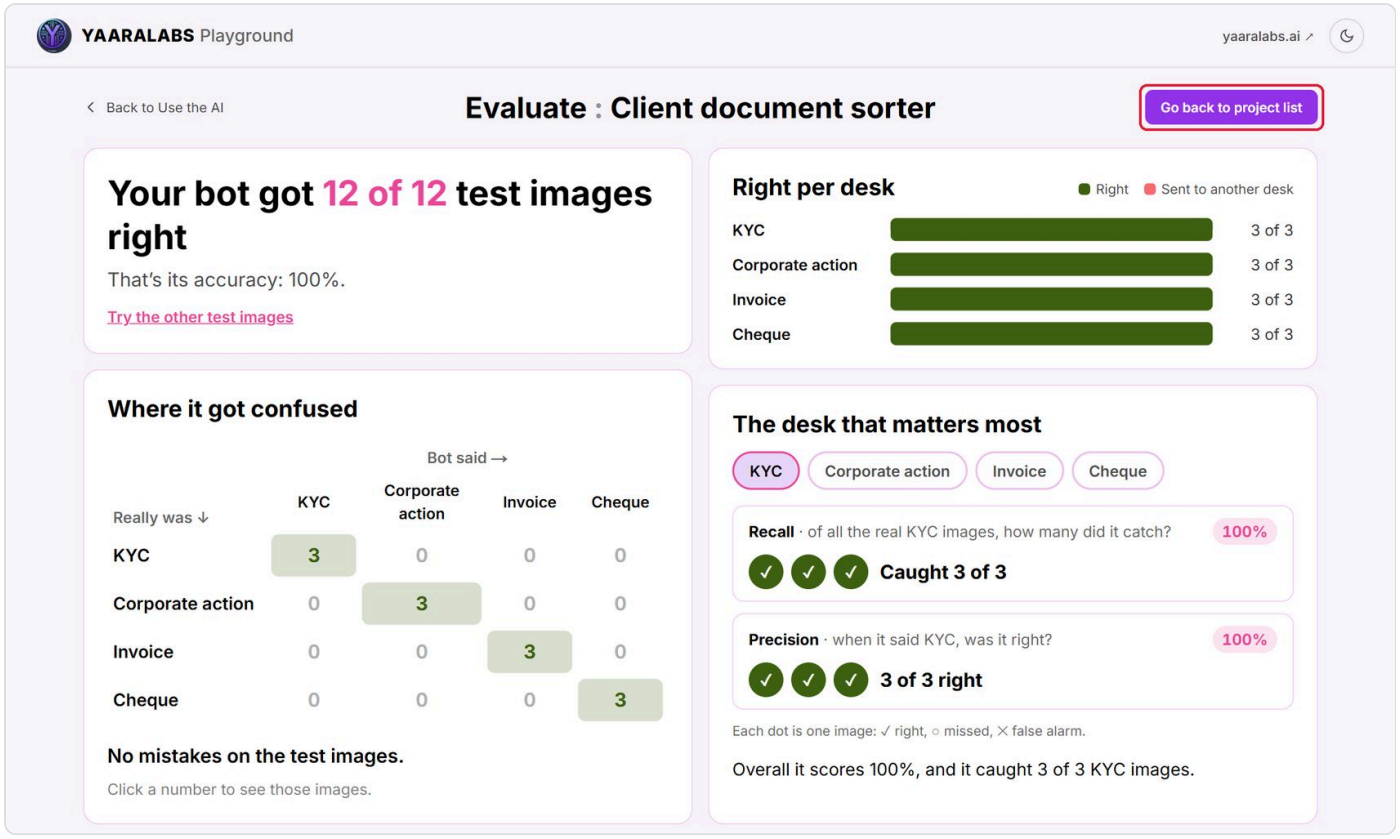
127 Exam results

The overall score on top, then Right per desk (each label) and Where it got confused (which labels it mixed up).



128 Back to the project list

Click Go back to project list.



129 The project list

You are back on the project list; Client document sorter is there.

